

Cup Anemometers and Slow Wind

A practical introduction

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Preface

I am Patrizia Favaron. My social function is supporting you in fulfilling your objectives, technical or research, using complex measurement instruments and techniques in the field of meteorology and environmental fluid dynamics.

I'm the main author of the MeteoFluxTM Core eddy covariance system, and other embedded devices.

I'm not an academic: hold no PhD, nor tenure at some university or research institution.

I got a job immediately after taking my Applied Mathematics degree, in 1987, in the field of heavy industrial automation, and stood there until circumstances made me change to the field of environmental measurements, around 1996. Mine has been a “typical M-shaped career”, with rises, and falls, and switches, mainly controlled by the accidents, happy and terrible, which constellate a human life.

The waves, and encounters, which shaped my professional and human path have been lucky and non-random enough to allow me to develop an experience, and I love sharing it in my work and life.

But maybe, the most important thing I had the opportunity to touch with my hands is the deep inter-relatedness on our complex technological work.

Think a moment. You are about to publish your PhD dissertation, or to communicate the important decision which will change the World: that's a

Preface

well devised moment in your life, a moment of intimate satisfaction worth being remembered and, maybe, some day told as a beautiful story.

In doing so, you maybe used data collected by some more or less simple system, and drew some interesting conclusion based on their values.

Behind each of the numbers you trustfully used there is an entire World: layers and layers of complexity, reflecting the ideas and work of thousands people. Even if the measurement looks “trivial” (a word which would be better banished from a scientist’s or decision maker’s vocabulary) you will end up relying over tens, hundreds of decision someone has taken before without asking directly to you.

The trust among the various responsibility layers is something we have to rely on, for our technological civilization to continue to function seamlessly. But there are cases in which one cannot be blindly accept things as they seem.

The fact that me, and people alike me (think to the numberless “lab technicians” and engineers of various specialization) hold no prestige or visible status adds to the problem: maybe we have manipulated you and your work, perhaps unintentionally, and would things go detectably awry you would have no way to even trace who, among us, made that terrible flaw, and whether is still alive with the material possibility to provide advice and some remedy.

There is a common misconception in science, that instruments make our sense sharper. They do not. Instruments, rather, *extend* our sensory system beyond the constraints millions of years of random mutation and natural selection have shaped for us. They give us information of entirely new kinds, without our minds possessing the instinctive knowledge necessary to evaluate its plausibility. And this demands on us all a special attention.

This is a technical book.

Preface

The reason I wrote it in the first instance is that I found a strange situation during one of my maintenance tours. I searched the scientific and technical literature looking for the field behavior of cup anemometers, but found nothing. There *was* indeed a lot of material, but not pertaining to the low wind speed case I was specifically interested in. I realized that, if I wanted such a reference work, I would have had to write it myself.

Which I did.

I cannot pretend my conclusions are the best, nor universal. What I've found was written after all by a Lombard, using data from Lombardy, and within the peculiar Lombard climatology. I've done my best, however, to explain what I have done and why, and anyone can in principle replicate the experiments in their own contexts.

The main subject is the *cup anemometer*, a wind speed sensor produced on mass scale in many variants, and used in many meteorological stations.

Wind speed data from cup anemometers are routinely used in applications as for example meteorological modeling, pollutant dispersion assessment, dams and large structures protection, air and marine navigation, road safety, and many others.

It frequently happens that things simple on the surface exhibit an intricate nature if we look closer. This is also the case with cup anemometers, whose behavior gives many surprises especially under conditions of wind below about 1.5 m/s.

As we shall see these oddities are mostly caused by the interaction of the sensor with local wind statistics and to operational constraints like maintenance history and other human factors.

This means by the way cup anemometers fall within the realm of the understandable and predictable, that is, at least in part *controllable*.

The aspect I, as a wind speed data user, am mostly concerned is their quality, that is their ability to depict the real wind accurately. My focus on

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slow wind data stems from the fact that in one of my professional fields (atmospheric pollutant dispersion) the most dangerous situations correspond to cases when wind is weak or almost absent, turbulence not enough to elicit a prompt chemical diffusion, but movement of the air sufficient to transport the high-concentration plume to some unaware “target”.

My personal curiosity, then, is to what extent a speed indicated by a cup anemometer can really represent the “real” speed and, if it does not, whether there exist ways for correcting it.

Of course the impact of using wind data of different quality in dispersion modeling has been investigated by the research groups I’m in, and this book is preparatory to further, more systematic work.

Besides research, however, what I’m interesting in is whether there exists some easy measure one may undertake when providing wind statistics (in particular means) for use in modeling applications. To an extent this is a technological problem, affecting the design and operational programming of meteorological data loggers: an aspect I, in my other role of measurements system engineer, am directly affected.

The idea of this work originated after I and my friends at Insubria University (professors Cristiana Morosini, Vincenzo Torretta and Elisabetta Zanardini) completed the data processing for a paper (which in the meantime has neared the end of the publication process), where we compared the results of Calpuff model applied to two data set, one from a conventional cup-vane anemometer, the other from an ultrasonic anemometer. The difference was really striking: the same-concentration maps obtained in the two case differed in many evident ways, to an extent we honestly did not expect. Why this? During our research work, after some hacking inside the Calpuff sources we discovered the cause was in the way Calpuff treats “calm winds”, basically by imposing a minimum turbulence intensity, but accepting the wind speed as it is (unless its value is exactly zero, in which case if I remind well it imposes a speed of 0.001 m/s). The combined effect of this model design choice, and the differences in wind statistics we observed by comparing the conventional and ultrasonic wind

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data, conspired in originating two entirely different spatial distributions of ground concentration.

The campaign we made for the paper, however, was quite short, and the conventional anemometer, as often happens in real-world plant-related installations, was of “amateur grade” (for certain types of non-dangerous plants their owners are not mandated by the law to install meteorological stations, but do nevertheless on a volunteering spirit which, in some cases, reveals greatly helpful when these data are needed in unanticipated ways).

So we wondered what could happen, would a professional grade conventional sensor be used along with an ultrasonic anemometer for an extended period.

We all reside in Lombardy, and then have access to the public data from the SHAKEUP micro-meteorological stations by the Lombard environmental protection agency: this was ideal, most SHAKEUP stations being equipped with both an ultrasonic and a professional-grade conventional anemometer, installed at same height above ground and not more than 2m apart.

In preparation of our possible next paper on the subject, I decided to make some preparatory work: this is the original reason I’ve written this booklet using part of the material.

During the book development it however became clear that the cup anemometer behaved in a much more complex and articulate way than I had initially envisioned, and the focus changed from just a scientific text to a more technically oriented one, like a sort of data processing and data quality and maintenance and ... What ended out being written is what you see: maybe, possibly, with some gap-filling, the draft of a draft standard.

Maybe, before committing to read this work you may be interested to know something more of me - from the professional side.

Preface

In 1987 I got a degree in applied mathematics at Milan university.

I then spent my first professional years working as programmer and later designer of “heavy” industrial automation systems, with a penchant for supervisory control and data acquisition systems applied to large scale thermoelectric power plants.

Around 1996 I changed my job and role entirely, and entered the field of lower atmosphere physics and its technical applications. I still continued designing things, this time complex data acquisition and processing systems for ultrasonic anemometry, SODAR/RASS and other devices. Technology changed widely in the meantime, and I have witnessed the shrinking of computing hardware from large refrigerator sized bulky devices to chips 1cm sized and less, without appreciable decrease of computing power.

To date I spend a part of my professional life building microcontroller-based measurement systems. The other part goes in data analysis and research.

My achievements include participating to the development of the first Italian eddy covariance system, and other “strange” devices. Traces of my work can be found from Antarctica to the Svalbard Islands, with many other sites in the middle.

Work, in my understanding, has an ethical value. And work on meteorological instruments in times of rampant climate change has, in my view, a big “frontier” value.

I love what I do, and would like doing it better and better. Knowing well that any honest work is first and foremost *an act of care*.

Dedication and acknowledgements

This book is dedicated to my parents Luisa and Sergio: without their love and support I would never ended being what, and *who*, I am. Professionally, and humanly.

As any human being I'm the result of an evolution process (both in phylogenetic and ontogenetic terms). This evolution would however not happen without the mental nourishment of continuous presence, conversation, and examples. In my specific case I am especially grateful to my "little brother" Massimo (the family's genius), and Maria Vittoria Cazzol, the sister I did not know to have until I met her at university (indeed sisters need not to be "genetic siblings", as I've serendipitously discovered). And, what to say of the love, and teachings, by my grandparents?

If one would take the little trouble to retrieve and check my curriculum vitae, they would presumably get the impression it is a movie script. But it can be summarized to few lines: all I am has been the consequence of an incredible and undeserved luck.

Many people, much more than a book can host, gave me their time, knowledge, help, dedication. That was not a-priori imaginable, but, did happen. And was a very good thing: in my feeling if one builds entirely on him/herself, then their achievement has a very high probability (say 1) of being trivial. For us to give, we all have to learn and this can only happen in that intricate net of exchanges we name "life".

So, please, forgive me for any name I am not mentioning: you *know* of my gratitude, and this book is a way I'm trying to give back all the good I received to who will come next.

Dedication and acknowledgements

Knowing this, I'll give here very specific acknowledgements: to Marta Salvati, Maria Ranci and Alessandro Loda, of the meteorological unit of ARPA Lombardia, who kindly provided the huge mass of data I've used in the analyses which constitute the hard core of this book. And also, to Daniele Fraternali and Roberto Sozzi by Servizi Territorio srl, who designed the SHAKEUP network which is the material source of the data.

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1.1. Why the cup anemometer?

Electro-mechanical (here also named “conventional”) anemometers are, as their name suggests, devices translating airflow (wind) into an electric signal which can then be translated into a number, whose value can then be stored, processed, transmitted.

Many types of conventional anemometers have been invented over a period spanning many decades, but in the end one type emerged as the “most popular”: the cup and vane anemometer.

By their very nature cup and vane anemometers are *two-dimensional* wind sensors, when the wind vector is three-dimensional. The question then arises why two of these three dimensions are privileged.

The answer is twofold. On the physical side, in many cases the vertical wind component has a mean much smaller than the two horizontal components, and sometimes even close to zero. From a more practical standpoint, in many use cases the vertical wind component, whatever its value, is considered not interesting.

Of course not always the vertical wind component can be ignored, and three-dimensional electro-mechanical anemometers have also been developed over time. These sensors were expensive and in demand of strict maintenance, and their use remained confined to research and specialistic applications.

The subject of this book is the cup anemometer.

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I have some reasons for this narrow focus.

To begin with, the horizontal wind speed is a valuable quantity. It relates to the kinetic energy contained in wind, which can be in principle extracted from it.

But wind speed also has to do with how quickly, or how slowly, natural phenomena evolve on the Earth surface.

In addition, the rotating cups provide an immediate feedback on the strength of wind, in a very intuitive way. Sometimes, intuition is at least in part misleading, and as we'll see this is just one of these cases.

1.2. Why not also wind direction?

In many applications the wind speed alone is not sufficient, and knowledge is needed for the direction also.

Many electro-mechanical anemometers are then designed as integrated cup and vane combinations. Some others are designed as separate couples, but are often installed side by side.

Despite their frequent use association, however, cup and vane are characterized by very different physical principles, each with very specific complexities. Dealing with both in sufficient depth would have made this book less agile, and longer to write.

Another reason for focusing attention on cup anemometer is that more research effort has been dedicated to it, and not to the vane. All types of research efforts are costly, and people tends to address one field instead of the other under the pressure of available grants and positive expectations about some form of return to investment.

Specific interest on cup anemometers has sparkled because of their direct connection with the financial yield of wind farms, and wind engineering is a very powerful and wealthy industry. Site-specific wind speed statistics

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can be used to select the best turbine, or to assess in advance the potential energy productivity of a location.

According to this viewpoint, wind direction is not as immediately important: most wind turbines automatically point their rotors towards the direction wind blows from, and some designs (for example the Darrieus turbine) are even totally direction-independent.

This unbalance of interest contributed to drive the bulk of research out of wind direction vanes, and so much less material is available about them.

Last, vane direction gauges are notoriously quirky: they are prone to resonances, especially in low wind conditions; their reading, often based on potentiometer transducers, is electrically noisy and difficult; the identification of failures may require weeks, or months, if based only on data plausibility checking. Vane anemometers could then be a marvellous subject for a book: another one, however.

1.3. Why not ultrasonic anemometers and other advanced wind sensors instead of cup anemometers?

In recent times the market has witnessed a flood of anemometers based on “new”, different physical principles. Sensors based on acoustic wave propagation (especially in the ultrasonic range) take today the largest share, with both two and three dimensional models.

It is possible that these ultrasonic sensors will sooner replace cup and vane anemometers in many applications: their most compelling engineering advantage is that ultrasonic wind sensors have no moving parts, and because of this are in demand of a much lighter maintenance than cup and vane anemometers (that this “lighter maintenance” is anyway needed may not be that evident to service companies accustomed to electro-mechanical

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sensors and their typically obvious failure manifestation, and this *is* a problem).

However, ultrasonic anemometers are not ideal: for instance there exist conditions (like rain and fog) in which their performance may be degraded. And, literally a jungle of models exist with extremely varied characteristics: choosing the “best for the application” could, today, be a purchasing office’s nightmare. Just as an example, in the same price range one may find ultrasonic sensors whose firmware emulates the response of conventional wind sensors, and others which exploit the peculiarities of ultrasonic measurements to their deepest. Ultrasonic anemometers could be the subject for another book.

In this book, a small subset of ultrasonic anemometers is anyway used, although indirectly. Three dimensional research-grade ultrasonic wind sensors will be used in the role of “reference sensors”, providing the data to which cup anemometers will be compared to. But that will be all.

1.4. Plan of this book

In very short: we’ll devote time and words to the cup anemometer, its proper use, and suggestions for improving its handling and maintenance.

In order to put this material in context, however, we’ll begin with a brief summary of what wind is: in Chapter 2 you will find what “wind” is, which important characteristics it has close to the Earth surface, and what “measuring it” really means.

In Chapter 3 we’ll then meet the cup anemometer, and analyze the facets of its physical principle, operating characteristics, and use.

Chapter 5 and Chapter 6, this book core, contain a detailed statistical comparison of cup and ultrasonic anemometer data. There we’ll see to what extent, and under which circumstances, does cup anemometer speed measurements differ from their ultrasonic counterparts.

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In Chapter 7 a simplified model of cup anemometer is presented, allowing to reproduce the behavior of hypothetical cup anemometer having known characteristics if exposed to the wind as measured by an ultrasonic anemometer. The models' possible used in support to maintenance and operation will also be explained.

Chapter 8 addresses the final practical questions about measurement sites. As cup anemometers exist in many shapes and with diverse characteristics, the problem arises on how to identify the subset of them better fitting the site peculiarities. This involves gaining a detailed knowledge on the site-specific wind statistical distribution before the cup anemometer is installed, by means of measurement campaigns made using a reference instrument. This chapter is very important both for wind measurement stations planning and design, and (with minimal changes) for assessing that a site once suitable for wind speed measurements still is today.

Finally, in Chapter 9 we shall draw conclusions. If in a hurry, you could jump to this chapter first of all, and then go back to the details chapter for a deeper understanding.

Apart from this quite liminar, executive-summary style access, the book is designed to be read start-to-end, each chapter building on the preceding ones.

1.5. Intended readers

In its layout and contents this work is a special case of “university level textbook”.

Its intended audience are undergraduate students in environmental engineering, physics and environmental sciences.

I made an effort to maintain the language at a simple level, sometimes at the expense of rigor (giving advice on this), and in my feeling and experience am confident the material is also accessible to technicians, man-

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agers and seasoned practitioners in the industry of meteorological measurements.

1.6. What this book will *not* do

This book is dedicated to the cup anemometer from a technical standpoint, that is, adopting the perspective of a prospective maintainer.

It will not provide detailed guidelines specific to some specific application (although applications will be sometimes used as examples).

In the book central chapters Chapter 5, Chapter 6 and Chapter 7 parts we'll use ultrasonic anemometers here and there, mainly as reference sensors to which cup anemometers will be compared. As a non-central subject, the ultrasonic anemometer will be assumed as it is, without delving in its functioning.

1.7. The role of cup anemometer in applications

1.7.1. An incomplete list

Knowledge of the wind is important in many applications, some of them surprising, and some others very new.

Because of this, any list of relevant applications of interest to date may not be valid tomorrow.

Nonetheless, it may be interesting to know in how many applications knowledge of the wind speed has importance. The following table is a very incomplete attempt in this direction.

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Table 1.1.: Examples of practical applications in which cup anemometers are commonly involved

Application	Role of wind speed	Notes
Air traffic management and safety	Together with wind direction indicates which lanes are best for landing-takeoff, and whether operations are safe	Wind speed of interest if higher than some significant threshold
Navigation and ship handling	In port: safety monitoring and help to operations Off port: safety, economy optimization	Wind speed of interest if higher than some significant threshold
Off-shore gas and oil plants	Operational safety	Wind speed of interest if higher than some significant threshold
Wind farms, inland	Anemometers positioned on wind towers are used in evaluating prospective sites for productivity; when plant is in operation, wind measurements are used for safety and economy	Wind speed of interest if higher than some significant threshold; vertical component also important (guarantees, maintenance planning)
Hazardous chemical and nuclear plants	Wind speed and direction used to predict fallout in case of hazard	All wind speeds relevant; low wind speed associated to most dangerous meteorological conditions

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Application	Role of wind speed	Notes
Air pollution monitoring and control	Wind speed and direction used in support to dispersion modelling and, at some sites, as a helper towards a “greener” operation	All wind speeds relevant; low wind speed associated to higher modeled ground pollutant concentrations
Civil engineering	Wind speed used to monitor safety of cranes, tall buildings and other facilities during construction	Wind is relevant only at very high speed
Defense	Wind speed and direction used to improve artillery hit accuracy	All wind speeds relevant
Odour management	Wind speed and direction often used to optimize data takes during olfactometric campaigns	All wind speeds relevant
Household management	Wind speed is often used within building automation systems for winding tents would wind speed be too large or variable	Wind is relevant only at high speed (threshold is tent-dependent)
Motorway safety	Wind speed often sensed at critical sites (for example viaducts)	Wind speed relevant if very high

As the list shows, the range of wind speed the various application experts could consider “interesting” shows a large variability, with some (but not all) application being focused specifically on slow wind.

1.7.2. An example application: atmospheric pollution dispersion modeling

1.7.2.1. The practical nature and needs of dispersion modeling

Although this book is not application-oriented, it is nevertheless useful to show some glimpses on a use case in which slow wind speed plays a paramount importance: modeling the dispersion of pollutants in the atmosphere.

I've chosen this particular example simply because I have direct experiences on it, but this does not mean dispersion modeling is the most important application, nor is an attempt to manipulate you.

Anyway. Suppose a tracer (a gas, or a particulate thin enough to be lifted and transported by wind) is released at some location, let's say for simplicity a point, and abandoned drifting in the air.

The nature and effects on health of this tracer are left entirely unspecified: it could be something harmful (a pollutant), but also something entirely else (for example a pheromone, a juvenile form of some animal, spores, seeds, plant nutrients like carbon dioxide, or an odor-bearing compound). The important is this tracer has a mass which is negligible respect to the mass of the whole atmosphere, and that it can mix more or less freely with air.

The tracer may then react chemically with other compounds present in the atmosphere, or, if a particulate, settle to the Earth surface. It may reach some "target system", for example receptors on fruit trees inducing synchronous maturation. Or other: these effects are all important, but not immediately relevant to this extremely simplified presentation, and we can put them aside.

Once the tracer has entered the atmosphere, the air will immediately capture and move adrift it, in proportion with its speed.

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Our tracer is passive, that is, it has no way of independent motion. So, it will follow the mean wind towards some eventual fate.

As the tracer is transported, however, it is also diffused. Air molecules are in constant agitation (the “Brownian motion”), and this alone causes some diffusion. In addition, the motion of air in the lower atmosphere is always turbulent, and turbulence induces a mixing too, usually four or five orders of magnitude more effectively than molecular diffusion.

To visualize what happens you could imagine a smoke plume released by an industrial stack, like the one shown in Figure 1.1.

Other emission geometries are possible as well (in Figure 1.2 an example of traffic over a motorway, but many other types of emission sources do exist).

And it need not be that emissions are necessarily artificial, as we can see in Figure 1.3.

Whatever the geometry and timing of these sources, and in particular of the industrial stack we’ll use in this example, they release into the atmosphere some chemicals and objects which are extraneous to “air”. This emission process can be described quantitatively by indicating the emission rate (for example in grams per second), its thermodynamic and mechanic properties (temperature, initial speed, ...).

We then are interested in assessing the *surface fallout*, that is, the amount of the emitted compound reaching the Earth surface. This quantity, expressed for example in concentration units like $\mu\text{g}/\text{m}^2$, will show different values for different positions, and if we limit attention to a single point on the Earth surface (a “receptor”), with time.

The mapping relating an emission $E(x, y; t)$ to a concentration field $c(x, y; t)$ is usually represented by means of a *dispersion model*.

We can imagine the dispersion model as a possibly intricate set of mathematical formulas rooted in physics.

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Figure 1.1.: Example of a plume from an industrial stack; in this case the white smoke is condensing water vapor

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Figure 1.2.: Traffic on a motorway; exhausts from vehicles is not visible due to the high temperature involved

To date, for practical reasons these formulas are delivered to users in the form of computer programs, but this is not a priori an imperative necessity. Indeed, in the Sixties of previous century the dispersion models were commonly delivered as a set of nomograms and numerical tables: what today is made by running an executable on a computer, in these years was obtained by hand-executing calculations with help of a slide rule, or for the few lucky ones by a desk calculator.

It is important to distinguish clearly between the physical and chemical dispersion phenomenon (Seinfeld and Pandis 2016) and its quite variegate modeling (Jacobson 2005; De Visscher 2014). This book is not the right place where to deal with the subject extensively, but it is important to observe the modeling of dispersion has been addressed using many different techniques, each demanding specific information in order to provide their results.

To date, these dispersion models share a common necessity: for them to

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Figure 1.3.: Plants, water bodies, animals, the ground, all are sources of compounds which enter the atmosphere; plants, animals, water bodies and the ground act also as sinks

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represent reality in their special approximations, they must be presented various input data:

- Data pertaining *emissions* (sources position, emission rates, ...).
- Data on the *meteorology*.

Some models may in addition need input data allowing to simulate chemical and photo-chemical reactions, dry and wet deposition, radioactive decay, and other chemical and physical processes. But in this example we consider only the transport and diffusion due to airflow, and can safely neglect these.

Now. If chemical reactions are neglected, then any existing model behaves linearly with respect to the emission rate. This means that, whatever the form of the surface concentration $c(x, y; t)$, a doubling of the emission rate E reflects immediately in the doubling of c .

The consequence of this is that any estimation error done on E can be recovered after model-based estimation of c has been made: it is sufficient to re-scale the concentration data proportionally to the change in emission rate.

Things are not as easy with meteorology. This happens with all types of models, but is particularly evident with “lagrangian” dispersion models (the most commonly adopted to date), whose outcomes depend not only by the individual meteorological records, but also on their history.

This problem is amplified by the peculiar nature of the meteorological data existing dispersion models require: in addition to measured data like wind, precipitation, air temperature and others commonly available from meteorological stations, modern dispersion models also demand data about turbulence and other factors affecting dispersion.

One among these is of special interest. Once emitted, tracers mix with air, and this mixing progresses over time at a rate which is a function of air turbulence. Would this process occur in complete freedom, without

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anything capping the emitted tracers, then dilution would soon become so strong that no detectable amount of the compound could be located after few hundreds of meters from the emission point.

But this capping exists, unfortunately. In engineering-oriented literature it is known as the “mixing height”, and is typically named “thickness of the Planetary Boundary Layer” by physically-inclined people. Below of it the mixing occurs rather freely, so that dilution may only occur in the crosswind direction once the whole Planetary Boundary Layer thickness has been occupied.

As a matter of fact, the Planetary Boundary Layer thickness changes with time even by orders of magnitude over one day. For example, at Lombard latitudes (around 45° North) the mixing height may range from 10-100m on night-time, up to 1-1.5km on a bright Summer day.

The effect of these changes on tracer fallout can be seen in Figure 1.4.



Figure 1.4.: Effect of thick (left) or thin (right) PBL over ground concentrations, assuming perfect mixing

The mixing height gives a huge problem: is not easy to measure directly. In the common case the only data available come from standard (or even advanced) meteorological stations, then the mixing height z_i is not measured directly, and its value can only be estimated (Batchvarova and Gryning 1991; Zilitinkevich and Baklanov 2002); the precursors of this estimate are turbulence parameters like Obukhov length and friction velocity (Stull 1988), whose values are very often themselves estimated starting from more common data.

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These estimates are made as a part of the “meteorological processing” which precedes the actual modeling step (Favaron 2022), maybe unawarely if an already available meteorological processor has been used.

1.7.2.2. Why slow wind can be nefarious for dispersion modeling

The overall picture we have seen until now seems not to depend on slow wind. But now it is the moment to consider it.

In this regard we’ll do a little mental experiment. Let’s imagine wind is strictly two-dimensional and constant in time and position, and that a tracer is instantaneously released somewhere at time t_0 , as we can see in Figure 1.5.

In this view, the emitted tracer “point” progressively dilutes while it moves. If wind speed is constant and uniform, at time t the center of the disk I’ve used to represent the diluted tracer will be at position $P = O + U \cdot (t - t_0)$, where I’ve used the symbols P and O to designate the final and initial positions respectively. The disk radius will increase with time, too, but will do with a function having a slightly different form, $r = \kappa \sqrt{t - t_0}$, where κ is a positive number depending on turbulence intensity.

This, under normal conditions.

If the wind is slow, however, the thing shown in Figure 1.6 may happen.

We have what is known as *back-diffusion* in this case: wind is slower than the diffusion-induced disk growth, and points *upwind* the source are affected by a non-zero fallout.

Or, *could*: what actually happens in this case depends on the model and the physical assumptions behind of it.

Let’s for example consider the case of ISC3. According to the US-EPA, this model is now deprecated. In its times, however, it was a paradigmatic example of stationary Gaussian dispersion model. As such, the physical

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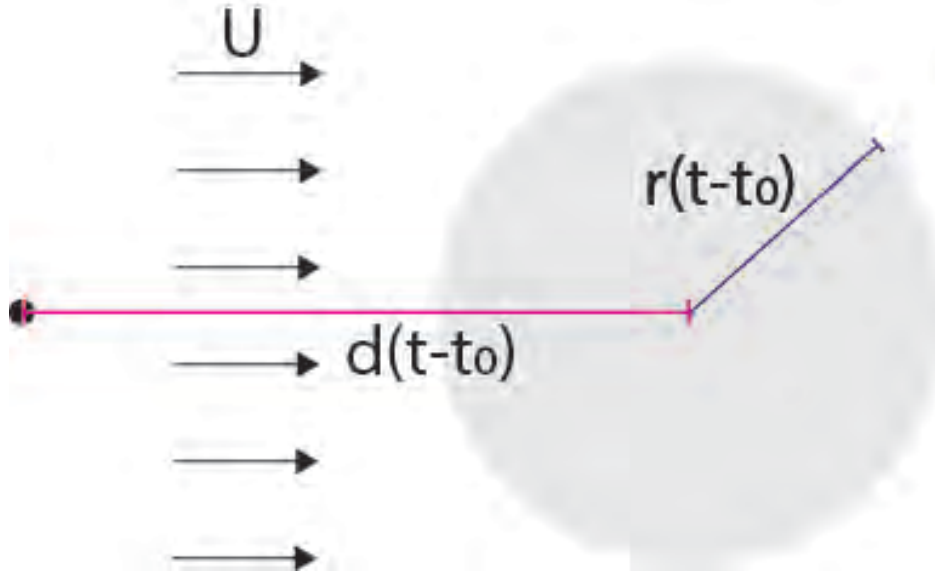


Figure 1.5.: After a given time interval the instantaneously emitted tracer, initially concentrated in a point, moves under the mean wind (assumed uniform and constant in time) and turbulent mixing (assumed constant with time and position) and dilutes

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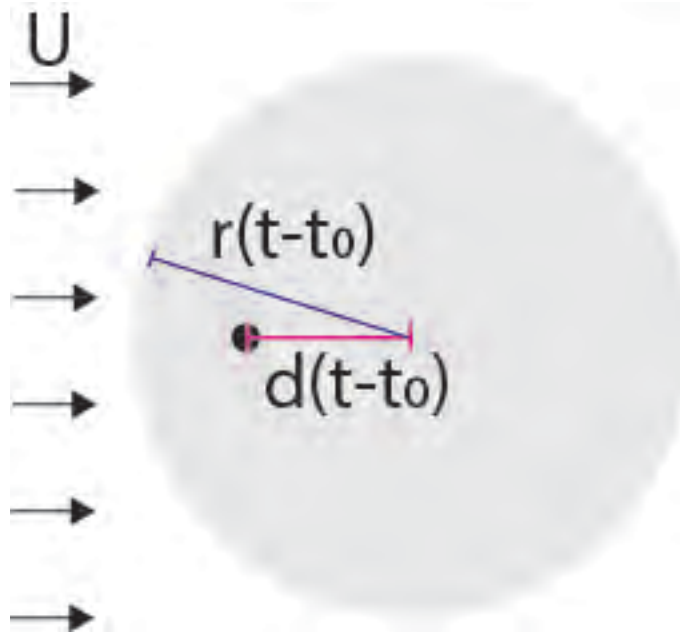


Figure 1.6.: In this case the wind is slower than the disk radius growth rate

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assumptions of the model, and in particular the back-diffusion does not occur, are violated and the Gaussian equations lose significance (Stockie 2011).

To prevent this, ISC3 clips from below the wind speed to a threshold $U_{\min} \approx 1$ m/s; by the way the model coefficients are defined, the case in which back-diffusion occurs are drastically reduced.

With Calpuff, a Lagrangian puff model highly popular when this book was written, “wind calms” are treated differently (Morosini et al. 2025). Wind speed is compared with 0.001 m/s instead of 1 m/s, and used as is (or set to 0.001 m/s in the rather uncommon case in which the wind is zero). But if the wind speed is smaller than a user-configurable threshold the turbulence-induced σ describing puff growth rate are set to a minimum reference value.

Other models may treat wind “calms” (situations in which the wind speed presented as an input to the model are smaller than some prescribed threshold) differently. But the commonality is, they *do*.

An interesting question is, why? The rationale is, dispersion objectively operates differently as wind speed tends to zero. Existing models account for this, each in their own parameterization scheme.

The detailed analysis of the impact of wind data quality in slow speed conditions has been accomplished only recently (Morosini et al. 2025), but the first results obtained indicate effects can be quite dramatic, and especially so in cases where the incidence of low wind speeds is large (a circumstance which turns to be very common in continental inlands).

1.7.2.3. Applications akin to pollutant dispersion modeling

The case of the atmospheric dispersion of “tracers”, pollutants in particular, is just one in which low wind speed can be a serious issue.

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Others exist, with their specific differences, and in this section we shortly mention some.

One is, *odor dispersion*.

In this case the effects of exposure to odors is often a reduction of the quality of life, not necessarily accompanied by a decrease of health (odorous compounds need not to be toxic).

Similarly to what happens with pollutant dispersion, odorous compounds are also subject to distance transport by the mean wind and diffusion by molecular effects and turbulence.

There are, however, some important differences:

- The effects of exposure to pollutants tend to manifest over long times, while odorous compounds can be detected almost instantaneously (more specifically, in a single inspiration-expiration cycle: circa 5-10s).
- Odorous compounds are often, although not always, released at ground level at ambient or close-to-ambient temperature: in many cases they stay close to ground.
- Odorous plumes may be narrow, and meander downwind because of low-frequency phenomena (Mortarini et al. 2016). Because of this, and the short time scale of relevance, odor concentration released by point-wise sources will tend to show extremely high peaks interspersed in more or less long zero-concentration pauses.
- Odorous compounds sometimes (although not always) are comparatively large molecules of biological origin, with physical properties somewhat in the middle between light gases and very small diameter particulates.

All these points, and possibly others, makes the modeling of odor dispersion particularly challenging, and demanding for the development of new, dedicated dispersion models.

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To date, however, in Lombardy odor dispersion studies are conducted using conventional models (both Lagrangian puff as Calpuff, or Lagrangian particle), and peak concentrations are estimated by the *peak-to-mean* approach (Piringer et al. 2012), with a 2.3 coefficient.

In literature, the requirement for a more sophisticated way has been expressed however (Lotesoriere et al. 2023), and research work is currently under way in view of this objective (Cava et al. 2017; Mortarini et al. 2016). This is an advanced topic however, and we'll not deal with its intricacies.

We only get the indication that data quality constraints are much more restrictive in this application, compared to pollutant dispersion modeling. As pointed out in (Lotesoriere et al. 2023), use of advanced micro-meteorological techniques could be recommended in this case, but this is possible only adopting high resolution wind measurements which are beyond of the possibilities of cup anemometers. The currently diffused adoption of cup and vane anemometers in odor dispersion studies, mainly due to lack of high resolution anemometer at odor emission sites, may in many case induce large and unforeseeable errors in odor ground concentration (Morosini et al. 2025). In the author understanding, however, this is a momentary condition likely to change soon, as higher resolution sensors will find their way into odor emitting plants and sites.

Another case in which low-speed wind is of special interest is the management of chemical hazards, due to releases (intentional or not) of toxic substances at concentration amenable to cause lethal or near-lethal level.

Situations like these include cases like the Bhopal disaster, on 3 December 1984, when a catastrophic release of methyl isocyanate occurred at a Union Carbide plant causing the death of more than 10000 people. Other circumstances may be imagined as well, like for example the intentional release of SARIN or another aggressive as an act of chemical warfare.

Overall, the same considerations developed for the odor case apply, with

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additional considerations from the possibly very high emission temperature (like one could experience during a massive fire).

We stop here in the list: you are advised to search for other application-specific uses of wind speed and meteorology, and find in literature or deduction the actual requirements on wind speed quality.

Part I.

This part contain chapters indicating what “wind” is, and some of its characteristics (turbulence for example) which may impact the cup anemometer operation.

Embarking in the exploration of a measurement instrument could be a very restricted experience without an awareness of the natural process it is supposed to sense.

This is true with wind too.

In this specific case, we humans have the disadvantage our sensory system is not equipped to perceive wind exactly. This may lead us to give shape to a mental image of “wind” which is far from the reality of Nature, and consequently to form expectations on measurement instruments which are too loose, or too restrictive.

Hence this part: as a help to better comprehend the elusive and strange “creature” we would like to measure with a cup anemometer.

2. A brief recap on wind

2.1. Definition

2.1.1. Descriptive definition

From a purely mathematical standpoint *wind* is just the name of the vector field

$$\mathbf{U}(x, y, z, t) = [u(x, y, z, t), v(x, y, z, t), w(x, y, z, t)] \quad (2.1)$$

representing the velocity of moving air at point $P = [x, y, z]$ and instant t , with $P \in E$ (here E is some finite region in the three-dimensional space) and $t_a < t < t_b$ (Holton 2004). According to a widely used naming convention, u is the velocity component directed eastwards, v northwards, and w upwards.

This definition is purely descriptive: it tells nothing of the wind properties, nor its actual nature (we'll see them later). Yet it is useful as a conceptual tool.

Indeed, according to the descriptive definition wind is allowed to change with position and time. We all know this really happens, and to get more convinced we may visualize the way clouds move in the sky or the breeze caresses our skin. In anemology, however, knowledge of wind is often restricted to the spatial point P where an anemometer resides, so it is customary to write $u = u(t), v = v(t), w = w(t)$. This view is partial, yet common, and in this book we'll adopt it. (For completeness, it is

2. A brief recap on wind

worth mentioning that “point-wise” are not the only possibility offered by measurement technology: for instance, various remote sensing instruments - like for example the SODAR or the LIDAR - are capable of measuring wind along a line.)

The descriptive definition of wind also says clearly the instantaneous wind has *three* components.

In many applications, however, the vertical component w is ignored and this is reflected by many anemometers sense only the horizontal wind. The wide diffusion of two-dimensional anemometers has in turn twisted people imagination towards assuming wind is only horizontal.

In the real World however $w \neq 0$, and in some specific occasions (for example a katabatic wind cutting the altitude level curves perpendicularly on a mountain slope) its value may be constantly large (Oke 1987).

2.1.2. Why wind occurs?

Knowing wind is a necessity which predates by centuries the development of meteorology. Without an even basic grasp of it, long-range navigation would have never developed, and likely the history of human civilization would have followed another path.

However one would have to wait until 1735 when George Hadley, in his study on trade winds (Hadley 1735), introduced the idea which is to date the one accepted that wind flows from higher to lower pressure areas, and these pressure imbalances are the inevitable consequence of the periodic changes of incoming solar radiation.

With time, and an increasing accumulation of discoveries, the field of meteorology evolved and specialized, and today articulates in many branches. The way wind is perceived has also changed from one user community to the other.

2. A brief recap on wind

Nevertheless, from a physical standpoint wind is the same: it originates from the tendency of air, as any gas, to move from areas of higher pressure to others where pressure is lower.

In Figure 2.1 we can see an example of the pressure distribution above Central Europe and Northern Atlantic, with values spanning the interval from 995 to 1030 hPa: quite a large gap.

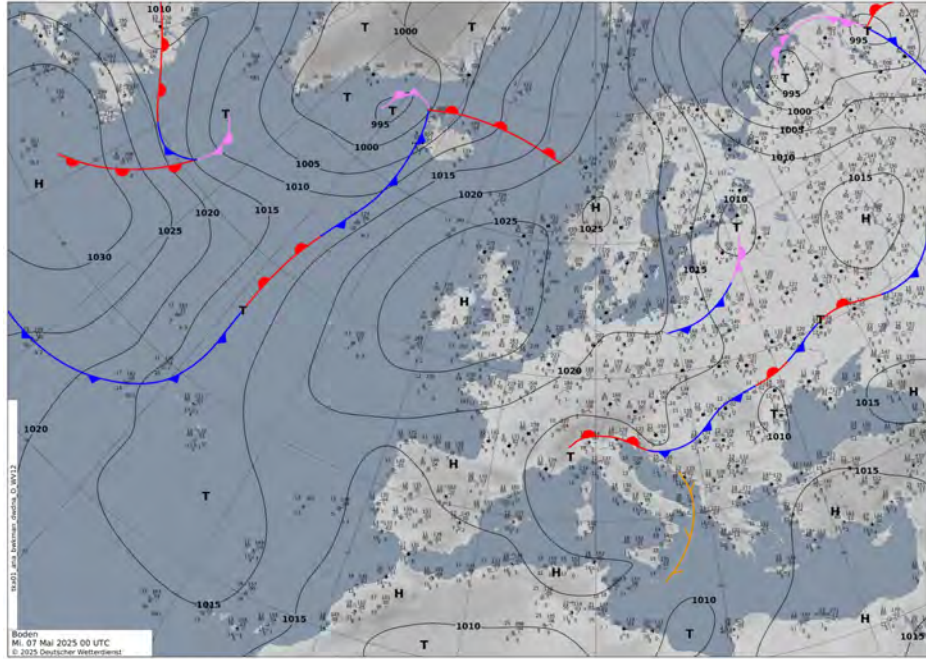


Figure 2.1.: A weather map of the European region, where we can see how large the pressure differences can be at a same reference altitude above mean sea level (Courtesy of Deutsche Wetterdienst)

The circulation of air masses does not occur freely however. For starter, the Earth revolution around its axis induces a Coriolis apparent force, intense

2. A brief recap on wind

enough to deviate the air masses. In addition the Earth atmosphere is thin enough for high mountain ranges and other “small” surface perturbations to interfere with its movement (R. Smith 1979).

Occurring simultaneously, these forcings conspire to provide a distribution of mean wind which is, to say the less, uneven. We can see this in Figure 2.2.

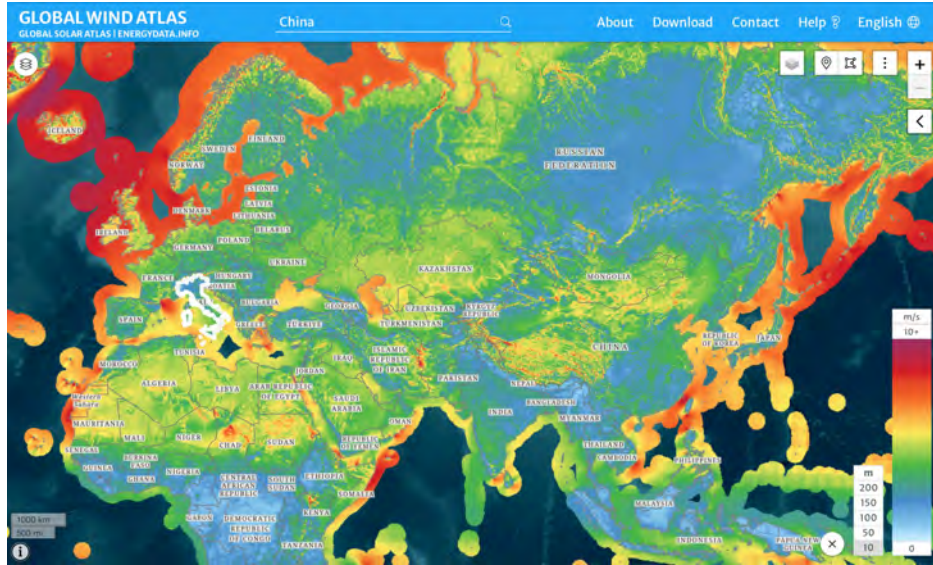


Figure 2.2.: Excerpt of the Global Wind Atlas showing the wind speed distribution in a part of the World, at 10m above surface

The figure scale, ranging from 0 to 10 m/s, is sufficient to see how vast portions of continents inland are characterized by a mean speed less than 3 m/s.

In Figure 2.3 we can see an enlargement over central Europe, where we can appreciate a significant local complexity.

The comparison of Figure 2.2 and Figure 2.3 visually suggests the impor-

2. A brief recap on wind

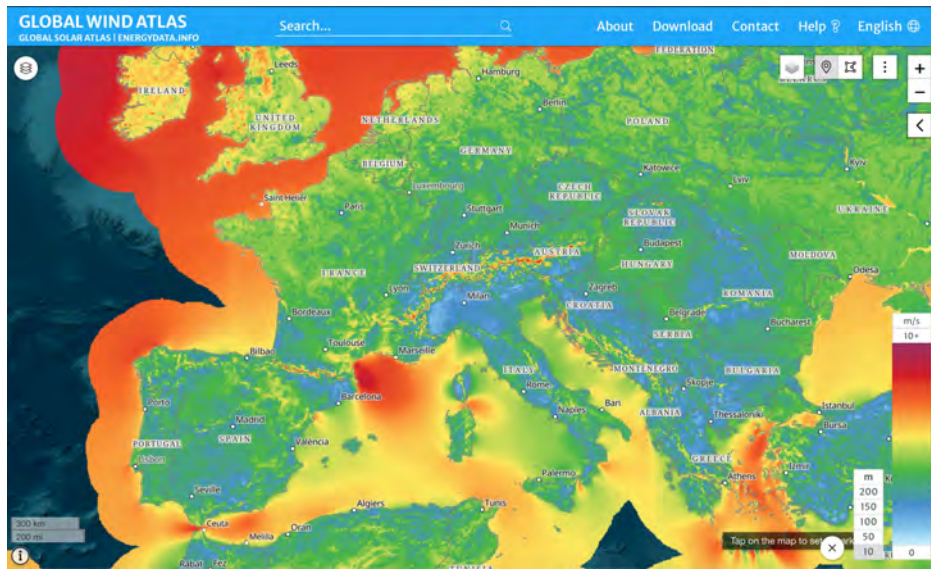


Figure 2.3.: Mean wind speed over central Europe, from the Global Wind Atlas; data at 10 m above surface

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tance of the observation scale. In this case we have seen the spatial side of the scale, with details emerging as we focus our view on a smaller area. But the same occurs in the time dimension, as we'll see soon.

2.1.3. A conceptual model of wind

2.1.3.1. Subject statement

Images as Figure 2.1, @fig-windatlas and Figure 2.3 makes one suspect the existence of some kind of deep regularity, which is amenable of a mathematical description.

And this is precisely the case, the physical and mathematical description of air circulation being the subject of *dynamic meteorology* (Holton 2004), whose key assumption is that air moves as any gas would, within the specific constraints established by the Earth geometry and structure.

Indeed, there is practically unanimous consensus that the air movements around our planet can be described using a form of Navier-Stokes equations (the same used in fluid dynamics to describe and predict the flow of liquids and gases in general cases) adapted to the atmospheric context (Holton 2004; Vallis 2006)).

In order to put wind in context it is my opinion that some exposure to the equations is useful, and I'll give here. We'll not explore the details of derivations however, but you can find them in the cited references.

2.1.3.2. NS equations in general fluid dynamics

In their most general settings the Navier-Stokes equations (abbreviated in NS) are the particularization to fluids of the Newton equation of motion,

$$\mathbf{F} = m\mathbf{a} \tag{2.2}$$

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with $\mathbf{F}$ the forces acting on the system of our choice, m the mass, and $\mathbf{a}$ the acceleration. If the system is composed by a discrete (not necessarily finite) set of “particles” P_i ($i \geq 1$), then one could write the particles individual equations as

$$\mathbf{F}_i = m_i \mathbf{a}_i$$

where

$$\mathbf{F}_i = \mathbf{f}_i + \mathbf{f}_{ij}$$

($\mathbf{f}_i$ here represents the force acting on the particle from outside the system, while $\mathbf{f}_{ij}$ is the force P_j exerts on particle P_i , be it caused by charge induced attraction or repulsion, gravitational attraction, impulsive forces following impacts, and whatever else) (Goldstein, Poole, and Safko 2002).

In this “classical” setting, whose details we’ll not pursue, we can imagine “particles” to be gas molecules. As their number per unit volume becomes large enough (and with real gases this is what happens), the system properties can be analyzed and described in statistical terms (Fermi 1937) as a system of *continuous* nature (Visconti and Ruggieri 2020; Kundu and Cohen 2002), whose properties can be described using a small set of macroscopic variables, the movement vector among them.

This, explained in non-rigorous terms, is the conceptual framework within of which the Navier-Stokes equations were originally introduced.

We’ll not analyze in depth the NS, but it is nevertheless useful to see them for illustrative purpose. If we consider a Cartesian reference, the general NS can be written as

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{Re} \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right]$$

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$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} = -\frac{\partial p}{\partial y} + \frac{1}{Re} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right]$$

$$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho uw)}{\partial x} + \frac{\partial(\rho vw)}{\partial y} + \frac{\partial(\rho w^2)}{\partial z} = -\frac{\partial p}{\partial z} + \frac{1}{Re} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right]$$

where $\mathbf{u} = [u, v, w]$ is the wind vector, t the time, ρ the air density, p the pressure, Re the Reynolds number (Kundu and Cohen 2002) and $\tau_{\alpha\beta}$ the stress tensor.

The NS can be rewritten more compactly using Einstein's index notation (Einstein 1916; Frankel 1997) as

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{1}{Re} \frac{\partial \tau_{ij}}{\partial x_j} \quad (2.3)$$

(As a remind, repeated indices in a term imply summation, and $[u_1, u_2, u_3] = [u, v, w]$, $[x_1, x_2, x_3] = [x, y, z]$.)

The leftmost side,

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j}$$

signifies the total derivative of the product of density and wind speed: if we assume a fixed (finite) control volume it is the *ma* written for a continuous fluid.

The rightmost side,

$$-\frac{\partial p}{\partial x_i} + \frac{1}{Re} \frac{\partial \tau_{ij}}{\partial x_j}$$

2. A brief recap on wind

represents the forces due to which the change of speed occurs, that is the $\mathbf{F}$ in Equation 2.2. How it is composed declares the kind of phenomena the NS take into account.

Its leftmost term

$$\frac{\partial p}{\partial x_i} = \mathbf{i} \frac{\partial p}{\partial x} + \mathbf{j} \frac{\partial p}{\partial y} + \mathbf{k} \frac{\partial p}{\partial z}$$

is the pressure gradient: in the NS it appears with the negative sign as the fluid moves from points with high to points with low pressure, that is, “against” or “downhill” the gradient.

The rightmost term

$$\frac{1}{Re} \frac{\partial \tau_{ij}}{\partial x_j}$$

represents the viscous forces operating as *diffusion*, with a positive sign in the NS. It could be written explicitly in terms of derivatives of velocity, remembering that (Visconti and Ruggieri 2020)

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \mu' \delta_{ij} \frac{\partial u_i}{\partial x_j} \quad (2.4)$$

where μ and μ' are the “primary” and “secondary” viscosity coefficients as introduced in (Visconti and Ruggieri 2020), and δ_{ij} the Kronecker symbol, $\delta_{ij} = 1$ if $i = j$, 0 otherwise.

If the fluid is incompressible then

$$\delta_{ij} \frac{\partial u_i}{\partial x_j} = 0$$

2. A brief recap on wind

and Equation 2.4 reduces to

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (2.5)$$

The usefulness of these formulae stems from the practical meaning they convey in a simple, unambiguous form: by critically analyzing them it is possible to reverse-engineer the NS and understand to which situation they do apply (in principle one could also proceed in a direct way, by deriving the NS starting from a set of assumed, reasonable principles: this is the way followed in (Kundu and Cohen 2002; Visconti and Ruggieri 2020) and other fluid mechanics books).

In particular:

- In the NS (@eq-basic-NS) the pressure gradient and viscosity forces are the only forces considered.
- Other forces, like Coriolis', are not taken into account: the NS are “general” from a mathematical standpoint, but apply to a *small* volume and a *short* time frame, so Coriolis and other external forces can be ignored.
- To solve (numerically) the NS an appropriate set of initial and boundary conditions are needed. They can be specified, however, in terms a Cartesian reference may easily describe.

In conclusion, the NS in @eq-basic-NS can be used profitably to describe or plan a laboratory experiment involving channels, pipes, wing profiles, fuselages, airframes, ships, steam turbine blades, and other similar “small” objects.

These same NS are not suitable to represent the whole atmosphere, or a part of it.

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2.1.3.3. NS adaptation to meteorology

In order to adapt the NS to the atmosphere two necessary additions are needed.

The first addition, of great importance if one desires to represent the atmospheric dynamics at a global scale or in an area so large that Earth curvature cannot be ignored, is in reality a formal change, that is rewriting the equations adopting a geographical coordinate frame based on latitude, longitude and height (Holton 2004), or another reference taking into explicit account the Earth curvature and overall (Geoid-scale) “local” shape (Doms and Baldauf 2013; Baldauf 2004).

In fact Equation 2.3 is not simple to apply to geometries which do not fit the idea of a box, or a pipe, or some other very simple shape.

The details and the developments related with this idea are quite far from the necessities of measuring wind at or close to the Earth surface, and we’ll not follow them: I’ll continue to illustrate NS and their applications in a rectilinear Cartesian coordinate system, with an awareness it can be used only for very small areas on the Earth surface, namely at the 10-100km “topographic scale” (Lisle, Brabham, and Barnes 2011; Svens 2023) or less.

Of more immediate interest to our case is the explicit addition of two terms addressing Coriolis and gravitational effects, whose effect can be found in wind measurements even at the smaller scales.

Following (Stull 1988), the NS adapted to the specific case of the atmosphere in Cartesian coordinates can be written as

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\delta_{i3}g - 2\varepsilon_{ijk}\Omega_j u_k - \frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} \quad (2.6)$$

where the term $\delta_{i3}g$ represents the effect of gravitation along the, and $2\varepsilon_{ijk}\Omega_j u_k$ is the Coriolis effect (ε_{ijk} is the Levi-Civita symbol, equal to 1

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if the indices ijk form an even permutation, -1 if ijk form an odd permutation, and 0 otherwise).

Again we'll not delve in the details (for these you can access the cited references), and use Equation 2.6 in a purely illustrative manner.

2.1.3.4. Describing wind variation mathematically

The NS form we have seen in Equation 2.6 is not bound to one specific time or length scale, although specific terms can become more or less important if observed from nearby or afar. Its solutions are allowed to literally contain “anything” their boundary and initial conditions may reproduce. And this is in particular true for phenomena whose existence is experimentally known since a long time, namely, the small scale irregular fluctuations collected under the umbrella term “turbulence”.

The question then arises whether a mathematical description of these fluctuations exists, and whether it can be separated from the “mean” flow.

The answer to this question was originally found in intuitive terms by Leonardo da Vinci who, in his *Codex Atlanticus*, first named turbulence and described it as superposing to a smooth flow.

Leonardo's intuition was based on qualitative observations made on water flowing in channels having a complex shape. The first quantitative observations (and confirms) came centuries later when the fluid-dynamicist Osborne Reynolds in the second half of Nineteenth Century studied systematically the transition between “laminar” and “turbulent” flow regimes (Reynolds 1883).

In his pioneering work, Reynolds first introduced the idea that the “superposition” of a regular *en-masse* flow and irregular fluctuations was actually an algebraic sum.

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The implications of this idea was later developed by various authors, and finally took an almost-axiomatic form in the seminal volumes on the statistical mechanics of turbulence by A.S. Monin and A.M. Yaglom, where it is named *Reynolds Decomposition* (Monin and Yaglom 2007). This description was then popularized, in a slightly simplified form, in later famous textbooks like for example (Stull 1988; Sozzi, Georgiadis, and Valentini 2002; Kaimal and Finnigan 1994). I summarize it here in non-rigorous terms, in preparation of later uses.

Let $x = x(t)$ any quantity sampled within the Planetary Boundary Layer (this always happens with any meteorological station).

Then we can write it as

$$x = \bar{x} + x'$$

with $\bar{x}$ indicating a “mean part”. One common misunderstanding is to imagine the mean part as a number, or a constant function.

This is not required however, and a more general way is to let the mean part be a smooth version of the original signal x , and the “averaging procedure” a smoothing operator.

If the mean part is known, the “fluctuating part” may be obtained by difference, setting $x' = x - \bar{x}$.

For this partitioning to make physical sense the averaging procedure cannot be chosen at will, but must fulfill a set of basic requirements modeled after the ensemble averaging of many experiments replica originally used by Reynolds (Reynolds 1883).

Later on, a more general definition was accepted among researchers, and took an axiomatic form in (Monin and Yaglom 2007).

In this two-volumes monograph Monin and Yaglom introduced a set of rules defining what a legitimate mean is. These rules are named “*Reynolds conditions*”, and may be written this way:

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- $\overline{x'} = 0$.
- $\overline{\alpha x + \beta y} = \alpha \overline{x} + \beta \overline{y}$ ($\alpha, \beta \in \mathbb{R}$: “linearity”).
- $\overline{\bar{x} \cdot y} = \overline{x \cdot \bar{y}} = \bar{x} \cdot \bar{y}$ (as far as I know this property has no name).
- $\overline{\frac{\partial x}{\partial \alpha}} = \frac{\partial \bar{x}}{\partial \alpha}$ (this property too has no name; it is used extensively, but standard boundary layer meteorology textbook like (Stull 1988) do not mention it explicitly).

Besides their use as “axiom-like” phrases stating what an acceptable averaging operator is, Reynolds conditions are also widely used as a modeling tool.

One famous use in this role is to derive the so-called “Reynolds Averaged NS” equations (RANS), a key building block in atmospheric modeling.

2.1.3.5. Removing turbulent fluctuations: the RANS equations

As mentioned in Section 2.1.3.3 the meteorological NS in Equation 2.6 specify solutions which are “too rich”, and mostly containing irrelevant details.

Surely, the mean wind is relevant to many practical applications (among these the meteorological modeling), but it is buried among anything else in any solution to Equation 2.6. But in line of principle it could be obtained as a solution to a set of equations obtained by applying Equation 2.6 the Reynolds averaging procedure described in Section 2.1.3.4. The resulting “equations for the mean wind” are named *Reynolds Averaged Navier-Stokes equations* (abbreviated in RANS).

The derivation of RANS from the meteorological NS is lengthy, and I omit the details here. The only interesting thing is, apart from the differential calculus tricks needed in some of the passages, the need of few physically-grounded assumptions.

Following again (Stull 1988) the RANS can be written as

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$$\frac{\partial \overline{u_i}}{\partial t} + \overline{u_j} \frac{\partial \overline{u_i}}{\partial x_j} = -\delta_{i3}g - 2\varepsilon_{ijk}\Omega_j \overline{u_k} - \frac{1}{\overline{\rho}} \frac{\partial \overline{p}}{\partial x_i} + \nu \frac{\partial^2 \overline{u_i}}{\partial x_j^2} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j} \quad (2.7)$$

The last term,

$$\frac{\partial \overline{u'_i u'_j}}{\partial x_j}$$

is additional and somewhat surprising: it indicates the mean airflow is affected by the covariance among the various momentum components. In other terms, turbulence concurs in determining the mean wind.

This transport of momentum is more important than it seems on a first glance: it implies a turbulence-driven interaction between any two nearby air volumes. A part of this interaction manifests as *mixing*.

Observations have shown molecular diffusion, although very important at the microscopic scale at which the massive interaction between plants and the atmosphere occurs (Monteith and Unsworth 2008; Novak 2012; Fitter and Hay 2002; Nobel 2009), in the atmosphere tends to be masked by another form of diffusion due to turbulent fluctuations (Stull 1988; Foken 2008).

2.1.4. Predictable consequences on wind measurements

In 1963, the meteorologist Edward Norton Lorenz (with help by the computer scientists Ellen Fetter and Margaret Elaine Hamilton) developed a simplified model describing the time evolution of a layer of fluid heated from below and cooled from above.

Computer simulations of the model (Lorenz 1963), consisting in the three ordinary differential equations

2. A brief recap on wind

$$\begin{cases} \frac{dx}{dt} = \sigma(y - x) \\ \frac{dy}{dt} = x(\rho - z) - y \\ \frac{dz}{dt} = xy - \beta z \end{cases} \quad (2.8)$$

revealed something strange: if physically sensible parameters σ, ρ, β were selected, the numerical solution exhibited a behavior which apparently switched unpredictably and randomly between two qualitatively different regimes.

Initially, Lorenz assumed an error perturbed the solution, but through code verification eventually convinced him the computer procedure was correct, and the strange behavior he and his coworkers saw was genuine.

This was the serendipitous birth of what is known today as “deterministic chaos theory” (McCauley 1993; Peitgen, Jürgens, and Saupe 2012).

This is very interesting to our specific case because Equation 2.8 is a particular case of the meteorological NS in Equation 2.6.

If the simplified NS yield a chaotic behavior, it is reasonable to expect the same will happen even more so with the full NS, possibly with amplified effects. And something even more complex may be visible with Equation 2.7, due to the momentum mixing implied in them.

The question is, whether these chaotic effects will manifest at a time scale one should worry of, or not. In a following section, when we will see graphs of instantaneous wind measured using a fast high resolution wind sensor, we’ll realize we have all reasons to.

2.2. The horizontal wind

2.2.1. Reasons for considering the horizontal wind only

Equation 2.7 and Equation 2.6 are three-dimensional, and this suggests the Cartesian wind vector $[u, v, w]$ may have a nonzero vertical component w .

In this book we'll focus on *horizontal* wind, that is we'll only consider the two-dimensional vector $[u, v]$.

If the wind vector has a vertical component, why then to ignore it?

There are two reasons:

- First, and more important in our case, the cup anemometer is designed to operate in the horizontal dimension.
- Second, but connected with incidental facts rooted in details of Planetary Boundary Layer climatology (Oke 1987), the vertical component w is usually close to zero if the wind measurement occurs over a smooth and flat area.

Our mainly instrumentally-driven neglect does not mean the vertical wind is not important. On the contrary, as investigators have discovered (Kristensen 1998; Kristensen, Hansen, and Højstrup 2003) the existence of $\bar{w} \neq 0$ can change the horizontal wind readings of cup anemometers.

But on the other side, if a meteorological station operates a cup anemometer which is not accompanied by another anemometer sensitive to vertical wind, the w component will remain unknowable and its effects unknown.

As we will see in a later chapter, knowledge of the site-specific statistical distribution of w can be gained through a *micro-meteorological characterization campaign*, in which the cup anemometer is flanked or precedes a cup anemometer by a three-dimensional ultrasonic anemometer or some

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other high-resolution wind sensor able to probe the vertical component. This is another story however.

2.2.2. Flow and provenance conventions for the instantaneous horizontal wind

Let's go back to definition Equation 2.11, and ignore for the moment any dependency on position and time. If $\mathbf{U} = [u, v, w]$ is the full wind vector, with w aligned to the local gravitational vertical, then the *horizontal wind* is defined as

$$\mathbf{u} = [u, v] \tag{2.9}$$

It is customary to interpret this definition according to the *flow* convention: the vector $[u, v]$ points towards the direction wind is blowing *to*.

The other, or *provenance* convention, assumes instead the vector $[u, v]$ points towards where wind is blowing *from*.

In most literature, when wind is expressed in component form the flow convention is implicitly assumed. But it is better to not imagine this choice mandatory, or otherwise universal.

2.2.3. The horizontal wind as a complex number

As any two-dimensional vector, the horizontal wind $[u, v]$ can be also interpreted as a complex number $Z = u + iv$. This change has no practical consequence on the horizontal wind itself as considered in this book.

It is, however, useful in allowing us to see the wind in a way closer to what is reported by cup and vane horizontal anemometers: the complex number Z can in fact be expressed in polar form, $Z = |Z| \exp(i\delta)$.

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The absolute value of the complex-form wind, $|Z|$, is the *instantaneous horizontal wind speed*, and is the quantity cup anemometers are exposed to.

The angle δ is related to the *instantaneous horizontal wind direction*, to which the vane reacts.

It can be noticed the instantaneous horizontal wind speed does not depend on whether the flow or provenance convention is adopted: its value is always

$$U = |\mathbf{u}| = |Z| = \sqrt{u^2 + v^2} \quad (2.10)$$

The wind direction δ , on the contrary assumes different values depending on whether the flow or provenance convention is used. Most, if not all, vane anemometer employ the provenance convention.

The wind direction is not our core subject, but nevertheless it is a good thing to express it explicitly. I do with help from the R programming language (Team 2023):

```
dir <- 180/pi * atan2(-u, -v)
if(dir < 0) dir <- dir + 360
```

In this code excerpt you may notice I’ve used the two-arguments arc-tangent, which is an implementation of the arc-tangent function not placing restrictions on the sign of its output.

In addition, by the way it is written the formula assumes:

- The $[u, v]$ wind vector is expressed in the flow convention, so that its provenance counterpart is $[-u, -v]$.
- The wind direction “dir” follows the provenance convention.
- “dir” is expressed in degrees, and not radians.

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- “dir” is 0° if wind blows from North, and increases clockwise.

It can be noticed the two-argument “atan2” function is also present, with different names, in all programming languages supporting basic trigonometry. The definition adopted is so that “atan2(x,y)” function designates

$$\tan^{-1} \frac{x}{y}$$

evaluated on the branch $-\pi \cdots +\pi$, then taking signs and zeros into account.

As complex numbers can be converted between the Cartesian and polar forms, the same happens with wind vectors:

$$\begin{cases} u = |\mathbf{u}| \cdot \sin \delta \\ v = |\mathbf{u}| \cos \delta \end{cases}$$

This transformation formula is convention-neutral: if the original wind is expressed according to the flow convention, then the transformed will also be, and the same for provenance convention. Change from one convention to the other in Cartesian coordinates may be accomplished using an appropriate sign change.

2.3. Mean wind

Equation 2.9, relevant to data collected by an anemometer over specific instants or small time intervals, defines as we saw in Section 2.2.3 the *instantaneous* time.

This may be collected and archived per se, and later transferred to some facility external to the meteorological station for offline processing.

2. A brief recap on wind

But in most cases it is used within some local data collection and processing device to compute, together with many other instantaneous wind measurements, some “mean values”.

We’ll deal with specific wind speed processing methods in a later chapter, but in the moment let’s imagine we have access to individual instantaneous “samples” of wind, that is

$$u_i = u(t_0 + i \cdot \Delta t), \quad v_i = v(t_0 + i \cdot \Delta t)$$

for an integer $i = 0, 1, \dots, n-1$. From these lists of numbers we can obtain (at least) two different “mean speeds”. The first is the *vector* speed,

$$U_v = \sqrt{\left(\frac{1}{n} \sum_{i=0}^{n-1} u_i\right)^2 + \left(\frac{1}{n} \sum_{i=0}^{n-1} v_i\right)^2} = \sqrt{\bar{u}^2 + \bar{v}^2} \quad (2.11)$$

The second is the *scalar* speed,

$$U_s = \frac{1}{n} \sum_{i=0}^{n-1} \sqrt{u_i^2 + v_i^2} = \overline{\sqrt{u^2 + v^2}} \quad (2.12)$$

Equation 2.11 and Equation 2.12 express two *very* different concepts. The vector wind speed in practice tells us something about how much space an hypothetical particle could have traveled in a specified time interval Δt :

$$\Delta P = U_v \cdot \Delta t$$

with ΔP designating the position displacement occurring in time step Δt .

The scalar speed, quite on the contrary, informs us on how much kinetic energy is available in horizontal wind that a turbine could extract:

2. A brief recap on wind

$$KE \propto \frac{1}{2}U_s^2$$

In a later chapter we'll see how a cup anemometer is built, but even an intuitive grasp is sufficient to realize wind speed sensors measure wind instantaneously, or on a small time interval.

If we (or an automatic data logger on our behalf) samples these speeds and averages them directly, it yields the value from Equation 2.12, that is the *scalar* speed. This may be wanted in some applications (e.g. wind energy studies), but not in others.

2.4. How we imagine wind, and how it really is

2.4.1. Wind perception

Our perceptions condition our judgement, and shape our imagination by means of complex physiological and psychological interactions.

In the same time, our perceptions are conditioned by the physical nature of our senses and the structure of the neural circuitry processing the information coming from them.

The perception can be direct (we can feel the wind on our skin and within our hairs), or indirect (we can also see a cup anemometer rotating).

Whatever its nature and timing, we may have the perceptual impression of some variation.

But overall, experienced in these way the wind seems to us a relatively smooth and “slowly” changing phenomenon. Sometimes, even completely absent.

2. A brief recap on wind

An interesting question, and argument for some physical reflection, is whether this steady perceptual image squares with the considerations exposed in Section 2.1.4 or not.

The answer is, broadly speaking, a “yes”. Our senses inform us that wind speed and direction change, and that to some extent these variations can be predicted in direction and magnitude. But also, that this prediction is highly unreliable, as wind changes often suddenly and without any prior sign.

This, however, conveys a vision which is largely incomplete: the consequences of deterministic chaos go well below the discriminating ability of our senses.

2.4.2. Wind, seen by a high-resolution sensor

2.4.2.1. Horizontal wind speed

The smoothness of wind speed turns out to be a perceptual illusion, as we can see in Figure 2.4. There is shown a sample of wind collected using a research-grade ultrasonic anemometer: in the leftmost plot you can see the wind speed variation in an hour, and in the rightmost attention is given to 5 seconds in the same hour. We can see a massive variation, but, looking at the 5s plot, one may wonder whether we humans can just acknowledge at least a little part of it.

The rightmost plot in Figure 2.4 appears less crowded than the leftmost. This is a direct consequence of the finite sampling rate of ultrasonic anemometers, 10 Hz in SHAKEUP stations; the model used allows sampling rates up to 40 Hz, and would it have been used the leftmost and rightmost plots would have resembled more.

Adopting a shorter sampling rate would be taxing in terms of resources (SHAKEUP stations archive raw ultrasonic data in addition to processed ones), but would allow to visually appreciate the phenomenon of *scale*

2. A brief recap on wind

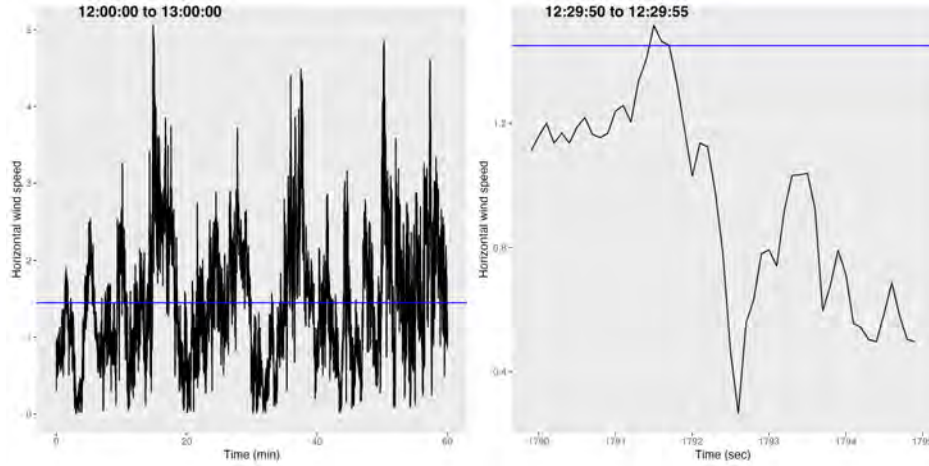


Figure 2.4.: Sample of wind speed collected at the SHAKEUP site “Cinisello Parco Nord” on 15 August 2016; the blue horizontal line indicates the hourly mean

invariance. This graphical property can be found in many natural phenomena and is considered a hallmark of their *fractal* nature (Mandelbrot 1982).

In general, phenomena manifesting under the control of a chaotic systems do not necessarily exhibit a fractal geometry (Perko 1996), but often these two aspects manifest together.

Scale invariance in wind plots is then not surprising, given wind can be modeled by solutions to NS equations (Stull 1988), which are in essence chaotic as we have seen in Section 2.1.4.

It is curious, in my opinion, that a possibly fractal phenomenon like wind appears to our senses “smooth”. It looks me a wonderful demonstration that our senses do not mirror reality precisely (Seth 2021), with the possible consequence of hampering our intuition. In Figure 2.4 we have seen a

2. A brief recap on wind

large amount of variation occurring at very small time scales, in fact much beyond the possibilities our sense offer to grasp it.

2.4.2.2. Vertical velocity

The fact most conventional anemometers do not measure the vertical component of the wind vector does not mean it is always zero.

In Figure 2.5 we can see the vertical wind component sampled over one day at a 10 Hz rate.

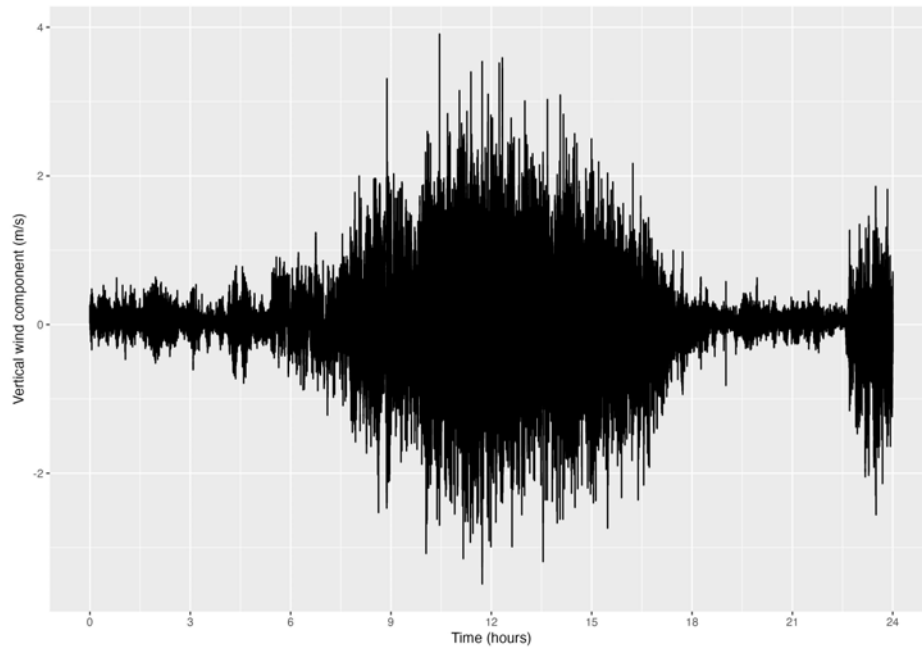


Figure 2.5.: Vertical component of the instantaneous wind sampled at 10 Hz on day 2016-07-16 at the Cinisello Parco Nord micro-meteorological station

2. A brief recap on wind

Compared to the time span, the sampling time is so small than the plot in Figure 2.5 reduces to a corrugated black band. But we can clearly see this band expanding in the central (sunny) hours, reflecting the presence of thermally-driven turbulence. It's curious the peak-to-peak amplitude of the band is about 7 m/s: so short-time fluctuations cannot be perceived by our senses directly.

Apparently, and this is particularly true on night-time when the band is narrower, the band center looks shifted towards positive values, possibly indicating a non zero mean velocity component.

That $\bar{w} \approx 0$ is one of the tenets of dynamic meteorology (Holton 2004), based on consideration on airflow in the deep troposphere, where conditions of non-zero values (convective storms, subsidence and others) occur seldom.

This is not necessarily true in the Planetary Boundary Layer, because of obstructions to wind and dishomogeneities in the surface nearby the measurement point. Indeed, within the PBL the existence of non-zero mean values for w occurs frequently, and can be seen in Figure 2.6.

Of the two plots in Figure 2.6 the second, portraying the angle of the wind vector to the horizontal, defined as

$$\phi = \frac{180}{\pi} \cdot \arctan \frac{\bar{w}}{\sqrt{\bar{u}^2 + \bar{v}^2}}$$

is especially informative. In the day selected the angle from the horizontal plane tends to be relatively small, yet distinctly non-zero from about hour 00:00 to 03:00, and 21:00 to 22:30. Particularly interesting are the two large spikes in the time range from 05:00 to 08:00, with absolute values exceeding or almost reaching 60° in magnitude.

Indeed, the occurrence of non-zero mean vertical wind is ubiquitous at measurement sites located on surface.

2. A brief recap on wind

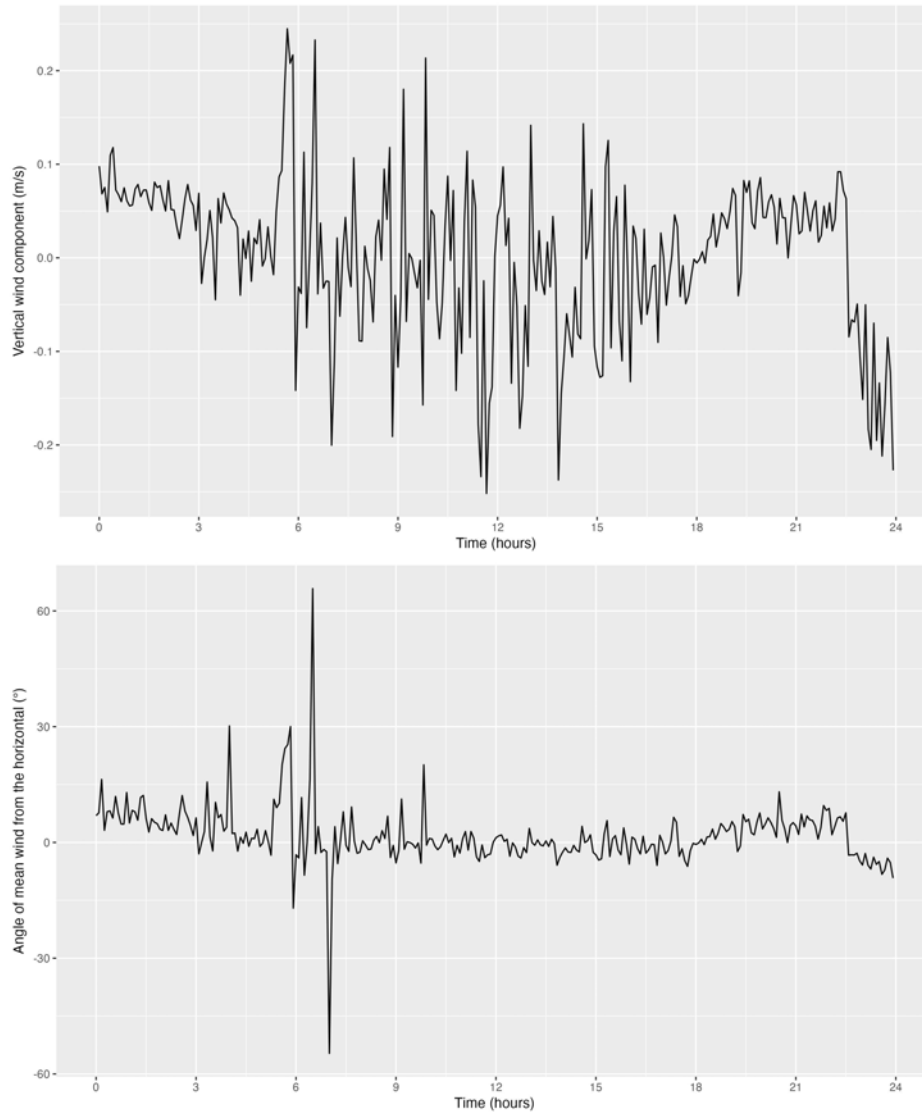


Figure 2.6.: Plots of the mean vertical component of wind and of the angle of the wind vector from the horizontal plane, as determined at the micro-meteorological station of Cinisello Parco Nord on day 2016-07-16; means have been computed with an averaging period of 5 minutes

2. A brief recap on wind

2.5. Change with altitude

Technically-oriented literature (for example (Burton et al. 2011)) indicates the wind speed is not the same with height above ground, and assume a dependency having form

$$U(z) = \frac{u_*}{k} \log \frac{z}{z_0} \quad (2.13)$$

where u_* is the so-called *friction velocity*, a proxy of mechanical turbulence (see (Stull 1988)), $k \approx 0.4$ is the so-called *von Karman constant*, z_0 is the *surface roughness length*, and z the height above ground.

(Sensible values of z_0 may be found in (Oke 1987) for different types of terrain and land use.)

Equation 2.13 is however just an approximation, and in time much research have been devoted to find more realistic formulas valid in the whole Planetary Boundary Layer or in parts of it (Basu 2023). These formulas, of semi-empirical nature, involve other information about stability (expressed as Obukhov length L), the PBL thickness z_i , and others.

However, for heights in the order of $z \leq 20$ m the simplified form Equation 2.13 departs little from these more sophisticate models, and can be used at least for descriptive purposes.

The most important consequence of Equation 2.13 is that the observed wind speed *changes with height above ground*. Because of this reason the anemometer height is accurately measured at meteorological station, and included in the metadata.

In case of the network whose data we'll use the anemometer height above ground is fixed to 10 m, and so comparisons among different SHAKEUP sites is possible without any need to have sophisticate micro-meteorological information (e.g. u_* , L) at hand.

2. A brief recap on wind

It is advisable to exert additional care if cup anemometer speed measurements are needed at heights above this limit, as often happens for example with wind energy.

Part II.

This part contain a description of the cup anemometer, from both a physical and a technical standpoint.

Indeed, a knowledge of an instrument physical principle is of essential importance in view of understanding how the sensor can couple with the wind occurring in Nature, and may help forming a first intuitive image of the quality of the data it can realistically produce.

This is, however, not enough to get a full understanding of the problems at stake. The umbrella term “cup anemometer” covers in fact a vast array of sensor characterized by huge design and build differences: understanding this engineering level variation is then of paramount importance not only to engineering oriented people, but to physicists and application experts too.

3. The cup anemometer

3.1. Premise: drag-only sensors

Before to analyze the cup anemometer in detail it is better to see a simplified version of it: in this section we'll realize it cannot work in practice, but will draw some important background information we'll build on later.

Our idealized wind sensor is shown in Figure 3.1.

On its point opposite to the disk the shaft connects to a junction allowing the shaft to be rotated in a plane parallel to the constant vector wind field, and the torque thus exerted can then be measured.

We start our mental experiment with the angle $a = 0^\circ$, that is with the disk exposed perpendicularly to the airflow. If we indicate with l the length of the shaft plus the disk radius, then the absolute value of the torque will approximately be

$$M = l \cdot F$$

where the magnitude F of the force, which we assume to be entirely due to drag, will be

$$F = \frac{1}{2} \rho U^2 C_d S \quad (3.1)$$

where ρ is the air density, U the wind speed, $S = \pi r^2$ is the disk area, and C_d a shape-dependent drag coefficient.

3. The cup anemometer

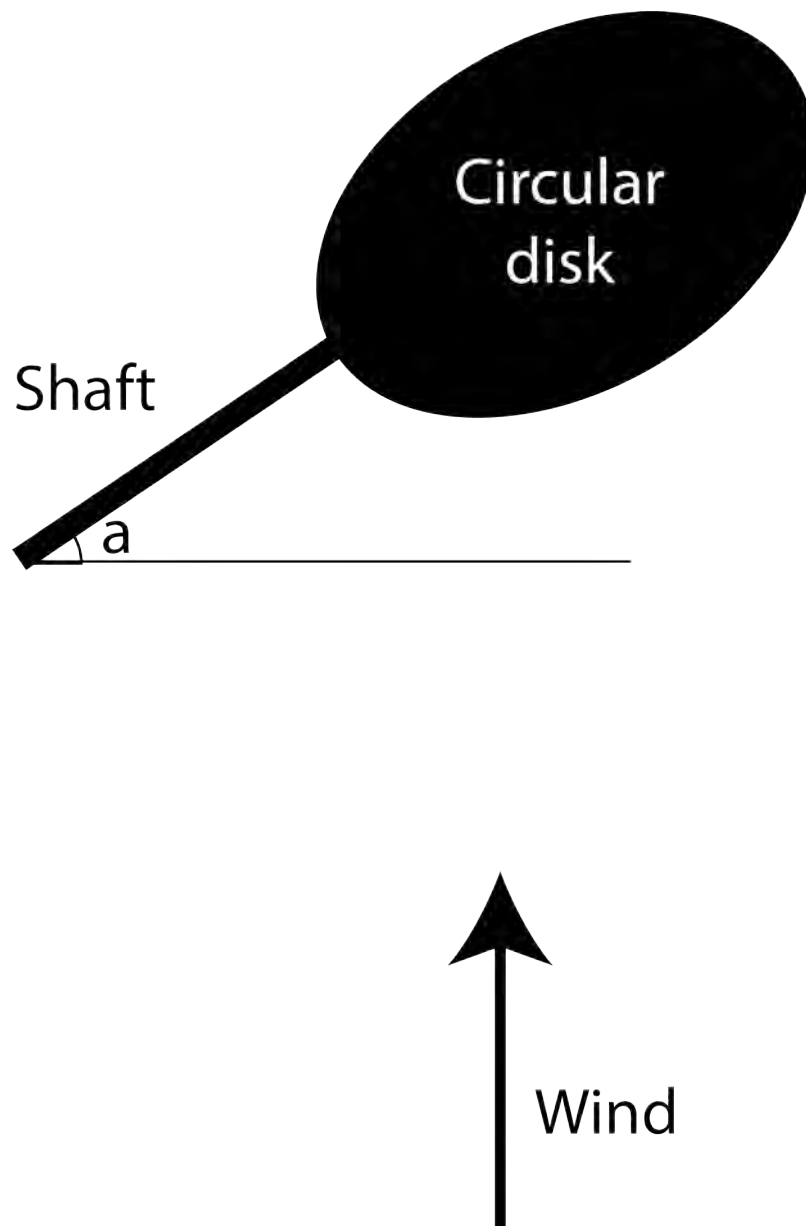


Figure 3.1.: Idealized wind speed sensor, composed by a shaft connected to a circular disk and exposed to a uniform airflow

3. The cup anemometer

Now let's imagine to rotate the shaft at an angle a as shown in Figure 3.1. The wind speed remains constant, but this time the force exerted on the now inclined disk will change, now the frontally exposed area $S(a)$ being $\pi r^2 \cos a$.

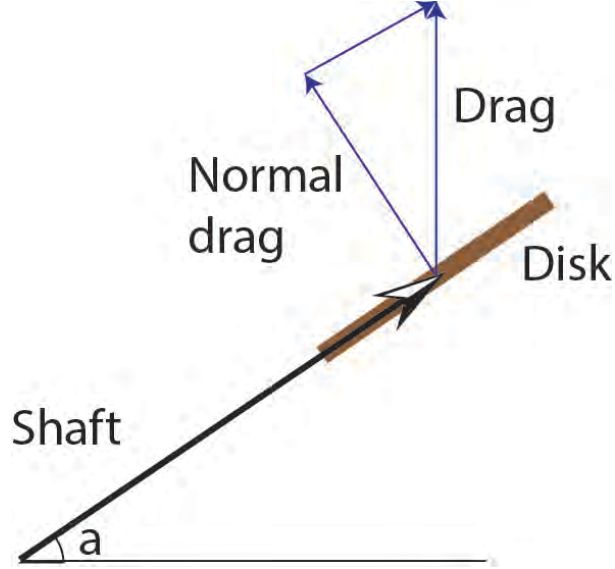


Figure 3.2.: Graphical representation of the forces generating the torque applied to the shaft

In Figure 3.2 we can see how the forces and geometry interact generating the torque. If we indicate with $\mathbf{F}_d$ the vector indicating the drag force, by $\mathbf{F}_n$ its component normal to the disk, and by $\mathbf{L}$ the vector from the junction point to the disk center, then we have

$$\mathbf{T} = \mathbf{L} \times \mathbf{F}_d$$

with $\mathbf{T}$ the torque vector, having magnitude

3. The cup anemometer

$$|\mathbf{T}| = |\mathbf{L}||\mathbf{F}_n| = |\mathbf{L}||\mathbf{F}_d| \cos b$$

where b is the angle formed by the two vectors $\mathbf{F}_d$ (“Drag” in Figure 3.2) and $\mathbf{F}_n$ (“Normal drag” in Figure 3.2). Now, with a bit of high-school geometry it is possible to prove $b = a$, so that the magnitude of the momentum can be written as $|\mathbf{L}||\mathbf{F}_d| \cos a$.

Let’s now introduce a signed torque.

The vectors $\mathbf{L}$ and $\mathbf{F}_d$ define together a plane (actually, the sheet Figure 3.2 is drawn on), and their vector product is then directed perpendicularly to it.

Then, assuming a three-dimensional reference system in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ lay on the sheet plane, we can write

$$|\mathbf{L} \times \mathbf{F}_d| = |\mathbf{L}||\mathbf{F}_d| \cos a$$

The angle a was defined as being 0 when the disk and shaft are perpendicular to the airflow, but this choice may not be convenient if we want to introduce the sign to the $\mathbf{k}$ component of torque.

A better choice is by adopting the complementary angle

$$\beta = \frac{\pi}{2} - a$$

and observing that if $\beta > 0$ the disk is placed on the “right” respect to the origin downwind, and on “left” if $\beta < 0$.

Now, by elementary trigonometry we have the identity

$$\cos a = \cos \left(\frac{\pi}{2} - \beta \right) = \sin \beta$$

3. The cup anemometer

This allows a sensible definition of the torque vector as

$$\mathbf{L} \times \mathbf{F}_d = \mathbf{k} |\mathbf{L}| |\mathbf{F}_d| \sin \beta$$

This allows to define the signed torque as the $\mathbf{k}$ component of the torque vector,

$$\tau = |\mathbf{L}| |\mathbf{F}_d| \sin \beta = \frac{l}{2} \rho U^2 C_d S(\beta) \sin \beta$$

The section $S(a)$ frontal to the wind flow is

$$S(\beta) = S_0 \sin \beta$$

where $S_0 = \pi r^2$, so we may finally write

$$\tau = |\mathbf{L}| |\mathbf{F}_d| \sin \beta = \frac{l}{2} s(\beta) \rho U^2 C_d S_0 r^2 \sin^2 \beta \quad (3.2)$$

where $s(\beta)$ is defined as the signum operator applied to $\sin \beta$.

Once we have the expression of the signed torque for one disk, we may generalize it to the case of $n > 1$ disks spaced by equal angles.

The total torque, then, can be written by summing Equation 3.2 over the partial torques:

$$\tau(\beta, n) = \frac{l}{2} \rho U^2 C_d S_0 r^2 \sum_{i=0}^{n-1} s\left(\beta + \frac{2\pi}{n}i\right) \sin^2\left(\beta + \frac{2\pi}{n}i\right) \quad (3.3)$$

If n is even, then for each disk presenting at some angle on the positive side there will be another presenting with the same cross-section on the negative side, so their contributions cancel out, and it is by necessity

$$\tau(\beta, 2m) = 0 \quad (m \in \mathbb{Z})$$

3. The cup anemometer

If $n = 2m + 1$, that is n is odd, then the summation $F = \sum_{i=0}^{n-1} s(\beta + \frac{2\pi}{n}i) \sin^2(\beta + \frac{2\pi}{n}i)$ does not cancel out, and we have situations like those shown in Figure 3.3.

If we integrate the total drag, or its coefficient F , over the interval from 0 to 360° we obtain zero: as mentioned, in the beginning of this section this abstract drag-only “disk anemometer” would have no practical utility.

It however gives us some practically useful information.

The first, maybe intuitive, is that would drag be the only force involves with a disk anemometer then the torque generated on it by wind would be always zero if the number of disks is even.

This is interesting, as historically the first cup anemometers had 4 cups.

A physical (and essential) reason four-cups anemometers can work despite their torque symmetry is that the torque-generating force in a cup anemometer is not only drag, but also *lift*.

In any cup anemometer the flat disks of our idealized “disk anemometer” are replaced with hemispherical or conical cups (other shapes are possible, by the way). Hemispherical or conical cups will offer a different drag coefficient C_d depending on whether they are exposed to wind front or back, and this may generate some non-zero torque also in presence of an even number of cups.

In addition, let's consider a single disk. If $\beta = 0$ or $\beta = 180^\circ$ then the disk is parallel to wind direction, and it is infinitely thin then no drag is generated. If $\beta = 90^\circ$ or $\beta = 270^\circ$ then the disk is perpendicular to wind, and drag is maximum. In either cases, however, the only force generating torque can be drag: because of the “flat” position of the disk respect to wind there cannot be any lift.

But in all other situations, that is if $\beta \neq \frac{k}{2}\pi$, then the disk has a non-zero and non- $\frac{\pi}{2}$ angle of attack, and *does* generate lift. The equation of lift magnitude L for a general airfoil is the same as Equation 3.1 with the drag

3. The cup anemometer

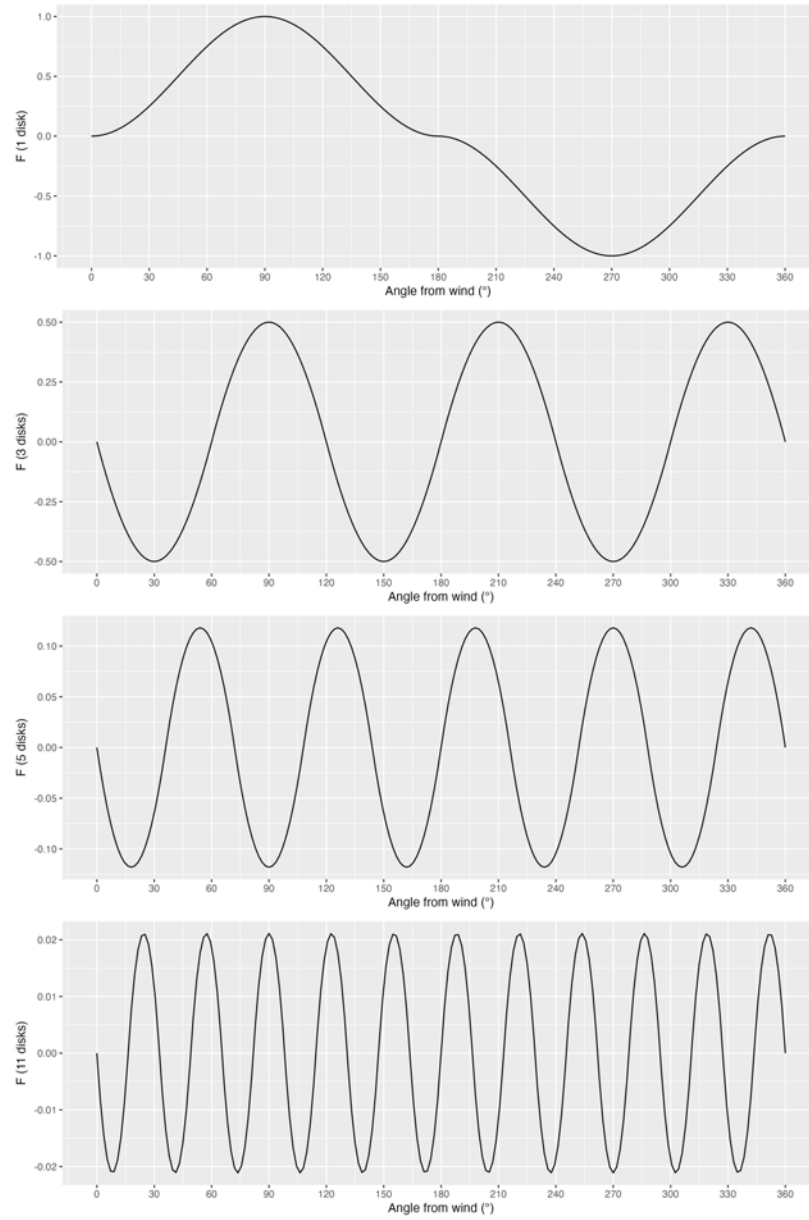


Figure 3.3.: Value of the drag anemometer coefficient F for the reference disk changing angle from current wind direction

3. The cup anemometer

coefficient C_d replaced by an experimentally determined lift coefficient C_l :

$$L = \frac{1}{2}\rho U^2 C_l \quad (3.4)$$

The lift coefficient C_l contains all dependencies on angle of attack, roughness and other factors, and is usually determined experimentally. NACA, the precursor of NASA, qualified this way many airfoil shapes including the case of flat surfaces, and data can be found in the World Wide Web. This level of detail is not important in this book, and is then not given here.

What's important, however, is that a flat airfoil as a disk generates lift only if the angle of attack is greater than 0, and up to a value $\beta_{\max}$ at which stall occurs; before than that, and for a wide range of possible angles of attack, airflow above the flat disk separates and all the lift is generated by the pressure increase generated below: as a consequence lift magnitude will be less than the one generated by a NACA wingfoil, in which case the suction generated above the wingfoil contributes to the total lift. The “low lift efficiency” of flat disks is, however, not relevant to our case: lift indeed occurs under the right angles of attack, that is, at some points in the disk rotation.

Again, because of geometric symmetry this effect will cancel out if the number of disks is even. It may not, depending on the details of C_l , with an odd number of disks; the details are complex and, as usually, not worth an in-depth analysis in this book.

But lift too becomes an important factor if a bent, and possibly curved, airfoil profile is adopted instead of a flat one, something certainly occurring as the disks are replaced with hemispheres or cones as it happens with real anemometers.

By the way, cup anemometer manufacturers soon abandoned the four cups design, and switched to the three cups design still in use in these days.

3. The cup anemometer

Figure 3.3 suggests one of the possibilities why they did so: if we consider drag forces only, we see the maximum torque soon decreases as the number of disks (cups) increases.

We also notice another thing: as the disk anemometer rotates the number of torque maxima also grows. The alternation of minima and maxima implies changes in the frequency of vibration a real cup anemometer rotor would be subject to, and it is likely this effect is larger when the difference between maxima and minima values is maximized, that is, when few cups are considered: selecting three cups instead of four may improve the anemometer's rotational efficiency, but may in the same moment diminish other performances: design, as always happen, is an art of finding the technical compromise which is best for a given use case, within specified budget constraints.

A point which has not been analyzed, but which is still important, is the amount by which disks, or cups, interfere one with each others.

Indeed, unless the number of cups is 1, there will be revolution angle values β at which at least one cup is downwind from another. At the high Reynolds numbers characteristics for air at mean sea level this “shielding” effect will have the consequence that one cup lies in part or totally within the turbulent wake of another, so that the optimistic interference-neglecting Figure 3.3 is likely to provide a wild overestimate of the total torque, especially if the number of cups is large.

Indeed, to my knowledge no one has ever built anemometers with 5 cups or more. But would one do, they might try to mitigate wake-related phenomena by adopting a shaft length $|\mathbf{L}|$ which is much larger than cup diameters. This possibility, besides worsening the anemometer's size clutter, would expose to another danger: as the cups are moved progressively away from the rotor center, they would be exposed to *different* wind speeds because of wind dis-uniformities. Ideally, a cup (or any other technology) anemometer should be as small as possible for both practical and measurement quality reasons, but once again this would increase aerodynamic

3. *The cup anemometer*

interference among different cups, and a compromise then needs to be taken.

The last thing we consider in the simple case of the “disk anemometer” is that if an instance of it is actually constructed starting from real-world material it would have a non-zero mass and moment of inertia (MOI). If for whatever reason the “disk anemometer” is put in rotation, its inertia will conserve and would induce rotation to continue, until some other forces (friction maybe) stops it. As we shall see this *is* a problem with real cup anemometers in conditions of slow gusty wind.

We could summarize this section by saying even an intuitively “easy” disk-based drag-only anemometer is complex - more than one could in the beginning suspect.

3.2. Design and physical principle

3.2.1. Outside appearance

Basically, modern cup anemometers have a simple appearance, as we can see in Figure 3.4.

Here we can also see the two most popular designs, one in which the cup and vane are combined into a unique integrated assembly, and another in which they are separate and intended for a nearby installation.

As you can see, the cups may be hemispherical or conical. In theory other shapes could be possible, especially as 3D printing is readily available, but hemispherical and cone designs are the only possibilities really considered.

As mentioned in Section 3.1, all the sensors produced and sold to date have three cups.

3. The cup anemometer

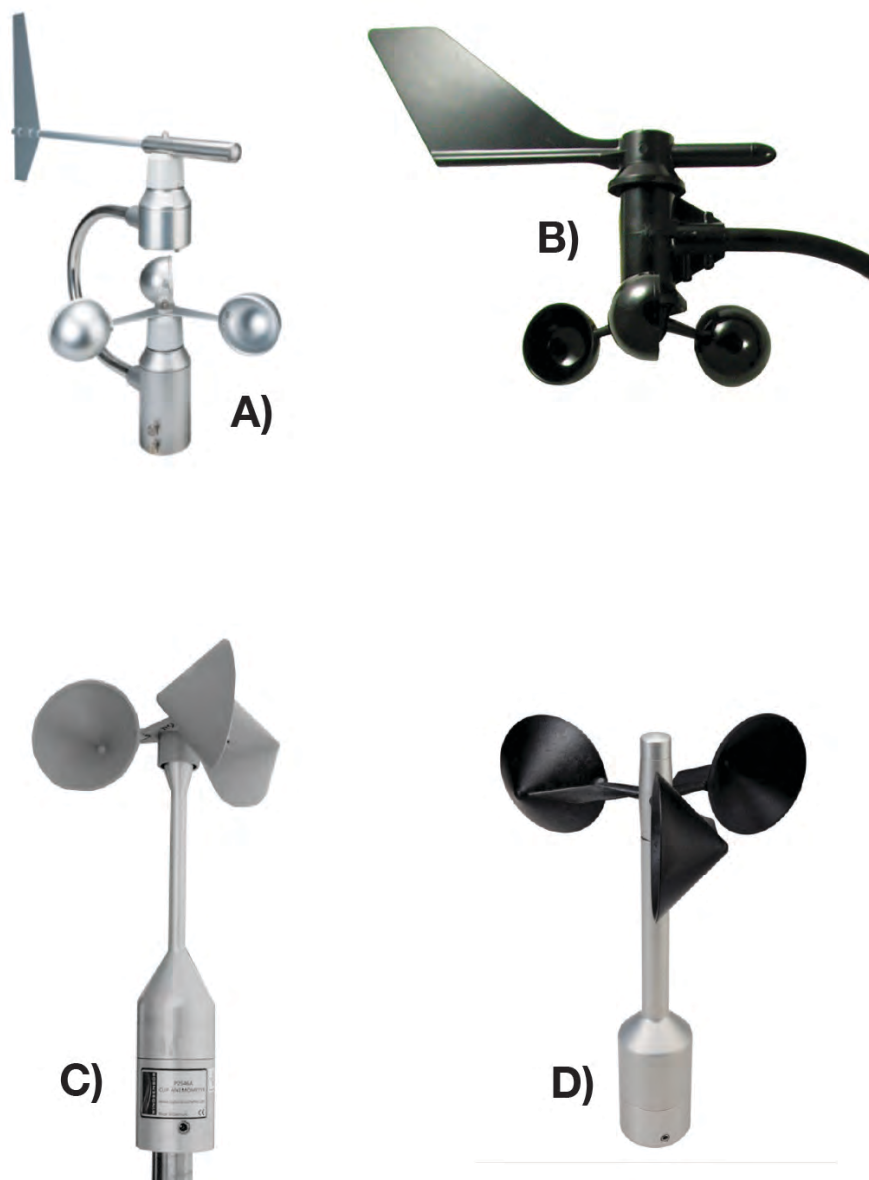


Figure 3.4.: Some examples of cup anemometers, combined with vane and self-standing: A) “Classic” model from Thies Clima; B) 7905 by Davis; C) P2546A Class 1 by WINDSENSOR; D) “First class advanced” model by Thies Clima; the sensors are not represented in scale

3. The cup anemometer

3.2.2. Cup anemometers classification

The various shapes and structures presented in Section 3.2.1 reflect the design decisions their manufacturers took in view of intended (or assumed most likely) uses.

As these uses are widely diverse (see Section 1.7) the need soon occurred of a classification and rating of the existing sensors.

Various classification schemes exist, and one in wide use is explained in considerable detail in (Dahlberg, Friis Pedersen, and Busche 2006), a report from the Danish RISOE institute.

As the development of classification schemes and the related facilities is expensive developments have been addressed towards high-value applications, which in case of cup anemometer actually means the wind power industry.

Any classification implies a value rating, but from what precedes it is clear this is application-dependent: a hypothetical model which is “class A” for wind power engineering may be worthless in other applications with different characteristics.

In the case of wind power engineering, the application-driven focus privileges the wind speed range at which medium and large size wind turbines can produce energy safely, that is indicatively from 4 to 20 m/s. No interest is normally allocated to “unproductive” speeds below 4 m/s, and this suggests to exert an extreme caution when interpreting the class indication often found in marketing materials and sensors data sheets. This is especially true in our case, where attention is given to low wind speed exclusively.

Nevertheless, classification schemes can provide highly valuable information, and consequently be worth a synthetic presentation.

3. The cup anemometer

In this section we'll employ the most relevant parts of the RISOE "ACCUWIND" scheme (Dahlberg, Friis Pedersen, and Busche 2006), whose articulation is summarized in Figure 3.5.

As can be seen, cup anemometer classification involves calibration and comparison to a reference theoretical sensor model, followed by a field test for refining or confirming the estimated anemometer's estimated performance parameters, the responses to real-world wind among them.

Calibration and field test, the way ACCUWIND scheme intends it, demand a laboratory facility equipped with complex and costly devices (like a wind tunnel for example), which is not our case. Yet, the analyses made in view of planning their use and making sense of test results may constitute a starting point for our considerations.

3.3. Cup anemometer dynamics

3.3.1. Introduction

In the ACCUWIND scheme the following reference equation for the cup rotor revolution is used, as a basis for further considerations:

$$T = I \frac{d\omega}{dt} = \frac{1}{2} \rho U^2 A R C_T \quad (3.5)$$

where T is the magnitude of torque, I is the rotor's moment of inertia, ω the angular speed, ρ the air density, U the wind speed, A the cup front area, R the radius to cup center, and C_T the torque coefficient. Some of the dimensions are shown in Figure 3.6 ($A = \pi \frac{D^2}{4}$).

In addition to Equation 3.5 an equation for the speed ratio λ is also given as

3. The cup anemometer

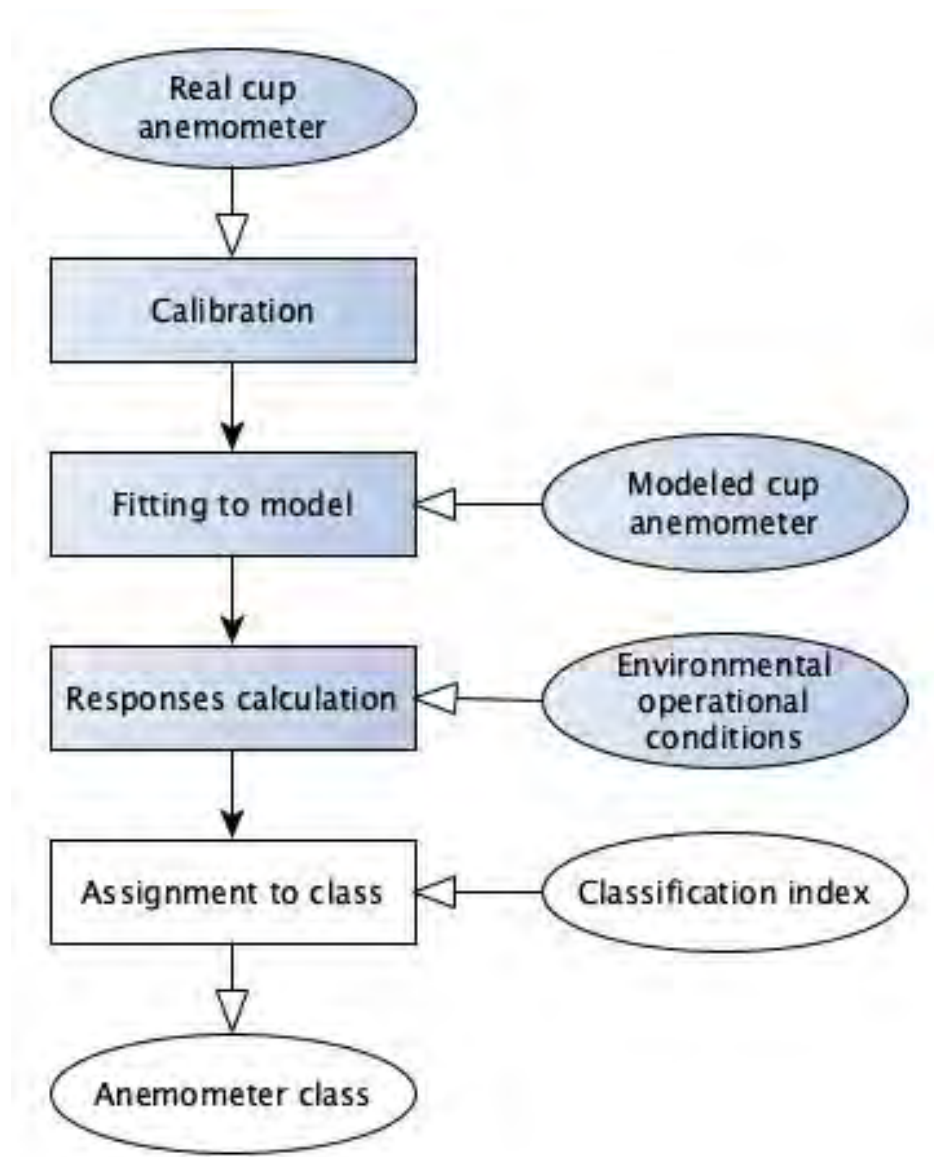


Figure 3.5.: ACCUWIND cup anemometer classification scheme (RISOE institute; the blocks in color are of immediate relevance to our cases)

3. The cup anemometer

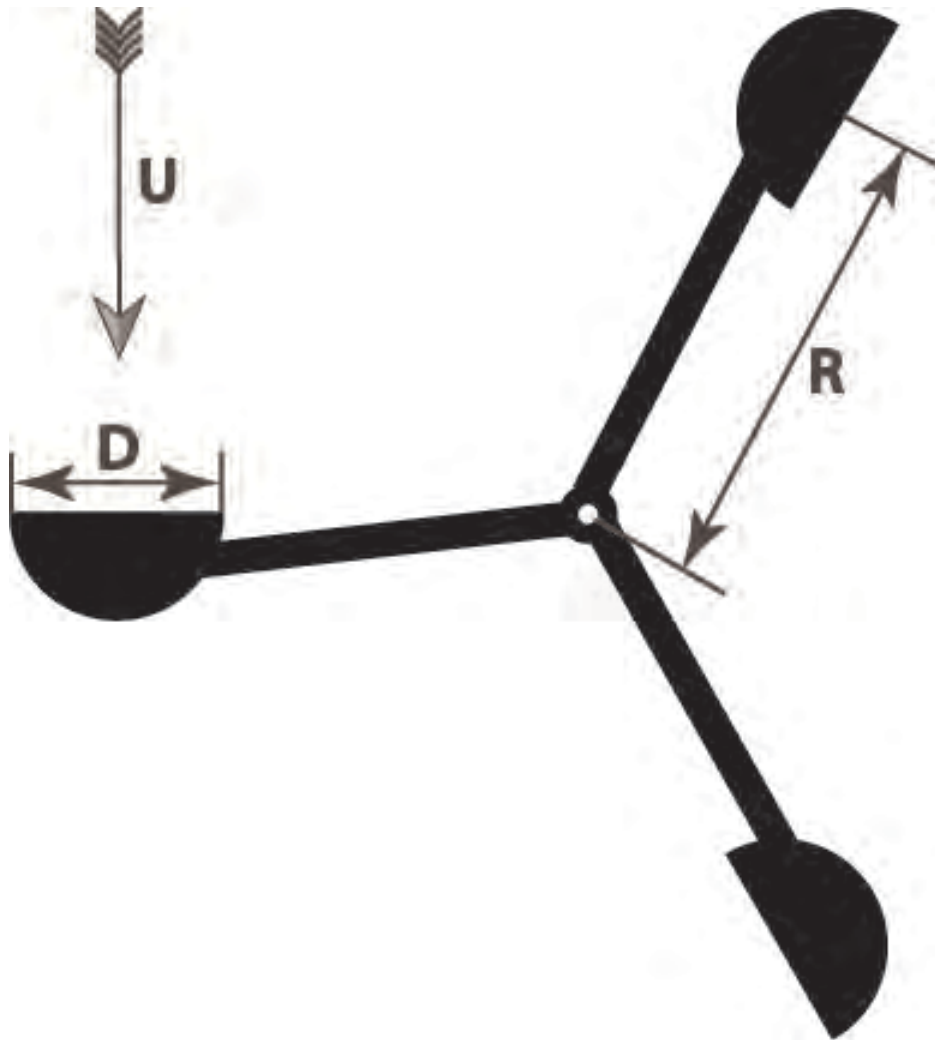


Figure 3.6.: Schematic representation of a cup anemometer with characteristic dimensions

3. The cup anemometer

$$\lambda = \frac{\omega R}{U - U_a} \quad (3.6)$$

with U_a the important “activation threshold speed”, that is the speed at which the anemometer begins its rotation. It is important to notice $U_a > 0$, due to the combination of bearings friction and aerodynamic effects (stall for example).

The torque coefficient is determined experimentally under diverse conditions, by solving Equation 3.5 after having replaced the derivative with an appropriate finite difference:

$$C_T = \frac{\Delta\omega}{\Delta t} \frac{2I}{\rho A R U^2} \quad (3.7)$$

Actually, the value of C_T is one of the main outputs of the calibration phase. As we have seen in Section 3.1 the torque coefficient is likely to contain information about both drag and lift forces, and any other mechanical effect a real object may have: in other terms we can expect its change with external conditions is complex.

Taken together, Equation 3.5 and Equation 3.6 permit to formulate a first-approximation dynamical model for the cup anemometer:

$$\begin{cases} \frac{d\omega}{dt} = \frac{1}{2} \frac{\rho U^2 A R C_T}{I} \\ \omega(U \leq U_a) = 0 \end{cases} \quad (3.8)$$

3.3.2. The cup anemometer as a first-order instrument

3.3.2.1. First-order transducers

A *first-order transducer* is a device mapping some time-dependent quantity $y(t)$, represented as a continuous band-limited function, into another

3. The cup anemometer

quantity $x(t)$ (possibly of an entirely different physical nature), whose response can be described through the first-order linear differential equation with constant coefficients

$$\tau \frac{dx}{dt} + x = y(t) \quad (3.9)$$

First-order sensors are quite common, thermometers being a paradigmatic example.

Knowing a sensor is first-order allows to predict a set of characteristic behaviors. For example, if the external forcing $y(t)$ is a step-function, that is $y(t) = 0$ if $t \leq t_0$, $y(t) = y_f$ otherwise, then the particular solution of Equation 3.9 with the condition $x(t_0) = 0$ is

$$x(t) = y_f \left[1 - \exp\left(-\frac{t - t_0}{\tau}\right) \right] \quad (3.10)$$

In Figure 3.7 we can see the shape of the curve $(t, x(t))$ for $t_0 = 0$ and $y_f = 1$, for some values of the time constant τ .

In Figure 3.7 we can see that as the time constant increases the time demanded by the response function $x(t)$ to reach a prescribed value after a step change occurred also increases: the time constant τ can be interpreted as a rough indication of the sensor response time.

First-order sensors have another interesting property: they act as low-pass filters on the input signal $y(t)$. To see this, imagine $y(t) = \sin(2\pi ft)$ with $f > 0$ a specified test frequency. The particular solution of the problem

$$\begin{cases} \tau \frac{dx}{dt} + x = \sin(2\pi ft) \\ x(0) = 0 \end{cases}$$

is

3. The cup anemometer

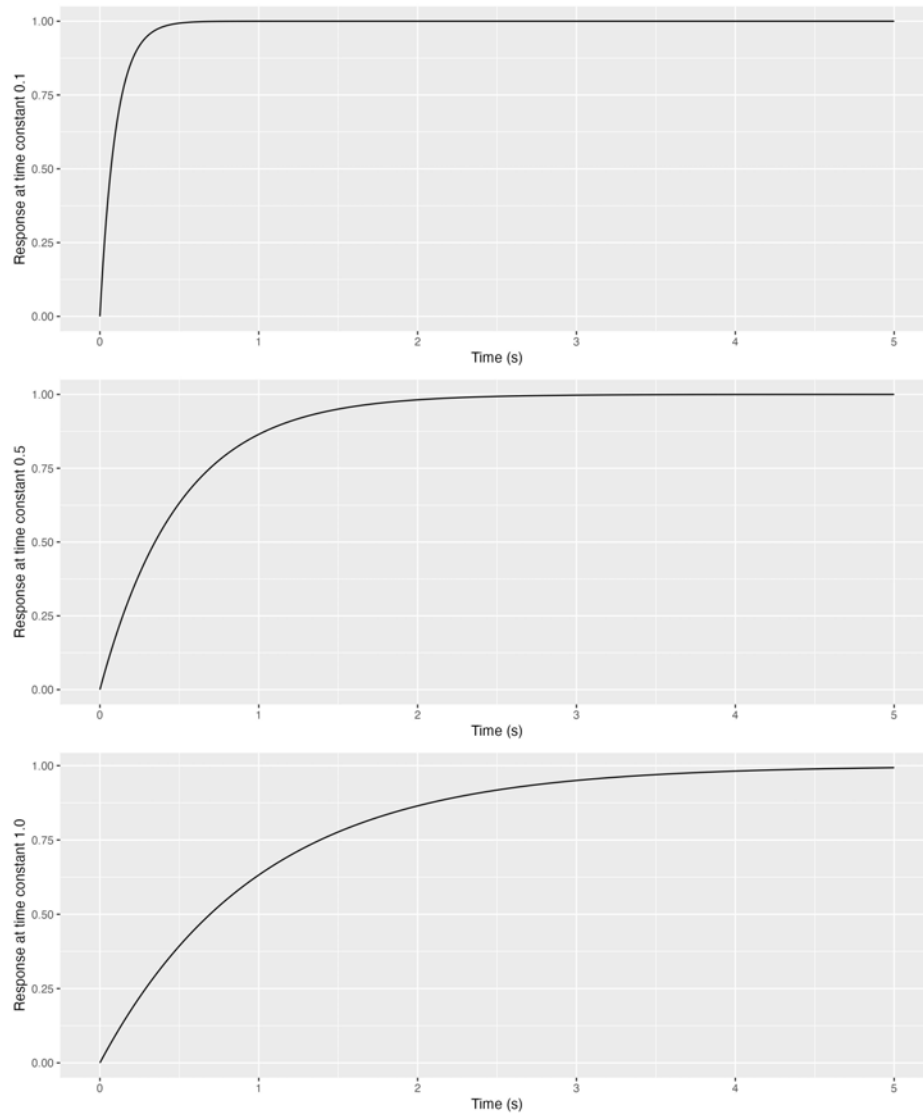


Figure 3.7.: First-order responses at different time constant values

3. The cup anemometer

$$x(t) = \frac{2\pi\tau f \exp\left(-\frac{t}{\tau}\right) - 2\pi\tau f \cos(2\pi ft) + \sin(2\pi ft)}{4\pi^2\tau^2 f^2 + 1} \quad (3.11)$$

For $t > 2\tau$ the term $2\pi\tau f \exp\left(-\frac{t}{\tau}\right)$ becomes so small it can be neglected in Equation 3.11. The remaining numerator part

$$-2\pi\tau f \cos(2\pi ft) + \sin(2\pi ft)$$

has amplitude

$$\sqrt{4\pi^2\tau^2 f^2 + 1}$$

and this allows to say the amplitude of Equation 3.11 is

$$A(\tau, f) = \frac{1}{\sqrt{4\pi^2\tau^2 f^2 + 1}} \quad (3.12)$$

Equation 3.12, seen as a function of either τ or f , and leaving the other parameter unchanged, tells the response tends to zero (attenuates) as the external signal frequency increases, with $A(\tau, f) \approx f^{-1}$. In other words, a first-order sensor behaves as a *low-pass filter*, whose damping effect is more and more evident, again approximately as τ^{-1} , with increasing values of the time constant.

3.3.2.2. Where the first order equation comes from in the case of cup anemometers

Equation 3.9 to represent the time behavior of a cup anemometer is used quite ubiquitously in literature, as we can for example see in (Sozzi, Georgiadis, and Valentini 2002; Kristensen 1998) and many others.

3. The cup anemometer

This makes sense in practical terms, but as we compare Equation 3.9 with the ACCUWIND equation @eq-dyna-model (Dahlberg, Friis Pedersen, and Busche 2006) we can easily see they differ significantly.

In particular, assuming (and this is quite a crude approximation) that in the differential equation Equation 3.8 the only changing part in the coefficient block is the wind speed, the differential equation can be written as

$$\frac{2I}{\rho ARC_T} \frac{d\omega}{dt} = U(t)^2$$

If we formally identify $\frac{2I}{\rho ARC_T}$ with the time constant τ , the former equation can be rewritten as

$$\tau \frac{d\omega}{dt} = U(t)^2 \quad (3.13)$$

which is quite different from Equation 3.9: the linear damping term, ω , is missing. If we blindly adopt Equation 3.13 as a dynamical model of the cup anemometer, its response to a step change or a purely sinusoidal oscillation would differ in important, qualitative ways from a first order sensor's. For reasons of space the detailed proof of this difference is not given here, but may be obtained quite easily applying elementary calculus (Lang 2002).

The core point here is in what I stated in parentheses: assuming

$$\tau = \frac{2I}{\rho ARC_T}$$

to be constant is a very crude approximation. Indeed, the only parts in the right member whose value is not changing with time are the factors depending only on the geometry and materials of the cup anemometer,

3. The cup anemometer

namely the moment of inertia I and the distance of the cup center from the rotor center R .

Anything else is subject to changes:

- The air density ρ depends on temperature, which is subject to turbulent fluctuations, and other factors (air pressure, relative humidity, ...).
- The frontal area A the cups present to airflow changes (predictably) as the cup rotate.
- The torque coefficient C_T depends on the cups angles of attack (whose value changes as the rotor revolves), the wind speed, the air viscosity, and many other factors.

If one of the preceding terms is treated as non-constant, the predicted solution $\omega(t)$ will change extensively, in a qualitatively relevant and complex way.

In short, if it is true that Equation 3.8 offers an accurate model for a cup anemometer, it can also be taken into account it is extremely difficult to use it in practice.

On the other side, one of the laboratory tests made during the ACCUWIND qualification process (Dahlberg, Friis Pedersen, and Busche 2006), the step response, originates a behavior which resembles the functions presented in Figure 3.7 superposed by a small oscillation due to the cups presenting to airflow at different angles as the rotor revolves. Now: if this small speed vibration is neglected, the remaining smooth curve can be approximated as a typical first-order response - interesting plots can be found in the freely downloadable ACCUWIND report (Dahlberg, Friis Pedersen, and Busche 2006).

If the experimental plots are suggesting a similarity with a first-order response, then it makes engineering sense to imagine the cup anemometer *is* a first-order sensor. This, however, is not a guarantee the first-order

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model is also universally useful and accurate: for instance, it does not take into account that a cup anemometer response is zero if $U \leq U_a$ for a sufficiently long time.

Said differently: the first-order approximation response is a powerful intuitive tool if one desires to develop a mental image of cup anemometers, but may lead to incorrect conclusions especially where it does not apply strictly, and particularly *in the low wind situations* where the condition $U \leq U_a$ is likely to occur.

Modeling simplifications are ubiquitous in engineering, and the adoption of a first-order response for cup anemometers is one of them. The experimental facts on which they are based are very interesting, and in most cases documented in detail in scientific papers or technical reports: for this reason I urge anyone having a practical interest to not step blindly to the model “all people use”, but to check thoroughly the foundational scientific assumptions the model are based on, and verify whether they are in line with their stated application objectives. Models are always a simplification of reality, and adopting one instead of another within a competing set is an exercise of responsibility, be one aware of it or not. Of responsibility and, would things go awry, *liability*...

In our specific case, it is very interesting and illuminating to see how the ACCUWIND step response of cup anemometers is measured.

As a first step, the anemometer is placed in a wind tunnel, and connected to a device which blocks its rotor.

Then the wind tunnel is started, and the airflow set to a constant $U > U_a$. A lot of care is taken to ensure the airflow is laminar upwind the anemometer. In this second step the cup anemometer rotor remains still.

The third step occurs instantaneously: the lock device is instantly removed, and the rotor, subject to a wind speed larger than the activation threshold, begins to spin.

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In the fourth step the angular velocity increases, superposed to small vibrations like those we have predicted in Figure 3.3. If vibrations are averaged-out the remaining smooth angular velocity converges from below to a value $\omega_{\max} = \alpha U$, where α is a factor.

From step two on, the angular velocity of the rotor is recorded at high speed; initially, its value is zero as the rotor is locked to still.

As anticipated, the experimental curve obtained has a shape closely resembling the ones in Figure 3.7. If this test is repeated at different set speeds U_i , $i = 1, 2, \dots, n$, then a sequence of corresponding settling times and terminal speed-to-angular-velocity conversion coefficients α_i are derived. These latter, once averaged out, yield the conversion factor α the anemometer manufacturers publish in their sensors' data sheets. What they typically do not show is the variation of the α_i values as a function of U_i , which would be an interesting element for deciding whether a specific cup anemometer model is suitable to one's own application or not; this information, anyway, should be available as part of the calibration certificate of the specific cup anemometer item, or at worst model.

We'll return to this important aspect later, but for the moment we can accept the sensor response, tested in this way, is well approximated by a first order response. The "modeling assumptions" we were forced to accept are:

- The upwind airflow is laminar.
- The first order behavior was tested at speeds not smaller than $U_{\min} = \min_{1 \leq i \leq n} U_i$; in the ACCUWIND procedure, originating from the wind energy industry, $U_{\min} = 1.5$ m/s.

As we'll see in later chapters, these ideal conditions do not hold in the field: in the Planetary Boundary Layer, above the first centimeter from the Earth surface, the airflow is always turbulent. And wind speeds in excess of 1.5 m/s seldom occurs, especially at sites placed deep in the Po Valley.

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Besides, what the ACCUWIND tests actually say is not “cup anemometers are first-order sensors”, but only their response seems so under extremely specific, non-real-World circumstances.

3.3.2.3. The terminal angular velocity

Figure 3.6 shows the characteristic dimensions of a cup anemometer. Among them the characteristic radius R , the distance between the sensor revolution center and the center of a cup, allows us to say that if the anemometer rotated at angular velocity ω the center of the cups will move at a velocity ωR directed, for each cup, perpendicularly to their rods.

Let’s now assume the rods are perfectly streamlined and offer no resistance to the passage of wind, and all the force exerted by wind occurs at cups, in the same direction as wind and with a magnitude depending on the cup angle to incoming airflow. We have then a situation like the one shown in Figure 3.8.

Although this conceptual model is highly simplified respect to reality, it nonetheless allows to make some qualitative prediction.

Indeed, the angular velocity tends to an asymptotic limit in the wind tunnel step change test. Besides the fine details of the transition from the initial $\omega = 0$ to the final $\omega = \omega_{\max}$, we can see an angular acceleration takes initially place, to wane out as the terminal angular speed is reached.

It makes sense to imagine this angular acceleration be a function not of the wind speed magnitude U itself (otherwise the rotor would keep accelerating indefinitely at a constant rate), but rather on the *difference* between U and the cup speed ωR in correspondence of some reference angular position of the rotor (for example when one cup faces wind perpendicularly).

As the angular speed of the accelerating anemometer increases, this speed difference decreases. In the limit $t \rightarrow +\infty$ this speed difference will have reached its terminal value ΔU .

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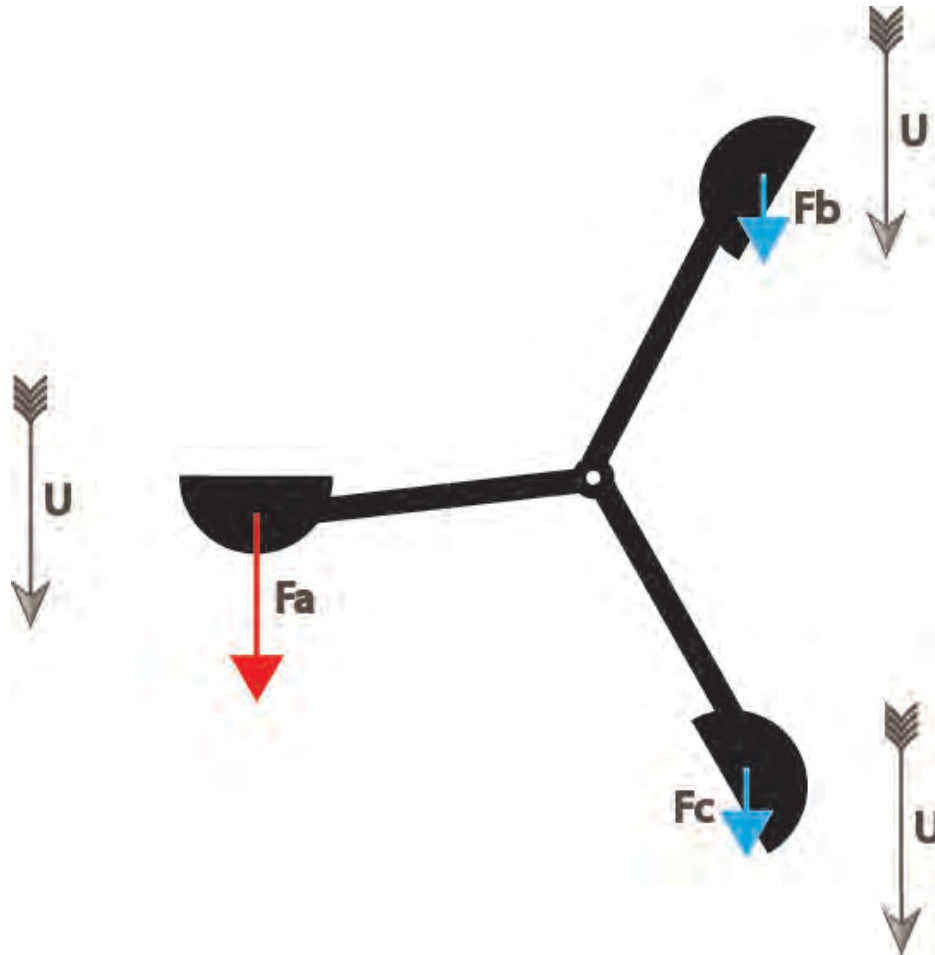


Figure 3.8.: Forces acting on a cup anemometer; only active forces are shown, with different colors depending on the sign of their contribution to the total torque

3. The cup anemometer

Intuition demands this terminal speed difference will be positive, because the cups rotating with wind will contribute the total torque with a value whose sign is opposite than the cups moving against the wind. This may not be necessarily true however, depending for example on the detailed shape of the cups, and the way it presents to wind as they rotate.

The visual consequence of the existence of a non-zero, and likely positive, terminal wind difference is that the speed of cups is in general different (and typically slower) than wind itself.

As I mentioned this model is quite primitive and inaccurate, and oscillations around the terminal angular velocity can be expected because of changes in the cup front shape during rotor revolutions. Nonetheless, the qualitative effect can be seen in the field.

3.3.3. Turbulence effects

Early investigators interested in studying the quality of data from cup anemometers concentrated on overspeeding effects, which were detectable since those times (Patterson 1926; Brazier 1914). The behavior of cup anemometer under field, and in particular turbulent, conditions came much later (Kristensen 1998), motivated by the ubiquitous presence of turbulent fluctuations within the Planetary Boundary Layer, the part of atmosphere within of which the cup anemometers are actually used.

Before to shortly expose the main conclusion of the seminal paper by dr. Leif Kristensen (Kristensen 1998), it is interesting if the current section is just an academic fixation by an unknown professor, or it is a real field issue with technical ramifications.

The amount of raw turbulence is typically expressed by the *turbulent kinetic energy*

$$TKE = \frac{1}{2} (\overline{u'u'} + \overline{v'v'} + \overline{w'w'})$$

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where the overlines represent means as defined in Chapter 2, and the prime symbols the fluctuations, also defined in Chapter 2. Upon applying the definition of mean of primed quantities, one can see that $\overline{u'u'} = \sigma_u^2$, and similarly for all other wind components - a thorough presentation can be found in (Stull 1988; Kaimal and Finnigan 1994; Sozzi, Georgiadis, and Valentini 2002).

As the turbulent kinetic energy is half the trace of the covariance matrix, linear algebra guarantees its value does not change as the reference system is rotated (Landi and Zampini 2018; Golub and Loan 1996). So, its value can be obtained if a set of high speed and high resolution wind vector measurements are available using directly the instrument frame of reference, without applying eddy covariance rotations (Aubinet, Vesala, and Papale 2012). The only precaution, when calculating the variances, is to remove any linear trend in advance, as it may lead to overestimates.

Would the TKE be zero, then we would have nothing to worry about. But if the TKE is non-zero, then the conclusions of Kristensen's paper apply, and it would be better to take them into proper account.

In Figure 3.9 we can see the value of TKE, evaluated every 30s on two days at a micro-meteorological station located close to Milan. The characteristics of the station, and the sensors hosted by it, are presented in detail in Chapter 5, and are not repeated here.

As can be seen in Figure 3.9, turbulence intensity does present. Its maximum value is smaller on Winter than in Summer (inbound solar radiation is much less in Winter at Cinisello Parco Nord site), and overall value is larger on daytime (with some nocturnal bursts). But besides the rich dynamics, the important thing is turbulence is present.

For comparison, in we can see the overall kinetic energy, defined as

$$MKE = \frac{1}{2} (\overline{u^2} + \overline{v^2} + \overline{w^2}) \quad (3.14)$$

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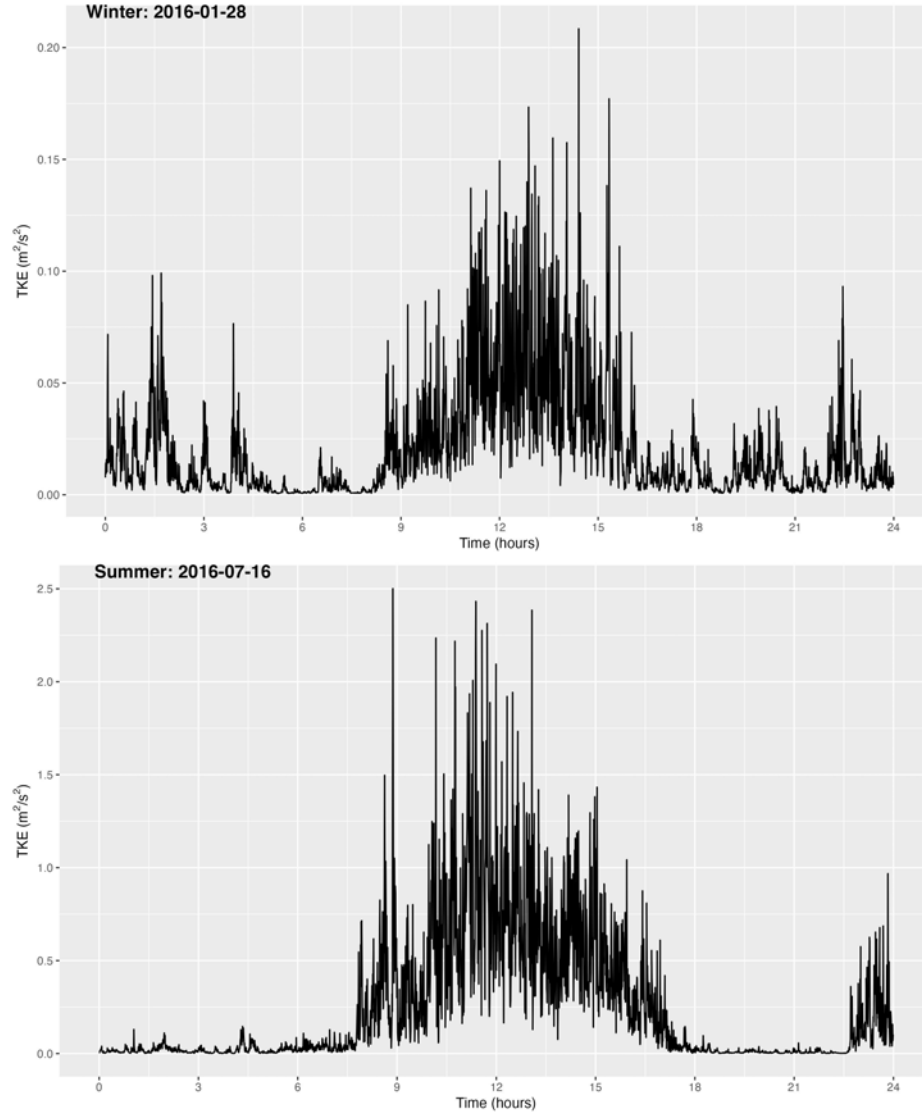


Figure 3.9.: Turbulent kinetic energy per unit mass on two days, representative of winter and summer, at the micro-meteorological station of Cinisello Parco Nord (courtesy: ARPA Lombardia)

3. The cup anemometer

(Oke et al. 2017). In Figure 3.10 we can see it on the same two reference days: higher values can be seen during day-time, as often happens at Cinisello Balsamo Parco Nord site.

Last, in Figure 3.11 we see the ratio of TKE to MKE. As we can see, in the Winter day we see most of the kinetic energy is stored in mean wind, but the opposite happens in the Summer day.

Individual days may not be representative of a mean situation, but this difference is consistent with the intuitive expectation that in Summer the turbulence of thermal origin is higher than in Winter.

So, in general, turbulence is present and then has an opportunity to perturb cup anemometer measurements.

In (Kristensen 1998) some conclusions are drawn on the effects turbulence has on cup anemometer instantaneous readings, and are shortly summarized here:

- Under turbulent condition the cup anemometer overestimates the real wind speed.
- The bias can be decomposed in three major contributions, respectively from the variances σ_u^2 along the wind direction, σ_v^2 along the cross-wind direction, and σ_w^2 along the vertical direction (these three reference directions can be obtained by appropriate rotation of the instrument reference system, as shown in (Aubinet, Vesala, and Papale 2012; Sozzi and Favaron 1996)).
- In addition to the biases due to variances, a smaller one due to $\overline{u'w'}$ is present; its value may be positive or negative.
- Various models can be defined for estimating the bias extent using only measurements from conventional anemometers; these models however demand detailed experimental data not usually collected

3. The cup anemometer

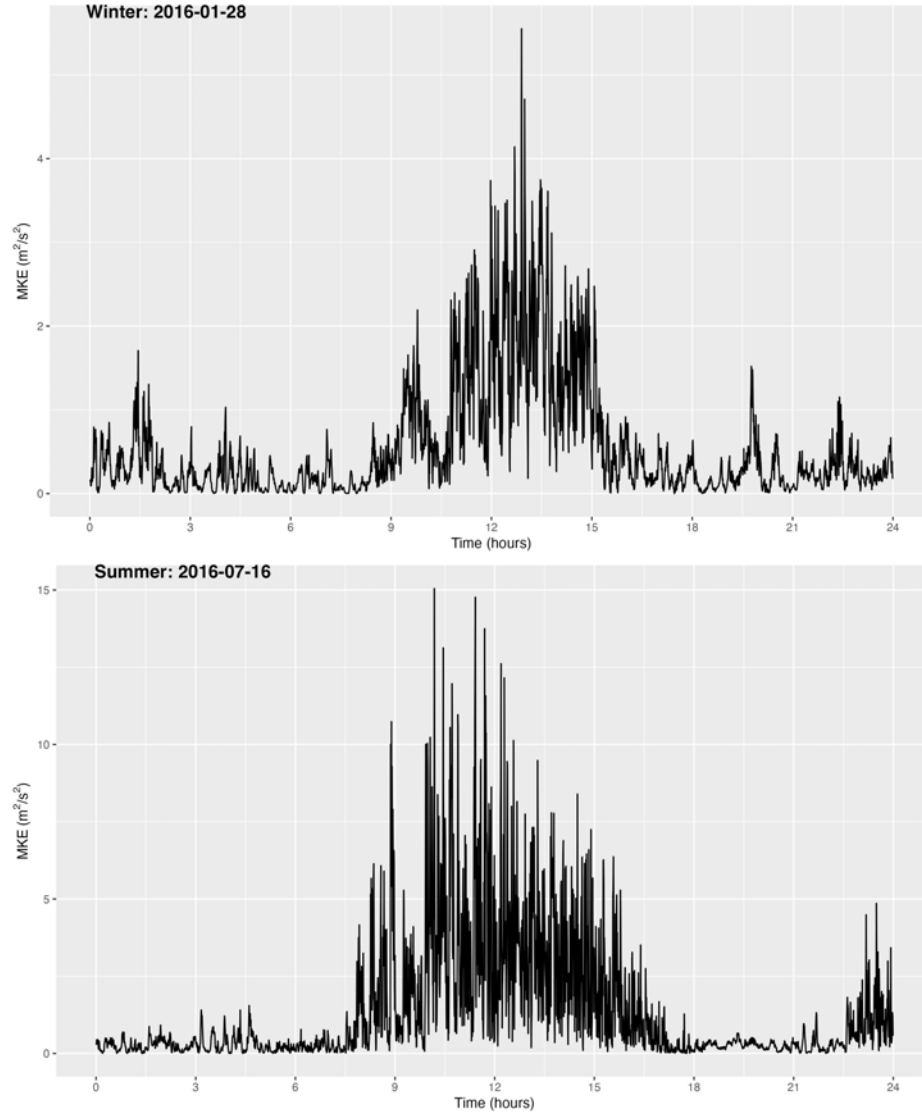


Figure 3.10.: Turbulent kinetic energy on two days, representative of winter and summer, at the micro-meteorological station of Cinisello Parco Nord (courtesy: ARPA Lombardia)

3. The cup anemometer

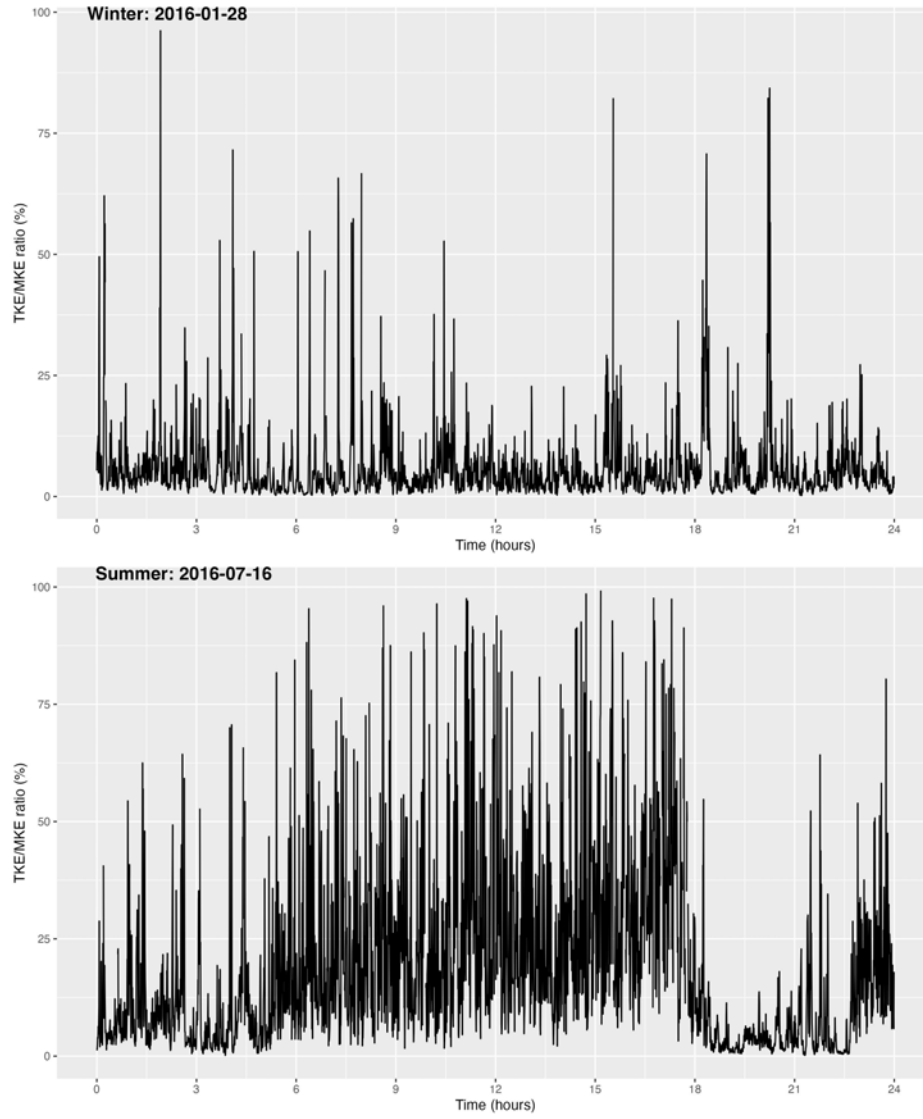


Figure 3.11.: Ratio of TKE over MKE on two days, representative of winter and summer, at the micro-meteorological station of Cinisello Parco Nord (courtesy: ARPA Lombardia)

3. The cup anemometer

during calibration and classification trials, and if available, they require sophisticated data processing. To the author knowledge, to date these corrections are not implemented in data loggers.

- In line of principle it is possible to design cup anemometers which, although overestimating the mean speed, are able to reproduce fluctuations along the wind direction (Frenzen 1988).

3.3.4. Changes in rotation speed due to tilt

The ACCUWIND classification procedure (Dahlberg, Friis Pedersen, and Busche 2006) contains specific tests for assessing the effect of a tilt in cup anemometer mount, or equivalently of a non-zero vertical wind component w .

That the instantaneous readings of the w component taken using a high resolution and fast sampling rate sensor can be different from zero, and even by a large extent, can be seen in Section 2.4.2.2.

That the vertical *mean* wind could be also non-zero is another question. An example addressing to the answer can be found in the same Section 2.4.2.2, where a non-zero mean vertical wind can be clearly seen (on some times reaching even 60° from the horizontal plane).

So in general the tilt effect on cup anemometer readings cannot be neglected when field measurements are evaluated. Graphs illustrating the measurement perturbations due to the cup anemometer tilt in case of a specific sensor can be found in (Dahlberg, Friis Pedersen, and Busche 2006).

3.3.5. Response under low wind speed

All the facts and considerations exposed in the preceding section of this chapter are strongly biased towards relatively high wind speeds.

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In the case of instantaneous wind speeds close to the cup anemometer activation threshold U_a an additional element should be added.

As we have seen in Section 3.3.2.2, the wind tunnel calibration procedure from which a first-order type response appears assumes the wind speed is first set to a constant value, and then the previously locked rotor is released.

As mentioned, the test is made at various pre-set speeds and a calibration line is then inferred. For the tests to be significant, these preset speeds should be larger than the sensor activation threshold U_a .

In practice, however, the exact value of U_a is unknown when a cup anemometer is first classified. The problem then arises to find it.

Experiments suggest that under laminar flow conditions, and neglecting first-order effects, the indicated speed U can be written (Yahaya and Frangi 2004) as

$$U = l \cdot \omega + U_0 \quad (3.15)$$

with l a length scale depending on the sensor geometry and U_0 a threshold speed. Equation 3.15 can be rewritten to

$$\omega = \frac{U - U_0}{l}$$

valid for $U \geq U_0$ (when $U < U_0$ it is $\omega = 0$): the angular velocity ω shows an approximately linear relationship with the indicated speed U , which is shifted from the ideal line by the term U_0 .

This so being, it is in principle possible evaluate the sensor response at some (no less than two) preset speed, fit the experimental points so found to these experimental data, and find the point where the regression line intercepts the U axis. This would allow finding the “activation threshold” U_0 without the need to set many low-speed experimental points.

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If these low-speed points are introduced, however, one finds the actual relationship between wind speed and the rotor angular speed is non-linear at low speeds, and the actual activation threshold U_a is larger than U_0 (Kristensen 1998), as can be seen in Figure 3.12.

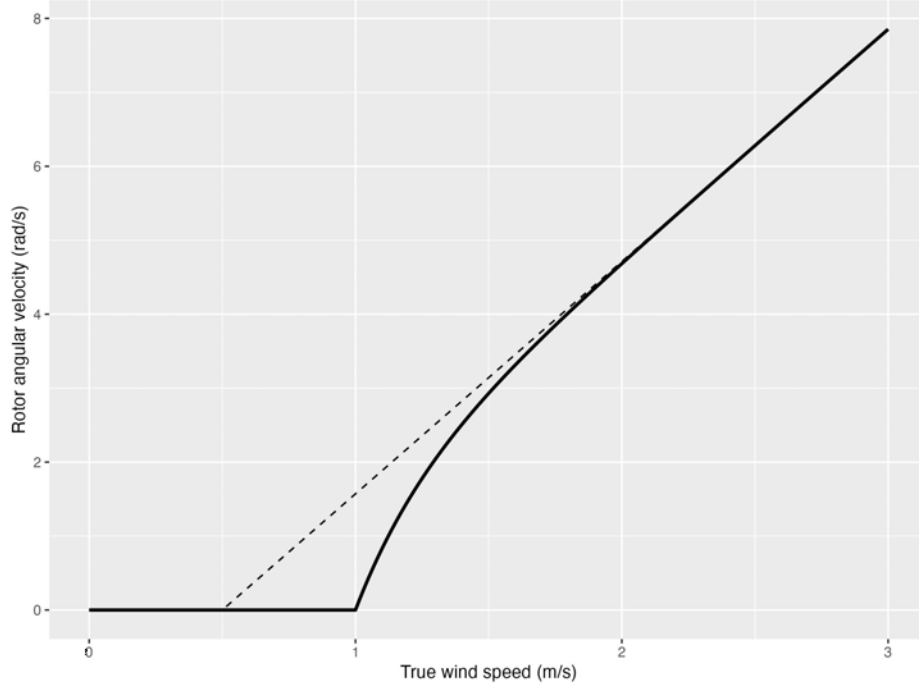


Figure 3.12.: Example response curve of a hypothetical cup anemometer with nominal activation threshold 0.5 m/s, actual starting wind 1 m/s, characteristic length 1/3.1415 and damping factor 4

The actual distance of U_a from U_0 depends in general on many factors, among which

- The air temperature (low temperatures may increase lubricant vis-

3. The cup anemometer

cosity).

- The cup anemometer natural wear-out, along with its maintenance state.
- The design details of the cup anemometer model considered.
- Ice and dirt deposits.

And more.

3.3.6. The distance constant

As mentioned in previous sections in this chapter real-world cup anemometers respond to changes in wind speed and overall conditions, and this response is not instantaneous.

When a cup anemometer is tested for speed step change as shown in Section 3.3.2.2, the response time τ (defined as the time necessary to the rotor to reach a given fraction - usually the 63% - of its terminal angular velocity counted from the instant in which the rotor is left free) is found to change with the value of the constant speed U set in the wind tunnel according to:

$$\tau = \frac{l_0}{U} \quad (3.16)$$

with l_0 the *distance constant*, usually reported in sensor data sheets. The larger the distance constant, the longer the response time.

The distance constant value depends mainly on the cup anemometer geometry and materials, which are fixed by design. Other dependencies may be expected from operating conditions (for example air temperature and maintenance practices), but there is some consensus on them being relatively small: it makes sense to consider l_0 practically constant. The

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inverse proportionality shown in Equation 3.16 implies the existence of a maximum response time

$$\tau_{\max} = \frac{l_0}{U_a}$$

The maximum response time is not normally published in sensor data sheets, but can be readily estimated by the user.

3.3.7. The cup anemometer as a low-pass filter

In cases when the mean wind remains constant or changes little with time, as we have seen in Equation 3.16 we can imagine the cup anemometer as an approximate first-order instrument having constant or almost response time τ .

This effect can be simulated (Sozzi, Georgiadis, and Valentini 2002) by a recursive filter (McMillen 1988) having the form

$$\bar{U}_i = (1 - \beta)U_i + \beta\bar{U}_{i-1} \quad (3.17)$$

where $0 \leq \beta \leq 1$ is a real constant whose value can be expressed as (Sozzi, Georgiadis, and Valentini 2002)

$$\beta = \exp\left(-\frac{\Delta t}{\tau}\right) = \exp\left(-\frac{\Delta t \bar{U}}{l_0}\right)$$

where Δt is the (constant) time spacing between U_i and U_{i+1} , and τ the anemometer's time constant defined in the previous sections in this chapter.

The time constant gets slower and slower as $\bar{U}$, the real mean wind, approaches to the activation threshold U_a from above.

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This fact is interesting: a cup anemometer's response time would be longest in correspondence of low, but detectable, wind.

Equation 3.17 can be seen as a smoothing operator mapping the real wind speed, including its short time scale fluctuations, to a less variable changing version of it.

We could say that Equation 3.17 transforms the original signal, and its entire spectral contents, into another in which higher frequencies are diminished: assuming Equation 3.17 represents the dynamic response of a cup anemometer accurately under constant mean wind, then it means the cup anemometer acts as a (complex) mechanical-aerodynamical low-pass filter.

Before to proceed it is worth mentioning low-pass filtering of wind may be highly desired in many applications, where short scale wind fluctuations are considered irrelevant.

And the fact the low-pass filtering becomes less aggressive as wind speed increases may even be beneficial in others, like for example wind turbine engineering, where turbulence (maybe induced by the intense wind distortion caused by other wind turbines upstream) may pose serious maintenance troubles at the high speeds when wind farms are productive, and the mechanical effects of sudden wind changes are maximized.

The opposite happens, however, in all applications in which the *slow* wind is of special interest, as in chemical hazard management or defense.

And last, it may be interesting to see what the effect of the smoothing operator Equation 3.17 can be on a real signal, depending on common cup anemometers' response times.

In Figure 3.13, together with Figure 3.14, we can see the effect on measurements and their power spectrum of low-pass filtering at two different wind speeds.

Figure 3.13 and Figure 3.14 show that the effect on the signal in the time domain and the spectrum is more pronounced in the first case, Figure 3.13

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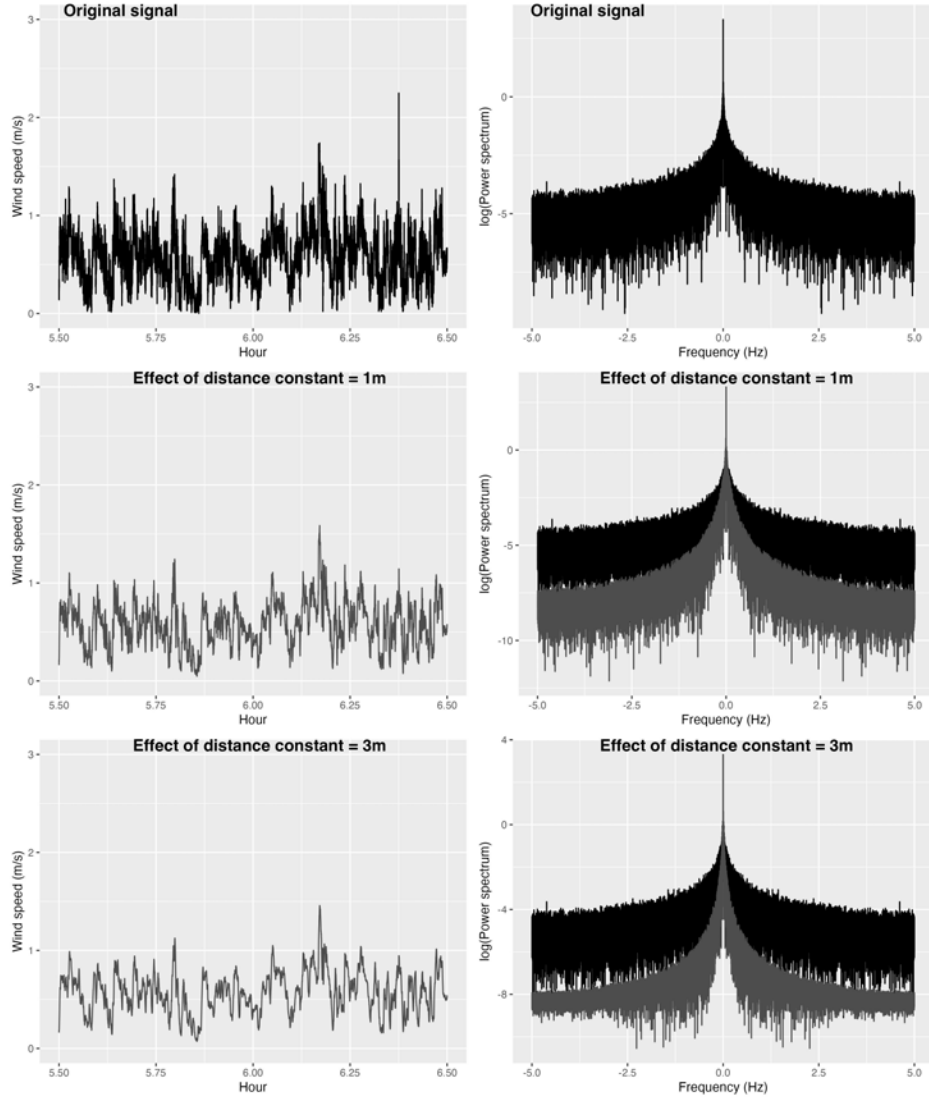


Figure 3.13.: Effect of low-pass filtering on an original signal and its power spectrum due to different values of the distance constant (mean wind speed is 0.58 m/s; signal and spectrum of perturbed signal in grey; data processed from ultrasonic anemometer measurements taken on 2016-07-16 at Cinisello Parco Nord)

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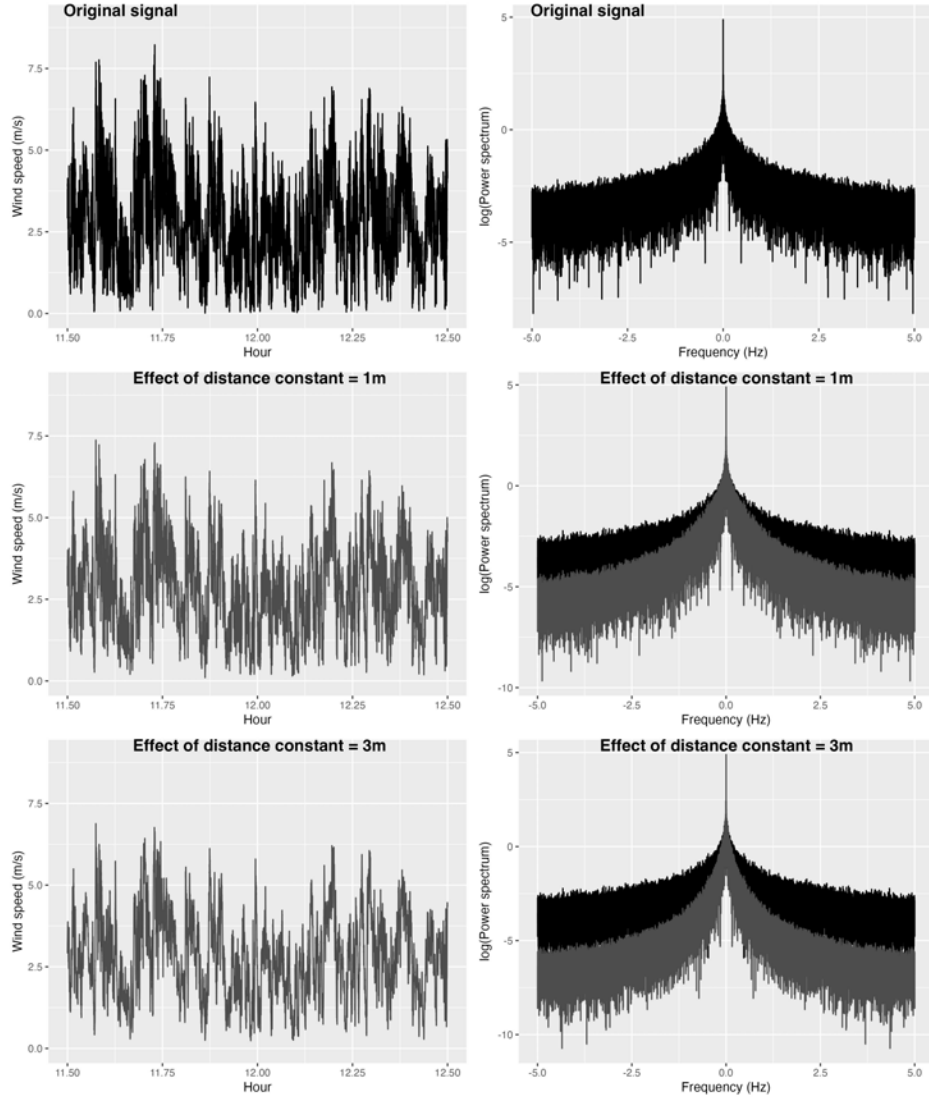


Figure 3.14.: Effect of low-pass filtering on an original signal and its power spectrum due to different values of the distance constant (mean wind speed is 2.83 m/s; signal and spectrum of perturbed signal in grey; data processed from ultrasonic anemometer measurements taken on 2016-07-16 at Cinisello Parco Nord)

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being characterized by a slower mean wind speed. The following table shows the values of β , and the response time τ , depending on the various conditions.

	$l_0 = 1\text{m}$	$l_0 = 3\text{m}$
$\bar{U} = 0.58 \text{ m/s}$	$\beta = 0.94, \tau = 1.73 \text{ s}$	$\beta = 0.98, \tau = 5.19 \text{ s}$
$\bar{U} = 2.83 \text{ m/s}$	$\beta = 0.73, \tau = 0.35 \text{ s}$	$\beta = 0.91, \tau = 1.06 \text{ s}$

The effect of a change in the mean wind speed on the response time is, as we can see, quite severe.

3.3.8. Response to a step-down

The cup anemometer response to a step-up function, whose experimental justification is exposed in Section 3.3.2.2, leads to consider cup anemometers as approximate first order sensors (with a response time which is not fixed, and rather depends on wind speed).

Investigators (Pedersen and Dahlberg 2024) have also posed the question of what happens if, instead of suddenly increasing, the wind speed decrease.

A real step-down is extremely difficult to reproduce in a wind tunnel, and experimenters oriented towards simulating a sinusoidally varying speed.

A diffused opinion is that at low wind speed a sudden decrease in wind velocity is accompanied by a decrease in the rotor angular velocity; it has been also hypothesized this decrease occurs slower than when the wind speed increases, possibly contributing to an increase of overspeed.

In the extreme of wind passing suddenly from a value significantly above the activation threshold U_a to another significantly below the rotor may keep rotating for some time before the small internal friction and the aerodynamic drag eventually stop it.

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In this specific case the speed overestimation may be large, and cannot in principle be neglected.

This example shows by the way the possibility that, under some circumstances, and slow wind in particular, the cup anemometer dynamics may be affected by phenomena whose occurrence is normally negligible under stronger speed. In particular, the (normally occurring) difference between sliding and rolling friction may need to be taken into account explicitly.

That ball bearings may be subject to significant sliding friction maybe sounds unusual (ball bearings are effectively employed to reduce friction among moving parts since many decades), yet the fine dynamical details of lubricants (possibly missing or degraded), seals and balls allow for it. Like cup anemometers, ball bearings are mechanical devices whose operation is intuitive on a first superficial glance, but which quickly reveal their complex nature as details are taken into account (SKF, n.d.). More later in this chapter on this important aspect.

All this said, the question remains on whether the *degree* of differences between step-up and step-down response is significant, or not.

In view of this, the author has done a little experiment using a cup anemometer (a nameless amateur-class sensor mass-produced by a Chinese manufacturer and readily available from major on-line retailers, shown in Figure 3.15) and a hair-drier.

This sensors, of the combed disk type, has 20 prongs and a rate of air path length to revolution equal to 1.75 m/rev; the measurement range is declared to be 0 to 70 m/s. No data is available at the distributors' sites for the sensor activation threshold, and other calibration data (the device, as often happens with Chinese products, is produced by one or few large factories in large quantities, and then sent to first-level distributors who may apply their own trademark - in this case the sensor I got remained unlabeled; this does not mean the device was not certified: maybe it has been, possibly by the original manufacturer, but it is not normally made available to users, especially in the amateur class).

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Figure 3.15.: The cup anemometer model used for visualizing the difference between step-up and step-down response times

3. *The cup anemometer*

The trial was made according to the following procedure:

- The anemometer is connected to the data acquisition system (an Arduino Uno R4 WiFi, equipped with a counter program yielding the number of up and down voltage changes every second).
- The hair-drier is switched to on, and the airflow is directed from about 70cm to the shaft of the anemometer, whose rotation is blocked by hand.
- Data acquisition starts, and the locked anemometer response (about 0 m/s) is recorded for some seconds.
- The anemometer is unlocked, while the airflow remains steady. Enough time is given to the anemometer to reach a stable speed reading.
- The hair-drier airflow is suddenly deflected so the anemometer is left rotating in still air.
- Data recording continues until a zero reading is stably attained.

As can be seen, the procedure is far from providing a real measurement of the sensor response times. Yet it gives a useful indication on the time scale of the effect mentioned in this section.

In Figure 3.16 we can see the sensor dynamics.

The speed increase phase ranges from second 3 to second 11, totaling 9 seconds. The airflow was deflected on second 17, and the zero count is reached on second 46, totaling 30s: the step-down settling time is about three times the step-up.

Figures may differ for professional and research grade sensors, and in a wind-tunnel test made as specified in literature, but a warning is given: the difference may be as large as a half order of magnitude.

The interesting fact stems from the fact that in a real first-order sensor, the response time would be identical for a step-up and a step-down. This

3. The cup anemometer

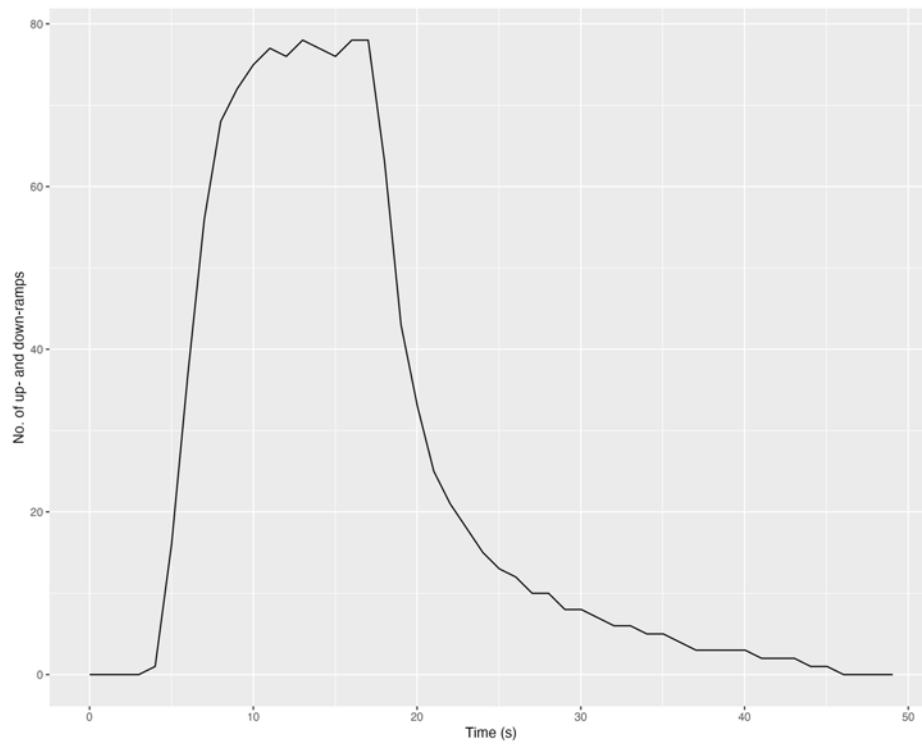


Figure 3.16.: Cup anemometer output recorded during the informal “hair-drier test”

3. *The cup anemometer*

is well exemplified by the time dynamics of thermometers, which are often used as a paradigmatic case (Sozzi, Georgiadis, and Valentini 2002).

That it does not happen with cup anemometers can not be considered unexpected, however. As mentioned in Equation 3.1, the drag forces to which an object exposed to airflow is subject vary in proportion to the square of the wind speed, and in the test procedure delineated above the two speeds between which the cup anemometer is forced are non-zero and zero respectively.

In the former case the torque initially acting on the sensor shaft is maximum from the beginning of measurement. In the latter, it is maximum as the airflow is deflected (the cups rotate at same speed they did under airflow, but the air surrounding them is now still), to decrease with the square of instantaneous shaft angular velocity, and eventually to wane out at very low angular velocity.

In Equation 3.1 this effect is somewhat visible: from second 17 to 22 the decrease rate is visually not dissimilar from the one observed in the step-up ramp. But since second 22 onward the decrease rate diminishes, indicating the aerodynamical forces tend to zero as the cups encounter less and less resistance from air.

Eventually, the only force stopping the anemometer rotation are the small bearings friction.

In the process shown the moment of inertia of the rotor plays an important role, higher values being likely associated to even longer step-down response times. This could be the case with heavier and more robust professional grade models whose purpose is measuring high wind speeds: the sensor used in this informal test is of lightweight build, and the rotor moment of inertia consequently small.

3. The cup anemometer

3.3.9. Summary of cup anemometer parameters

Summarizing, then, we can say that the behavioral description of cup anemometers demands knowing two main parameters:

- A *starting speed* $U_a > 0$ depending on parts friction and the anemometer aerodynamic properties. Its nominal value is typically stated in the anemometers' data sheets, and its real value may grow from that ideal condition because of parts wear-out, maintenance state, and local conditions. The equal sign occurs rarely (in practice never); the starting speed increases with time. Being it dependent on operating story and conditions this parameter is *not* reported in data sheets.
- A *distance constant* $l_0 > 0$, depending mostly on the inertia and other design choices. This can be reported in the data sheet (this parameter may not be declared, however; in case, users can conservatively assume the response time was not measured during the sensor qualification phase: better to ask directly the manufacturer in this case, and if the distance constant is not available, then in case your application is sensitive to response time it is advisable to not take the risk and look for some other models).

4. Engineering aspects

4.1. Mechanical aspects

4.1.1. Primary and secondary transducers

In instrument engineering, the sensors intended for automatic use in connection with a data-logger must provide on output an electrical signal.

The production of an electrical signal related in a predictable way to the quantity to measure (horizontal wind speed in our case) is performed in two stages within the sensor, as shown in Figure 4.1.

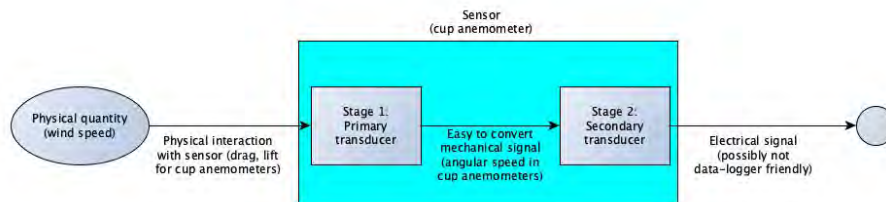


Figure 4.1.: Sensor functional blocks and interfaces

The first stage, known as “primary transducer”, is the device converting the physical quantity of interest to a form (mechanical in case of cup anemometers) which is easier to measure.

The secondary transducer “reads” the mechanical quantity provided on output by the primary transducer, and transforms it to an electrical quantity.

4. Engineering aspects

In line of principle many types of “electrical” signals are possible in theory: they could encode the measured quantity in terms of voltage, or current, or resistance.

The signal type of choice may present a time variation pattern resembling a smooth continuous function, or a sequence of square waves, or a sequence of short pulses, or other. Electrical signals are in addition inherently noisy, the signal-to-noise ratio may be close to 0 (very small noise), but also significantly larger: the secondary transducer is designed to cope with all these sources of variation behaving reliably and, to the extent possible, predictably.

In any existing sensor the primary and secondary transducers are chosen to match one with the other, and this is true also with cup anemometers: with the additional caveat that in this case the name designating the instrument coincides with the primary transducer description.

So let’s explore it in more detail.

4.1.2. Structure

All cup anemometer models existing share the same basic structure: a rotor is connected to a shaft allowed to freely rotate around its axis through some kind of device connecting it to a fixture solidly attached to a mast, or some other supporting structure.

The shaft is also solidly connected to a device whose purpose is to convert the angular speed to some easily detectable electrical signal, as we saw in Section 4.1.

This common structure is schematically shown in Figure 4.2.

The anemometer shown in Figure 4.2 does not exist, but reflect various aspects which can be found in many real-world models.

4. Engineering aspects

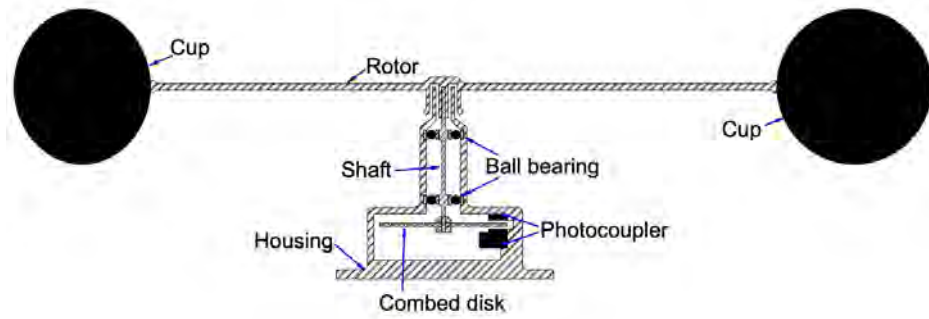


Figure 4.2.: Schematic anatomy of a cup anemometer

4. Engineering aspects

The materials may be different (the one imagined in the drawing is UV-resistant plastics, a material often found in amateur models, but sometimes also used in professional-grade sensors due to its low specific weight, able to reduce the momentum of inertia of the rotor), and the proportions may not be the same.

Also, the position of the bearings may be different respect to the combed disk (which may be between them), and they may not be ball type (liquid mercury bearings are used for example in some research-grade sensors).

Overall, some sensors may be designed for servicing, while the one in Figure 4.2 is clearly not (the ball bearings are enclosed in the housing, and maybe fixed by glue or soldered). Serviceable cup anemometers are common among professional- and research-grade models, while non-repairable ones are the rule among amateur sensors (in this case, when the anemometer fails it is replaced with a similar one).

4.1.3. Ball bearings and their failures

In Figure 4.2 we can see the cup anemometer described holds two ball bearings.

Ball bearings, or alternatives to them like liquid mercury bearings, allow an as free as possible rotation of the shaft around its axis while the cup anemometer structure is fixed to some mounting.

Maintenance history, and sometimes the data sets, reveal the ball (and other) bearings are a critical element in cup anemometer, and their failure may lead to catastrophic consequences on data.

Ball bearings are also a long-standing presence in mechanical engineering, and their operation and failure modes are extensively dealt with in literature, usually in the form of easily accessible technical reports or product selection web sites released by major bearings manufacturers.

4. Engineering aspects

One example is SKF, a manufacturer to which the author is not connected in any way, and whose mission is to design, produce and sell bearings of any size, from the tiniest used for cup anemometers and objects alike, to the multi-meters giant ones.

In report (SKF, n.d.), SKF presents and discusses the failure modes of ball bearings, and gives a procedure for estimating their life time.

These are some of the most frequent (in general use cases):

- 1/3: fatigue.
- 1/3: lubrication problems (wrong lubricant, wrong quantity, wrong lubrication interval).
- 1/6: contamination (ineffective seals).
- 1/6: other reasons (improper handling or mounting, loading different than anticipated, wrong or inadequate fits).

The “SKF formula” expressing the expected lifetime of ball bearings is

$$L_{nm} = a_1 a_c \left(\frac{C}{P} \right)^p \quad (4.1)$$

with a_1 a life adjustment factor, a_c a corrective factor, C the basic dynamic load rating (in kN), P the equivalent dynamic bearing load (in kN) and p the life equation exponent.

In Equation 4.1 the lifetime L_{nm} is expressed in millions of revolutions.

This unit, compared with the operation of cup anemometers, seems large on a first instance. But assuming a sensor will made one revolution per second on average, in an hour it will make 3600 revolutions, and in one even year $8760 \cdot 3600 = 31536000$ revolutions.

Before comparing this number to estimated lifetime, it is important to consider these estimates have been devised under “industrial” operating

4. Engineering aspects

conditions, which are by far less harsh than encountered in the open field where cup anemometers operate.

In open field conditions, with the ball bearings in direct contact with outside air as shown in Figure 4.2, contamination by dirt and lubricant degradation under extreme temperatures can be reasonably expected. It is then safe to consider the estimated ball bearings lifetimes as a very optimistic upper limit.

This given (essentially, assuming the ball bearings of a cup anemometer will operate in an open-field environment orders of magnitude more severe than the “indoor” industrial settings), a highly reasonable suggestion is to rely on estimated lifetimes only as a preliminary indication, and systematically collect failure data from maintenance reports.

This will allow to define site- and condition-specific reliability values, which can be used to identify appropriate maintenance practices.

What happens, when bearings fail?

Many failure modes are possible.

In extreme cases the anemometer may seize up, and yield from then on an infinite sequence of zero readings. This event is very easy to detect.

Or, the bearing begins exerting friction and the actual number s of wind path length per revolution decreases significantly from the value declared by the manufacturer. The activation threshold U_a will also increase in this case: this offers an opportunity to detect the failure occurrence by inspecting the statistical distribution of measurements, and identifying significant changes in it, especially towards the slower speed values. A rationale for acting this way is given in Chapter 5.

Preventive maintenance may also be useful, replacing the bearings or the whole anemometer based on some conservative schedule (Andrews and Moss 2002).

4. Engineering aspects

4.1.4. Common cup sections

To date the two most common cup sections (Pedersen and Dahlberg 2024) are the hemispherical and the conical, shown in Figure 4.3.

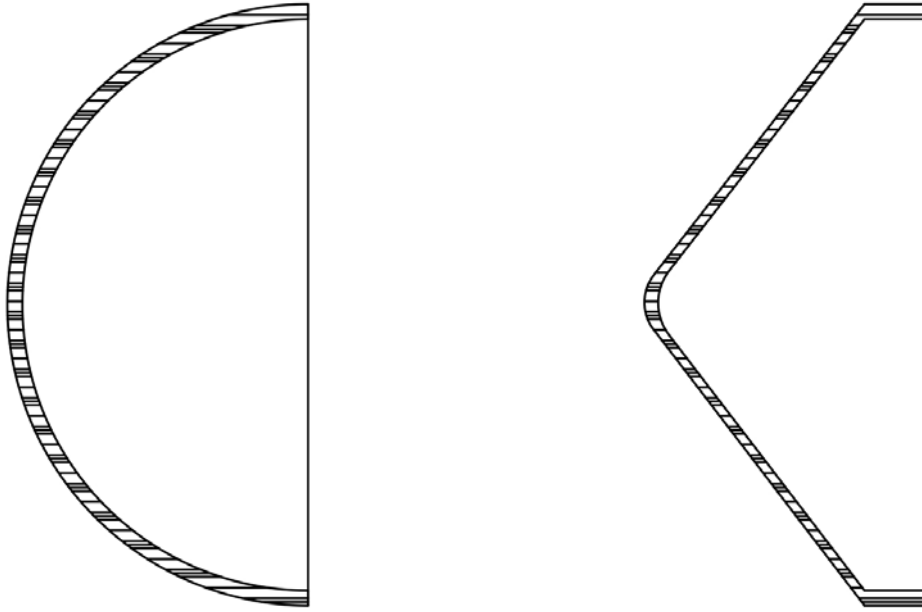


Figure 4.3.: Schematic representation of hemispherical (left) and conical (right) cup sections; dimensions, highly model-dependent, are not given

Both the layouts are common in sensors, despite some difference in their performance, as the conical design has been found slightly more efficient (Pedersen and Dahlberg 2024); however, the performance differences have been verified for wind speeds in excess of 1.5 m/s, and few data exist for low-speed cases.

4.2. Electrical aspects

4.2.1. Magnetic flux circuits

As explained in Section 4.1.1 cup anemometers provide on output an electrical signal which is produced by a secondary transducer, and whose value informs about the angular velocity of the primary transducer's shaft.

Various circuits have been used over time to perform the secondary conversion, and here we'll start from the ones often used in engineering to convert a rotating motion into electricity: the dynamo and the alternator.

As both these devices operate by the interaction of a set of conductor coils in a magnetic field, we'll name them "magnetic flux" circuits.

In case of a dynamo, the most common magnetic flux device used for anemometers, the desired output voltage is proportional to the rotor angular velocity. This is not possible with a bare dynamo, whose output is shown in Figure 4.4.

In order to obtain a smooth signal, some form of filter should be applied. In Figure 4.5 we can see one possible simple circuit made with discrete components. The example is for illustration only, real designs being likely realized using integrated circuits.

Figure 4.5, in its crude simplicity, allows to understand the problems associated with the dynamo transducer. First and foremost, the RLC filter is characterized by a fixed response time τ : frequency of the dynamo output voltage changes in direct proportionality to the shaft angular velocity, and is consequently minimum in correspondence to the activation threshold U_a .

During sensor design the dynamo topology and the values of R, L and C are selected accordingly to the cup anemometer intended use. The choice is dictated by a technical compromise among the response time to a sudden

4. Engineering aspects

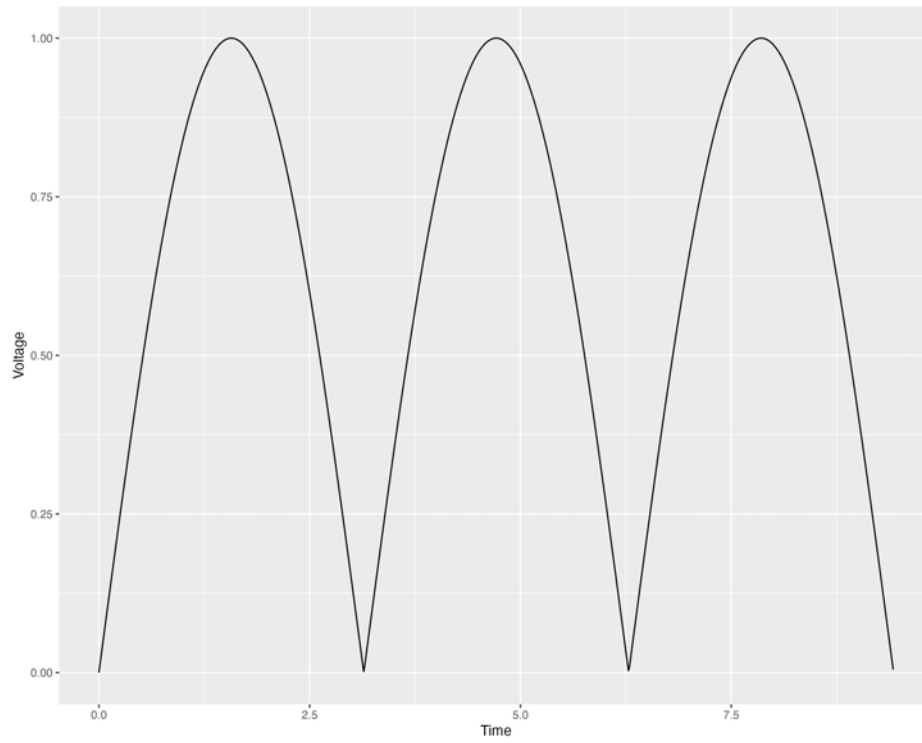


Figure 4.4.: Typical response of a dynamo: both the output peak voltage and frequency are proportional to angular velocity

4. Engineering aspects

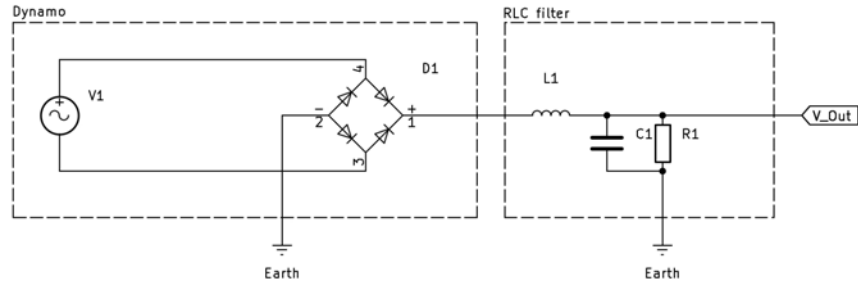


Figure 4.5.: Example of dynamo-based secondary transducer; the actual behavior depends on the values adopted for resistance, inductance and capacity in the RLC filter part, and the number and disposition of coils in the dynamo

speed change, the filter smoothing ability, the sensor toughness, and other market-driven factors.

Given an all-purpose sensor is not possible, cup anemometer manufacturers will privilege the wind speed range most profitable to them. To date this tends to mean wind energy and flight safety, that is, relatively fast speeds.

At low speed these sensors, for the reasons explained above, may underperform: a thorough examination of technical documentation, and possibly some tests, could be of great interest in this case.

4.2.2. Relay-based counters

4.2.2.1. Operating principle

This solution is not common among cup anemometers designed for measuring wind, but can be found in models whose main purpose is to act as

4. Engineering aspects

an on-off switch in crane and building automation systems.

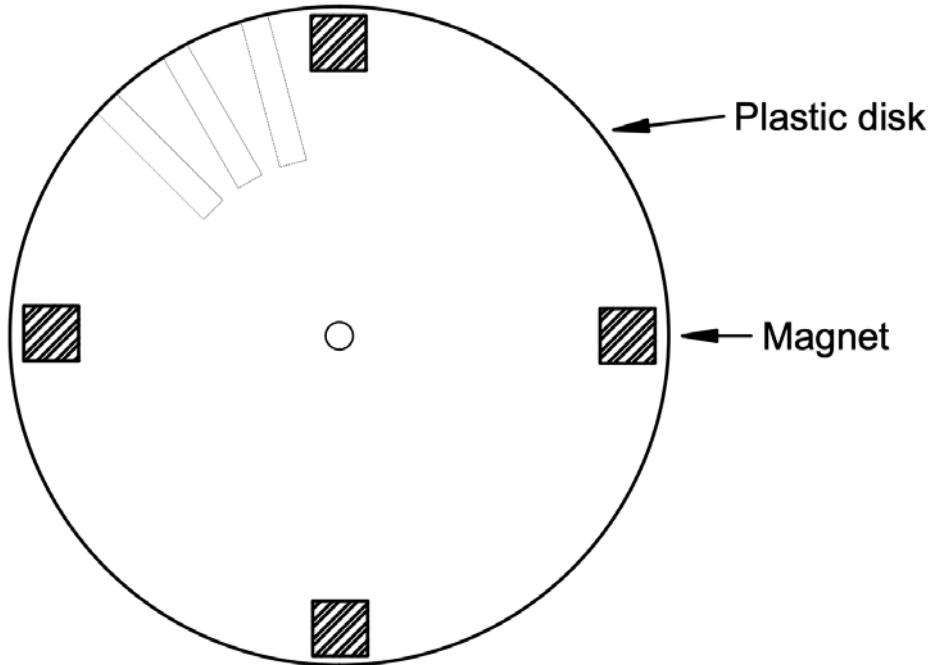


Figure 4.6.: Disk with magnets used in cup anemometers designed for crane and home automation

One such disk, housing four magnets, is shown in Figure 4.6. The number of magnets may be as low as 1 in lower-cost solutions.

The concept of this solution is similar to the wheel revolution counters used in cycle computers: as the disk rotates the magnets pass close to a reed relay, forcing it to close the conducting circuit.

Magnetic disks have the advantage of a relatively simple operation. This is counterbalanced by the possibility of bouncing which the data-logger is then obliged to filter out.

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4.2.2.2. De-bouncing circuitry

Reed relays, as a special case of mechanical switch, are subject to “bounce”, fast alternations between the open and close states occurring over about one millisecond (Horowitz and Hill 2015).

If bounce can be considered “quick” in human perspective, it is “very slow” at the time scale of the many-megahertz clock rates of existing micro-controllers: the careless read of a switch closing or opening by e.g. an interrupt mechanism may yield many spurious counts.

This behavior must be prevented, and this can be made in hardware, in software, or (better) both.

The simplest debouncing circuit may be an RC filter as shown in Figure 4.7.

Software debouncing may be implemented in many ways, a common one by waiting some time after the first rising or falling transition.

It is worth noticing the debouncing circuit

4.2.3. Photo-couplers

4.2.3.1. Definition

In this book photo-couplers are defined in a broad sense as any device which is able to emit a different voltage depending on whether a light beam emitted continuously at one end is received or not at the other.

In cup anemometers the photo-couplers may exist in two different shapes, transmissive and reflective.

In a transmissive photo-coupler the transmitter and receiver ends are typically an IR LED and an IR photo-transistor, mounted on the opposite sides of a light switching combed disk as can be seen in @fig-combed-disk.

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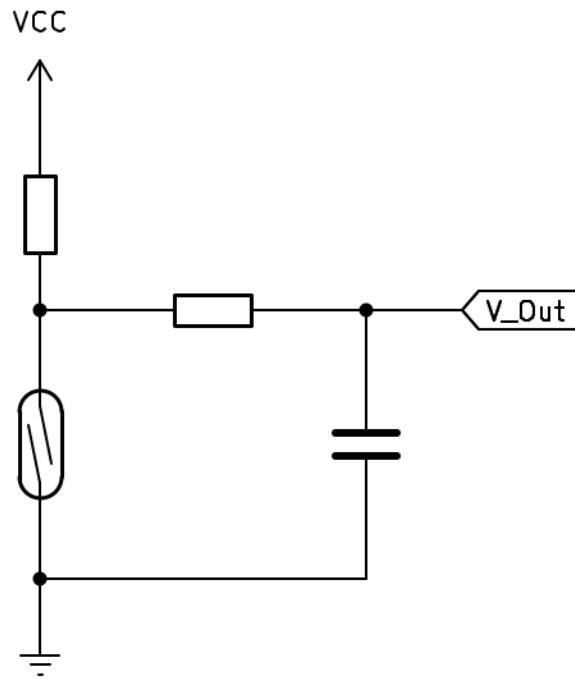


Figure 4.7.: A simple debouncing RC filter; the resistor on top is the usual switch current limiter used to convert the reed switch state to a logical signal, while the R and C values are chosen to ensure a response time matching the sampling rate expected on the Vout end

4. Engineering aspects

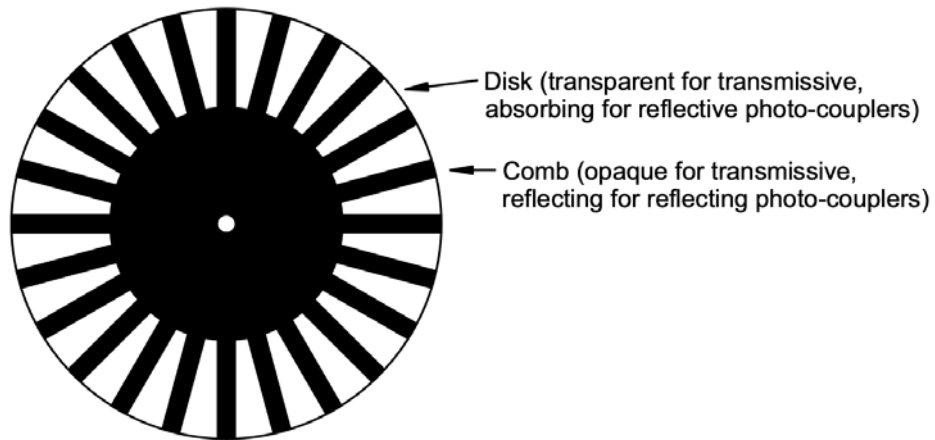


Figure 4.8.: Example of combed light-switching disk; the comb, represented here in black, is typically a thin pigmented layer superposed to a disk which is transparent for transmitting-type photo-couplers, and adsorbing for reflecting-type photo-couplers: the terms “transparent” and “adsorbing” refer to the photo-coupler light wavelength actually used

4. Engineering aspects

An example of conversion circuit can be seen in Figure 4.9. Other designs, based on discrete or integrated components, are also used, but the overall circuit operation is the same: an output voltage is produced, having different values depending on whether the light beam is interrupted or not. The details of the actual waveform differ as a consequence of the details of the components used.

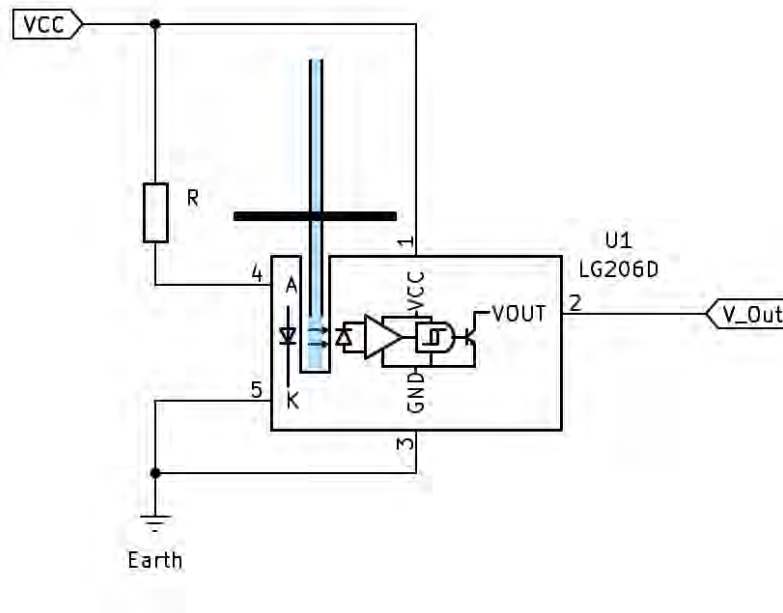


Figure 4.9.: Example of photo-coupler secondary transducer circuit; the blue rectangle represents the combed disk in section, rotating with the rod (black rectangle)

4.2.3.2. Theoretical waveform

The “photodiode” in Figure 4.9 may operate in two ways:

4. Engineering aspects

- As an on-off switch, and in this case the output voltage will be a variable-period sequence of square waves.
- As a continuous 0-1 dimmer, where the response will be proportional to the fraction of the beam section not intercepted by the light-stopping prongs in the disk comb.

Of these two solutions the first is likely to provide very sharp on-off transitions, while the second is more gradual. Because of this the first is often preferred by cup anemometer manufacturers.

The adoption of a sharp or smooth electrical signal has little practical consequences on the receiver end if the signal is “read” as digital value.

In fact, within a data-logger’s “digital input” port the input voltage is compared to a reference level, and encoded as “low” (0) if lower, “high” (1) if higher than it. This comparison is practically instantaneous, and independent of the story of the signal before the signal reading occurred.

So, at least in line of principle, the data-logger is capable of detecting the moment in which the reference voltage is crossed in both directions, and take appropriate measures when this happens (for example incrementing a counter).

In Figure 4.10 we can see the shape the sharp signal variant could take under changing wind speed: the time difference between two subsequent high-low or low-high transitions is equal to the time taken by one disk prong, or void, passing through the light beam as the combed disk rotates.

4.2.3.3. Practical aspects of decoding photo-coupler output

Taking the time of these passages may seem a cozy way to get a fast-changing estimate of instant wind speed, but as we’ll see in Section 4.3 the current technology makes this all but easy, and other solutions are used instead.

4. Engineering aspects

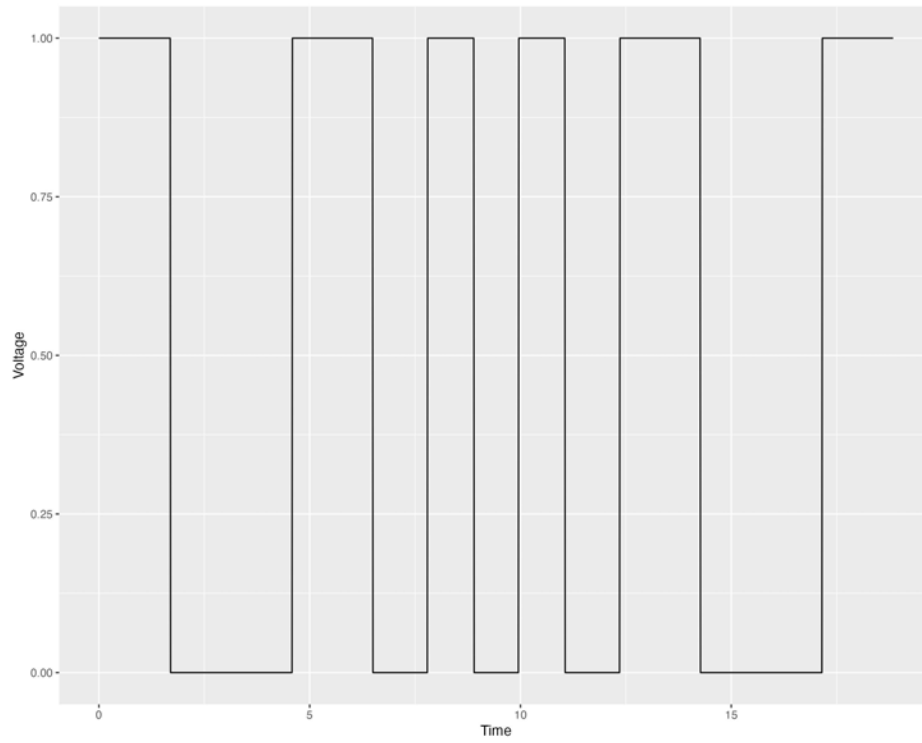


Figure 4.10.: Example voltage on output from a cup anemometer photo-coupler as wind varies

4. Engineering aspects

Photo-coupler based designs offer various advantages respect to others: among them, they do not perturb the rotation and do not add their contribution to the activation threshold.

In change of these advantages, photo-couplers are *active* components and require an external power source to function.

Photo-coupler based designs have not been invented specifically for cup anemometers: they are used since long ago for measuring the angular rotation of shafts in machines, and constitute the basis of tachometers and many other applications in mechanical engineering. The engineering novelty in the case of cup anemometers is the combed disk: the shafts used in machines are normally large enough to allow the light adsorbing/reflecting bands to be directly impressed on the rod. This is not easy at low cost with the very thin and delicate shafts used in cup anemometer, the disk allowing for a convenient lightweight size expansion.

4.3. Conversion of electrical signals to numeric values

4.3.1. In general

The secondary transducer has made its job, and produced an electrical signal of some type: this needs to be “read” using a data-logger, that is, converted to a number and stored somewhere.

The conversion to number form can be accomplished in many ways, depending on the nature and span of the signal, and the possibilities offered by the data-logger.

In order to better appreciate the subtleties of the conversion process to our case, the choice has been made to show it assuming the stand-point of a data-logger builder, in charge of developing the data acquisition system using off-the-shelf microcontroller (MCU) technology.

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To allow a discourse which is not centered on a specific MCU model, with the danger of focusing attention to irrelevant details, the Arduino platform has been selected as reference: its programming model is MCU-agnostic, and hides many model-specific details behind a common interface.

Although possibly less efficient than processor-tailored code, the Arduino approach is very effective as a communication and teaching tool. If one then has to design their own data logger, they can tailor the “universal” concepts presented here to the specific platform they chose.

The direct connection of the cup anemometer’s secondary transducer output to the data acquisition system is not always possible. As explained in Section 4.1.1 the electrical quantity on output from a sensor secondary transducer may be of any kind, but micro-controllers only can read voltages, in a processor-specific range.

On Arduino, the channels able to read voltage exist in two varieties:

- “Digital” channels (on-off, depending on an MCU-dependent threshold, and configurable as input or output).
- “Analog” input channels (configurable as analog, or “digital” in the sense above).

This situation is, by the author knowledge, true for all MCU currently on the market. The threshold separating the logical on from the off in digital input channels, the upper admissible voltage for analog and digital input, and the resolution associated to analog inputs (more on this later) depends on the MCU supply voltage, which is loosely correlated with technology: 8 bit MCUs tend to be powered at 5V, while more recent 32 bit MCUs are more commonly powered at 3.3V (although 1.8V and 5V are also not rare).

Whatever the case, the Arduino programming environment permits dealing with the specific situation: the Arduino user must know (from the data sheets) which MCU is used on their actual board, and the corresponding

4. Engineering aspects

voltage rate and resolution. The API is almost identical despite the processor differences, but the programmer has the responsibility to understand the actual effect of the unified calls.

4.3.2. Signal conditioners

Use of digital or analog voltage input implies the signal from the cup anemometer secondary transducer must be adapted to voltages in the MCU range 0 to V_{cc} .

This is the task of the hardware *signal conditioner*. Its nature depends on the secondary transducer output, which in case of cup anemometers can be assumed to be either a continuous voltage proportional to the shaft angular speed, or a voltage changing more or less sharply between 0 and V_{out} .

So, three cases are possible:

- $V_{out} = V_{cc}$ (in practice, the equality sign may also be a less than sign, $V_{out} < V_{cc}$, provided the difference is small enough - that is, in case of a “digital” signal, V_{out} is large enough to safely exceed the threshold separating a logical low from a logical high). If this happens, the conditioner may not even exist, or be just an emitter-follower circuit.
- $V_{out} > V_{cc}$, in which case the conditioner has the role of an *attenuator*.
- $V_{out} \ll V_{cc}$: in this case the conditioner is a voltage *amplifier*.

4.3.3. Emitter-followers

An *emitter-follower* is an active circuit having the purpose of replicating the voltage received on input (less one diode drop, about 0.5V), but with

4. Engineering aspects

the ability to drive a load - possible when for example there is a long data line.

Of the possible implementations of emitter-follower, in Figure 4.11 a simple one based on a transistor (Horowitz and Hill 2015) has been shown for illustrative reasons.

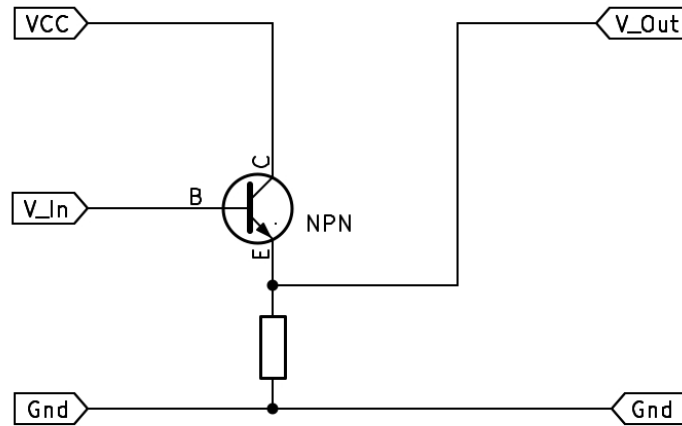


Figure 4.11.: A simple example of emitter-follower circuit

This circuit has no voltage gain in the strict sense; the output voltage V_{out} in this case is equal as mentioned to $V_{in} - \Delta V$, where $\Delta V \approx 0.5$ (the precise value can be found in the data sheet of the transistor selected).

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4.3.4. Amplifiers

An example of voltage amplifier based on an operational amplifier (Horowitz and Hill 2015) is shown in Figure 4.12.

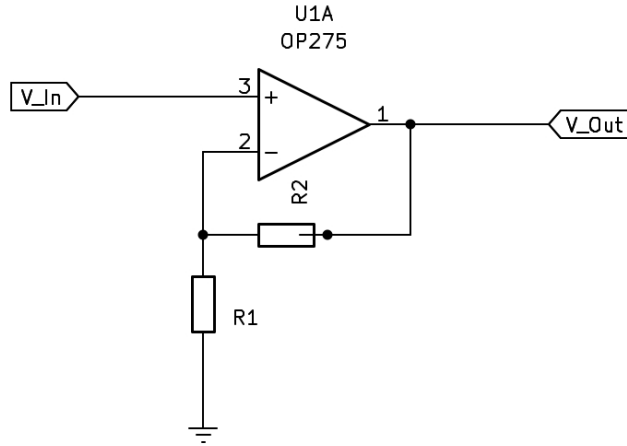


Figure 4.12.: Simple amplifier based on an op-amp

The output voltage is

$$V_{out} = V_{in} \left(1 + \frac{R_2}{R_1} \right)$$

4. Engineering aspects

4.3.5. Attenuators

An attenuator can be thought as a special case of amplifier with a gain less than 1.

A common implementation, entirely passive, is shown in Figure 4.13.

The output voltage is

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

4.3.6. Problems associated with converters

Active or passive, all the voltage converters presented introduce some electrical noise which compose with the noise already present on output from the secondary transducer and the interesting signal.

The additional noise magnitude depends on the quality of components, the layout of the PCB hosting them, the soldering and other construction details.

As a general rule, the best approach is to adopt the least aggressive signal manipulation possible; in the specific case of cup anemometers, however, the need of transforming the sensor output extensively occurs seldom.

This fact, essentially probabilistic, should not be overlooked however, and a close evaluation is always advisable on a per-case basis.

4.3.7. Modes of conversion

The conversion circuits we have seen so far may be classified in two broad groups according to the type of variation the voltage signal at their input exhibits with time:

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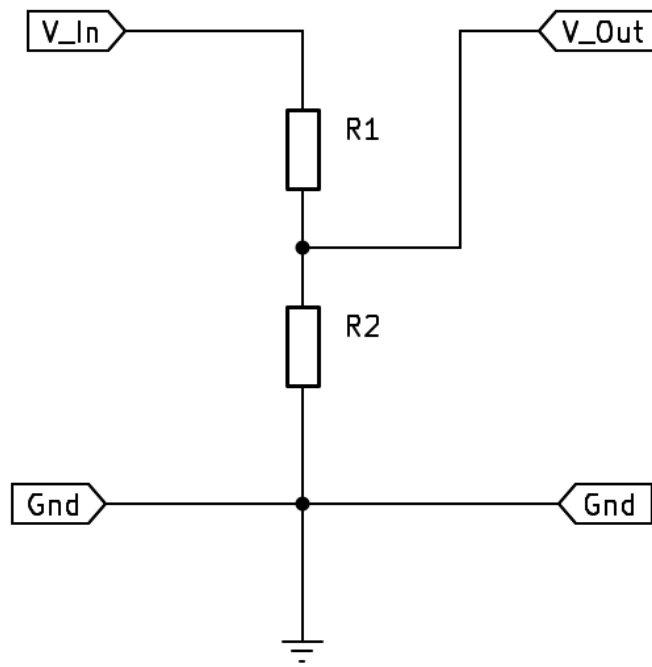


Figure 4.13.: Voltage divider, an example of simple attenuator

4. Engineering aspects

- “Analog” sensors: a continuous voltage $V(\omega)$, monotonically increasing with the angular velocity ω ; the wind speed is then obtained as $U = aV + b$ with a a positive multiplier and b an offset). In some cases a current A is used instead of the voltage, but the practical meaning is unchanged: a continuous electrical variable is mapped directly to a wind speed.
- “Count” sensors: a sequence of voltage pulses, or continuous waveforms, whose frequency $f = \frac{n_v}{\Delta t}$ is proportional to the wind speed according to a formula having the form $U = ef$ where e is a multiplier.

Currently the second type is more common, and will be considered later in more detail. Nevertheless, knowing the first, and the difference between the two, is an important thing when selecting the right instrument for one’s own application, and when writing the data-logger program better fitting the characteristics of the specific model selected.

4.3.8. Sampling vs Counting: an important difference

4.3.8.1. Analog anemometers

Sensors providing on output a continuously-changing voltage do not, strictly speaking, require a conversion. Rather, the necessity exists of adapting this signal to the data-logger input characteristics (for example the voltage input range).

The two possible situations requiring a conditioning circuit are when the voltage from the cup anemometer is larger or much smaller than the data-logger’s input range. In the former case, some form of attenuation is needed. In the second, an amplifier is due.

Dealing first with the first, it is to be noticed that in principle an analog signal is continuous, but to date its reading will occur through some kind

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of digital equipment, and current digital equipment may only perform readings at fixed time instants, and at finite resolution.

If we denote the sampling period by Δt , and imagine the time δt needed for taking an individual reading is small compared to Δt , then we have a time span $\Delta t - \delta t$ during which no reading is taken. In this interval the cup anemometer rotor may change its angular speed many times, but none of this variation leaves some trace in the readings.

In other words we could have a situation similar to the one we can see in Figure 4.14.

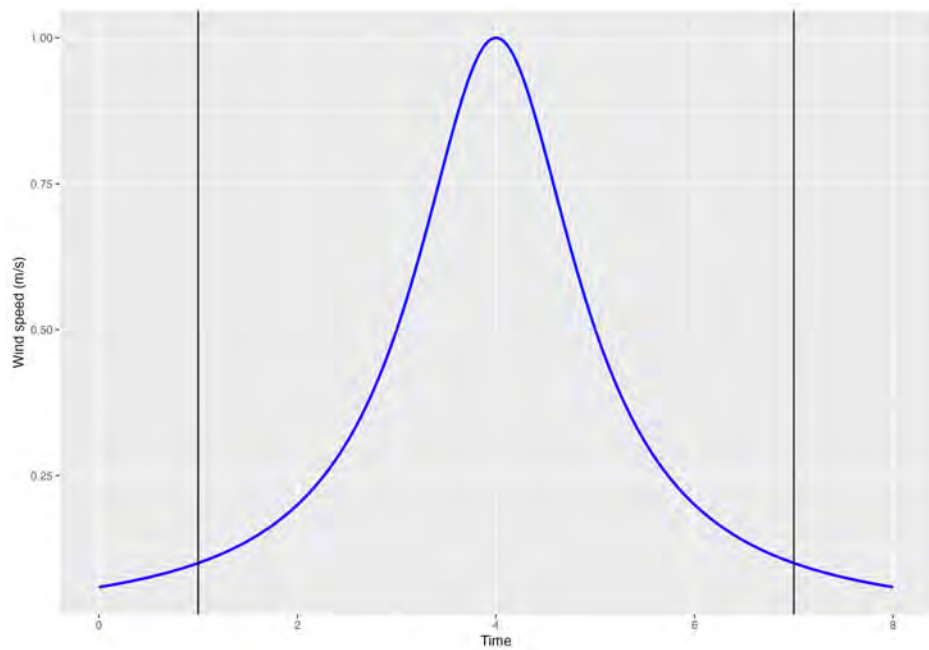


Figure 4.14.: Hypothetical instantaneous wind speed in which a large spike has been missed because the sampling times did not occur in correspondence of it

4. Engineering aspects

The effect shown in Figure 4.14 can be prevented by adopting in the data-logger a sampling rate fast enough to capture all the variation likely present in the “mechanical” signal presented on output by the primary transducer (instantaneous angular velocity).

4.3.8.2. Count anemometers

As previously mentioned this is by far the most common case to date. The electrical output from the sensor is a signal oscillating between a “low” and a “high” states, the number of up- or down-rising fronts being proportional to the number of spins the anemometer did between the two measurements considered.

If as usual we designate by $\omega(t)$ the instantaneous rotor angular velocity, then the total angular displacement is given by

$$\Omega(t, \Delta t) = \int_t^{t+\Delta t} \omega(t) dt$$

A count anemometer provides an approximation to this integral. The mean angular speed can then be obtained by considering the ratio

$$\bar{\omega}(t) = \frac{\Omega(t, \Delta t)}{\Delta t}$$

4.3.9. Actual method for counting revolutions per time step

As we have seen in Section 4.2.3 the level changes of a combed disk type cup anemometer occur depending on the erratic changes in the shaft angular velocity, that is, at instants distributed randomly in time.

By the way the circuit is conceived any transition from low to high state is followed by another from high to low, so to quantify the number of

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revolutions and fractions it is sufficient to count the number of rising (or falling) fronts.

This count can be in principle obtained by sampling the signal at a fixed rate, and counting all cases in which the current reading is a logical high and the previous a low.

If the sampling rate is high enough no transition is missed, and one has an opportunity to express the rotation angle easily. Assuming a combed disk with n prongs, a wind speed equal to the maximum value U_{max} rated for the sensor, a wind path length per revolution s and a counting time interval Δt , then the number of rising transitions can be estimated to

$$1.75 \text{ m/rev} = (1.75/n) \text{ m/step}$$

$$70 \text{ m/s} = 70/1.75 \text{ rev/s} = 20 \cdot 70/1.75 \text{ step/s}$$

$$p_{max} = n \cdot \Delta t \cdot \frac{U_{max}}{s}$$

For example, in the case of the anemometer mentioned in Section 3.3.8 we have a maximum speed U_{max} of 70 m/s, a number of prongs n equal to 20, and a response rate s equal to 1.75 m/rev. Replacing in the previous expression, we have $p_{max} = 800 \cdot \Delta t$. Assuming the time step Δt equal to 1 s, we have a number of transitions per second equal to 800. For comparison, a reasonable clock rate for an STM32L476RG MCU is 80 MHz, 100000 faster than the transitions per second. So, in theory, there would be plenty of clock time to poll for measurements and count the transitions.

This method, often used in embedded engineering introductory classes, is not the one most commonly used, as it demands an intense CPU work which is wasteful of the meager resources available on a typical MCU.

The standard way is through a clever use of the *interrupt* mechanism all MCU have.

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Assuming the Arduino API mentioned in this chapter, use of an interrupt triggered by a logical state change on a digital channel (in this case the pin labeled “D2”) demands configuring the pin as an input, and then associating it with an interrupt

```
int inputPin = 2;

void setup() {

    Serial.begin(115200);

    pinMode(inputPin, INPUT_PULLUP);
    attachInterrupt(
        digitalPinToInterrupt(inputPin),
        onChange,
        FALLING
    );
}
```

In this case the interrupt is triggered on any falling front. The rising front could have had been used as well in this case (the output from the cup anemometer is very clean, and visually identical in both transition directions); in the general case the choice is made depending on which of the two fronts looks better at the oscilloscope.

The symbol “onChange” designate an *interrupt service routine*, a short and fast-executing function whose purpose may, for example, be to increment a counter variable:

```
int passages = 0;
void onChange()
{
```

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```
    if (digitalRead(inputPin) == LOW) passages++;  
}
```

The counter variable, “passages”, is declared outside any program function, and initialized to zero before the “setup()” function shown above is invoked. From then on, it is incremented as soon as a high to low transition occurs, that is, on passage of a prong over the reading edge of the photo-coupler, or on every $2\pi/20$ angular movement of rotor.

In the interrupt service routine, as you can see, the number of passages is incremented conditionally, only if the pin state is *still* low. From the time the falling edge was detected by the interrupt-serving circuitry of the MCU to the actual instant of the invocation of the interrupt serving routine a tiny interval has passed (quantifiable in microseconds, the actual value depending on the MCU, the clock frequency it’s used on it, and other higher-priority interrupts served in the meantime).

Given the way the cup anemometer under investigation works, it is extremely unlikely that after few millionths of a second after the high to low transition occurred another, low to high, ensued. But the pin may have been driven high by an electrical disturbance (do-it-yourself assemblies are notoriously prone to electromagnetic compatibility issues), and the code shown, defensively, checks for this.

Interrupts are served asynchronously from the main code, and if the interrupt servicing routine is written carefully their processing demand very little CPU load (orders of magnitude less than the polling strategy mentioned above).

What an external observer perceiving the system evolution in time would see, is that the variable “passages” from time to time, magically, increases.

This is indeed the Arduino “loop” code in the case:

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```
unsigned long timeStep = 0;
unsigned long smpTime = 1000;

void loop() {

    if ((millis() - timeStep) > smpTime) {

        timeStep = millis();
        float speed = passages * 0.0875f * (smpTime/1000.f);
        Serial.print(timeStep); Serial.print(", ");
        Serial.print(passages); Serial.print(", ");
        Serial.println(speed);
        passages = 0;

    }
    delay(1);
}
```

In the code above “avgTime” represents the time step in milliseconds (and “timeStep” the current time, also in ms).

The loop, in itself, contains an “if” block executed whenever the current time reaches the next sampling instant. In this case the current time is updated, and the speed averaged along “smpTime” is computed by multiplying the “passages” count by $1.75/n = 0.0875$, thus obtaining an estimate of

$$U = n_p(\Delta t) \cdot 0.0875$$

Afterwards, the number n_p of prong passages through the opto-coupler is reset to 0, allowing to start a new counting period.

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As mentioned above, to date MCU clock frequencies typically exceed 1 MHz, and the time necessary to the slowest imaginable Arduino to execute the code block inside the “if” block is by far smaller than 1 ms. Would no specific measure be taken, this would imply a probability of almost 1 that the calls to “millis()” executed within two consecutive “loop()” invocations yields an identical value, and the block within the “if” would repeat over and over.

This is the reason for the tiny 1 ms delay placed outside the “if” block: because of it, successive invocations of “millis()” will yield by necessity different values, so no spurious lock-in can occur.

1 ms may be a small time grain for a human, but is a very long time for an Arduino: because of this, the MCU will spend the very most part of its activity waiting for anemometer-driven logical state changes on pin D2 and waiting for wake-ups from the delay.

On a normal microprocessor this would make no serious change on the CPU activity (it would execute some form of do-nothing loop during the waiting times). On some MCU, on the contrary, entering a wait state gives an interesting opportunity to switch the MCU, or parts of it, to a lower-power state, so saving valuable energy, and better resisting to potentially dangerous environmental condition changes.

4.3.10. “Resolution”, in the case of combed-disk cup anemometers

In a comb-based design we have seen that the number of revolutions in the sampling interval Δt is

$$\frac{n_p(\Delta t)}{n}$$

with n_p the value of the number of rising or falling wave fronts found in the sampling interval, and n the number of prongs on the combed disk.

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We notice two things:

- The value of n_p is by construction a non-negative *integer*. So changes in the reported number of revolutions in the sampling interval will occur step-wise, and not continuously. This means the cup anemometer has a finite intrinsic resolution which is completely independent of the resolution of the data-logger digital to analog converter. Improving the data-logger resolution beyond some sensor-dependent level yields no advantage.
- The (fractional) number of rotor revolutions in the sampling period can in principle be converted to a mean angular velocity whose absolute value expressed in radians per second is $\varphi = 2\pi n_p (\Delta t) / n \Delta t$. This, however, because of the finite resolution of the comb, should be considered with some caution: the uncertainty comes from the fact that the actual rotation angle was chopped to a value corresponding randomly to a large fraction of a whole angle (for example $2/n \cdot \pi$ - in case of the test anemometer used in this chapter $n = 20$, so that the angular resolution is 18° , quite a coarse value). This is not an issue for fast wind speed, but can well be for slow wind.

Cup anemometer data sheets often report the “speed resolution”: this is the angular velocity resolution seen above scaled by the wind path length, so to get a value expressed in m/s. Its value is not strictly connected with the anemometer’s dynamic response, but just an indication of the number of prongs on the combed disk: the higher, the finer resolution. A high resolution may be an indication of a special care in the design and construction of the sensor’ secondary transducer, but to date, with the current state of serigraphic print and the miniaturization of electro-optical component, might not be something to overvalue.

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4.3.11. How long should Δt be?

In the preceding section we often assumed Δt to be 1 s. This is convenient for calculations, but is not a sampling time very commonly used in practice.

Meteorologists, when programming data loggers, often opt for $\Delta t = 10$ seconds, or even more.

Choosing a “long” sampling time does not change appreciably the MCU workload, but has a distinct advantage: it contributes to reduce the error in determining the speed.

To see where this solution comes from, let's consider a fractional revolution angle

$$r(\Delta t) = \frac{n_p(\Delta t)}{\Delta t}$$

But what, if the value $n_p(\Delta t)$ is in error? It could have had been larger or smaller of one unit, just because of the position of the prongs in the moment when the sampling interval began.

If so we could write

$$r_+(\Delta t) = \frac{n_p(\Delta t) + 1}{\Delta t}$$

and

$$r_-(\Delta t) = \frac{n_p(\Delta t) - 1}{\Delta t}$$

We could then introduce the error

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$$E(\Delta t) = r_+(\Delta t) - r_-(\Delta t) = \frac{2}{\Delta t}$$

If the sampling time is 1 s this error estimate is 2. But if it is 10 s, the estimated error drops to 0.2. Larger sampling times yield even smaller errors.

This is why “long” sampling times are preferable when using cup anemometers.

4.4. Accumulating individual readings

4.4.1. Scalar and vector speed

Individual (“raw”) wind speed and direction readings may be stored as they are (and this is desirable for diagnostics and fine scale study), but most often they are aggregated into proper averages and other statistics, and stored in this form.

As mentioned in Section 2.3 there is not a unique definition of the mean speed: two different statistics are the “vector speed” defined by Equation 2.1, and “scalar speed” in Equation 2.12.

These statistics are usually computed simultaneously. In case this is not, the data logger programmer or manufacturer should state clearly which speed definition is used.

Of course, the vector speed can be computed only if a wind direction sensor is available: in case the only sensor is a cup anemometer then the scalar speed is the only possibility.

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4.4.2. Computing common statistics for scalar speed

Let's assume that in addition to a sampling period Δt we also have an *averaging period*, S_a . For reasons of simplicity we'll assume S_a is integer, and Δt divides it exactly.

So, within a whole averaging period we'll have

$$N = \frac{S_a}{\Delta t}$$

wind speed data items, which we'll designate as

$$U_1, U_2, \dots, U_N$$

Of course we could calculate the most frequently used statistics based on their definition:

$$\bar{U} = \frac{1}{N} \sum_{i=1}^N U_i$$

$$\sigma_U = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (U_i - \bar{U})^2}$$

$$U_- = \min_{1 \leq i \leq N} U_i$$

$$U_+ = \max_{1 \leq i \leq N} U_i$$

The first and second of these indicators, mean and standard deviation, are typically calculated using floating point arithmetics. This poses a potential problem, especially evident with the standard deviation, with

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cancellations due to the adding of quantities having very different orders of magnitude (Stoer and Bulirsch 1993).

A summation approach presented in (Knuth 1981) arrives to mean and standard deviation by means of a recurrence formula. The initial step is given by the two assignments

$$M_1 \leftarrow U_1 \quad S_1 \leftarrow 0$$

with U_1 the first wind speed measured.

The following steps can be written as

$$M_k = M_{k-1} + (U_k - M_{k-1}) / k$$

and

$$S_k = S_{k-1} + (U_k - M_{k-1}) \cdot (U_k - M_k)$$

The recurrence terminates once reached the last measured wind speed in current block, U_n ; the final standard deviation is given by

$$\sigma = \sqrt{\frac{S_n}{n-1}}$$

and the mean by

$$\bar{U} = M_n$$

Most data-loggers to date do not make the standard deviation of individual readings available, but making it accessible to users (possibly on demand) yields an important failure detection hook. For example, a zero standard deviation indicates all readings were found identical, and this, along with

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a mean speed always equal to 0, may be an indication of a gripped sensor in need of a replacement.

The next code excerpt illustrates the concept of the simultaneous calculation of wind mean speed and standard deviation, employing as usual the Arduino API as a communication tool.

```
unsigned long smpTime = 10000UL; // (10000 ms = 10 s)
unsigned long avgTime = 60000UL; // (60000 ms = 1 min)

float Mnew = 0.0f;
float Mold = 0.0f;
float S     = 0.0f;
float U     = 0.0f;
float sigma = 0.0f;

void loop() {

    if ((millis() - timeStep) >= smpTime) {

        // Get raw mean speed
        timeStep = millis();
        float speed = passages * 0.0875f / (smpTime / 1000UL);

        // Compute block index
        unsigned long int k = (timeStep / smpTime - 1) % (avgTime / smpTime) + 1;
        if(k == 1) {
            // Initialization
            U     = Mold;
            sigma = sqrt(S/(avgTime / smpTime - 1));
            Mold  = speed;
            S     = 0.0f;
        }
        else {
```

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```
// Update
Mnew = Mold + (speed - Mold) / k;
S    += (speed - Mold) * (speed - Mnew);
Mold = Mnew;
}

passages = 0;

}
delay(1);
}
```

This code is of course not realistic (for instance it does not associate readings with time stamps from a real-time clock unit, nor it contains provisions for saving somewhere the data gathered and processed).

Yet it shows what could be done in actuality on a real programmable data logger, or when designing one.

The code also highlights an important concept: variable ‘speed’ contains the individual (*raw*) readings (the mean speed encountered in the sampling period just concluded), while ‘U’ and ‘sigma’ hold respectively the mean speed after end of last averaging period and its standard deviation, which can be collectively named *processed* data.

In normal meteorological practice only processed data are stored and disseminated among users. But storing and disseminating raw data may provide precious data for understanding the wear-out state of a sensor, and should definitely be considered by data-logger designers.

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4.4.3. Calm treatment

As shown in Section 3.3.9 any cup anemometer is characterized by a finite activation speed $U_a > 0$.

Ideally, would cup anemometers have zero inertia, if the real wind speed is below or equal to U_a the cup anemometer would yield a 0 speed reading.

As we have seen in Chapter 3, however, this does not happen and the anemometer keeps rotating for possibly a long while as wind speed falls below the activation speed.

If this happens, the anemometer instantaneous wind speed will decelerate due to the braking effect of the tiny friction exerted by the bearings, and its value will consequently be smaller than U_a .

The most common cup anemometers present in the market, combed disk models, do not yield instantaneous speed, but rather a mean over a sampling time period decided at the data-logger level.

Then it may well happen to find a situation where this mean is below the activation speed, but the actual wind for some time exceeded it.

In other words, on a combed disk cup anemometer an indicated mean speed whose value is less than the sensor's activation threshold is not a sure indication of a "full calm". What can surely be stated was that the mean speed was affected by some values below the activation threshold, and this contaminated the mean to some extent (quite large by the way, the mean having been found less than the activation threshold).

Circumstances like this, in which a deterministic test cannot indicate with absolute confidence whether a condition has uniformly occurred or not, are common in engineering practice. They are called "plausibility checks", and the detection of wind calms is one of these.

One possible approach for plausibility checking cup anemometer means to contain at least one individual "calm" is given in the code excerpt

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following. Of course there exist alternatives: for example, the calms which contributed to an average can be actually counted, and the results assumed as a “calmosity” indicator for a mean speed.

```
unsigned long smpTime = 10000UL; // (10000 ms = 10 s)
unsigned long avgTime = 60000UL; // (60000 ms = 1 min)

float Mnew = 0.0f;
float Mold = 0.0f;
float S     = 0.0f;
float U     = 0.0f;
float sigma = 0.0f;

const float Ua = 0.5f;
bool calm = false;
bool nocalm = false;

void loop() {

    if ((millis() - timeStep) >= smpTime) {

        // Get this step's prong passages count and convert it to mean speed
        timeStep = millis();
        float speed = passages * 0.0875f / (smpTime / 1000UL);

        // Compute the within-averaging block index, and use it to distinguish between
        // an initialization or update in Knuth's averaging
        unsigned long int k = (timeStep / smpTime - 1) % (avgTime / smpTime) + 1;
        if(k == 1) {
            // Initialization
            U = Mold;
            sigma = sqrt(S/(avgTime / smpTime - 1));
            Mold = speed;
            S = 0.0f;
        }
    }
}
```

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```
    calm    = (speed < Ua);
    nocalm  = !calm;
}
else {
    // Update
    Mnew = Mold + (speed - Mold) / k;
    S    += (speed - Mold) * (speed - Mnew);
    Mold = Mnew;
    calm |= (speed < Ua);
    nocalm |= (speed >= Ua);
}

Serial.print(timeStep); Serial.print(", ");
Serial.print(passages); Serial.print(", ");
Serial.print(speed); Serial.print(", ");
Serial.print(calm); Serial.print(", ");
Serial.print(nocalm); Serial.print(", ");
Serial.print(k); Serial.print(", ");
Serial.print(Mnew); Serial.print(", ");
Serial.print(S); Serial.print(", ");
Serial.print(U); Serial.print(", ");
Serial.println(sigma);

    passages = 0;

}
delay(1);
}
```

In this version two boolean variables, ‘calm’ and ‘nocalm’, are introduced along with the activation threshold ‘Ua’ (here assumed to be a constant). Both ‘calm’ and ‘nocalm’ are initially assumed to be false, but as soon as

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an individual measurement will become available either ‘calm’ or ‘nocalm’ will become true (‘calm’, if $U < U_a$, and ‘nocalm’ if $U \geq U_a$).

This check is repeated and accumulated with a logical “or” operator over an entire averaging period, and so at end of period ‘calm’ will be true if and only if at least one individual speed was smaller than the activation threshold. Similarly, ‘nocalm’ will be set to true if at least one speed was larger or equal to U_a .

If these definitions are adopted, then there may be situations in which both ‘calm’ and ‘nocalm’ are true, with the meaning that the average contains both calm and non-calm measurements. Other actual possibilities are ‘calm’ is true and ‘nocalm’ is false (all data have been found in calm), and ‘nocalm’ is true while ‘calm’ is false (no data in calm). The fourth possibility, with both ‘calm’ and ‘nocalm’ false, will never occur in practice as soon as the first speed measurement becomes available.

Following the method indicated above, the data acquisition program “improves” the mean speed (and possibly standard deviation) values by complementing them with a calm-nocalm indicators which users may use later to decide whether an individual row, or the whole data set, is acceptable for their application-specific uses or not.

This “value-preserving” approach is of course not the only one possible. Another, more aggressive, consists in replacing the values in calm with a suitable “improved” value. Unfortunately no “best” solution exists, and it is the user’s responsibility to choose among the infinite possibilities.

Just for illustration, here is a short list with some of the possibilities:

- We could set $\overline{U}$, the mean speed, to 0 m/s.
- Or, we might set $\overline{U} = U_a$.
- Or, one could set $\overline{U} = f(U_a)$, where f is some appropriate function: for example $f(U_a) = \frac{1}{2}U_a$ as often happens with chemical measurements below the minimum detectable value.

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- Or, one could remove any zero reading from the mean.

Neither of these will in general yield a mean which is identical to the mean of the real wind. But the error associated to one or the other method will be larger or smaller, depending on site- and sensor-dependent conditions, and possibly in a direction which is desirable to the user application profile.

Indicating a sensible method for finding a site-tailored could be the objective of a “wind characterization” phase, conducted periodically on cup and vane anemometers at critical sites: more in following chapters.

4.5. How to read a cup anemometer data sheet

4.5.1. General considerations

This chapter introduced the cup anemometer, its physical principle and its characteristic parameters. I conclude it with a section giving specific advice on how to use all this knowledge in immediately useful terms.

In this section I will adopt the viewpoint of a designer of a meteorological station, assuming she or he has the power of deciding which anemometer model to select.

A model selection process can be acted in many ways. In my personal experience the best way to conduce it is driven from data, within prescribed budget constraints, and with a knowledge of the specific application requirements for the case.

Of course it is possible the application is “general unspecified use”, the most difficult among all. And, it is also possible that no known cup anemometer model can satisfy the application requirements at the budget level considered.

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The designer, because of this role, has the responsibility to deal with these difficult cases with their clients in a rational and effective way: I give this for granted, and spend no word about.

The information needed to choose the “best” cup anemometer within a set of competing alternatives is usually delivered by sensor manufacturers in the form of a *data sheet*. This is an official document stating the characteristics of the model described, and is made easily accessible by the manufacturer (in most cases it can be found by a quick web search).

Not all cup anemometer models are accompanied by a data sheet. Those which are not, or for which the in-theory-existing data sheet cannot be easily accessed, should be treated with extreme caution since the very beginning (my personal advice is to avoid them altogether, if the application at stake involves some risk for its end users and other stakeholders).

It is also very important to check the version of the data sheet is the most up-to-date. Data sheets are searched automatically in the web by bots able to download them and distribute automatically through collection sites: this is an especially nasty case, as old versions remain in these collection for an inordinate time, and discovering the last version may require months, or even years: if interested, search the *manufacturer* site. Consider the collection sites real purpose is to collect and distribute advertisement, and gather your personal data.

4.5.2. The essential parameters

As mentioned in this chapter any cup anemometer distinguishes from the others by the value of their specific characteristic values. Two of them are of paramount importance:

- The activation threshold (also named “starting speed”), U_a .
- The distance constant, l_0 .

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The following table presents data for some models, as I have found (or missed?) them in the manufacturers' data sheets.

Model	Manufacturer	Activation threshold	Distance constant
Wind Transmitter "First Class" Advanced	Thies Clima	$< 0.3 \text{ m/s}$	$< 3 \text{ m}$
WAA151	Väisälä	$< 0.5 \text{ m/s}$	4.0 m
Gill 12102	Young	0.5 m/s	2.3 m
WindSentry 03101-L	Campbell Scientific	0.5 m/s	2.3 m
DNA202.1, DNA301.1	LSI-Lastem	0.5 m/s	Undeclared
VV-20	CAE	0.5 m/s	Undeclared
t031 TVV	SIAP+MICROS	Undeclared (min meas. range 0.25 m/s)	Undeclared
Vantage Pro2 cup anemometer	Davis	3 km/h (about 0.83 m/s)	Undeclared

4.5.3.

These two numbers should always be present: if not, then the model under consideration may not be intended for measurements; this can legitimately happen: for example there exist in the market "toy" anemometers meant for educational purposes, and designed to make the operation intuitive, but not intended for real measurements). Or, many cup anemometers are built to feed an on-off controller of a crane, or the retracting device used for curtains in home automation systems.

Part III.

This part presents a statistical comparison of ultrasonic anemometer and cup anemometer data, with emphasis on slow wind speed cases.

The actual response of a cup anemometer under real-world condition can only be obtained from field tests, where the full span of natural phenomena overlapping on mean wind can be appreciated, and slow wind naturally occur besides the stronger.

Compared to this, the tests conducted in a wind tunnel under closely controlled conditions, and at speed ranges higher than those considered interesting in this book, can only give an incomplete view.

The chapters in this part are statistical in nature, and show the behavior of six identical cup anemometers located at sites of strongly different characteristics: this allows to fully appreciate site-dependent effects, and begin to draw consequences of application value.

5. Wind speed statistics

5.1. Chapter outline

The objective of this chapter is to present the wind speed statistics one might compute at “macroscopic” level, i.e. starting from wind speed averages as computed and stored by a data logger.

Many authors (especially in the wind energy field) have spent a significant effort on the understanding of the statistical behavior of the horizontal wind speed (Shi et al. 2021). Up to this review paper the most frequently adopted wind speed probabilistic model is the Weibull distribution (Rinne 2009) (Burton et al. 2011), but other distributions have been proposed as well (among them the log-normal, mixed-Weibull, exponential polynomials, and others) (McWilliams, Newmann, and Sprevak 1979) (Wang, Liu, and Qian 2019).

Many authors have observed the proposed distributions tend not to fit well under zero or slow wind conditions, and have proposed alternatives.

To my knowledge, however, none of them tried to relate the mis-fits with the slow-wind response of the wind speed measurement chains used in the studies.

This makes sense however: the studies mostly originated in the wind energy field, and focused on the *strong* winds necessary to allow wind turbines to produce energy (minimum productive values range from 4 to 8 m/s in typical large-scale turbines; pushing this limit down is not rational, as the

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kinetic energy content of slow wind is too small to guarantee a positive return on investment).

In this chapter we will start with the popular Weibull distribution as a comparison, and see that it (or any other continuous distribution) do not fit well with cup anemometer data; we'll discover that wind speed data collected using a three-axial ultrasonic anemometer exhibits a somewhat less evident visual similarity with a Weibull distribution, but (much more important) we'll discover that the empirical distribution of cup and ultrasonic speeds differs mainly in the slower wind speed ranges.

We'll then explore the behavior of cup speed classified by their corresponding ultrasonic anemometer speeds, and realize this naive comparison is quite misleading, especially in the low speed range.

The path we'll follow can be summarized as follows:

- First I'll present the data sets used in this chapter and the micro-meteorological stations they come from.
- Then we'll come in contact with the Weibull distribution, a popular wind speed statistical model which, would it provide an acceptably accurate estimate of wind speeds distribution, could ease the comparison work.
- Then we'll check using real cup anemometer data whether the empirical statistical distribution obtained from their measurements resembles a Weibull distribution of some type. (Spoiler: it does not.)
- After that we'll check whether the cup anemometers' wind speed empirical distributions are "genuine", or affected by artifacts. In order to do this we'll compare the distribution of cup and ultrasonic speeds: assuming the ultrasonic anemometer can be used as a reference for wind speed data, we'll realize there is a difference.

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- We'll then introduce a simple statistical model for comparing cup and ultrasonic wind distributions and assess their similarity and differences. In the process we'll introduce a *minimum affected speed* at which a cup anemometer begins to report speeds statistically comparable to ultrasonic anemometers.
- We'll then try a “direct” comparison of cup and ultrasonic speed, by averaging all the cup speeds corresponding to any specific ultrasonic speed means (the storage resolution of the data set used is 0.1 m/s, and fits ideally with this intention). In the process we'll find that this method, although simple and intuitive to formulate, is affected by a speed-dependent bias which at low speeds makes it uninformative.
- Then we'll do a step back, and study the difference between same-time-stamp cup and ultrasonic speeds from a statistical standpoint.

In order to accomplish this task we first of all need real-world data. I'm lucky in this regard, as I live in Lombardy where comparison data are readily and publicly available.

5.2. The data set: the SHAKEUP micro-meteorological network

5.2.1. The network

The SHAKEUP micro-meteorological network, property of the Lombardy Environmental Protection Agency (ARPA Lombardia), is composed by some stations realized using uniform technology.

A subset of these stations are equipped with a three-dimensional ultrasonic anemometer and a cup and vane anemometer operating in backup mode, plus other sensors (namely a rain gauge, a thermo-hygrometer, a pyranometer and a net radiometer).

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The anemometers are on top of a mast at same altitude, as shown in Figure 5.1. The rod supporting the anemometers is aligned to West-East axis, corresponding to the direction of the dominant synoptic winds, to reduce inter-sensor interference.



Figure 5.1.: Anemometers of a SHAKEUP station; the sensors are always placed in the same position; the lateral rod supporting the three-dimensional ultrasonic anemometer (right) and the cup-vane anemometer (left) is about 2m long

The VV20 cup anemometer is an example of professional-grade wind speed sensor, with characteristics in data sheet (CAE n.d.). Its activation threshold U_a has a nominal value of 0.5 m/s; the real value may be different because of wear-out and non-nominal operating conditions.

Wind speed data are collected by the data logger (an OpenData20 by CAE SpA) on every 10s, and 10-minutes averages are built from these readings. Conversion from revolutions to electrical readings is performed by counting the comb-induced rising fronts in the sensor output: this

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means, in meteorological terms, that the readings on every 10s reflect the *mean* speed on that period, and not an instant value.

Values equal to zero because of wind calms (that is, $U \leq U_a$) are considered valid and included in averages; calm winds are also used to set the *direction* measurements invalid.

The USA-1 ultrasonic anemometer is a research-grade wind and temperature sensor with activation threshold of 0.01 m/s on all the three axes u, v, w . This is also the nominal resolution of the wind readings.

Data from the ultrasonic anemometers are sampled at rate of 20 readings per second; every couple of elementary readings is then averaged, providing a continuous data stream of 10 measurements per second.

It is worth noting the ultrasonic anemometer readings refer to the time instants they are taken at, and this differs conceptually from the way cup anemometers data are gathered: with an ultrasonic anemometer the mean wind must be recovered by some form of low-pass filtering.

Ultrasonic anemometer data are gathered by a MeteoFlux Core V2 eddy covariance system, like the one shown in Figure 5.2.

The MeteoFlux Core V2 system processes the data by applying 2-rotations eddy covariance (Aubinet, Vesala, and Papale 2012), followed by calculation of turbulence parameters using Monin-Obukhov surface layer similarity (Sozzi and Favaron 1996). A subset of the processing result is then sent to the OpenData20 data logger, whose main task in their regard is attributing them a time stamp comparable to the backup anemometer's data, and transmitting the stamped results to the Agency database.

5.2.2. Stations actually considered

The next table presents the stations whose data have been considered in the research phase preliminary to the writing of this book.

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Figure 5.2.: A MeteoFlux Core V2 real-time eddy covariance system (Servizi Territorio srl)

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Table 5.1.: The SHAKEUP stations considered

Station code	Station name	Position (UTM32, WGS84)	Altitude (m above msl)	Area type
AC	Arconate	488071E 5043901N	187	Peri-urban, high-plain
CL	Caiolo	561206E 5111570N	277	Grassland in W-E narrow valley
PN	Cinisello Parco Nord	516096E 5043218N	147	Grassland and mid-height deciduous trees, surrounded by dense urban areas
ML	Mantova Lunetta	643347E 5002120N	23	Grassland and scattered mid-size buildings
FC	Passo del Foscagno	592636E 5149610N	2319	Barren soil and alpine tundra
SG	Pieve San Giacomo	594004E 4997177N	36	Grassland

5.2.3. Some words about data validation

In strict terms the horizontal wind is a complex number

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$$W = U \exp i\theta$$

with U the speed, and θ the wind provenance direction in radians expressed according to the geometrical frame (that is, $\theta = \frac{\pi}{2} - \delta$, where δ is the wind provenance direction in the geographic frame).

The convention adopted for the SHAKEUP network is to apply (in)validation *to the complex number*.

In more detail:

- Invalid horizontal wind is notified by setting the wind direction (normally expressed in degrees) to the conventional value (777).
- The wind speed U is reported as they have been calculated, even if the corresponding wind direction is labeled invalid.

This allows, if needed, to invalidate *both* the horizontal wind speed and direction.

But also, if needed, to consider the wind speed alone: this is the approach I've followed in this work.

5.3. The Weibull distribution

5.3.1. Definition

The density probability function of the Weibull distribution is

$$f(x) = \frac{\gamma}{\alpha} \cdot \left(\frac{x - \mu}{\alpha} \right)^{\gamma-1} \cdot \exp \left[- \left(\frac{x - \mu}{\alpha} \right)^{\gamma} \right]$$

where γ is the shape parameter, μ the location parameter, and α the scale parameter (Rinne 2009), with $x \geq \mu$ and $\alpha, \gamma > 0$.

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In many practical applications the location parameter μ has value 0: this is the “two-parameter Weibull distribution”, the one most popular among wind energy practitioners: we’ll refer to it.

In Figure 5.3 we can see some examples of the Weibull density, all sharing the same location and scale.

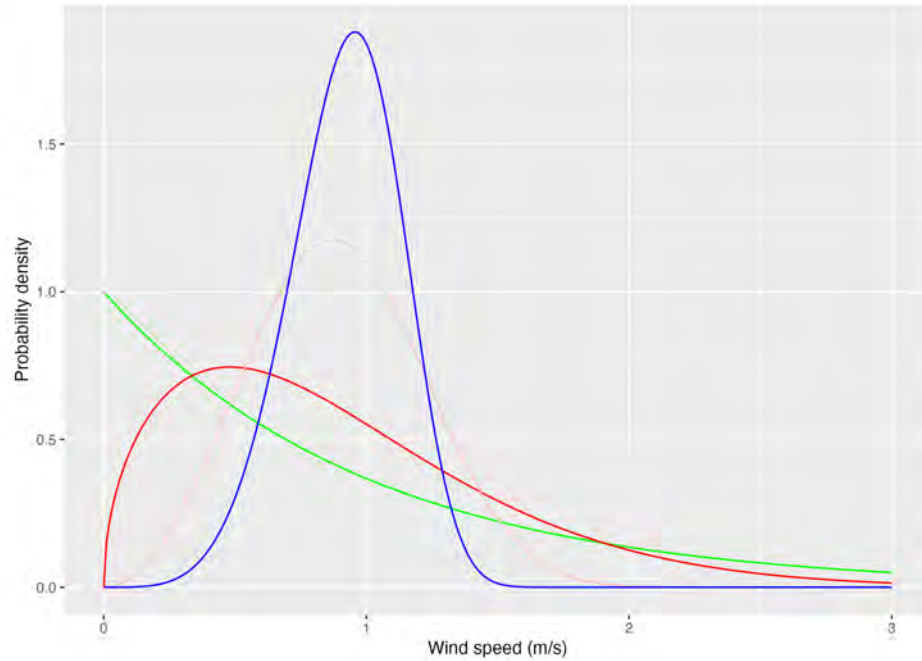


Figure 5.3.: Some Weibull probability density functions, with location parameter 0 and scale parameter 1; scale parameter is 1 for the green curve, 1.5 for the red, 3 for the pink, and 5 for the blue

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5.3.2. Estimating the shape and scale parameters from data

As a preliminary step towards actually using the Weibull distribution it is important to find a simple and robust way to estimate the scale and shape parameters from available data.

The context, in this case, is that of a “large sample”, that is, in which the number of data considered is larger than, say, 50: this could be used when deciding the approach.

To date many ways exist for estimating the Weibull parameters. Some are based on the method of moments, but the ones simpler and most popular to date are based on Maximum Likelihood Estimates, abbreviated in MLE (Ugarte, Militino, and Arnholt 2008).

In this section we’ll employ the bias-free modified MLE approach introduced in (Jokiel-Rokita and Pia 2024).

The method applies to the two-parameter Weibull distribution,

$$f(x) = \frac{\gamma}{\alpha} \cdot \left(\frac{x}{\alpha}\right)^{\gamma-1} \cdot \exp \left[- \left(\frac{x}{\alpha}\right)^{\gamma} \right] \quad (5.1)$$

and proceeds step-wise.

In the first step the shape parameter is estimated by solving the non-linear equation resulting from the application of log-modified-MLE,

$$f(\gamma) = \frac{n-1}{n} \sum_{i=1}^n v_i^{\gamma} + \frac{1}{n} \left[\sum_{i=1}^n \log(v_i^{\gamma}) \right] \cdot \sum_{i=1}^n v_i^{\gamma} - \sum_{i=1}^n v_i^{\gamma} \log(v_i^{\gamma}) = 0 \quad (5.2)$$

The presence of logarithms in Equation 5.2 demands strictly positive wind speed values, a condition which may be enforced prior to solving the equation by purging any $v_i = 0$. Once done, and then allowing us to assume

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$v_i > 0$, we notice Equation 5.2 is an equation in the single variable γ , with a unique real solution as explained in (Jokiel-Rokita and Pia 2024). As such it can be solved for example by an iterative numerical method.

The shape parameter γ is not free: it must satisfy the condition $\gamma > 0$. But under the stated condition on the wind speed we're sure that $v_i^0 = 1$ so that

$$f(0) = \frac{n-1}{n} \sum_{i=1}^n 1 + \frac{1}{n} \left[\sum_{i=1}^n 0 \right] \cdot \sum_{i=1}^n 1 - \sum_{i=1}^n 1 \cdot 0 = n-1 > 0$$

This allows to assume $\gamma = 0$ for the lower bound of the search interval.

An upper bound, on the other side, can be obtained by setting its provisional value to 1 then, if $f(1) > 0$, by doubling it. If also $f(2)$ is positive then the value 4 is explored, and so on checking the $f(2^k)$ for k increasing. As the function $f(\gamma)$ has one only root sooner or later this search will end with $f(2^k) < 0$: this will be assumed as the upper search bound.

Once the lower and upper bound are located, the actual root search may be accomplished by any numerical method.

Once the root $\hat{\gamma}$ is found, the scale parameter (Lei 2008) can be estimated as

$$\hat{\alpha} = \left(\frac{1}{n} \sum_{i=1}^n v_i^{\hat{\gamma}} \right)^{\frac{1}{\hat{\gamma}}} \quad (5.3)$$

It must be pointed out that both $\hat{\gamma}$ and $\hat{\alpha}$ are *estimates*, and their value may differ from reality.

The following table, obtained by estimating the scale and shape parameters from samples of various sizes taken from a Weibull distribution with $\gamma = 2.4$ and $\alpha = 1.2$.

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Table 5.2.: Modified MLE estimates of shape and scale parameters from a Weibull distribution having scale 1.2 and shape 2.4

Number of trials	Mean $\hat{\gamma}$ (Std.dev)	Mean $\hat{\alpha}$ (Std.dev)
10^1	2.58 (0.73)	1.18 (0.16)
10^2	2.43 (0.20)	1.20 (0.05)
10^3	2.40 (0.06)	1.20 (0.02)
10^4	2.40 (0.02)	1.20 (0.01)
10^5	2.40 (0.01)	1.20 (0.00)

If the modified MLE is applied to arbitrary (or measured) data instead of artificial then the method depicted will find anyway values for $\hat{\gamma}$ and $\hat{\alpha}$: necessity then arises (*whatever the case*) to test whether the empirical distribution is really Weibull up to some appropriate level of confidence, for example using a Chi-square or Kolmogorov-Smirnov test (Bartoszynski and Niewiadomska-Bugaj 1996).

5.4. The cup anemometer wind speed empirical distribution

In Figure 5.4 the yearly histograms of wind speed at various stations of identical built are shown for year 2023.

As we can see, no histogram in Figure 5.4 resembles any Weibull distribution.

The most striking departure from the situation presented in Figure 5.3 is the spike corresponding to values equal and nearby to zero: its magnitude depends on the station, but is nevertheless present,

In addition, the classes closest to zero show a tendency towards a gradual rise starting from a positive density value.

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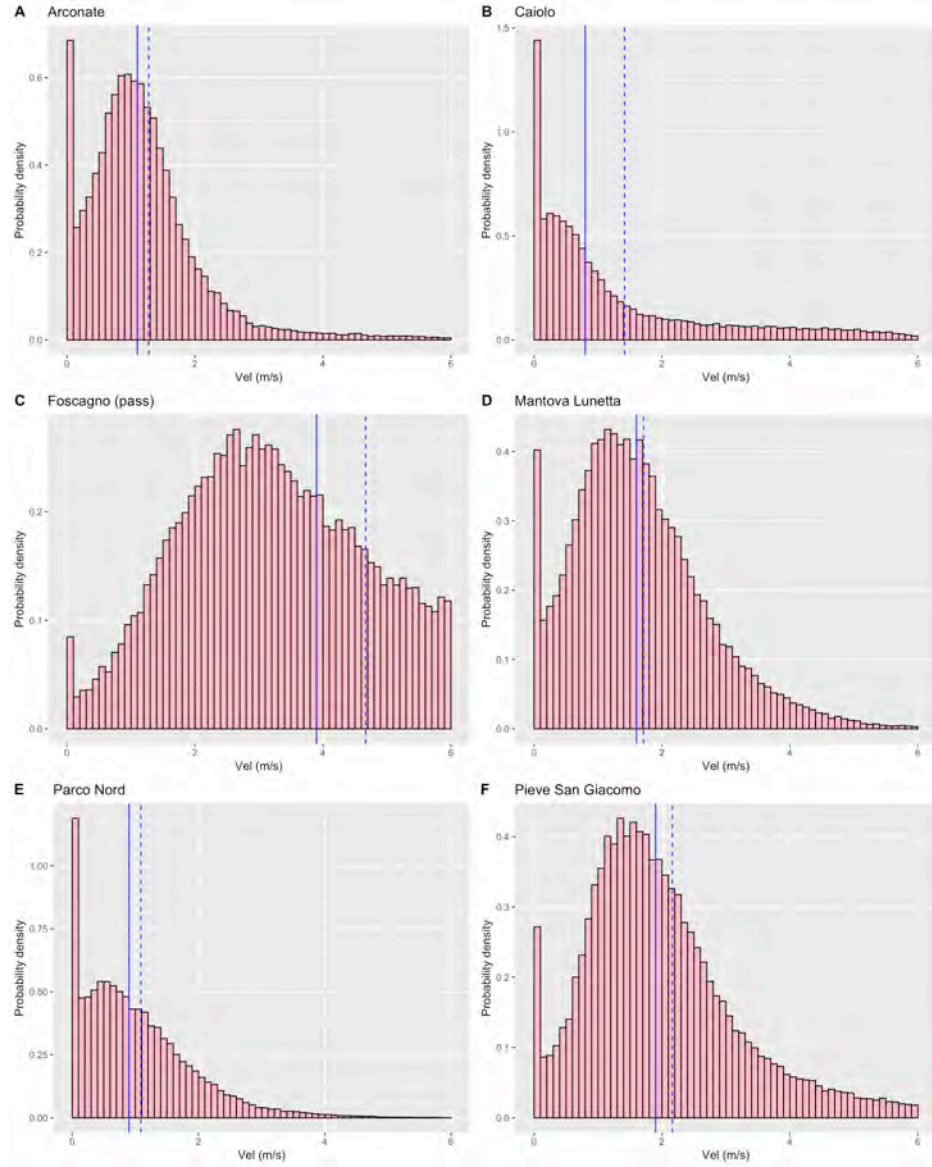


Figure 5.4.: Histograms of the wind speed measured by the cup anemometers located at six SHAKEUP sites; the blue vertical continuous line is located at the median, the blue dashed line at the mean

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But maybe there is something peculiar in year 2023: let's then, have a look to some other years.

Comparing Figure 5.5, Figure 5.6, Figure 5.7 and Figure 5.4 we can easily detect differences. But the pattern stays unchanged: spike at zero speed or close, and gradual upwards from a positive value nearby.

5.5. Genuine, or artifact?

5.5.1. A sensible question

Given the known fact that cup anemometers have a non-zero response only if the wind speed exceeds the activation threshold U_a , the question arises whether the histograms shown in Figure 5.4 were affected by this peculiarity, and to what extent.

In order to attempt answering this question, a comparison with co-located ultrasonic anemometer data has been made at the SHAKEUP sites mentioned in this chapter. This comparison makes practical sense, as ultrasonic anemometers are known to be able measuring any wind speed accurately within their measurement ranges. Use of ultrasonic anemometers as reference sensors for cup anemometers is also common in literature (Yahaya and Frangi 2004), so we can consider the comparison appropriate and, hopefully, revealing.

5.5.2. Comparing with a reference sensor

Figure 5.8 is the same as Figure 5.7, with wind speed data coming from the ultrasonic anemometer (UA) instead of the cup anemometer (CA).

The difference between the two figures is striking: no more spikes! Overall, the histograms suggest the ultrasonic anemometer data visually resemble a Weibull distribution more than cup anemometer's do.

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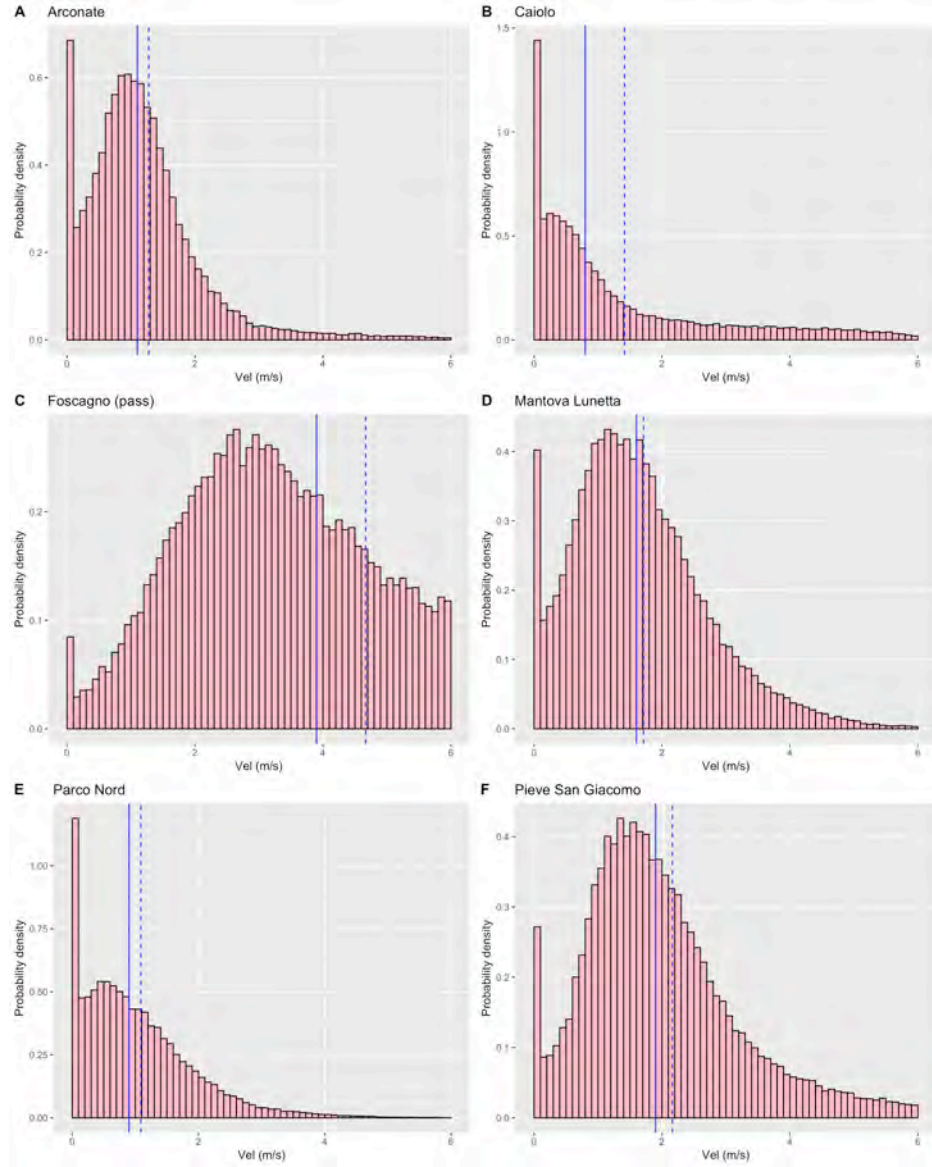


Figure 5.5.: Histograms of the wind speed measured by the cup anemometers located at six SHAKEUP sites; the blue vertical continuous line is located at the median, the blue dashed line at the mean

5. Wind speed statistics

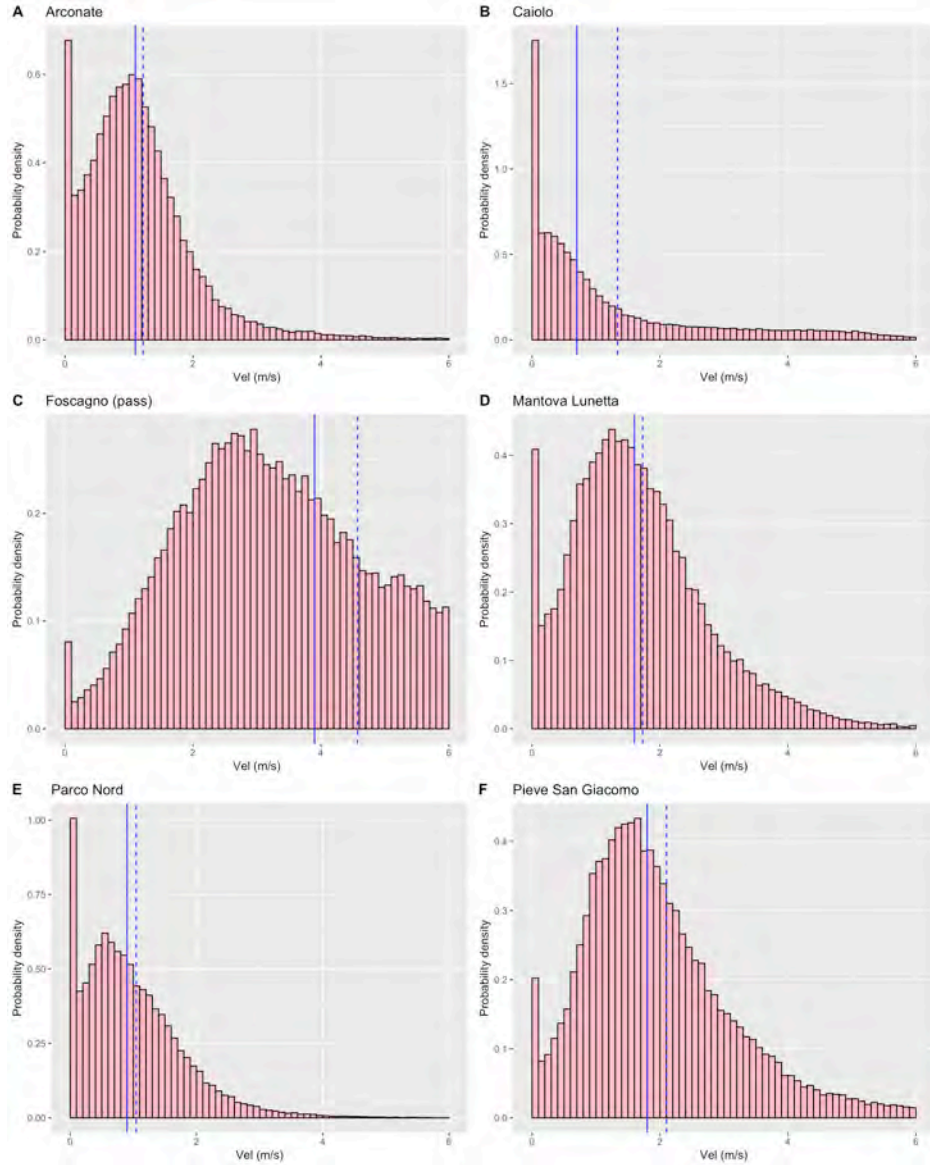


Figure 5.6.: Histograms of the 10-minute averages of wind speed measured by the cup anemometers located at the six SHAKEUP sites on year 2021; vertical lines as for 2023

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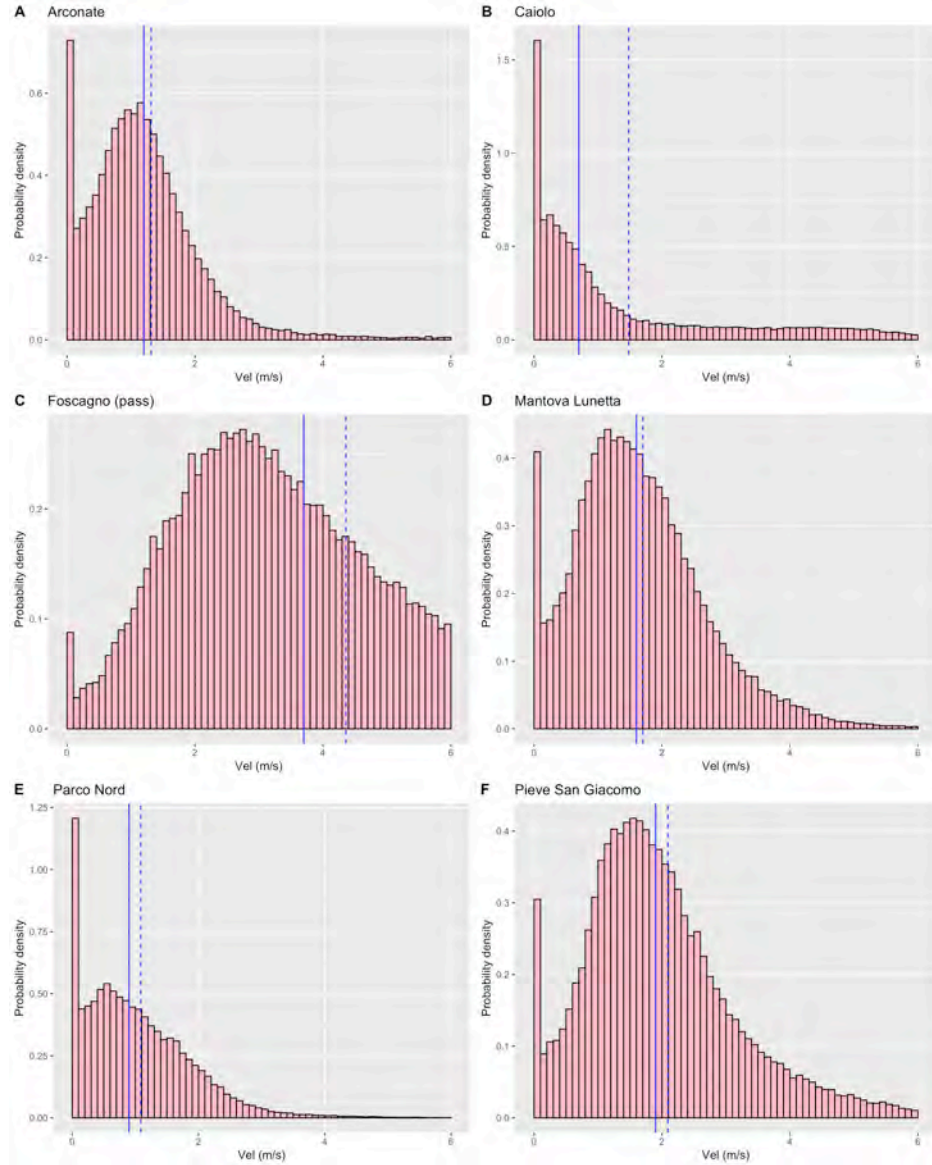


Figure 5.7.: Histograms of the 10-minute averages of wind speed measured by the cup anemometers located at the six SHAKEUP sites on year 2022; vertical lines as for 2023

5. Wind speed statistics

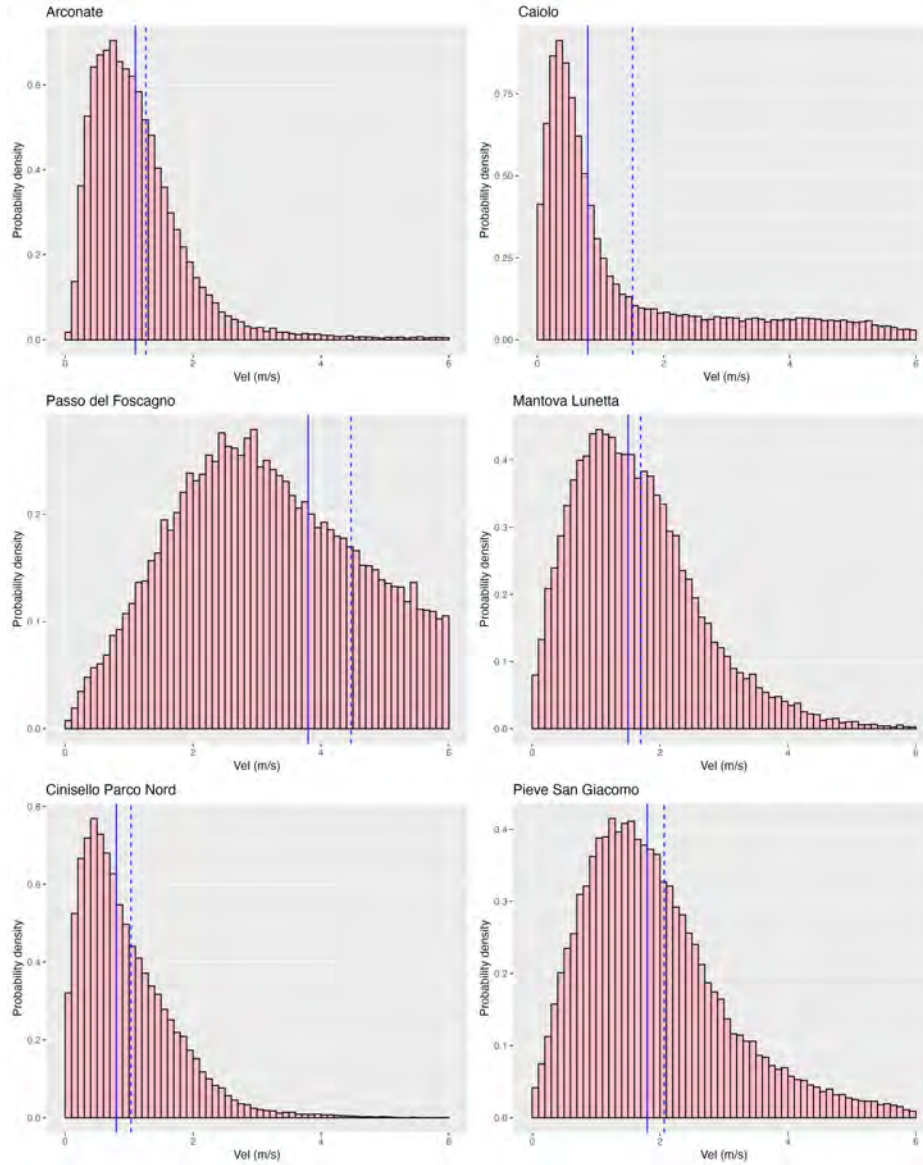


Figure 5.8.: Histograms of the wind speed measured by the cup anemometers located at six SHAKEUP sites; the blue vertical continuous line is located at the median, the blue dashed line at the mean

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However, if we compare the empirical distributions with their best fitting Weibull densities using the modified MLE approach described in Section 5.3.2 on valid wind data with $U \leq 3\text{m/s}$, we get results similar to what seen in the following table.

Table 5.3.: Best-fitting Weibull parameters for 2022 data and their corresponding chi-square statistics

ID	$\hat{\alpha}(UA)$	$\hat{\alpha}(CA)$	$\hat{\gamma}(UA)$	$\hat{\gamma}(CA)$	χ^2_{UA}	χ^2_{CA}
AC	1.26	1.36	2.03	1.93	1606.94	4119.28
CL	0.92	0.95	1.34	1.26	6056.38	3449.85
FC	2.33	2.24	3.24	3.28	5096.01	10035.56
ML	1.66	1.71	2.16	2.22	1822.43	6298.00
PN	1.09	1.22	1.60	1.59	1160.76	2977.54
SG	1.81	1.88	2.43	2.56	2439.42	8563.88

The p-value associated to the χ^2 statistics, computed with 29 degrees of freedom (corresponding to the new 30 wind classes starting at zero speed and increasing regularly by 0.1m/s), is 1 in all cases, indicating the null hypothesis that data are distributed according to a Weibull should be always rejected.

If no site-related data set follows a Weibull distribution, we can yet observe there is a speed beyond which the wind speed empirical distributions from the ultrasonic and cup anemometers are visually very similar.

Assuming the ultrasonic anemometer is the reference sensor for cup anemometers, determining this speed allows to see until which value the 10-minutes mean speed are somewhat perturbed by the activation threshold U_a of the cup anemometer.

In this analysis it is important to take into account the problem independent variables include, but are *not limited* to the activation threshold. In particular, the local wind frequency of 10s readings below the activation

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threshold (a climatological parameter) should be taken into account: in other words, the quality of data provided by a cup anemometer *depends on its climatological context*.

It may then be that a sensor of good characteristics on paper does not fit well with local environmental conditions: the proposal of a sensor or the other by a manufacturer responding to a bid request should consider not only budget constraints, but consideration of local conditions. Would these be not known, it would be rational the manufacturer actuates a preliminary survey measurement campaign allowing them to select the proper sensor. The customer, on their side, should state as clearly as possible which applications data will be used for; would data be released to the general public in open form, the manufacturer should imagine the most severe operating condition, for example high accuracy under slow wind (this is the explicit focus of this book, but other criteria are equally valid - provided they are *stated explicitly*).

So, let's extend the comparison to the slow wind behavior. The question, in this case, is whether there is some wind speed U_e such that the fraction of wind speed data below or above this threshold speed is the same regardless of the fact that the anemometer is ultrasonic or cup.

The rationale behind this question is that if the fractions differ by anemometer technology, then there one of the two induces some form of systematic perturbation large enough to be detected; if this perturbation is concentrated towards lower wind speeds, then as U_e increases sooner or later any difference will become so small to be undetectable, or at least not statistically significant at some stated confidence level. Because of this, what we are interested to is not one value of U_e , but rather their lower bound $\inf U_e$, or an approximation to it.

Of course, if there is some systematic deviation between the cup and ultrasonic data then it could well be the fraction mentioned above will differ by technology whatever the separation speed.

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In order to analyze the problem mathematically, let then $U_s \geq 0$ a separation speed, and let

$$N_{UA}^- = |\{U_{UA} \leq U_s\}|$$

$$N_{UA}^+ = |\{U_{UA} > U_s\}|$$

$$N_{CA}^- = |\{U_{CA} \leq U_s\}|$$

$$N_{CA}^+ = |\{U_{CA} > U_s\}|$$

Clearly,

$$N_{UA}^- + N_{UA}^+ = N_{UA}$$

and

$$N_{CA}^- + N_{CA}^+ = N_{CA}$$

If we divide both members by N_{UA} , respectively N_{CA} , and write $p^- = \frac{N_{UA}^-}{N_{UA}}$, $p^+ = \frac{N_{UA}^+}{N_{UA}}$, $q^- = \frac{N_{CA}^-}{N_{CA}}$ and $q^+ = \frac{N_{CA}^+}{N_{CA}}$, with

$$p^- + p^+ = q^- + q^+ = 1$$

So, we can say $p^+ = 1 - p^-$ and $q^+ = 1 - q^-$, that is, we may restrict our attention to p^- and q^- only.

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Finally, we can ask whether $p^- = q^-$: this is our null hypothesis; if it is fulfilled, then we can automatically conclude by the relationships above that $p^+ = q^+$, which is much more expensive to compute.

I will not follow this line, however, and leave its exploration to you as an exercise, if you like (spoiler: it *is* interesting).

5.6. A simple method for comparing cup and ultrasonic speeds

5.6.1. The speed proportion test

As we have seen, the distribution of homologous (same site and same time span) of CA and UA speed data differ, as shown for example by Figure 5.7 and Figure 5.8.

The same figures also suggest the difference is concentrated towards the left of the wind speed range, while moving to larger speeds the difference becomes apparently smaller.

This may be seen even more clearly if frequency lines are used instead of histograms, and their values are overlapped on the same plot, as we can see in Figure 5.9.

On plots we see the wind speed statistics tend to redistribute as the sensor changes from CA to UA: the lower speeds migrate towards right where the CA underestimates the UA speed, and intermediate speeds where the CA overestimates the UA speed migrate leftwards.

But there is a speed, in correspondence of the larger speeds tail, where the two curves visually overlap.

In order to quantify this effect we could introduce a *cut speed* U_c . Given the data sets

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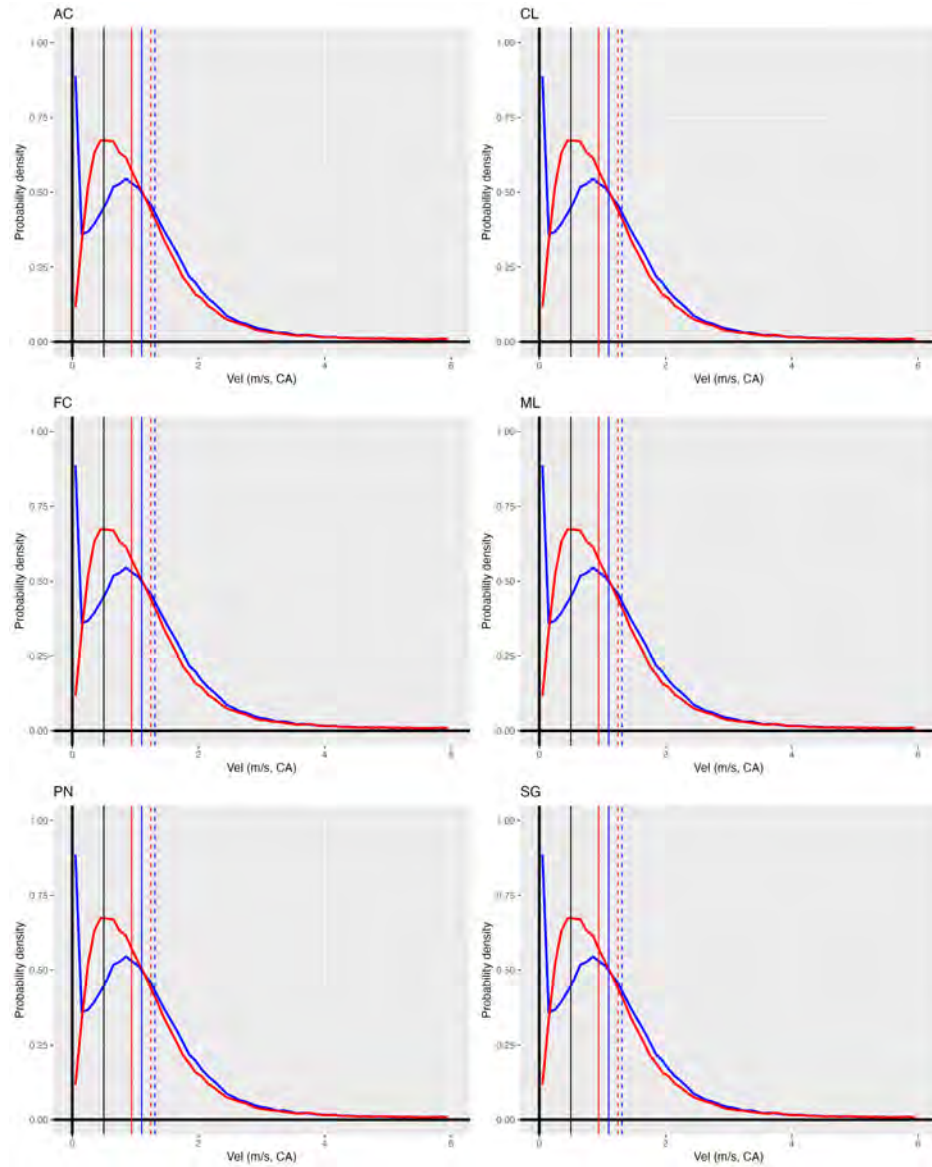


Figure 5.9.: Empirical frequency diagrams of CA (blue) and UA (red) 10-minutes mean speed (the continuous vertical line indicates the median, the dashed the mean; the continuous black line represents the position of the VV-20 cup anemometer activation threshold)

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$$\xi_{CA} = \{t \in E | U_{CA}(t) \leq U_c\}$$

and

$$\xi_{UA} = \{t \in E | U_{UA}(t) \leq U_c\}$$

we can set

$$f_{UA}^- = \frac{\|\xi_{UA}\|}{\|E\|}$$

and

$$f_{CA}^- = \frac{\|\xi_{CA}\|}{\|E\|}$$

where $\|S\|$ represents the number of elements of the finite set S . In a similar way we could define f_{UA}^+ and f_{CA}^+ by considering the wind speed values *larger* than the cut speed U_c , but this is not that necessary as, by construction,

$$f_{CA}^+ + f_{CA}^- = 1$$

and

$$f_{UA}^+ + f_{UA}^- = 1$$

so that

$$f_{CA}^- = 1 - f_{CA}^+$$

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and

$$f_{UA}^- = 1 - f_{UA}^+$$

As f_{CA}^- and f_{UA}^- are proportions their values can be compared directly. We do by verifying the null hypothesis that

$$f_{CA}^- = f_{UA}^-$$

at a given confidence level $\alpha = 0.95$.

Under this conceptual frame it is known the test statistics

$$Z = \frac{f_{CA}^- - f_{UA}^-}{\sqrt{\frac{f_{CA}^-(1-f_{CA}^-)}{\|\xi_{CA}\|} + \frac{f_{UA}^-(1-f_{UA}^-)}{\|\xi_{UA}\|}}} \quad (5.4)$$

follows approximately a standard normal distribution (that is, with mean 0 and standard deviation 1). So we will reject the null hypothesis at $\alpha = 0.95$ if the value of the statistics is smaller than -1.96 or larger than +1.96.

5.6.2. Applying the test to real data

The test devised in Section 5.6.1 was applied to the data available, and Figure 5.10 shows the results.

The interpretation is, however, not simple. Assuming the Z statistics does *really* follow a normal distribution, what the test says is that for some cut speed values it makes sense to reject our null hypothesis that the two proportions are equal.

The converse, however, is not necessarily true: if the Z statistics is within the test acceptance bands, then the two proportions *may* be equal *in the*

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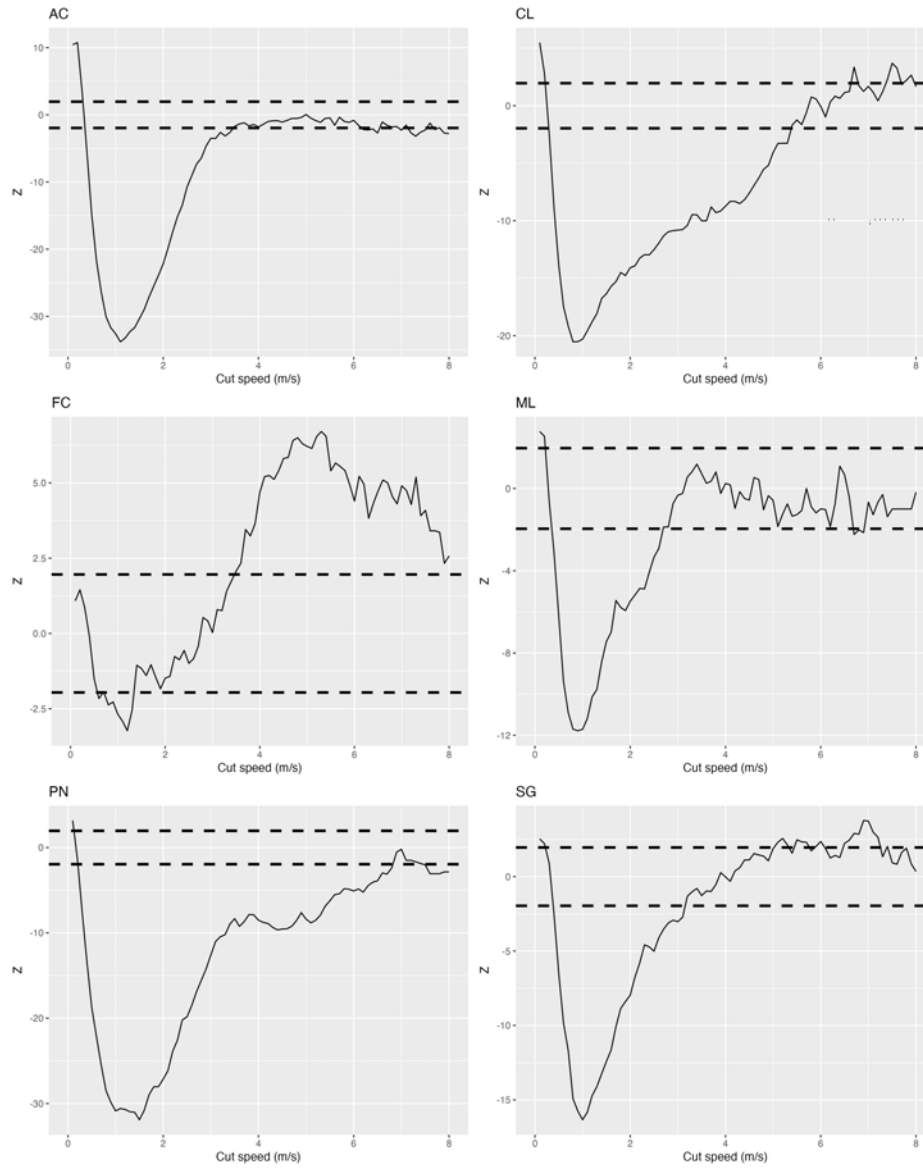


Figure 5.10.: Change of Z statistics with change of the cut speed (year 2022 data)

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whole population. We did not define it explicitly, but this whole population is just the set of all data, including the ones which the data collection procedure labeled as invalid and excluded from the sample. Is this form of sampling “random”? Likely not. And besides, the sample sizes may be extremely large, almost comparable to the whole “population”.

There is another thing to notice. If we look to the plots we discover that in all cases (with the remarkable exception of the mostly-high-wind-speed Passo del Foscagno, FC in the figure) at the minimum cut speed $U_c = 0.1$ m/s the values are marginally outside the null hypothesis acceptability band. Then they move quickly towards 0, pass it remaining for a short while in the acceptability band, and overshoot well outside the acceptability band. After then, the value of the Z statistics bend again and move towards 0, sometimes entering the acceptability band, some other times staying outside but close.

Overall, as expected, as U_c increases it is true the two proportions tend to become more similar. But the strange behavior at low cut speeds is, frankly, unexpected. And, the large “roughness” at higher wind speeds makes one suspect something is going quite wrong.

The possible cause can be found in Figure 5.11.

Here we can see that the band at which both the CA and UA proportions are within the interval 0.1-0.9 is not that extremely large. At Arconate (AC), for example, it spans the interval from 0.8 to 4.0 m/s, while at Caiolo (CL) the range is 0.7 to 5.7 m/s. At Passo del Foscagno (FC) the upper limit was not even reached by the explored range.

The proportion band from 0.1 to 0.9 indicates the speed range at which neither of f_{CA}^- and $f_{CA}^+ = 1 - f_{CA}^-$ (respectively f_{UA}^- and $f_{UA}^+ = 1 - f_{UA}^-$) gets below the 10% of available data, at the risk of becoming statistically not significant.

Figure 5.11 serendipitously indicates another very interesting thing. If we look attentively to the blue and red curves in each plot we see that they

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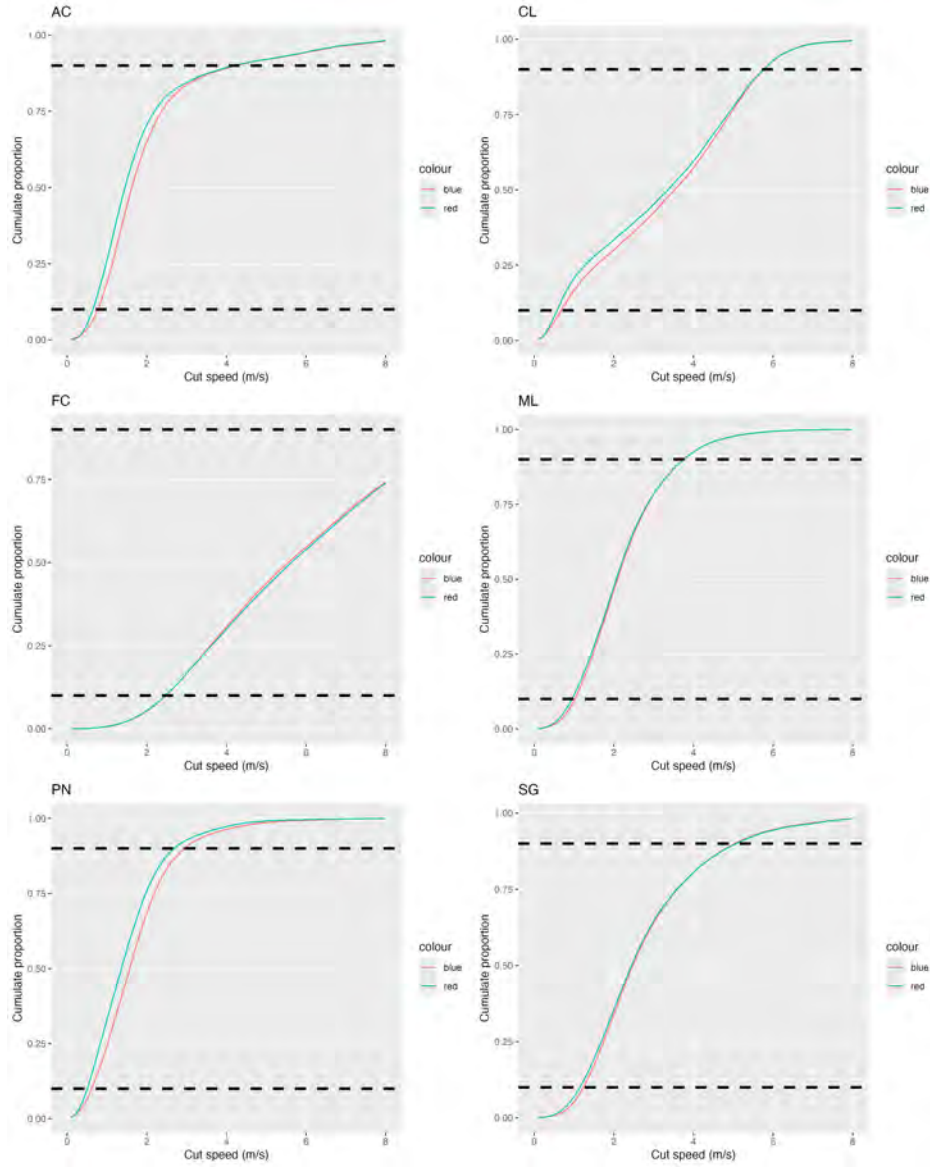


Figure 5.11.: Proportions of CA (blue) and UA (red) values not greater than the cut speed; the dashed lines indicate the 10% and 90% of total data in both cases

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indeed differ at intermediate speeds (the difference at $U_c = 0.1$ m/s, although appearing macroscopic due to the large number of zero CA speeds compared to none or close to none for UA, becomes very small if we weigh it by the *total* number of data). This difference wanes away as U_c increases, and becomes visually indistinguishable from 0 at a surprisingly high speed.

This means two things: first, the CA and UA speed distributions really become closer as the measured wind speeds increase. But second, the CA speed underestimation likely due to the presence of instantaneous speeds below the activation threshold U_a is present also at surprisingly high speeds.

The second effect is not entirely unexpected: as we saw in Section 2.4, near the Earth surface the airflow occurs in turbulent regime, and turbulent fluctuations may be massive (Stull 1988). As pointed out in (Oke 1987), for example, thermally induced convection (an important source of turbulence on daytime) may exist even at speeds of 4.5 m/s, and possibly more.

The behavior of cup anemometers under turbulence was studied by early investigators (Kondo, Naito, and Fujinawa 1971), and found significant. That it is observed at sites where low speed wind, and fully developed thermal convection, are widespread is indeed all but surprising.

This suggests adding the study of turbulence intensity to the characterization of sites where acceptable precision wind measurements are to be made using electro-mechanical anemometers. More in the dedicated chapter at end of book.

The same fact also suggests to consider the difference more directly, without scaling it by the estimated sample variance occurring in the square root at denominator in Equation 5.4.

In order to make understanding more intuitive, however, I've chosen to express the difference in relative terms:

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$$q = 2 \frac{f_{CA}^- - f_{UA}^-}{f_{CA}^- + f_{UA}^-}$$

In Figure 5.12 we can see the result.

Figure 5.12 shows a large positive spike for speeds close to zero: this is an artifact, explained in Section 5.8.2. What's really interesting is the way the indicator q approaches zero as U_c increases: fast, but not as fast as one would imagine - or hope.

Surely much more than the VV-20 cup anemometer declared activation threshold $U_a = 0.5$ m/s. This fact should not be overemphasized however: it may depend on instrument maintenance practices, but could also be influenced by the site-specific frequency of speed values below U_a as well.

Table 5.4.: Speed at which the scaled difference between CA and UA becomes smaller than 1%

ID	Station	Speed at which the CA and UA speed distribution become equal (m/s)
AC	Arconate	3.0
CL	Caiolo	5.1
FC	Passo del Foscagno	2.2
ML	Mantova Lunetta	2.6
SG	Pieve San Giacomo	3.8
PN	Cinisello Parco Nord	2.9

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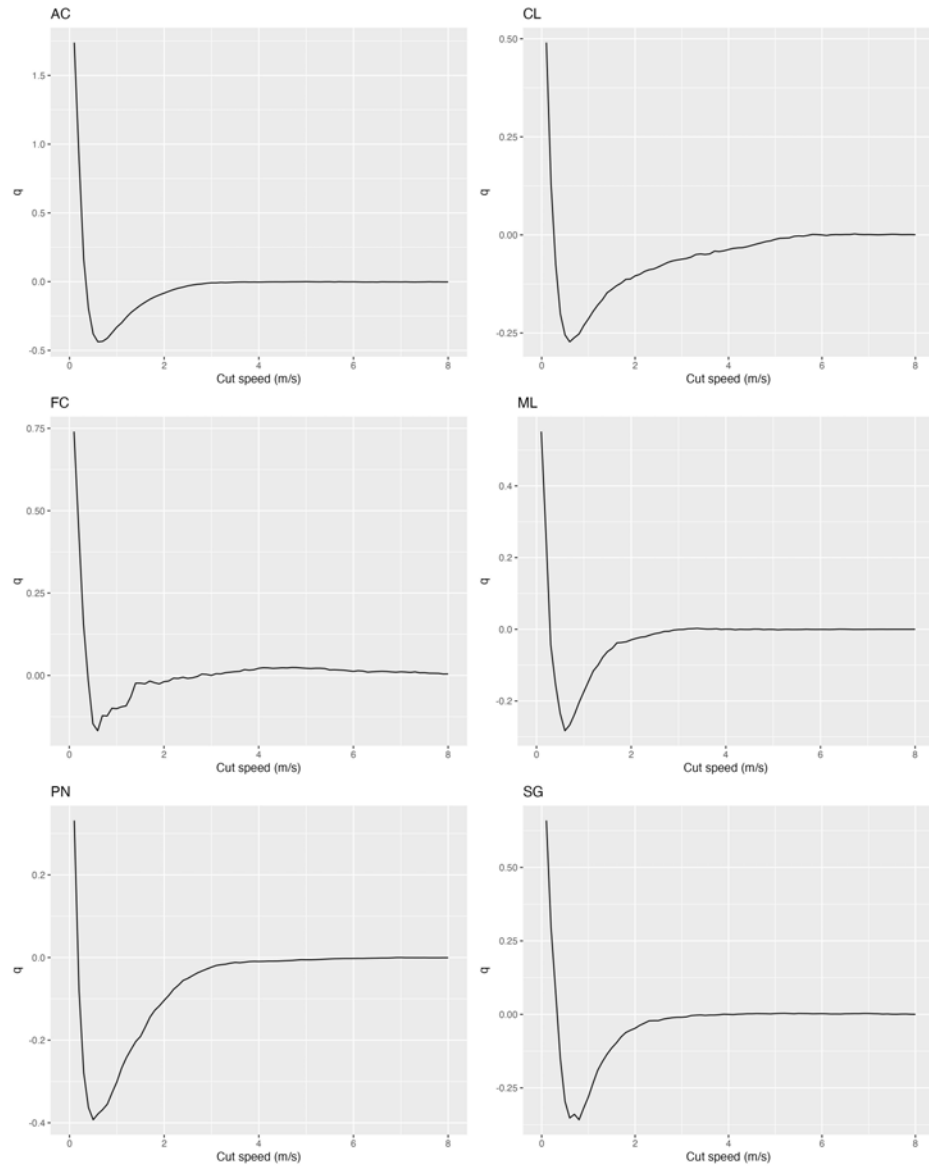


Figure 5.12.: Value of the scaled difference at SHAKEUP sites, year 2022

5.7. Quantile-quantile comparison of CA and UA speeds

Another way to compare the two distributions of CA and UA speeds is by the graphical method of “quantile-quantile plots” (see (Cleveland 1993); quantile-quantile plots were first introduced in (Wilk and Gnanadesikan 1968)).

Figure 5.13 and Figure 5.14 show the qq-plots of the CA and UA distributions focussing attention to different wind speed scales; the scaled appearance depends on the 0.1 m/s resolution in original data.

5.8. To what extent do conventional and ultrasonic wind speed sensors (dis)agree?

5.8.1. Beyond individual wind speed distributions for CA and UA data

In the approach followed until now we investigated the statistical distribution of wind speed observed using a cup or ultrasonic anemometer, and realized they differ when wind is slow.

This wasn’t really surprising, as cup (and more generally electro-mechanical) anemometers have a finite activation threshold U_a , that is, a wind speed below which the sensor stays as immobile as a statue because of mechanical friction, aerodynamic effects (e.g. stall) and maybe magnet-induced braking. While ultrasonic anemometers, having no moving parts, are free to sense wind at whatever speed it blows.

The difference in the CA and UA speed distribution, then, is all but surprising. Maybe, an indication of both sensor mechanics and the fine-scale statistics of low speed winds, if using the magnifying glass of ten minutes or hourly means.

5. Wind speed statistics

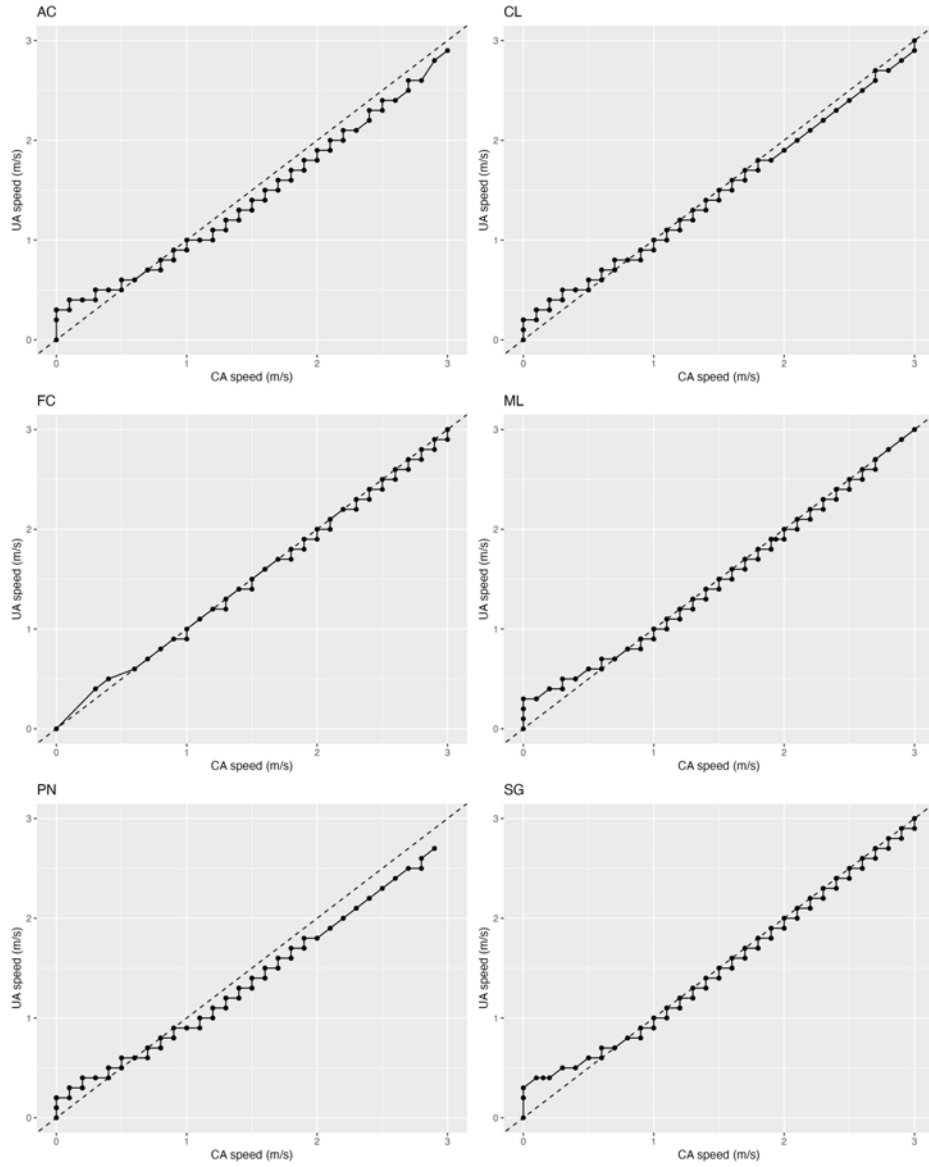


Figure 5.13.: Quantile-quantile plots of CA and UA speed distributions at different SHAKEUP stations for year 2022; focus on lower speeds; the dashed line represents the first quadrant bisector, indicating ideal distribution equality

5. Wind speed statistics

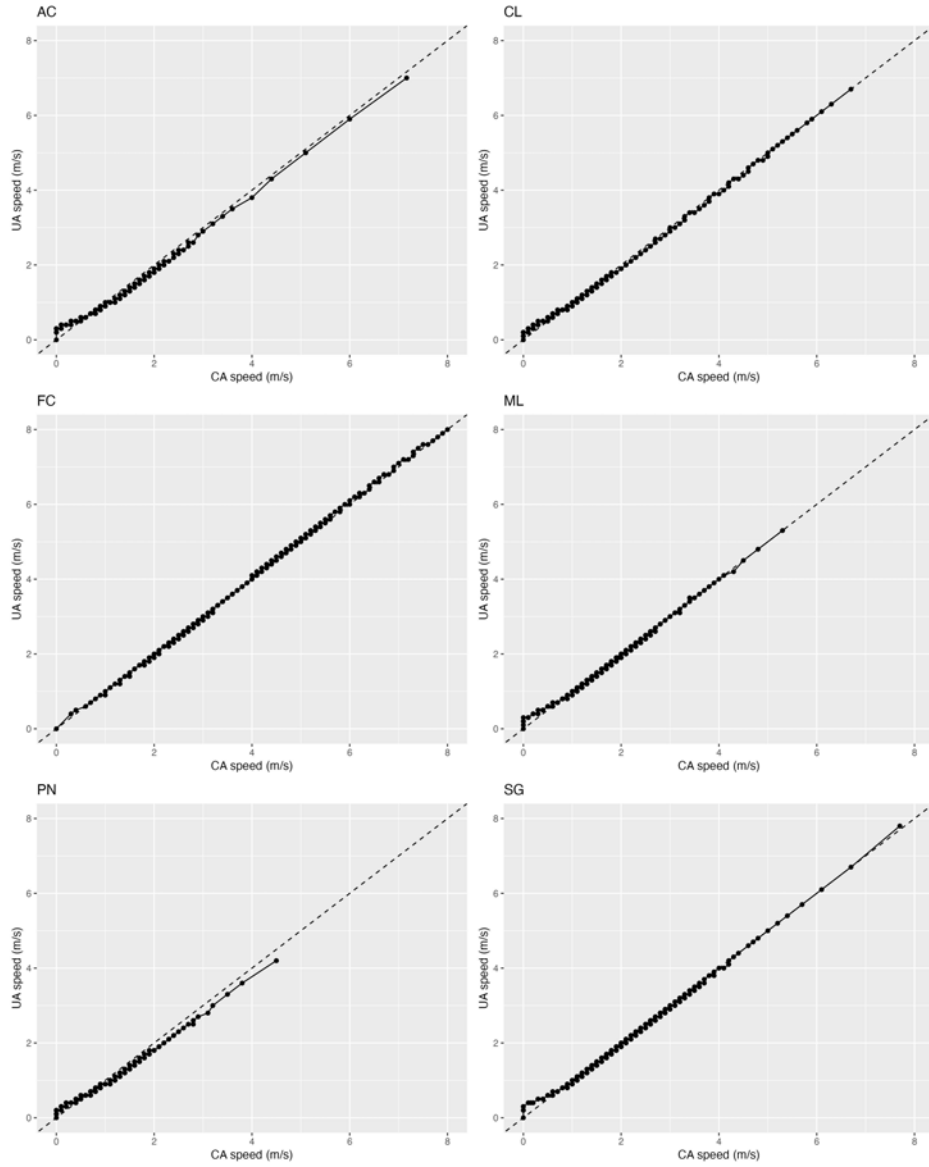


Figure 5.14.: Quantile-quantile plots of CA and UA speed distributions at different SHAKEUP stations for year 2022; focus is on extended speed range; the dashed line represents the first quadrant bisector, indicating ideal distribution equality

5. Wind speed statistics

These distributions are, of course, a consequence of the different behaviors sensors exhibit in front of the same reality. Let me say, a way to look at it a birds' altitude.

The synchronized 10-minutes means released by the SHAKEUP stations offer however an opportunity to push the limit of the comparison, by treating same-time-stamp CA and UA observations as *paired*: this offers an unique opportunity to understand how things do at quite a fine scale, possibly the smallest at which CA and UA data can be compared directly - behind of that, and towards finer resolution, there are the raw data, with their peculiar formats and sampling times, and there any direct comparison becomes impossible.

Figure 5.15, showing all the individual couples (U_{UA}, U_{CA}) , exhibit a wide data dispersion in all cases. The bulk of points however tends to group around the first quadrant bisector (black sloping line), but with a suggestion this “hopeful” correspondence breaks somewhat in cases of low wind speed.

Figure 5.15 just represents the lower speed side of the more general “total” scatters, shown in Figure 5.16. The latter shows the speed data tend to concentrate towards the lower tail of their distributions, with the only exception of the high-mountain saddle of Passo del Foscagno; in addition, data in Figure 5.16 appear strongly correlated - an idea Figure 5.15 does not support.

Said differently: the situation is quite confused. We need to use another, more direct approach.

5.8.2. Mean speed difference as a function of UA speed

Let's consider the difference $U_{UA} - U_{CA}$. In order to check whether it depends on U_{UA} we may aggregate the differences with respect to the possible values of U_{UA} in some appropriate interval, for example between 0 and 3 m/s.

5. Wind speed statistics

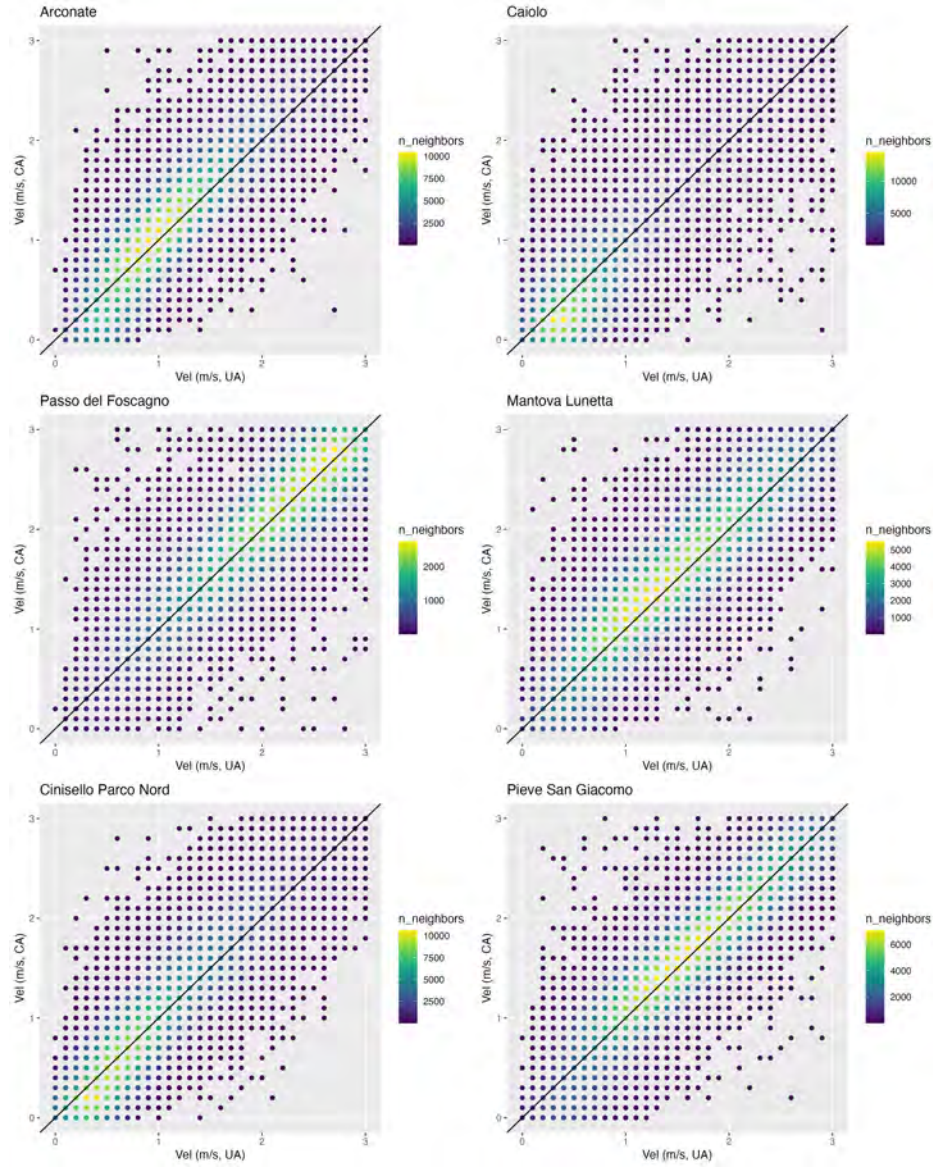


Figure 5.15.: Scatter plots of UA and CA data for year 2022 at the selected SHAKEUP sites; focus restricted to speeds from 0 to 3 m/s. The gridded aspect depends on the resolution data are retrieved from the database

5. Wind speed statistics

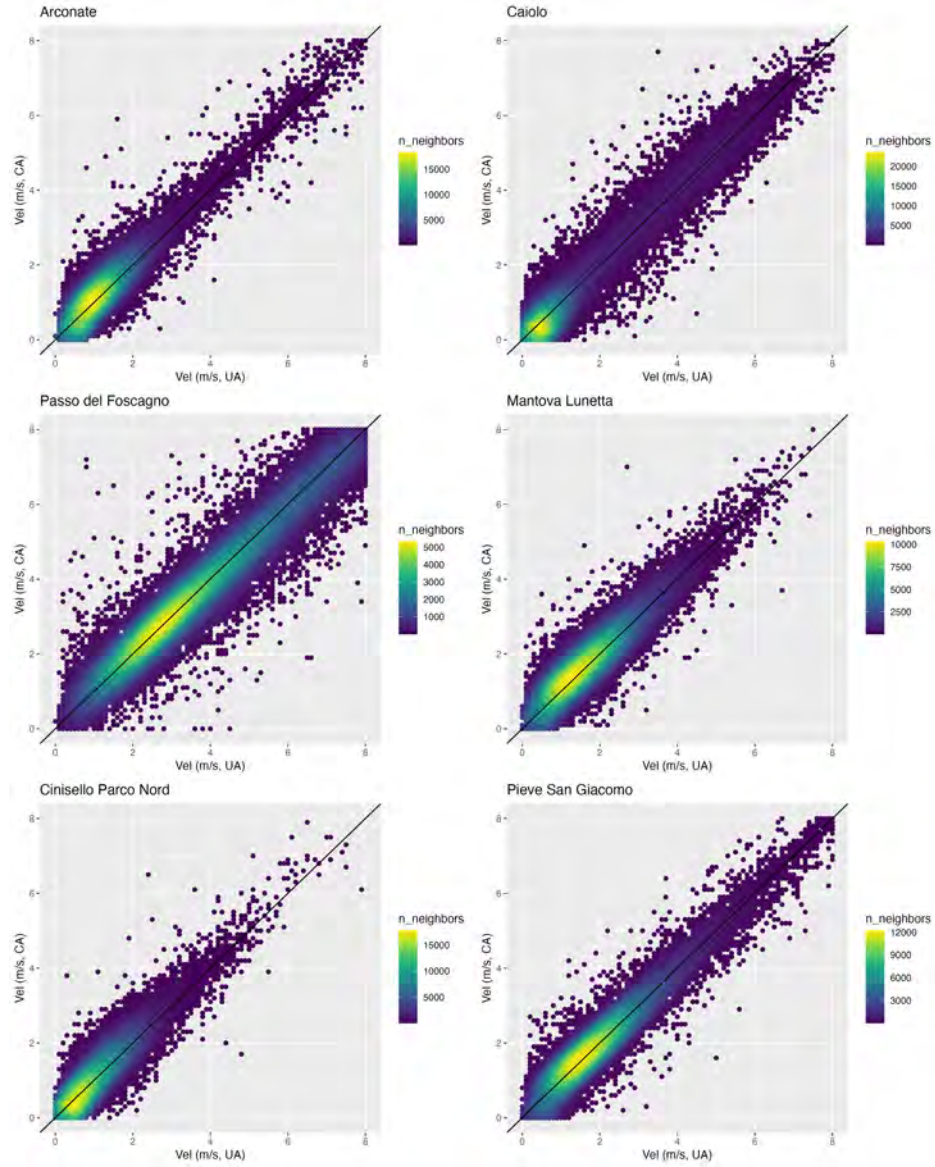


Figure 5.16.: Scatter plots of UA and CA data for r year 2022 at the selected SHAKEUP sites

5. Wind speed statistics

This makes sense because the speeds gathered from the ARPA database have one representative digit past the comma (that is, U_{UA} can be thought as “discrete ordered variable”). As in Figure 5.15 we restrict the attention to wind speeds $U_{UA} \leq 3$ m/s, thus allowing for 31 wind speed classes $U_{UA} = 0.0, 0.1, 0.2, \dots, 3.0$ m/s.

Before to begin, however, a caveat is to be made: comparing same-time-stamp wind speeds is deeply different from comparing distributions in non-coupled way.

The main reason is, for coupled data to be considered both couple members must be valid: coupling is a much more restrictive criterion than considering data separately, and may give different suggestions. This diversity, rather than contradictory, has the potential of being illuminating on other sides.

Plots in Figure 5.17 show some comparison between speeds sampled from CA and UA. Figure 5.18 tells the same story, using a slightly different scale. Error bars indicate the standardized sampling variance $\sigma/\sqrt{n}$, with n the sample size, that is, the number of CA speed data for which the UA speed equals the value indicated in abscissas, and σ the sample standard deviation: at same standard deviation, their value is larger when the sample is small, and this happens for “small” values of UA speeds, as indicated in Figure 5.19.

So, does a value above the first quadrant in Figure 5.17 necessarily indicates an overestimation? On a first glance this is the meaning common sense suggests. But if we relax and consider some ramifications more, we see this is indeed not.

For instance: let’s consider the way we selected the samples. First of all we swept out all couples where one or both their members have been labeled as invalid in the original data sets. This is not that special, and is quite necessary for any comparison statistics to be calculated. Yet it has some consequence, as you can see from the numbers.

5. Wind speed statistics

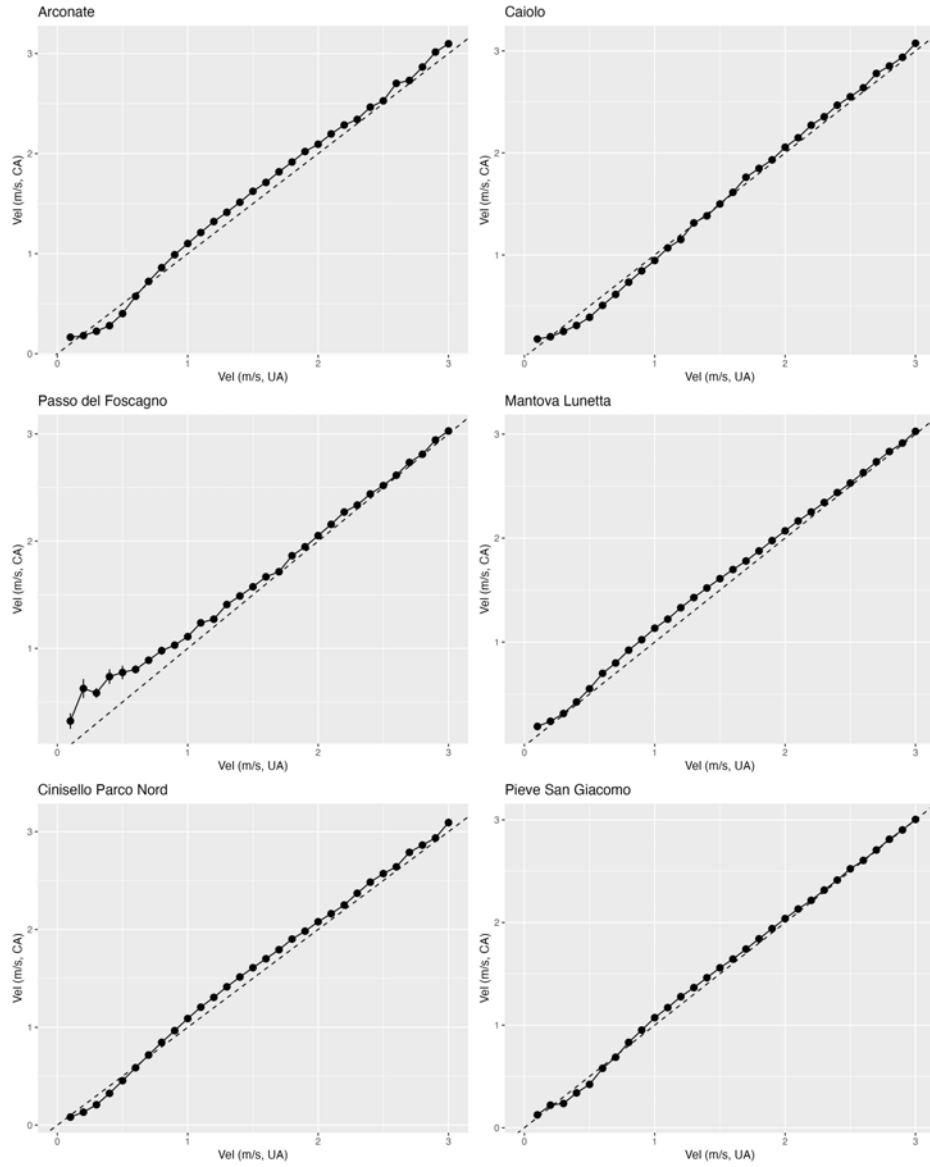


Figure 5.17.: Mean of CA speeds corresponding to specific UA values for the selected SHAKEUP sites; under ideal conditions all data points stay exactly on the first quadrant bisector (no difference)

5. Wind speed statistics

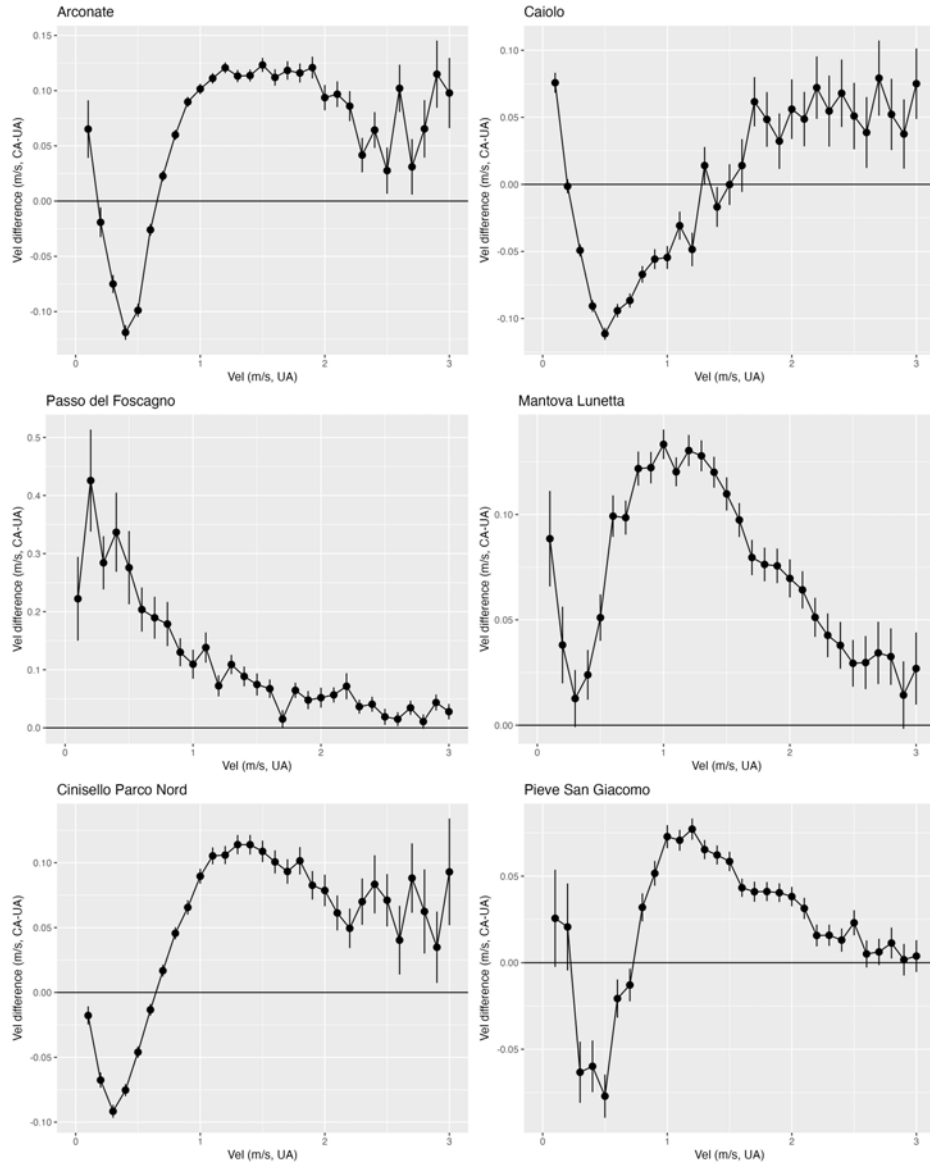


Figure 5.18.: Absolute difference between CA and UA averages instead of UA

5. Wind speed statistics

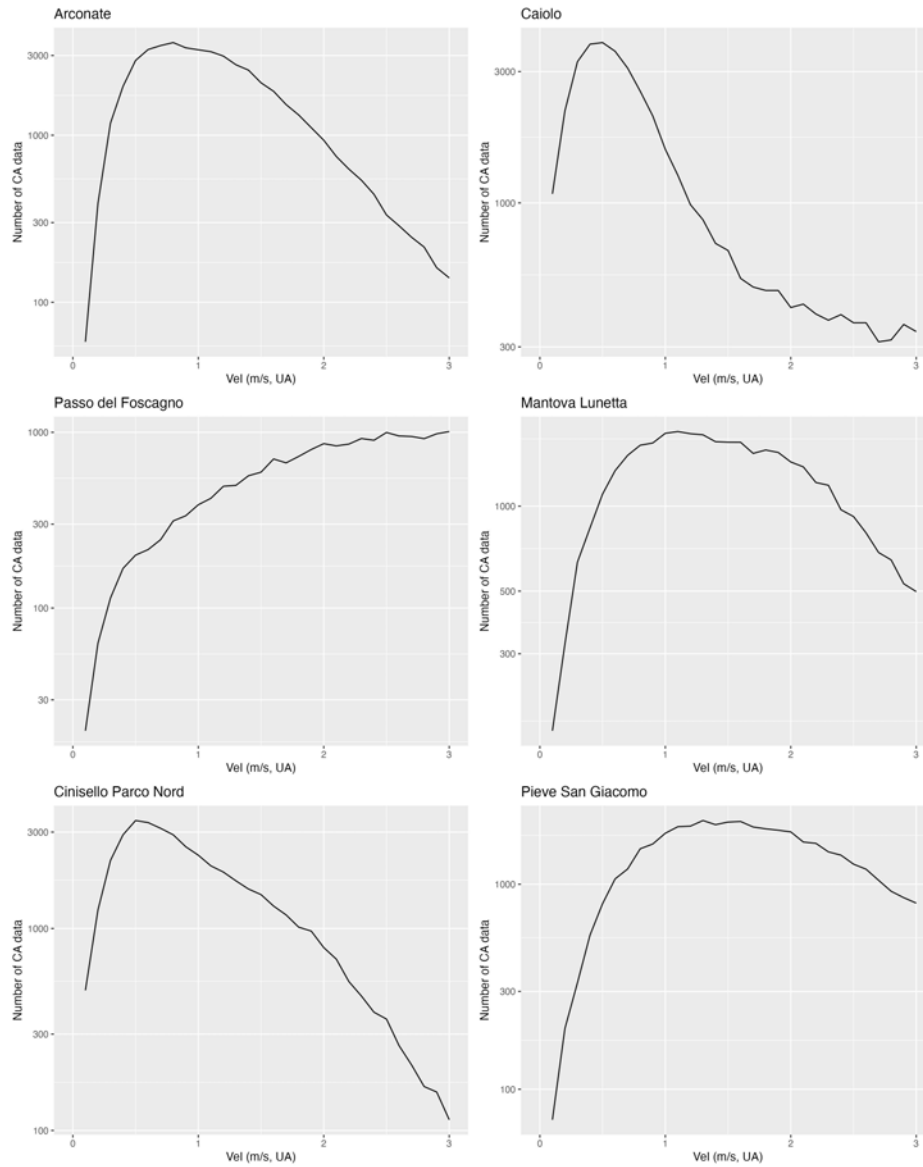


Figure 5.19.: Sample sizes corresponding to the UA speeds indicated in the abscissa (year 2022); zero UA values are not considered, their numbers being extremely small

5. Wind speed statistics

Let's consider for example the station at Arconate in year 2022. The CA data set contains 2421 speeds whose value is valid and equal to 0 (1348 with U_{CA} equal to 0.1 m/s), which become 2393 (1336) if we restrict attention to couples in which both values are valid.

The real problem occurs when we fix the UA speed, and use it as a sample selection criterion. Let's then take the subset of all the CA speeds whose corresponding UA speed is, say, 0.1 m/s.

Ideally, would the CA and UA behave exactly the same, this subset will contain many CA speed values, all equal to 0.1 m/s.

But if (as it happens) they are not, then the CA speeds in subset may be 0, which is smaller than the UA speed. Or 0.1, which is equal. Or 0.2, 0.3, ..., which are all larger than 0.1. The subset arithmetic mean then has a large likelihood of being > 0.1 m/s, which actually happens in many of the plots of Figure 5.17.

So, the large values of mean CA value corresponding to $U_{CA} = 0.1$ m/s do *not* indicate a tendency of the cup anemometer to overestimate the UA value.

What this fact really indicates is that a comparison based on subsampling and sample averages is not the best way for the slower speeds: it contains a massive bias which distorts any sense.

What about larger UA speeds? Also in these cases the CA speed is bounded from below (it cannot be negative) and not bounded from above. But the "width" is large, and the sample size also quite big as we can see in Figure 5.19. So, as UA speeds grows the plots become more and more informative, and it makes sense to say that in many cases cup anemometers slightly overestimate ultrasonic anemometer speeds, consistently with literature indications.

5. Wind speed statistics

5.8.3. The distribution of the difference between paired UA and CA speeds

In order to better understand the ramifications of the difference between CA and UA data at a “macroscopic” level, and do some step beyond the naive comparison attempted in Section 5.8.2, it is worth studying the statistical distribution of the variable

$$\delta_u = U_{CA} - U_{UA} \quad (5.5)$$

The question is, whether the distribution of this difference is normal or not.

In the former case traditional statistical tools can be used in analyzing it. In the latter, caution is to be exerted.

Figure 5.20 visually hints normality, so let’s test this assumption is true, using a quantile-quantile plot. Results are in Figure 5.21.

As Figure 5.21 quite convincingly shows the distribution of the CA-UA speed is far from normal. We could obtain the same response by using a formal test like Shapiro-Wilk normality test (see for example (Shapiro and Wilk 1965)): I’ll not follow this path, which in my experience on the large samples we are dealing with will surely confirm the null hypothesis of normality should be rejected with a so small p-value to make the test useless and uninformative.

I’ll rather focus on the *why* normality is not followed, even if a distract glance to Figure 5.20 would suggest the contrary. The following table presents the value of the moments for the empirical distribution of the CA-UA speed differences:

5. Wind speed statistics

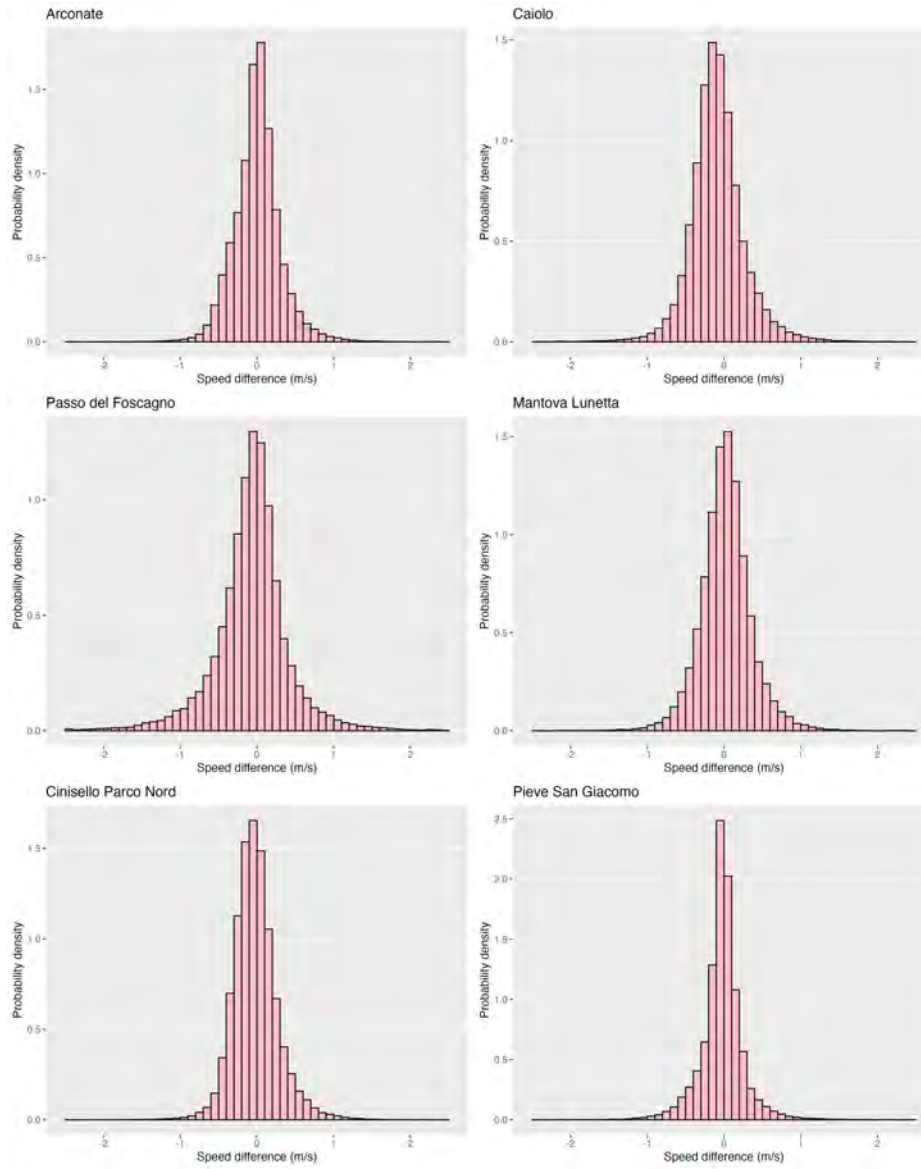


Figure 5.20.: Histogram of the CA-UA speed difference at SHAKEUP sites in year 2022

5. Wind speed statistics

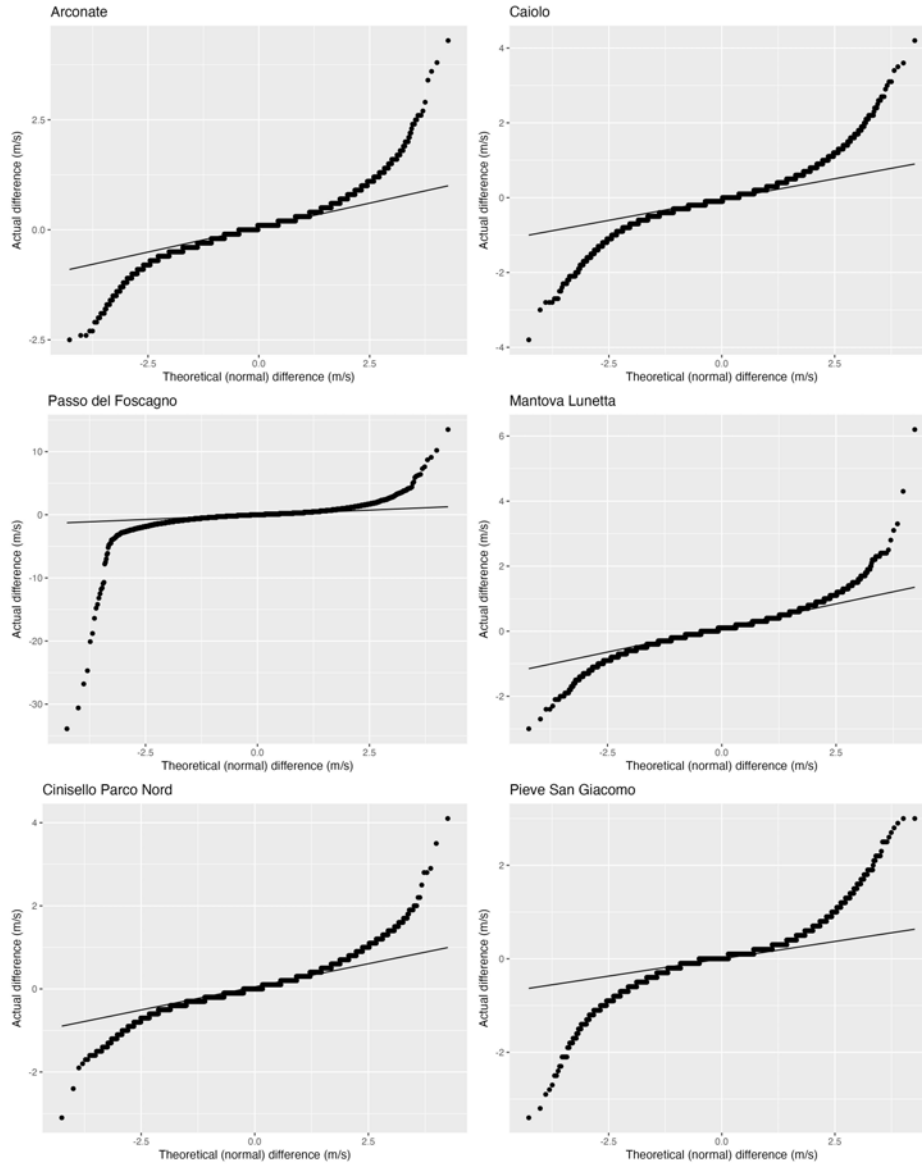


Figure 5.21.: Quantile-quantile plot of CA-UA speed differences at SHAKEUP sites in year 2022

5. Wind speed statistics

Table 5.5.: First moments of the CA-UA speed difference distribution at SHAKEUP sites during 2022

Station	Mean	Standard deviation	Skewness	Shifted kurtosis
AC	0.054	0.314	0.479	5.493
CL	-0.037	0.364	0.400	5.790
FC	-0.025	0.640	-10.558	467.699
ML	0.074	0.343	0.376	6.718
PN	0.029	0.298	0.648	4.897
SG	0.023	0.290	0.077	8.620

As a remind, given a sample $\{x_1, x_2, \dots, x_m\}$ its skewness is defined as

$$Sk = \frac{\frac{1}{m} \sum_{i=1}^m \left(x_i - \frac{1}{m} \sum_{j=1}^m x_j\right)^3}{\left[\frac{1}{m} \sum_{i=1}^m \left(x_i - \frac{1}{m} \sum_{j=1}^m x_j\right)^2\right]^{3/2}}$$

and the kurtosis as

$$Ku = \frac{\frac{1}{m} \sum_{i=1}^m \left(x_i - \frac{1}{m} \sum_{j=1}^m x_j\right)^4}{\left[\frac{1}{m} \sum_{i=1}^m \left(x_i - \frac{1}{m} \sum_{j=1}^m x_j\right)^2\right]^2}$$

The shifted kurtosis is finally defined as

$$Ku_s = Ku - 3$$

so that for normally distributed data it is $Sk = 0$ and $Ku_s = 0$. The table indicates the distribution of the difference is almost symmetrical ($Sk \approx 0$) for all stations but Passo del Foscagno, and that all differences distributions are “leptokurtic” (more concentrated around the mean than a normal

5. *Wind speed statistics*

distribution) to varying degree, the case of Passo del Foscagno again peculiarly large.

This is quite interesting in itself: no normality, so we need to exert caution. Or, go deeper in our analyses.

5.9. Relationship between number of zero speeds and mean wind speed

Our findings may be summarized by presenting the relationship between count of zero CA speeds and mean UA speed.

Figure 5.22 shows the overall result.

Figure 5.22 suggests a dependency resembling an inverse proportion, with some visible difference from station to station, season to season.

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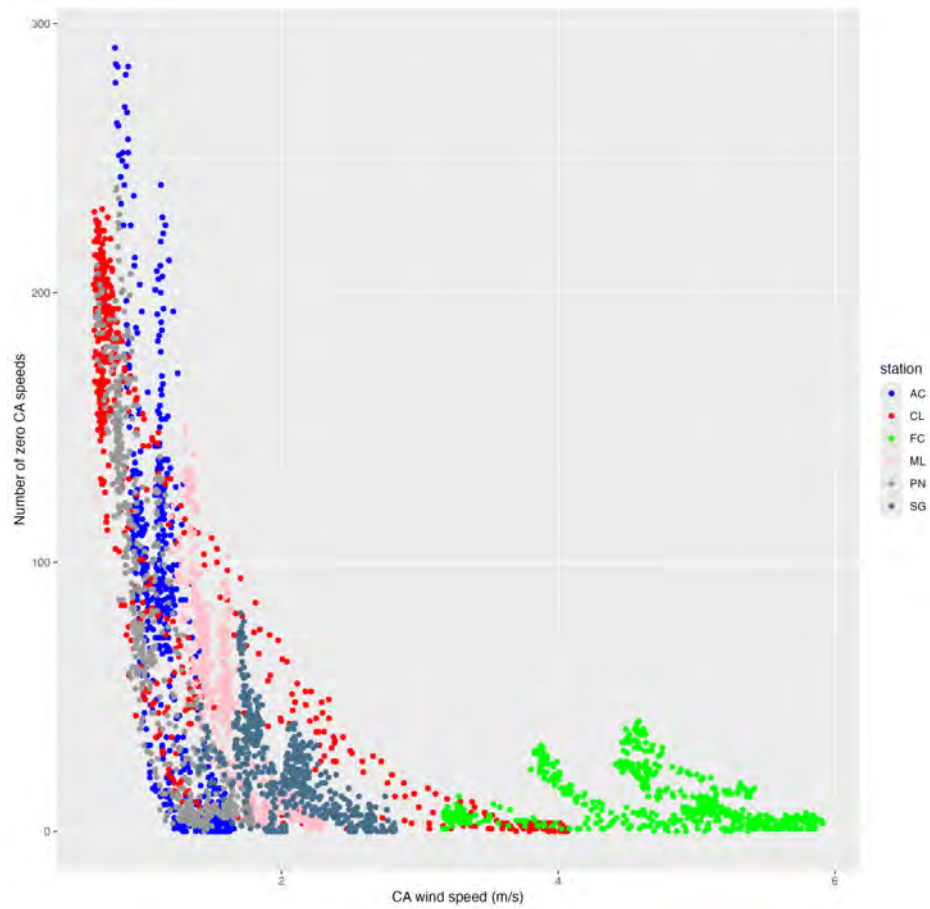


Figure 5.22.: Scatter plot of number of zero CA speeds vs UA speed

6. Cup anemometer performance in the time domain

6.1. Introduction

In Chapter 5 we have compared the wind speed statistics from CA and UA without taking time explicitly into account. The farthest point we explored was in just pairing data according to time stamp equality, but this didn't allow to study the time *variation* (and stability) of our findings.

This chapter extends the view considering the time dimension. Reasons for this are quite multiple, and here are some of them:

- The wind speed may change over time to a significant, even large extent. Because of this, the overall differences we found in Chapter 5 may get amplified or damped according to wind speed and turbulence ebb and flow.
- Changes may exhibit detectable time patterns, which allow to draw conclusions about, for example, the minimum time span of a characterization campaign or other practical points.
- Time variation in speed may show a more or less strong (or weak) site-dependence, and knowing this may yield interesting practical suggestions.
- Finally, knowing how speed changes with time, and to what extent this change is really captured by a cup anemometer, is of great importance by itself.

6. Cup anemometer performance in the time domain

The way we'll investigate this time change articulates in two basic parts:

- First, we'll establish a (site-dependent) “micro-climatology” of wind speed using ultrasonic anemometer data. Micro-climatology can be described as the application of climatological methods to phenomena occurring in the Planetary Boundary Layer (see (Stull 1988) for a definition), at possibly a time scale shorter than the “climate norm”; to understand what micro-climatology is, the books (Oke 1987) and (Geiger, Aron, and Todhunter 1995) provide a very good and thorough introduction.
- Second, we'll see to what extent the cup anemometer data match the micro-climatological conclusions found in the first part.

In so doing, we implicitly adhere to the convention already adopted in Chapter 5, that the ultrasonic anemometer can be used as reference instrument. This assumption is reasonable, in the light of the findings reported in Chapter 5, and here is used more extensively and confidently.

The *averaging* time scale used in this chapter is still the same employed in Chapter 5, that is, 10 minutes. As usual, we'll use the abbreviations CA and UA to designate respectively the cup anemometer and the three-dimensional ultrasonic anemometer.

Before to begin, let me anticipate a bit of terminology. By its nature this chapter deals with what is usual and what is not.

Laypeople often confuse “usual” with “normal”. We, on the contrary, know the difference: “normal” applies to a random variable, or a set of observation, to signify it follows a Normal distribution $N(\mu, \sigma)$ with unspecified mean μ and standard deviation $\sigma > 0$. A single data item, or an individual realization of a random variable, is not “normal”. At best may be close to the distribution mean, but this is quite a different thing.

What laypersons call “normal” actually means “matching their personal perception of what's *usual* or, on the contrary, *unusual*”.

6. Cup anemometer performance in the time domain

“Normal” is objective, “usual” is subjective. Intrinsically.

This chapter deals with what is usual, from the perspective gained accessing SHAKEUP data.

6.2. Time statistics of wind speed

6.2.1. Changes in wind speed yearly statistics

6.2.1.1. Assessing the significance of wind speed differences in a simple way

Once a measurement point is fixed at some altitude over the Earth surface, wind speed recorded at it is likely to change due to many reasons. Among them:

- Changes in the synoptic-scale circulation occurring randomly from one year to the next.
- Variation in nearby obstructions, for example as new constructions are added and old removed.
- The progressive growth of tall trees in more or less close proximity to the station.
- Change of crops occurring on yearly time scale at rural sites.

And of course many more.

Because of these multiple causes it makes sense to expect some variability of wind speed on yearly scale: the question is, to what extent does this occur.

In order to answer this question, some definition of “yearly wind speed” must first be introduced.

6. Cup anemometer performance in the time domain

Such a choice could be simple, if only yearly wind speed data are normally distributed: would this be, we could just adopt the arithmetic mean of the available data.

But as we discovered at the SHAKEUP sites in year 2022 presented in Chapter 5 the yearly distribution of wind speed resembles much more a distorted Weibull than a Gaussian. A (more or less distorted) Weibull distribution is characterized by a strong asymmetry not visible with the Normal distribution, and because of this the mean may be a poor indication of the central tendency of the data set (see for example (McPherson 2001)).

The most direct consequence of this is that the arithmetical mean may differ, and possibly vastly, from other “central tendency” indicators like for example the median. Indeed we have seen this phenomenon already in chapter Chapter 5, where many histograms contain vertical lines indicating the position of mean and median, and the two differ.

Generally speaking, when dealing with data whose distribution is decidedly asymmetrical (the case with wind speed) using the median and other quantile statistics is the preferred way (see for example (McDonald 2014) and (Ugarte, Militino, and Arnholt 2008)): in this chapter we’ll then rely on it.

If the purpose is to assess yearly *differences* in wind speed median then we cannot just look at the values and limit to an “Ok, they are different” - they always will. The real question is to gain an understanding about whether this difference is statistically significant, or not.

We could then perform some appropriate statistical test for equality, given some agreed-on confidence level. This would be surely appropriate in a research paper, but the long table of numbers which would almost surely ensue may detract from the perceptual clarity a practically inclined work should in my opinion have.

6. Cup anemometer performance in the time domain

So, I've opted for use of an enriched form of box plots which allow to detect the significance of differences visually. This, however, demands some preliminary explanation.

For starters, let's introduce a single population box plot, as we see in Figure 6.1.

The data set Figure 6.1 is based on is synthetic (actually a sample of 1024 numbers following a Weibull distribution with shape=2 and scale=1).

The original idea behind box plots was anticipated in a famous book ((Spear 1952)), and later made more explicit and perfected in (Tukey 1977). One year later the paper (McGill, Tukey, and Larsen 1978) introduced some useful improvements, and (Chambers et al. 1983) contains the now final version. The box plots in my Figure 6.1 have been generated using the **ggplot2** package ((Wickham 2016)) in its R ((Team 2023)) implementation.

Box plots, especially in their enriched form, are very useful in practical statistics, as they allow to visually get a lot of information about the distribution of data without actually estimating it in detail: this justifies some explain.

First of all you can see a “box”, containing a thick line. The upper and lower limits of the box are respectively placed at the 75th and 25th percentiles ($q_{0.75}$ and $q_{0.25}$), so that the box height immediately tells us the magnitude of the *interquartile range* ($IQR = q_{0.75} - q_{0.25}$, an alternative to standard deviation not demanding normality). The two whiskers extend from $q_{0.75}$ to $q_{0.75} + 1.58IQR$ and from $q_{0.25} - 1.58IQR$ to $q_{0.25}$, and give an indication of the range where “most data are found”. The outliers, finally, are placed in correspondence of all data above $q_{0.75} + 1.58IQR$ and below $q_{0.25} - 1.58IQR$: they indicate “exceptions”, and may not exist; on occasions (for example when very many data are present) they may be not represented intentionally - this is what we'll do.

Finally the notch, visible in the enriched box plot only, is placed at

6. Cup anemometer performance in the time domain

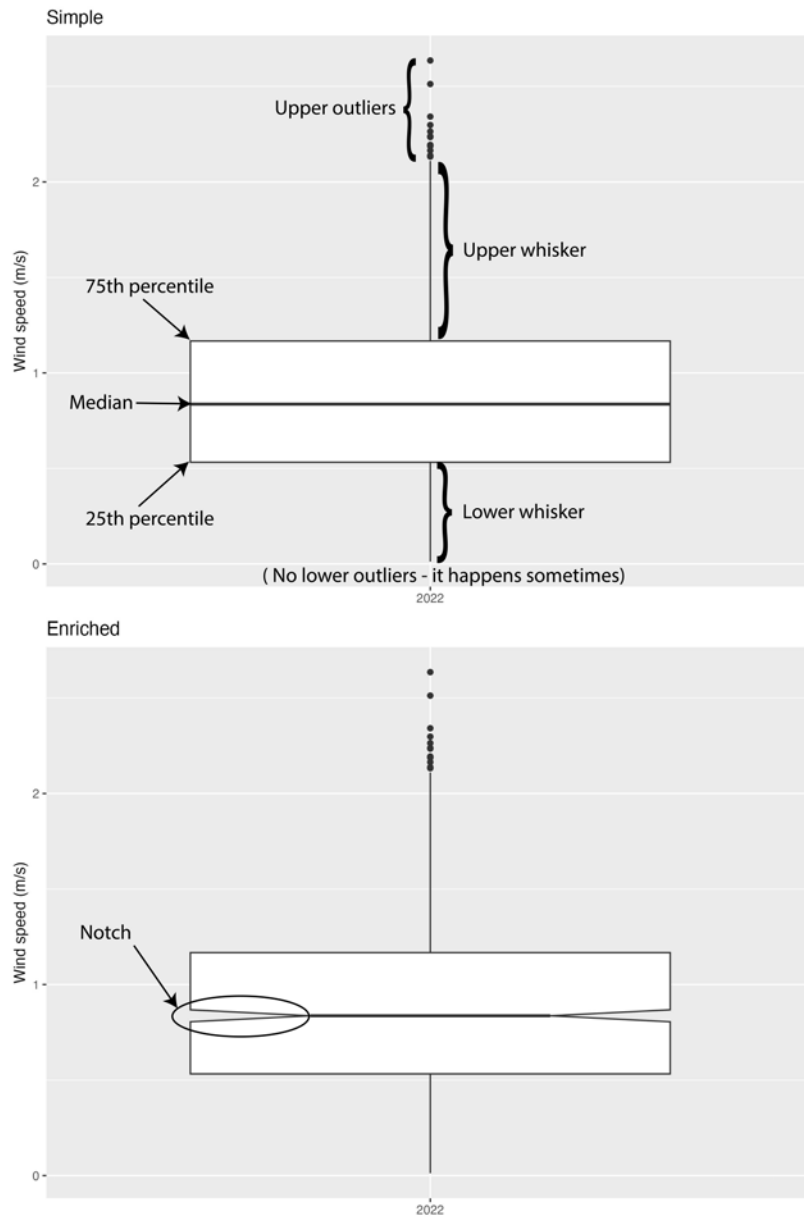


Figure 6.1.: Boxplot examples

6. Cup anemometer performance in the time domain

$$\pm \frac{IQR}{\sqrt{n}} + q_{0.50}$$

with n the sample size.

The rule for comparing two medians is very simple: if in a boxplot containing both of them their notches overlap, then the difference between the medians is not statistically significant: a lazy, simple alternative to formal testing.

6.2.1.2. Actual yearly variations in wind speed

In Figure 6.2 we see the variation of yearly median of wind speeds. In many cases the median remains outright “equal” - notches overlap. But also when they do not, values are remarkably similar, and the same can be said for the interquartile boxes.

The fact that little change occurs on yearly basis means we can investigate smaller timescale phenomena by focusing on only one specific year.

By looking the box plots it is apparent that 2017 could be such a year: it is one of the “most similar” to others at any sites with the only exception of the Passo del Foscagno high mountain station.

6.2.2. Monthly variation

6.2.2.1. Median speed

In Figure 6.3 we can see the variation occurring at monthly scale. As you can see there is much larger variation at monthly scale than at yearly.

But in the same time there is not a discernible common pattern, and stations tend to exhibit different behaviors.

6. Cup anemometer performance in the time domain

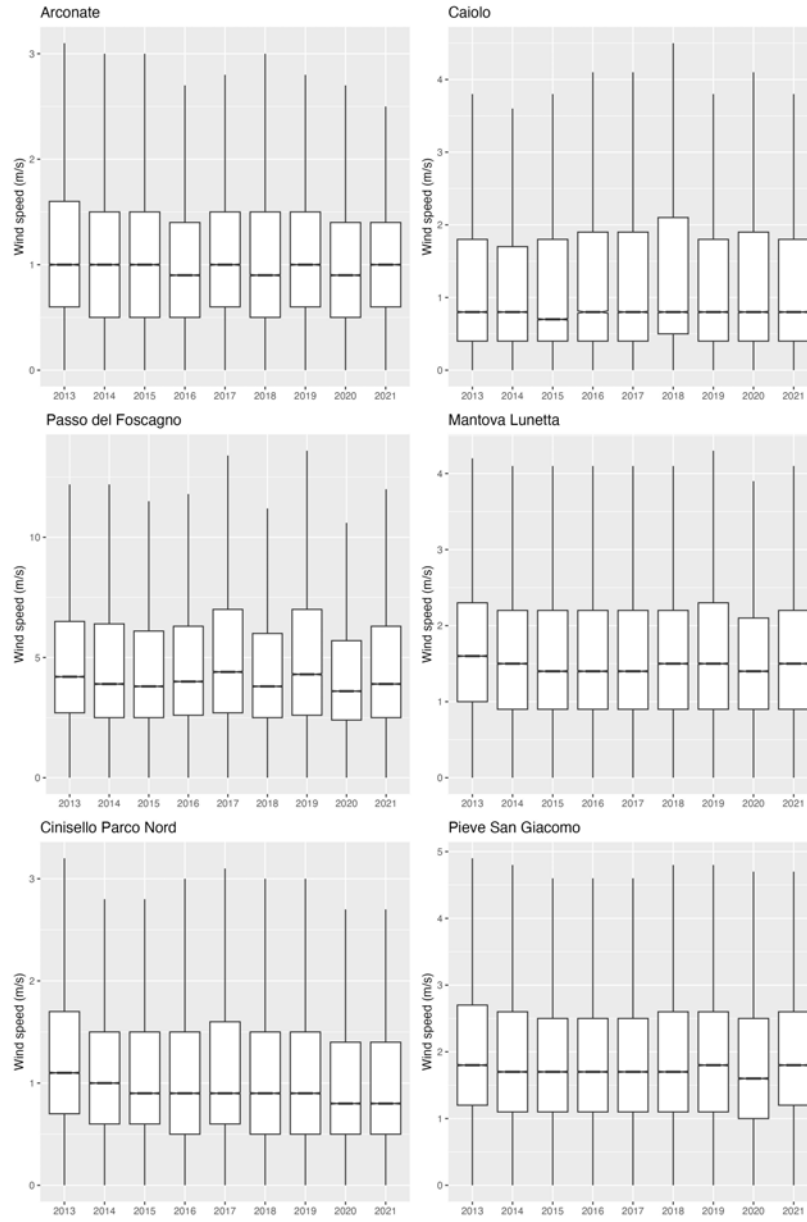


Figure 6.2.: Variation of yearly median of 10 minutes averages at SHAKEUP sites

6. Cup anemometer performance in the time domain

Broadly speaking we may group stations in two clusters, the “spring-fast” where median speed tends to be higher in the central part of the year (Arconate, Caiolo), and others where this tendency apparently overlaps with a “winter-fast” distribution.

But, is this one-year picture somewhat revealing, or years are much too different one from the other? We could test directly by selecting another year, for example 2021, and check the results are similar (I’ve done it in Figure 6.4).

As we compare Figure 6.4 and Figure 6.3 we can see a certain degree of similarity from one site to the other, but we can also see important differences between individual years. This can be reasonably expected on other years (and I’ll not do).

The existence of some degree of similarity between the same month data in different years suggests to introduce the concept of “median year”, a graph where data from all January, February, ..., from all years are considered together, and the median of their specific set is computed along with other statistics.

Figure 6.5 confirms how, on the basis of the available data, there exists a monthly variation pattern in wind speed.

This variation pattern however is highly site-dependent, with considerations similar to those exposed for Figure 6.3.

An interesting question arises, on whether the differences observed in the patterns tell some story about the way local (Planetary Boundary Layer) and synoptic scale phenomena affect the measurements at SHAKEUP sites. The task of finding an answer is outside the scope of this work and will not be pursued, but the question is objectively interesting and worth a more precise formulation allowing a scientific investigation.

6. Cup anemometer performance in the time domain

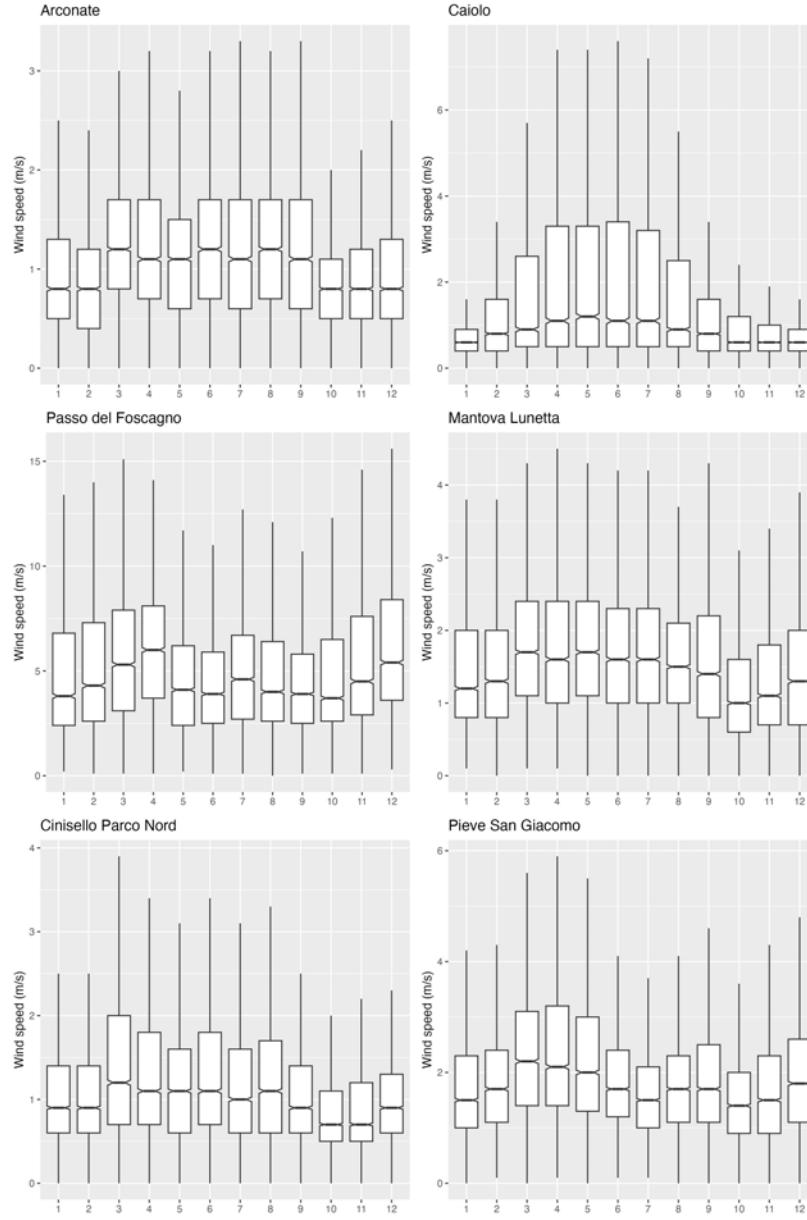


Figure 6.3.: Changes in wind speed monthly median at SHAKEUP sites (year 2017)

6. Cup anemometer performance in the time domain

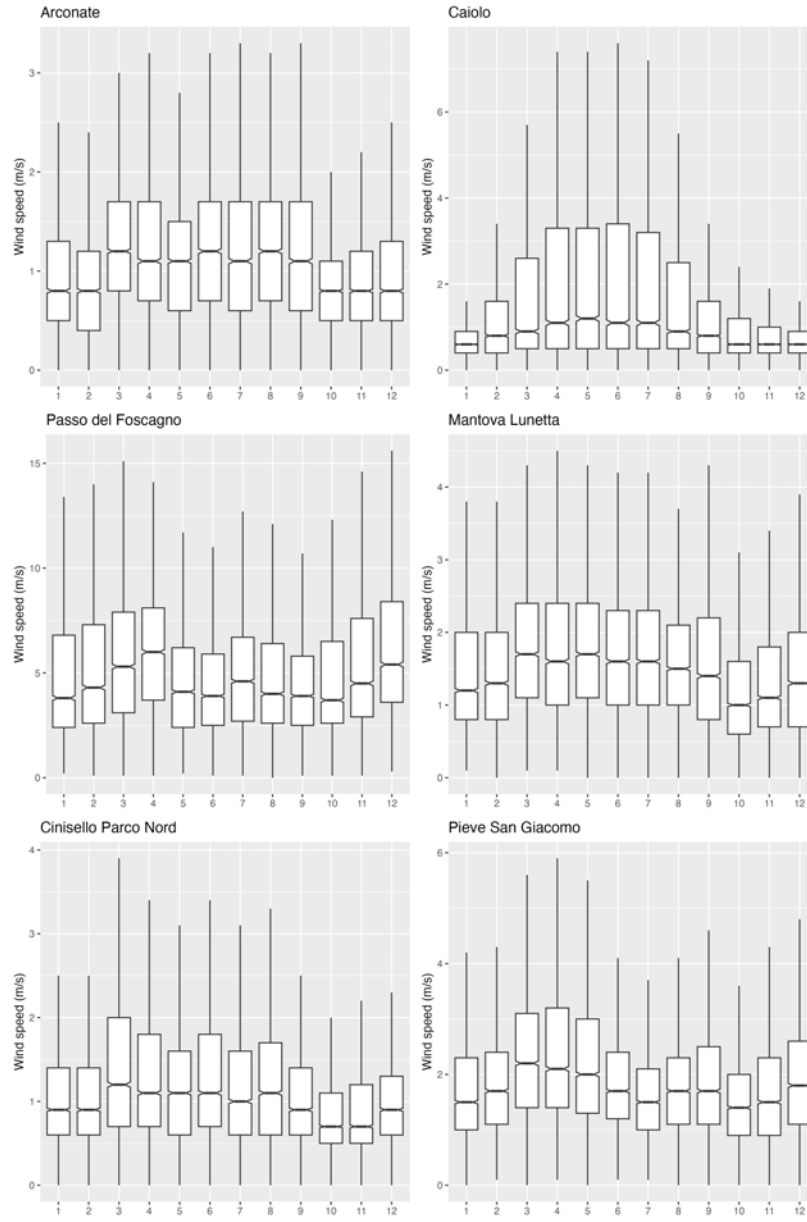


Figure 6.4.: Changes in wind speed monthly median at SHAKEUP sites (year 2017)

6. Cup anemometer performance in the time domain

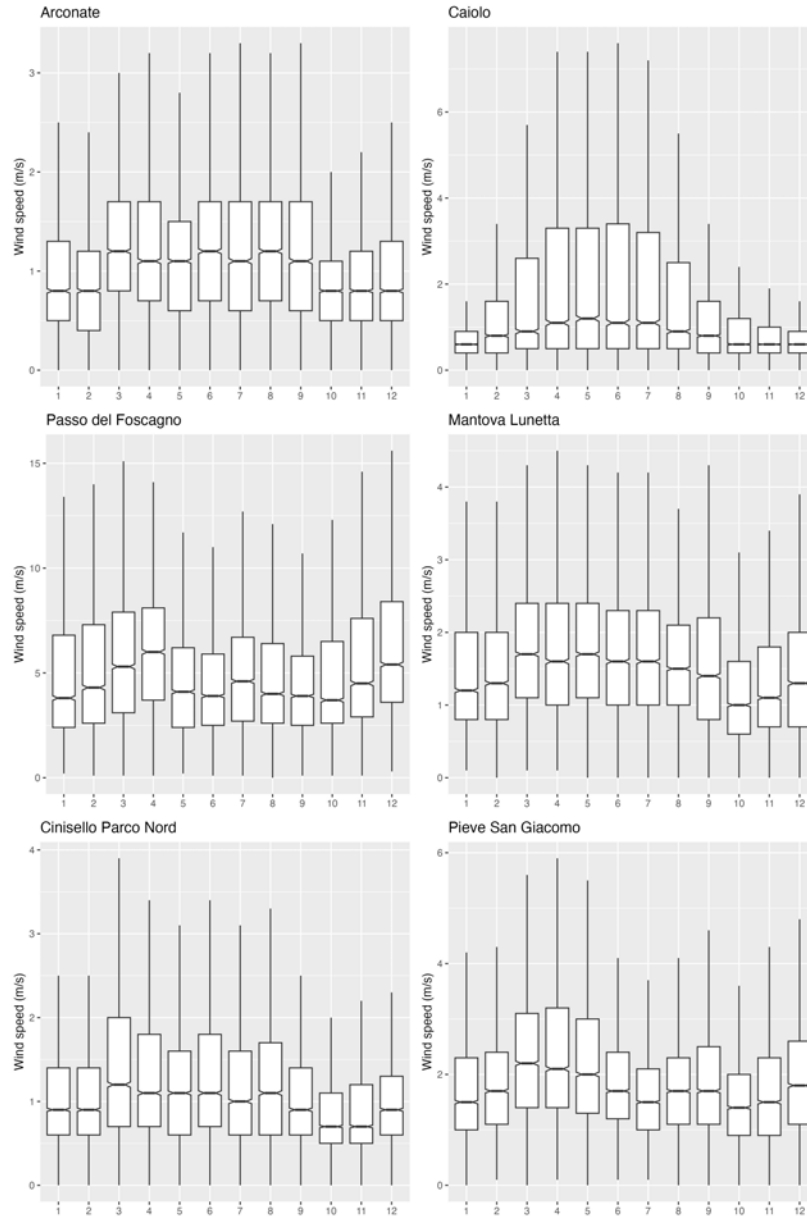


Figure 6.5.: Changes in wind speed monthly median at SHAKEUP sites (year 2017)

6. Cup anemometer performance in the time domain

6.2.2.2. Monthly fraction of low-speed winds

As we found in Chapter 5, ultrasonic anemometer speed data seldom report 0 ten-minutes means. The number of extremely slow speeds, for instance smaller than 0.5 m/s (the activation threshold of VV-20 cup anemometers), was however found significant. This, together the monthly pattern observable in Figure 6.5, suggests to check whether this fraction changes with month.

The presence of such a monthly change would be an indication that a VV-20 cup anemometer could encounter difficulties, proportional to the number of wind speeds frequently below the sensor detectability limit.

The same calculation can be repeated with other speed threshold, namely 1 m/s (a common limit for the robust professional-grade anemometers designed with flight safety in mind) and 1.5 m/s (an activation threshold common among amateur-grade anemometers not subject to regular maintenance). Results are given in Figure 6.6.

Figure 6.6 is quite interesting in practice. First, it confirms there is a strong monthly variation in the count of slow wind speed data. This is not unexpected as median speed itself did show a monthly variation.

Second (and this is the part I find quite surprising) most SHAKEUP stations exhibit a *large* fraction of wind speeds below the 0.5 m/s activation threshold (larger even, of course, for 1.0 and 1.5 m/s). This effect is less apparent at sites like Passo del Foscagno, and to a lesser extent Mantova Lunetta, characterized by higher wind speeds.

The implication is, well, quite worrying. If we know a cup anemometer has an $U_a = 0.5$ m/s, and we see that on a month the fraction of wind speeds below 0.5 m/s is the 20% or more, then we may expect that a very large fraction of wind speed data will likely be affected to some extent by anemometer-induced artifacts. This concern grows in importance as the fraction increases (like for example at Caiolo, where January and December fractions are close to 50%).

6. Cup anemometer performance in the time domain

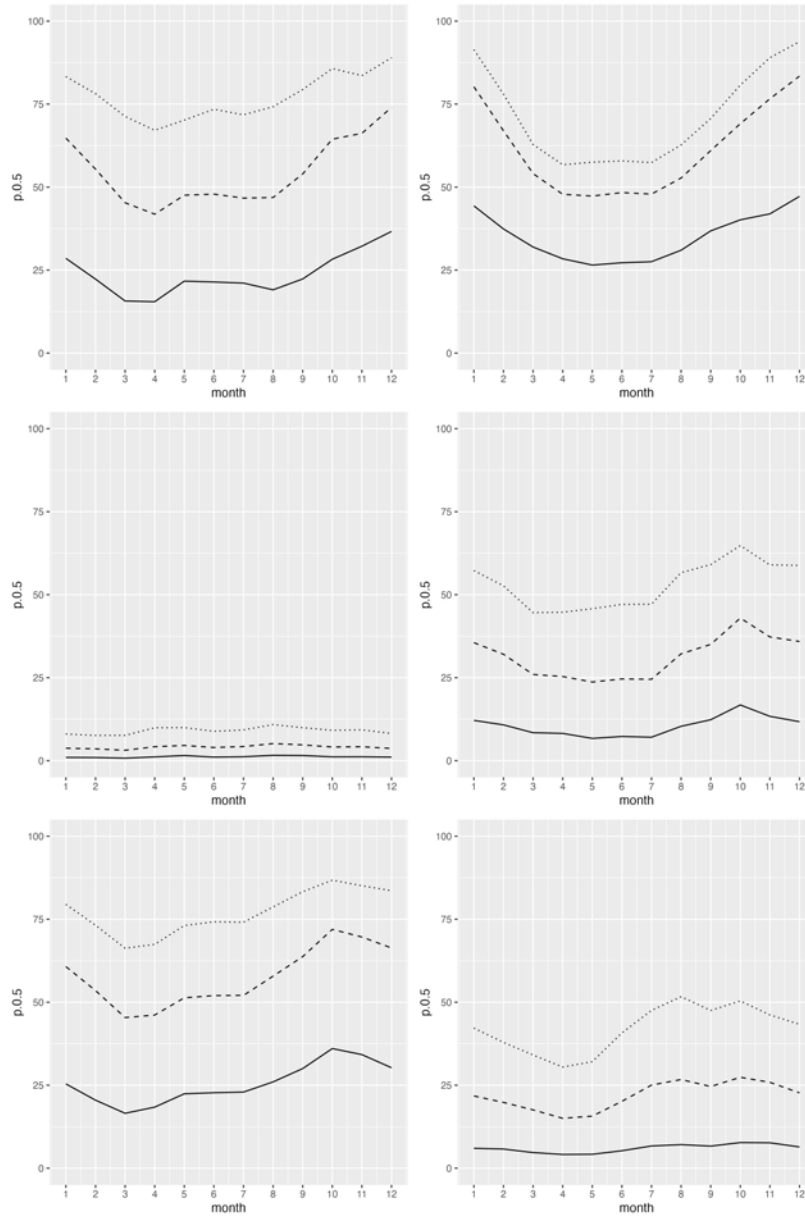


Figure 6.6.: Slow wind fraction per month (whole data set; solid line: limit speed 0.5 m/s; dashed line: limit 1.0 m/s; dotted line: limit 1.5 m/s)

6. Cup anemometer performance in the time domain

As the activation threshold of the sensor increases (by design, maintenance interval or both) the fraction of potentially distorted data increases more and more. In extreme cases (for instance amateur-grade sensors located in slow-wind areas) the question arises whether the speed reported by them can be believed or not; this suggests adopting in these case a very strict maintenance practice if data are to be used for some scientific or application purpose.

An important aspect in regard of the possible wind distortion is that as far as we have seen in this section *it may change with time*, following maybe a seasonal pattern.

The actual magnitude of the distortion experienced at any specific measurement site will of course depend on the amount of *instantaneous* sub-activation speeds within any 10 minutes average, and the magnitude of the speeds above U_a in the same interval. That is, it will depend critically on the *statistical distribution* of wind speeds within a 10 minutes averaging interval.

In case of the SHAKEUP sites the distortion magnitude will be investigated later in this chapter, when cup anemometer data will be explicitly taken into account.

6.2.3. Average day statistics

In addition to the wind variation within the months of one year, there could be a hourly one.

In Lombardy this could be of special interest, as most Lombard territory is affect to a more or less extent by orographic effects, as for example mountain and valley breezes. Large internal water bodies like the Maggiore, Como and Garda lakes, together with alpine peaks and ridges close nearby, add complications.

6. Cup anemometer performance in the time domain

These phenomena are connected intuitively with large, and possibly sudden, changes in wind *direction*, but this does not mean speed is affected in some way.

Changes may also be due to chance, and to the frequency synoptic scale circulation takes the helm in her large hands. So, it is reasonable to expect wide changes as we observe on a short time scale (say, daily or less).

Indeed, Figure 6.7 shows changes in wind speed are likely to occur, with magnitudes and position in time to change with the season.

In Figure 6.7 I've used the mean instead of the median, with the objective of having less cluttered plots. But for completeness I also report the median day in Figure 6.8.

Compared, Figure 6.7 and Figure 6.8 give the same qualitative information: there is a clear daily dynamics, whose details are to some extent influenced by the season.

We can also see that the extent and type of the dynamics is strongly affected by the geographical position. For instance the two sites placed deep in the Po Valley exhibit less difference from nocturnal to diurnal wind speed than the two “pre-alpine” sites of Arconate and Cinisello Parco Nord. Not having considered the wind direction we have not here the elements for attributing this difference to the stronger effect mountain breeze have at these sites, but this cause is surely tempting.

Finally, the two mountain sites of Caiolo and Passo del Foscagno stand out. In the former the nocturnal wind speed is almost constant and season-independent, while the diurnal speed follows a time evolution which closely resembles that of solar global radiation (see (Geiger, Aron, and Todhunter 1995) and (Oke 1987)). We have not considered wind direction, but this exact matching reflects a very strong local effect due to the mountain slope close nearby (Caiolo station is located within the Caiolo airport, in the hinterland of Sondrio, on a flatland at less than 1km from the steep regular South-facing slope of the mountains delimiting Valtellina).

6. Cup anemometer performance in the time domain

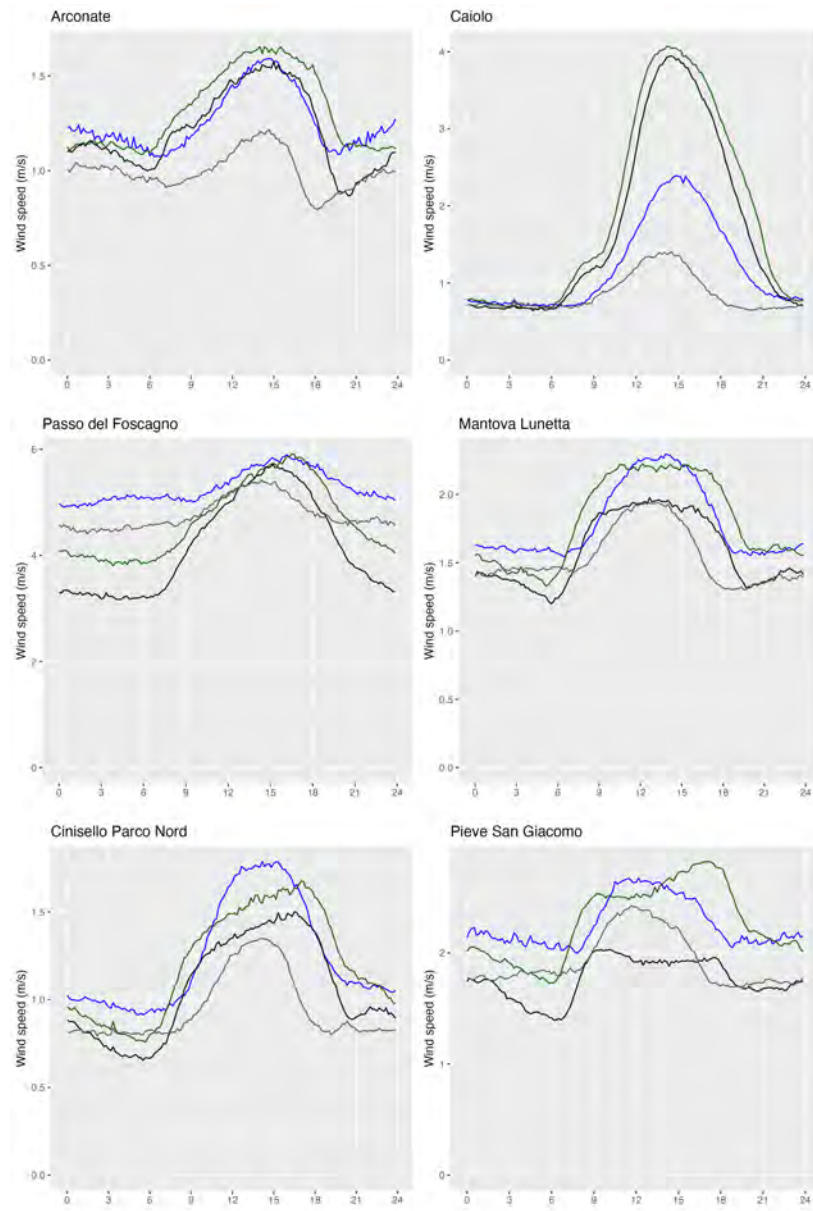


Figure 6.7.: Mean wind speed days per season (m/s; Blue = months 1, 2, 3; Green = months 4, 5, 6; Black = months 7, 8, 9; Gray = months 10, 11, 12) 223

6. Cup anemometer performance in the time domain

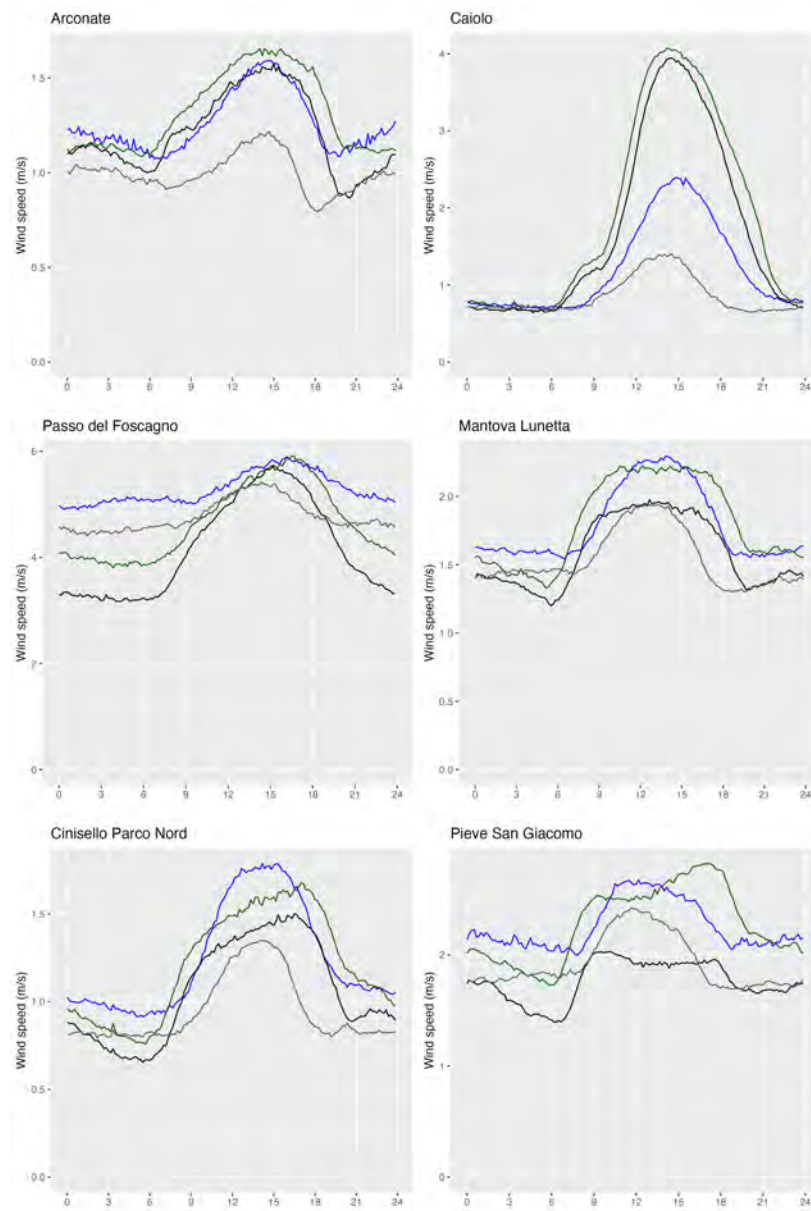


Figure 6.8.: Mean wind speed days per season (m/s; Blue

6. Cup anemometer performance in the time domain

At Passo del Foscagno it is the nocturnal wind to exhibit strong seasonal variation, suggesting an important contribution by synoptic-scale circulation. This is not unreasonable, Passo del Foscagno station being located in correspondence of a saddle with surrounding mountains facing West and East, at a high altitude.

6.3. Wind speed statistics in the time domain according to the cup anemometer

6.3.1. Changes in yearly speeds

Using UA data we have seen that yearly medians of wind speed are about the same (although, due mostly to the large data number, the small differences are labeled as statistically significant). Does the same happens if CA speed data are used instead? We can see the result in Figure 6.9.

Again we do not see extremely large variations over years. But now, compared to Figure 6.2, we have the impression of a somewhat larger variability at the Arconate site. In all other cases the conclusions drawn using UA data are substantially respected without appreciable modification.

To the author's best knowledge nothing special has changed in 2016 and 2017 at that site, but I have no specific information. The station is adjacent to a cultivated field, and in the "high Milanese plain" it is customary to alternate crops between corn and low-standing herbaceous plants, and this may have happened outside my knowledge (high crops interfere with wind in different way as low ones), so I cannot say anything conclusive.

6. Cup anemometer performance in the time domain

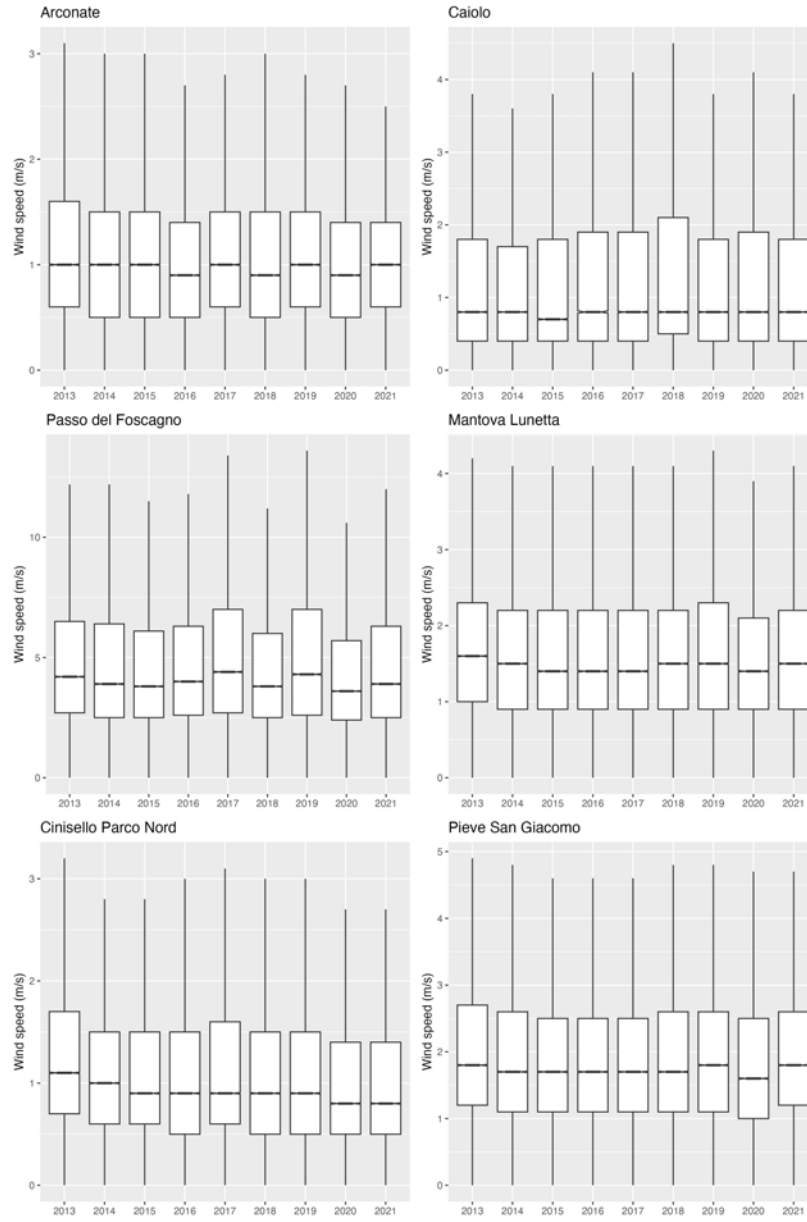


Figure 6.9.: Variation of yearly median of 10 minutes averages at SHAKEUP sites

6. Cup anemometer performance in the time domain

6.3.2. Changes in monthly speeds

6.3.2.1. Median speed

Here we apply the same processing convention adopted for ultrasonic anemometer speed data to the cup anemometer.

Figure 6.10 can be compared visually to Figure 6.5, and any differences could be spotted directly. To ease this step (requiring access to figures on different pages) I've also asked my computer to compute the difference.

The result can be seen in Figure 6.11. Overall, the CA data exhibit a tendency towards *overestimation* (positive difference), but the interquartile range is quite wide in all case.

At the Caiolo site, the Winter and Autumn months see a tendency towards underestimation: these are the months of longer night-time insolation, and as we have seen in Section 6.2.3 the nocturnal speeds are already very slow, and more vulnerable to below- U_a speeds.

6.3.2.2. Monthly fraction of low-speed winds

We proceed as already made in Section 6.2.2.2, but this time we show both the UA and the CA counts, made evident using different colors.

#fig-slow-ca is *very* interesting: it says that the CA sees more speed slower than 0.5 m/s than the UA does, to an extent which depends on site but is (not surprisingly) larger in correspondence to months characterized by lower speed on yearly basis.

This effect is visible even at the mostly high-speed site of Passo del Foscagno.

But as we consider higher cut speeds the effect becomes less clear: the tendency of the CA to underestimate UA speed data at slow velocity, and

6. Cup anemometer performance in the time domain

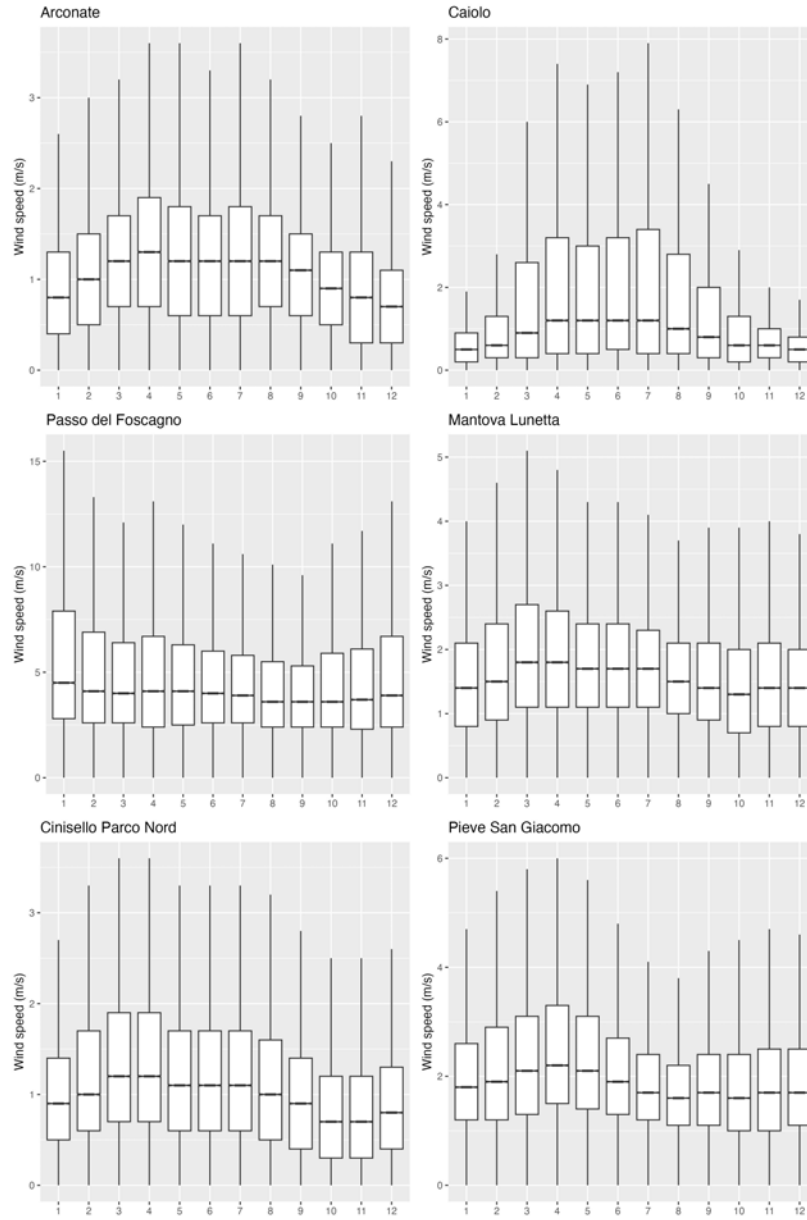


Figure 6.10.: Changes in CA wind speed monthly median at SHAKEUP sites (full data set)

6. Cup anemometer performance in the time domain

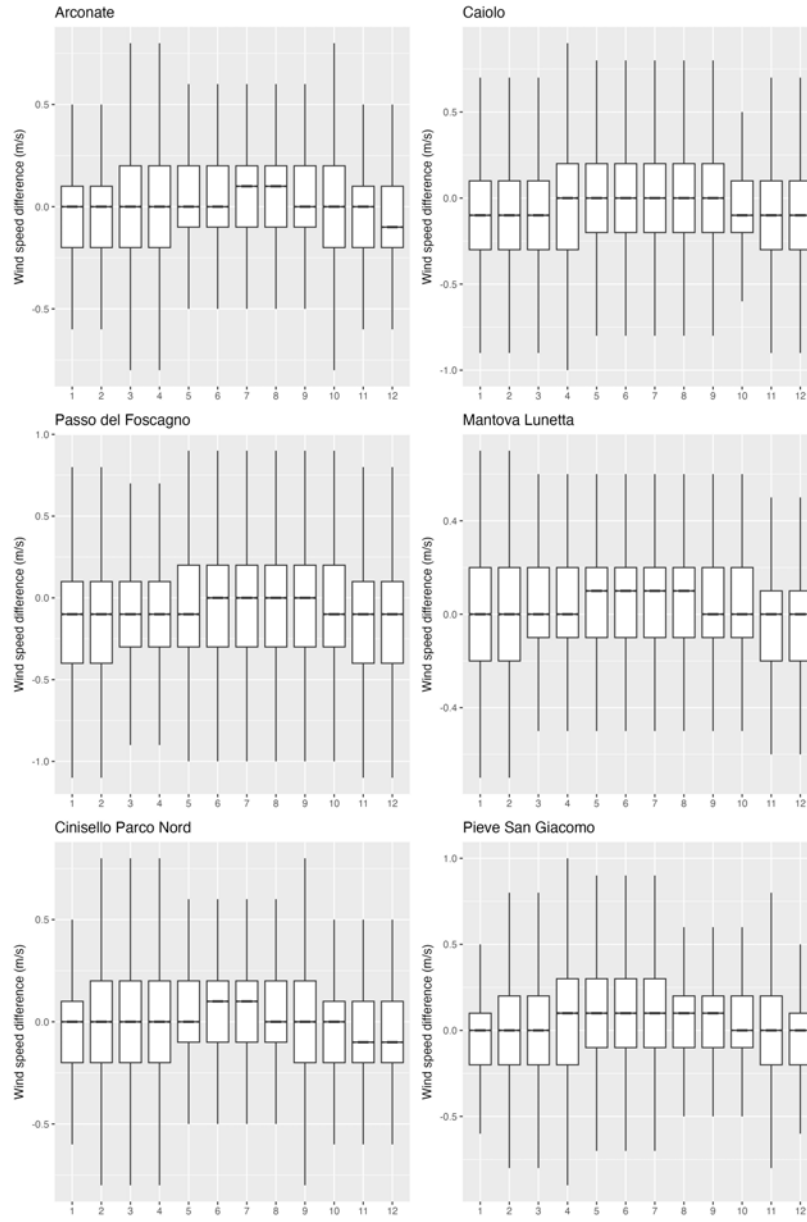


Figure 6.11.: Monthly difference between CA and UA data (full data set)

6. Cup anemometer performance in the time domain

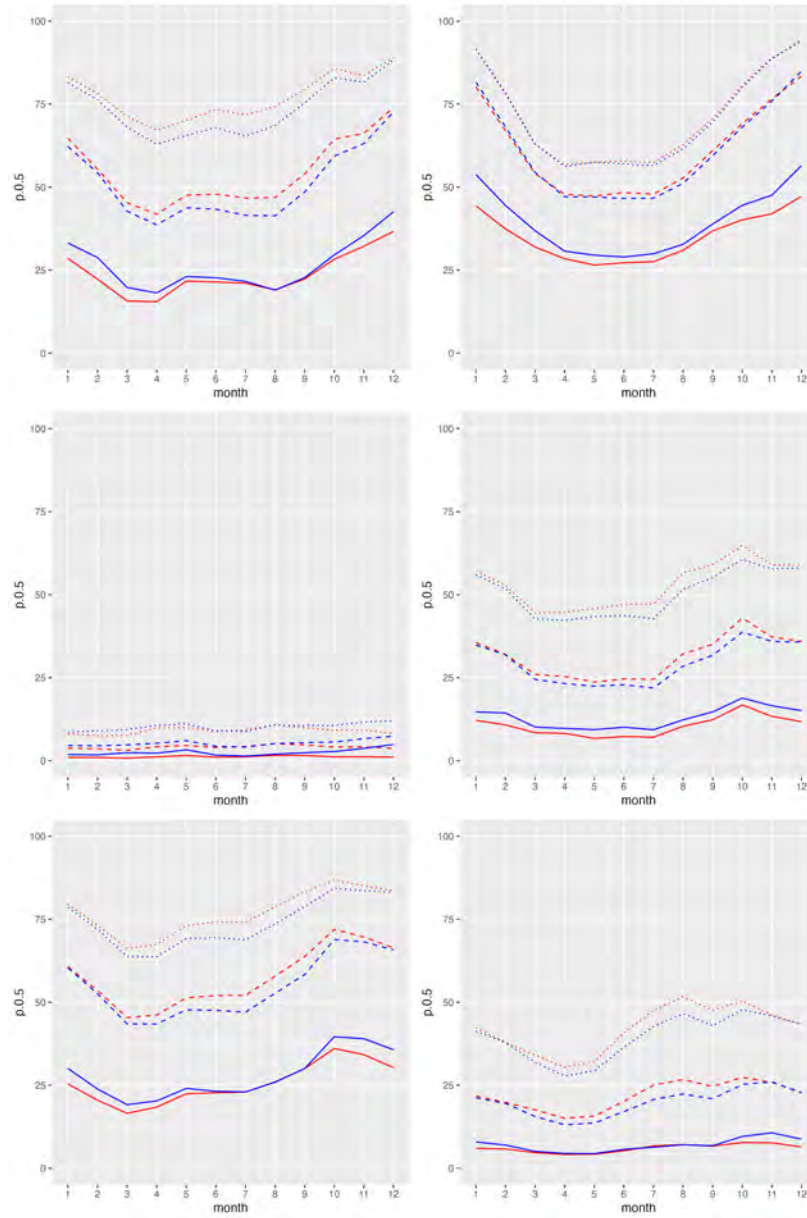


Figure 6.12.: Slow wind fraction per month (whole data set; solid line: limit wind 0.5 m/s; dashed line: limit 1.0 m/s; dotted line: limit to 1.5 m/s)

6. Cup anemometer performance in the time domain

overestimate at high, tend to balance with a prevalence of the CA towards overestimation.

However, *this effect depends on the CA model*, whose declared U_a is 0.5 m/s. Things would be likely much different if an amateur-grade cup anemometer with $U_a = 1.5$ m/s is used instead.

Considered together, Figure 6.10 and Figure 6.12 suggest that at the SHAKEUP sites considered the effect of having more very slow CA speeds is more than compensated by the tendency of the CA to overestimate UA speed at high wind velocity.

The higher number of CA speed values smaller or equal to 0.5 m/s, and lower number at higher speeds, suggests the monthly distributions of CA and UA wind speed exhibit a pattern similar to the one considered in Figure 5.9. This verification has not been made however, and is left to anyone willing to proceed further.

6.3.3. Average day

The last time-related question is, whether using CA impacts on the daily changes. As already done with UA data we'll use the less clutter-prone mean instead of the median.

Results are shown in Figure 6.13.

As we can see in Figure 6.13 there *is* a systematic impact in using a CA. The sign and magnitude of the difference respect to UA data depends on both the site and time in the day.

Overall, it is apparent that at low velocity values the CA tends to underestimate the UA, while it overestimates the UA at higher speed.

6. Cup anemometer performance in the time domain

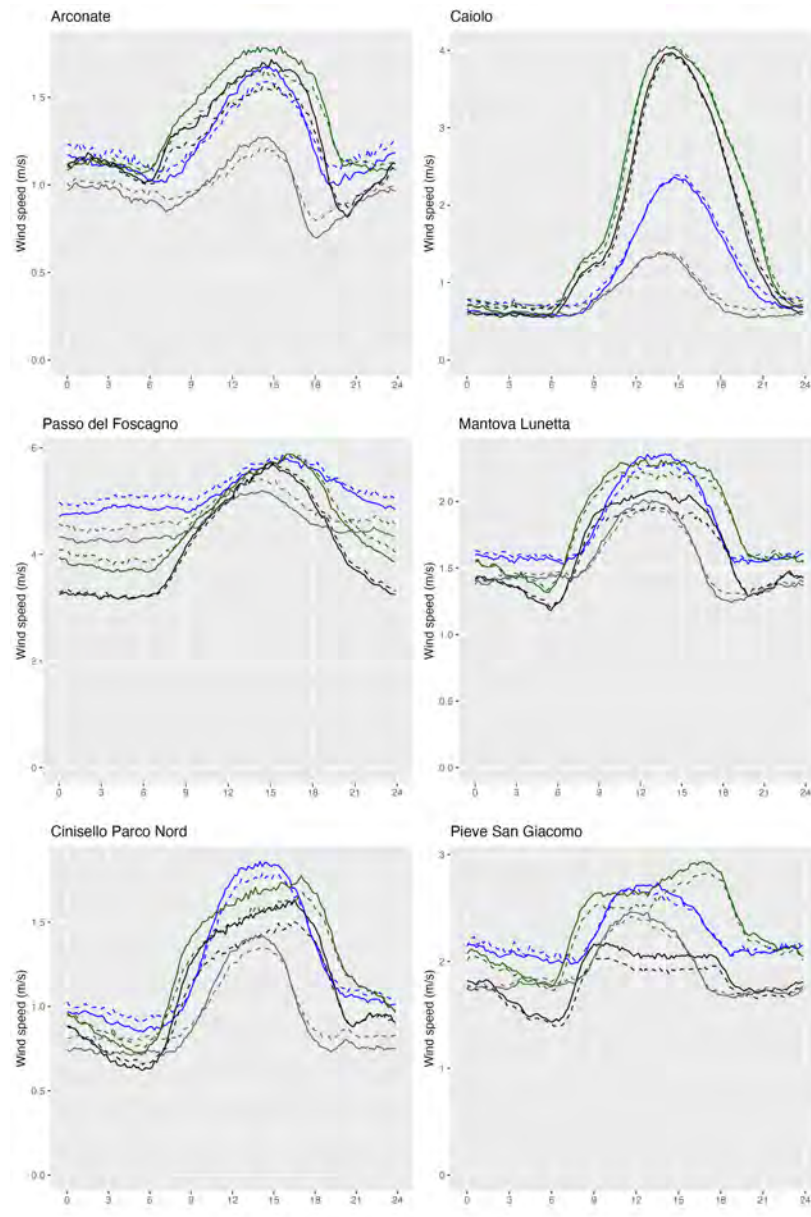


Figure 6.13.: Mean wind speed days per season (m/s; Blue = months 1, 2, 3; Green = months 4, 5, 6; Black = months 7, 8, 9; Gray = months 10, 11, 12; continuous: CA; dashed: UA)

Part IV.

This part introduces a simplified model of the cup anemometer, and applies it to the slow-wind case.

In the descriptive part where we had a glance of what a cup anemometer is we met the intricacies of the sensor dynamics. Models have been proposed in literature to reproduce these accurately, but their routine use with data readily available outside a wind tunnel facility is somewhat problematic.

This is the rationale for the simplified model presented in this part.

7. Simplified model of cup anemometers

7.1. Rationale for a cup anemometer mathematical model

The findings made so far indicate the readings from a conventional cup anemometer may differ from the ones released by an ultrasonic anemometer exposed to the same wind.

As we have seen, a part of these differences depend on the wind statistics. But the *way* a cup anemometer responds to this statistics is a consequence of the value of some instrument parameters: among these the activation threshold U_a , the linearity threshold U_l and the (speed-dependent) first-order response time.

These parameters are fixed once a specimen of a given sensor model is assumed. Until now this step has been done implicitly, the SHAKEUP stations hosting a unique type of cup anemometer maintained by the same company according to a single specification.

Using a single cup anemometer has proven useful in pinpointing interesting effects, but does not allow to determine to what extent these effects depend on the sensor parameters.

This level of knowledge, however, is essential if we want to not only understand the general lines of the measurement process irks and quirks, but

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also to control them - for example in view of some form of “predictive maintenance”.

Of course, in theory it would be possible to complement the extant SHAKEUP stations with additional cup anemometers characterized by different parameter values than the existing one. But quite admittedly this practice would likely be not sustainable in terms of cost, and maybe also able to perturb measurements as the sensors in the array interfere with each other.

So, having the possibility of using a mathematical *model* of cup anemometer would provide a reasonable alternative.

7.2. Requirements for a cup anemometer model

According to this work main purpose - finding more clever maintenance practices - we can expect our model will provide us with insight on what it happens would U_a change.

We could then neglect first-order effects.

In addition, we may assume the linear threshold equals the activation threshold, that is $U_a = U_l$: this is an approximation, but may be sufficient to address our immediate needs.

7.3. Is the model feasible in practice?

In case of SHAKEUP station it surely is: we will try to realize the model starting from the ultrasonic anemometer *raw* (unprocessed) data instead of the 10 minute averages we’ve used until this moment.

In general, however, the answer could be negative: normally, meteorological stations equipped with a cup anemometer have not an ultrasonic

7. Simplified model of cup anemometers

anemometer in their sensor array. But even so the considerations I'll provide are still useful if we can devise a “characterization campaign”, made for example using a vehicle-based ultrasonic anemometer station.

In case a good quality three-dimensional ultrasonic anemometer is already available at site, or deployed there as a temporary station, then it become possible in principle to *collect its raw data*. Once they are, any sort of detailed comparison becomes possible: among these, comparing the actual cup anemometer response with various possible modeled responses corresponding to different model parameters.

Maybe (and this is just an example of the possibilities offered) to check whether the actual activation threshold U_a matches the value declared by the manufacturer, or increased due to sensor wear out.

(Besides, I just say that this way of thinking, that is “data driven maintenance”, is not an original idea: it is widely employed in many industries where it takes names like “predictive maintenance”, and as far as I know even some famous manufacturers of meteorological sensors use some form of it.)

7.4. Some preliminary considerations

7.4.1. What we already used: 10-minutes UA mean speeds

Until now we have used ultrasonic anemometer data which have been already averaged, with an averaging interval equal to 10 minutes, and clipped to one decimal during archiving to the ARPA database.

These “averages” include, among the other things, the components u and v of the wind velocity vector along the x and y components of the instrument reference system (supposedly aligned to the geographical coordinates and the gravitational vertical).

7. *Simplified model of cup anemometers*

From these averages, individual samples of mean horizontal wind speed are obtained by using the formula

$$U = \sqrt{u^2 + v^2}$$

As we have seen these averages are immune to calms (sonic anemometers, by their very construction, have an activation threshold practically equal to zero) and overspeeds maybe due to the presence of non-zero vertical wind (incidentally a three-dimensional ultrasonic anemometer measures the vertical component of wind velocity vector with great precision).

Of course ultrasonic anemometers are not ideal sensors, and have their own problems (among these, they too require care and maintenance, although in many cases the absence of moving parts and the robust electronic operation may give a superficial observer the feeling they normally work).

But they are not affected by calms and overspeeds, and this is what matters in cup anemometers modeling.

7.4.2. **Simulating CA response**

Now. As mentioned in Chapter 3 existing cup anemometers sense the wind speed in what could be said “cumulative measurement”: the instantaneous wind blowing through the sensor transfers to it some energy, due to which the rotor spins (or not if this energy is insufficient to overcome the intrinsic friction of the sensor), and it is this spin count which is actually recorded by the data logger - typically over 10 seconds time step.

This “instantaneous CA reading” is surely affected by turbulence (as shown for example in (Kristensen 1998)), but we’ll not consider this level of detail, and will only take into account the speed itself.

Now, consider our 10 seconds CA sampling rate. If (as it happens in SHAKEUP stations) the UA is operated at 10 Hz sampling rate (this

7. Simplified model of cup anemometers

figure may be increased to 20 or 40 Hz by the way) in a 10s time interval we have 100 samples

$$U_i = \sqrt{u_i^2 + v_i^2}$$

and a “full data” mean

$$\bar{U} = \frac{1}{100} \sum_{i=1}^{100} U_i$$

More generally, for an arbitrary sonic sample rate, we’ll have a 10s mean equal to

$$\bar{U} = \frac{1}{n} \sum_{i=1}^n U_i$$

with n the number of data released by the UA in 10s.

Now, let us consider the “diminished mean”, defined as

$$\hat{U} = \frac{1}{n} \sum_{i=1}^n Q_i \tag{7.1}$$

where

$$Q_i = U_i \cdot \chi(U_i \geq U_a)$$

(the conditional function $\chi(B)$ is equal to 1 if the predicate B is true, 0 if it is false). This is our *simulated CA response*.

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7.4.3. The “defect error”

By construction we know

$$0 \leq Q_i \leq U_i \tag{7.2}$$

and then, by the well-ordering property of real numbers (see for example (Rudin 1953)), we can conclude that

$$\hat{U} \leq \overline{U}$$

The *defect error*, defined as

$$e = \overline{U} - \hat{U}$$

will then depend on both U_a , the assumed activation threshold, and the actual wind speeds U_i . By Equation 7.2, and once again the properties of the sum of non-negative real numbers, we see that

$$0 \leq e \leq \overline{U}$$

What’s interesting here is that the activation threshold U_a , on which the defect error e is computed, can be assumed at will to be any non-negative number.

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7.4.4. Simulated 10 minutes CA mean speed vs exact mean (“UA speed”)

Now, let $\bar{U}_j$ and $\hat{U}_j$ be the j -th 10s mean in the 10 minutes overall sampling interval. From these we may form the 10 minute averages by applying for example the formula

$$\tilde{U} = \frac{1}{60} \sum_{j=1}^{60} \hat{U}_j \quad (7.3)$$

Similarly we can calculate the exact 10 minutes mean

$$\bar{U} = \frac{1}{60} \sum_{j=1}^{60} \bar{U}_j \quad (7.4)$$

From now on we’ll name Equation 7.3 the *crude mean*: as we’ll see later this is just one of the ways available for aggregating simulated CA readings $\hat{U}_j$ into a “mean” speed, and we’ll indeed explore some more. Equation 7.4 indicates what we already know as “UA speed”.

Clearly the total defect error will be

$$E = \bar{U} - \tilde{U} = \frac{1}{60} \sum_{j=1}^{60} (\bar{U}_j - \hat{U}_j) = \frac{1}{60} \sum_{j=1}^{60} e_j \quad (7.5)$$

Simulated CA 10 minutes averages, UA exact 10 minute averages and total defect errors can be computed over a sample of data long enough, and histograms and other statistical indicators may then be derived.

7.5. Application of the basic model

7.5.1. The effect of a non-zero activation threshold

In Figure 7.1 we can see what happens to the 10 minute averages as a simulated CA with $U_a = 0.5\text{m/s}$ is compared in the time domain to UA data.

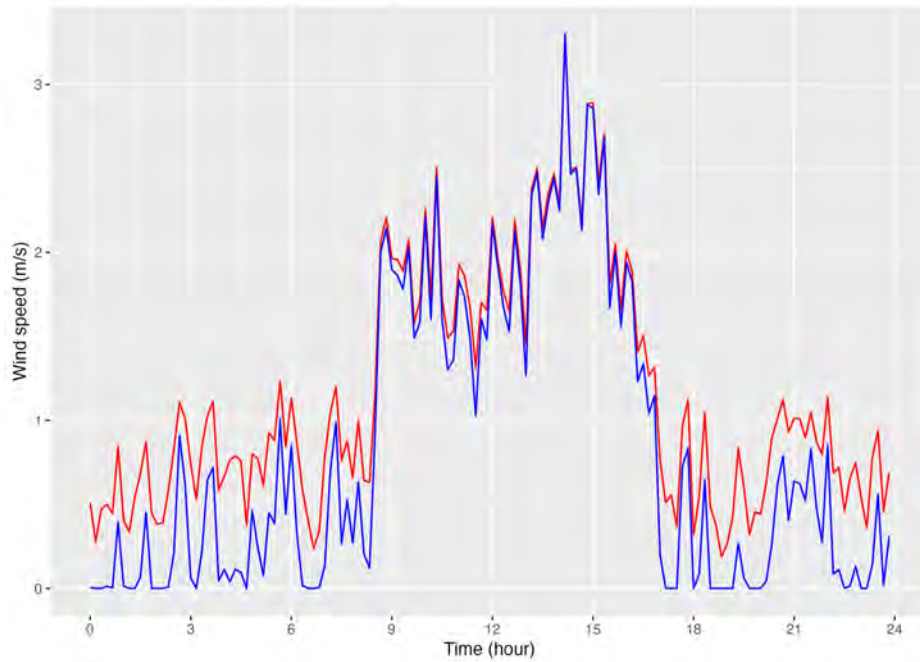


Figure 7.1.: Time evolution of 10 minute means of wind speed at Cinisello Parco Nord (red: UA; blue: simulated CA with activation threshold 0.5m/s ; data from 08. 03. 2016)

As we can see, in original UA averages there is no zero-valued data, but

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zero valued data occur in simulated CA data, and are more numerous where the UA wind speed is lower.

We may also see a general tendency towards underestimation by the CA, and again this is more pronounced if UA speed is low. This is consistent with the way we used when defining the model, and does not capture the tendency of the simulated CA to overestimate wind data when UA speed is higher than the activation speed.

This is however desired: the overestimation effect, likely a design and construction related characteristic, will change from cup anemometer to cup anemometer, and once built-in it likely cannot be changed (although it would be recommendable manufacturers provide correction tables, a thing I've never seen in cup anemometers data sheets).

On the contrary, the activation threshold is both an intrinsic cup anemometer feature *and* a maintenance subject: it might change as sensors deteriorate during field use.

7.5.2. A reality check

In Figure 7.2 we can see how the data appear if we get them from the ARPA database.

We can see some resemblance, but we also important differences respect to Figure 7.1.

These are quite interesting:

- Times are visibly not in synch! There is an about-ten minutes difference, the UA's red line occurring later than the CA's blue.
- When speed is "high" in Figure 7.2, namely between hours 10 and 18, the two plots roughly resemble each other if we imagine to smooth them a lot. Detail, however, fail to overlap. In Figure 7.1 they do quite exactly.

7. Simplified model of cup anemometers

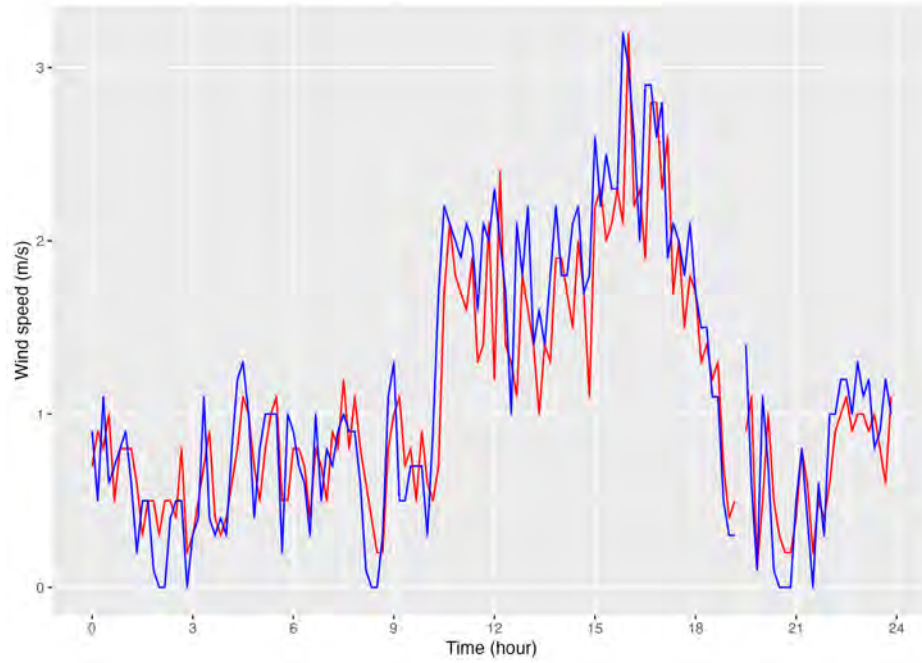


Figure 7.2.: Time evolution of 10 minute means of wind speed at Cinisello Parco Nord (red: UA; blue: CA; data from 08. 03. 2016, as gathered from the ARPA database)

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- Both CA and UA data have been declared invalid at the same nominal time, about 19:20.

Both effects can be attributed to artifacts induced by the quite peculiar hardware architecture the manufacturer did select.

In order to decide whether these differences are predictable or not, it is necessary to remember some characteristics peculiar to SHAKEUP data integration process, and try figuring out their possible consequences.

7.5.3. **Data treatment at SHAKEUP sites**

Some information about SHAKEUP stations was given in @sec-shakeup: before to get into details it may be worth a refresh there.

In this section we will go deep into a data processing model which is rooted into the (recent) history of SHAKEUP stations.

The thing to notice is that ultrasonic and conventional data are collected by two independent processes operating concurrently:

- Sonic processing, operated using a dedicated eddy covariance system (a MeteoFlux Core V2).
- Conventional data processing, done by a datalogger of quite “standard type” about 15 years ago.

What I’m about to say comes from the fact I am the designer-and-coder of the eddy covariance system, and took part to the meetings with ARPA and the manufacturer to agree the data transfer protocol.

In very short summary the SHAKEUP network pre-dates even ARPA Lombardia: it was designed within a contract by Regione Lombardia in the late Nineties of the last century by two colleagues of mine, dr. Roberto Sozzi and eng. Daniele Fraternali.

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In that occasion I had a minor role, but had the occasion to mass-deploy the MeteoFlux eddy covariance system which I participated to develop within the LIFE/Mo.Ni.Qa. EU project (Wow! That was the *first Italian eddy covariance system*, and I was in charge of the development of the computing kernel! This, automatically, gifted me the dubious honor to be “the first Italian woman to...” - guaranteed, it would be much more fulfilling to be “the *zillionth* woman who...”, but this, you may agree with me, is quite another story).

And, mass deploy it was, for an eddy covariance system: after having found its way to Antarctica, the Minatitlán refinery, one measurement site at Svalbard Islands, and various others around the globe, the MeteoFlux (now in its second version) finally found a place in my own *Heimat*, in Italy, and not too close from the place I live in (in Lombardy). Quite a dream.

The first edition of the SHAKEUP stations, in which “my” MeteoFlux operated as an embedded system within a PC-based conventional data logger by another manufacturer, operated flawlessly for some years.

Then, more or less around 2010, ARPA Lombardia made a new bill for the meteorological network maintenance, which included the SHAKEUP as sub-network. The winning bid arrived from another third party who, as often happened in these cases, changed the whole data logger set. We got the occasion to also revamp the MeteoFlux hardware and software, now in its “Core V1” incarnation.

But a problem arose very soon. The new conventional data logger was not based on a PC, but rather on a dedicated “close” machine, with limited programming facility. Porting the MeteoFlux complex logics on it was out of question, and so I and the other manufacturer engineers agreed on a data transfer protocol.

Unfortunately there were various hard-wired constraints. The most important was that on these times the objects designed as sold as “data loggers” dealt with simple sensors with an analog interface: the MeteoFlux Core

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V1 was *heavily digital*. (Today, most sensors are the “intelligent” type, with a digital interface and, if their manufacturer felt in good spirits on the product conception day, with an optional analog output.)

The only way in theory possible for transferring data from the MeteoFlux Core V1 to the conventional data logger was by a serial line operated by a separate specially built module, whose task was to translate the values coming from the MeteoFlux from its native to the data logger’s own format, and, in the process, by labeling it as “available for read”.

Once the data logger completed one of its averaging steps, it interrogated the RS-232 decoding module checking for not-yet-transferred data, gathered them, and *attributed them the data logger’s time stamp*. Then it labeled the data as “already read”.

I knew the software engineer who programmed the RS-232 decoding module, and is one of the myths in my personal Pantheon: he made the whole job in assembly - people my age know what this means...

But honestly the solution was quite a kludge: the RS-232 connection was uni-directional, and so did not allow to propagate the data logger RTC reading to the MeteoFlux.

The obvious, necessary consequence of using two systems with time bases taken from different hardware sources which are not possible to synchronize is, the two clock will diverge.

And I’m not speaking of seconds: random real-time-clock deviations may reach many minutes per year and even more if a PC clock not assisted by a quality RTC chip is used.

Anyway, the workflow occurring in the two parts constituting the “nerve system” of SHAKEUP stations is illustrated in Figure 7.3.

The consequence of the way data are collected and time stamped by the data logger prior than their transfer to ARPA premises may still not be immediately clear, but I feel a glance to Figure 7.4 will help forming a first intuitive picture.

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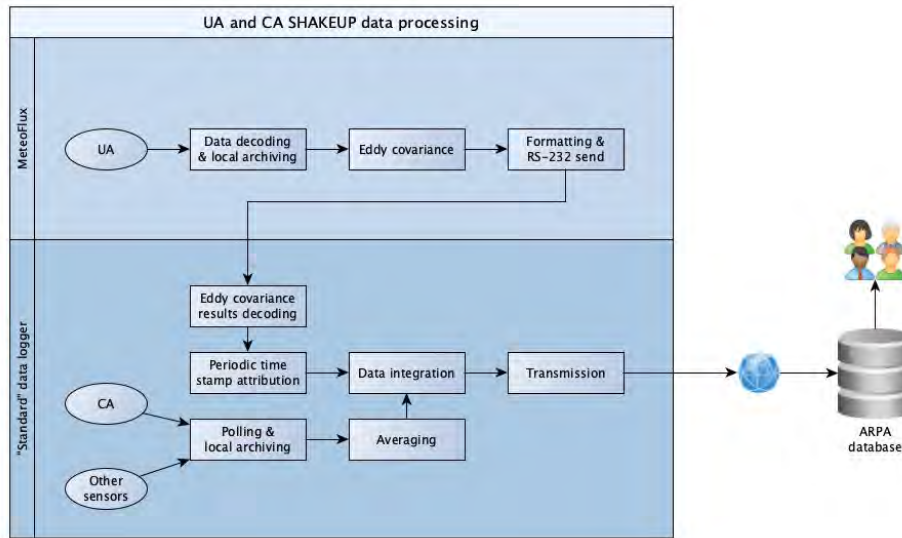


Figure 7.3.: Workflow occurring within and between the two independent parts composing a SHAKEUP station (the eddy covariance system MeteoFlux Core, and the conventional data logger)

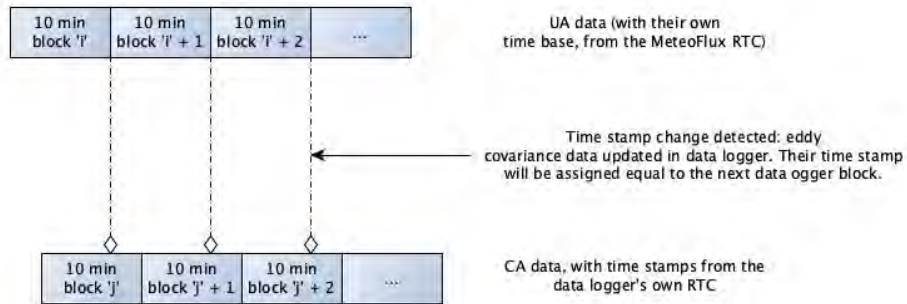


Figure 7.4.: Schematic presentation of the time stamping process performed by the SHAKEUP stations' data logger.

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In practice: the data logger detects a change in the time stamp of the MeteoFlux Core processed data (sent over and over every 5s), and from this fact infers a “new” data is available.

This is not integrated immediately, however. What actually happens is that later on, when the data logger will decide another averaging time has filled, it will attribute this time stamp to ultrasonic anemometer data.

But (and this is the essential point) in the meantime the two clock have diverged somewhat. Because of this, the beginning of a new averaging interval according to the data logger will *not* coincide with the start of a MeteoFlux averaging interval. The dire consequence is that in general the CA and UA 10 minutes averages *may not refer to the same time*. And if the time intervals do not overlap perfectly, then the indicated speeds will not be homologous.

Indeed we *can* simulate this effect, for example by shifting the time of raw UA speed data. The effect is shown in Figure 7.5.

As we can see, the black and blue graphs yield a similar impression than we got by comparing UA and CA speeds in Figure 7.2: the appearance of a time shift, a large resemblance, and in the same time differences.

Why that? The origin of this perceptual difference can be traced back to what we have seen in Section 2.4: wind can be imagined as the result of the overlap of a smooth “laminar” flow and a fluctuation due to “turbulence”. The two intertwine, in continuous change, occurring at time scales in part well apart from the possibilities of our senses and the resolution of conventional anemometers. As we change the time base, we implicitly observe different slices of this highly variable signal, and the consequence is an apparent change in 10 minutes averages.

Of course in Figure 7.5 we do not have any spurious speed change as a CA would induce, and because of this fact we get two signals more similar to those visible in Figure 7.2.

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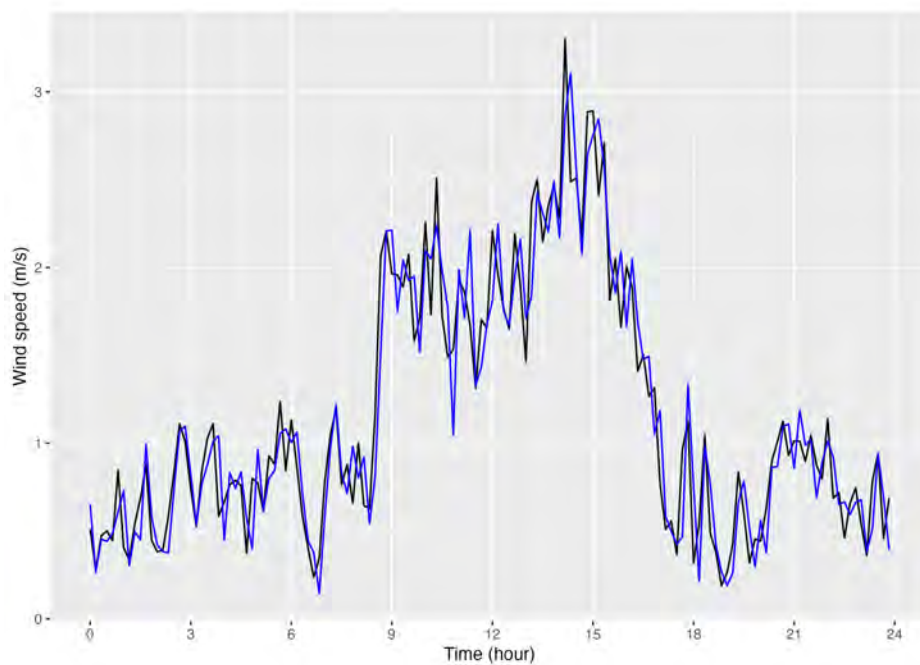


Figure 7.5.: Simulated effect of time stamp re-attribution (black: data averaged with no time shift; blue: data averaged after time shift of 5 minutes)

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But this also means that at least some of the difference we see in Figure 7.2 is due to time relabeling made by the conventional data logger. This is quite a minor effect statistically, but adds some noise which could have been avoided if a different and more permissive hardware architecture was adopted.

(Of course a shift in time labeling does not change the intrinsic statistical nature of wind: after all the kinetic energy stored in the airflow is the same whatever misalignment the clock we base our averaging on has. So, long-term statistics may end up preserved - this, by the way, is an interesting research subject: I leave it to anyone wishing to spend time and energy on this subject.)

7.5.4. Statistical effects of increasing the activation threshold

In Figure 7.1 we have seen that introducing a non-zero activation threshold does change the wind speed signal in a way which apparently gets more and more evident as the wind speed decreases.

A first question could then be, which is the number of zero speeds as the activation threshold U_a increases? The answer for the day we're considering can be seen in .

7.5.5. Using the model in support of predictive CA maintenance

Figure 7.6 is also a good illustration of what models can be used for in a practical setting. Suppose you have a CA data set, and in it you find n speed values which are equal to zero, out of a total N .

If in the same time you can collect UA data, then you may set up the model the way we had, and use it to find the value U_a at which the modeled number of zeros matches the one reported by the model.

In mathematical terms this means finding the solution of

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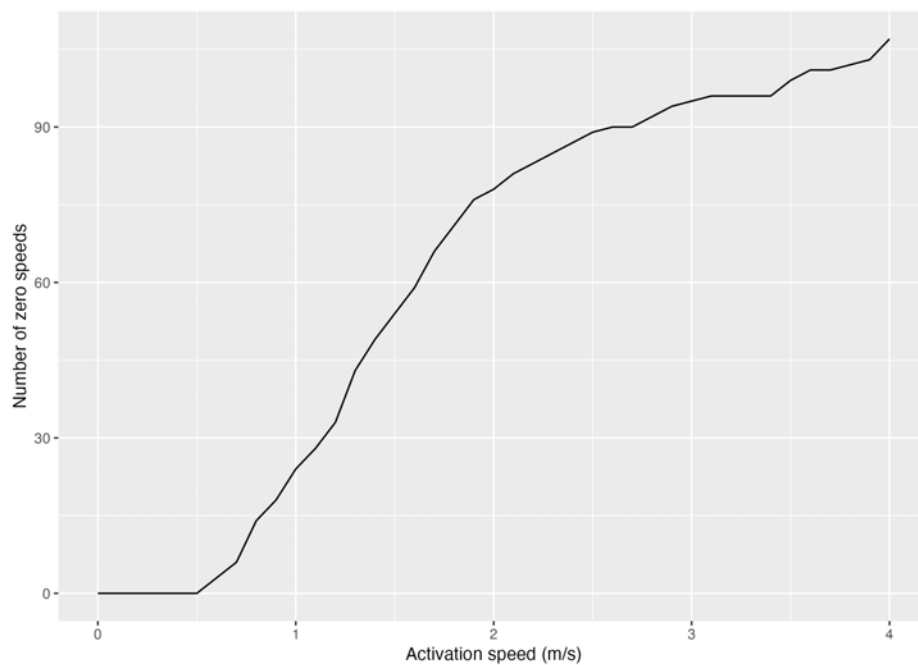


Figure 7.6.: Number of zero speeds as a function of activation speed (data for 08. 03. 2016 at Cinisello Parco Nord SHAKEUP station; means computed using the “crude” approximation)

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$$n_z(U_a) = n$$

where the model zero count $n_z(U_a)$ is a monotonic function (which unfortunately, given its nature of “count”, is discontinuous and then, automatically, not differentiable: so, sorry, no immediate way of using a nicety like Newton-Raphson solver).

Counting “real zeros” using the model is easy, but may not match exactly the way zeros are generated with CA data. As a remind, they are delivered as 10-minute averages, with values characterized by one decimal figure. So, what a CA value of “0.0” really signifies is any value provided by averaging sensor data whose value, once rounded, yielded 0.0.

That is, a CA non-rounded value $0 \leq u_{CA} \leq 0.5$ m/s.

Would we not take into account this fact, and blindly compare the CA “zero” count with the “really-zero” modeled values, we would likely obtain a zero-matching solution U_a^* which *overestimates* the real activation threshold of the CA.

But with the model in our hands we can change things at will, and introduce a revised zero counter $n_z^*(U_a)$ defined as the number of modeled values in the selected time interval satisfying the relationship $u \leq 0.05$ m/s.

As Figure 7.7 confirms, the revised model counting the CA speeds below 0.05 m/s is a fairer comparison term.

As an example application on day 08. 03. 2016 the number of CA zero means is 9.

According to the model counts (Figure 7.7) the crude exact zero count is 6 for $U_a = 0.7$ m/s, and 14 for $U_a = 0.8$ m/s. A count of 9 could then be reached somewhat in the middle, say at $U_a^* = 0.75$ m/s.

7. Simplified model of cup anemometers

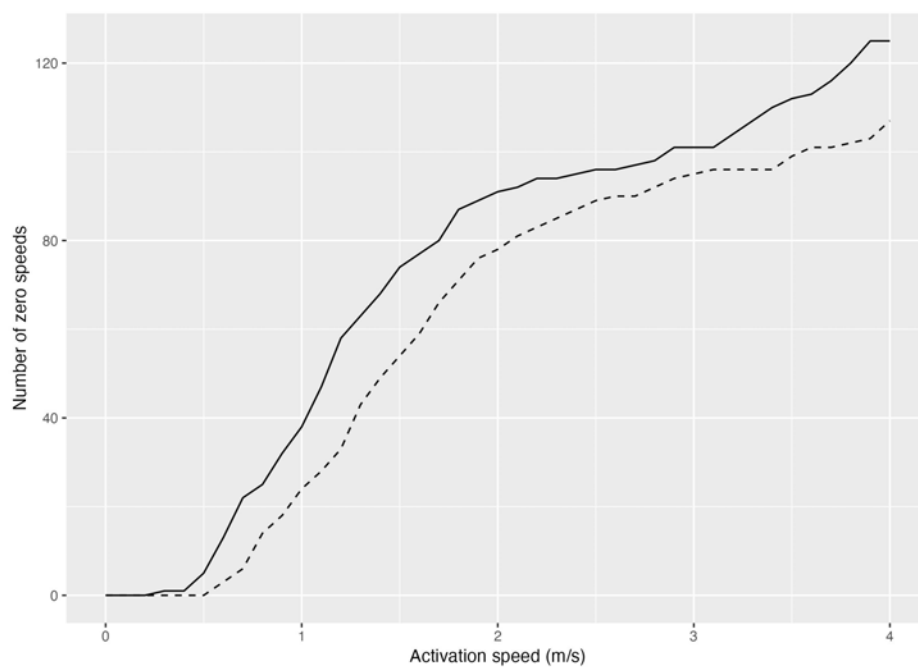


Figure 7.7.: Plots of the counts of exact zeros (dashed lines) and zeros-by-rounding (continuous line); data for 08. 03. 2016 at SHAKEUP station of Cinisello Parco Nord

7. Simplified model of cup anemometers

Using the crude counts “as rounded” we have 5 zeros at $U_a = 0.5$ m/s and 13 at $U_a = 0.6$ m/s, so the estimate becomes something like $U_a^* = 0.55$ m/s: a much lower, and likely less biased, value.

Now, imagine this calculation is repeated day after day for an extended time period. On every day, a different estimate of the activation threshold is calculated.

Once a sufficiently long series is available it becomes possible to see whether a trend is under way, and in this case it is not difficult (for example using a linear regression line) to see whether the trend is rising, or not.

In the former case, i.e. a rise, one could fix a threshold beyond which to start a maintenance action on the CA.

And this, depending on the threshold choice, could happen much beyond a dramatic sensor failure, with minimum impact on the data production process. This idea is the essence of *predictive maintenance*, a process management paradigm in increasing use on these years.

7.6. Data averaging

7.6.1. The raw data set

In the preceding section of this chapter we explored the possibilities coming from a simplified CA model by using a single day: this may be a nice approach when a method is in its initial development phases, or when short time-scale questions are posed, but in general it is not enough if the need is to get solid conclusions.

In all these “other” cases a larger sample is necessary. This is especially important when addressing issues like checking whether a CA over- or underestimates UA data: in cases like these the wind speed statistical distribution comes into play, and a too small sample allows going nowhere.

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If a single day is too small a sample to conclude anything relevant, then a “large” sample could be the solution.

In order to avoid contamination from spurious phenomena, however, this sample can not be entirely “random”: ultrasonic anemometers, as good and informative as they are, still are human-made instruments and for this very reason they are *not* ideal.

As ultrasonic anemometer data sheets mention, there are conditions in which these sensors may give less-than-usually accurate readings.

Most recently designed UAs host extensive plausibility checks and corrections in their firmware, and are then able to perform reasonably well even in these anomalous conditions (this is also the case with the Metek’s USA-1 sensors deployed at SHAKEUP stations).

Plausibility checks do not come without cost, however: the check may fail, and the usual response by the UAs in these case is by labeling the suspect data as invalid, or lower quality.

Dealing with these anomalies is one of the responsibilities allocated to eddy covariance systems like the MeteoFlux Core used in SHAKEUP stations, and its typical implementation is by not including erroneous data in averages: if the number of these “excluded data” is not too large (a threshold in wide use among UA users is the 25%) one assumes the not-so-random-sample mean and other statistics are very close to the whole population’s, and in very most cases this is supported by evidence.

But in the case of CA models we have to start from *raw* UA data, and so we should deal with anomalies on our own.

We could do, of course. But the approach followed in this work has been to accept days in which *no* anomalies were detected by the UA, and in which the meteorological conditions were not conducive to *possibly* measurements at risk of anomalies - in particular all rain and “foggy” days have been ruled out in addition to those in which invalids were found.

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The data set used (the same described in (Favaron, Zintu, and Morosini 2025)) is then a censored subset of 2016, and its statistical distribution is shown in Figure 7.8.

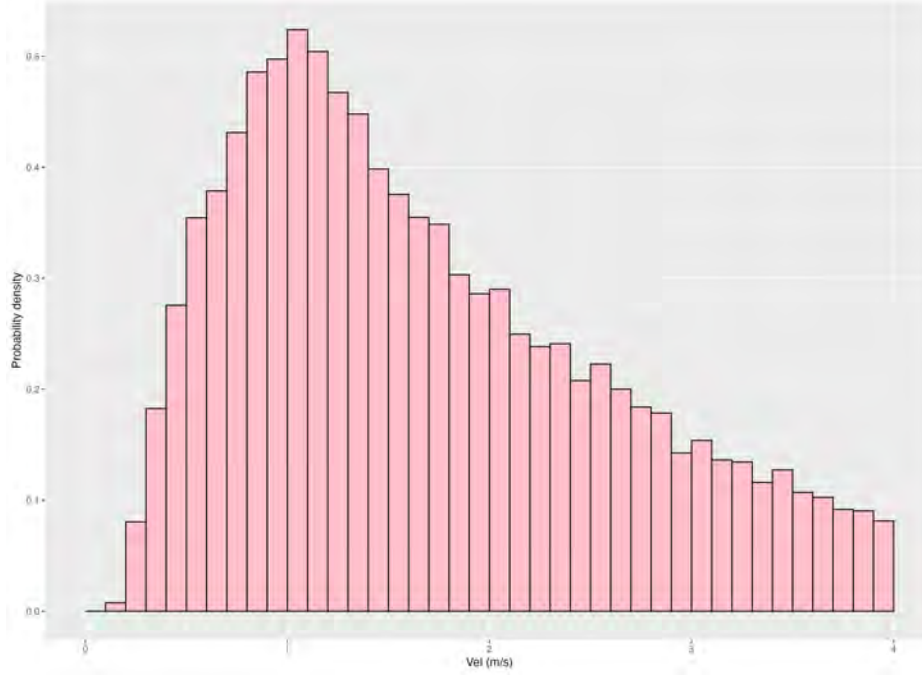


Figure 7.8.: Histogram of UA 10-minutes averaged speeds gathered in a subset of days in year 2016 at SHAKEUP station of Cinisello Parco Nord (the days considered are characterised by no missing data and no rain; additional information in the referenced paper)

As you can see in the paper cited, the number of data retained for analysis is about one half of the total calendar days present in 2016: the censoring process was extremely restrictive.

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Because of this, we can safely assume the statistical distribution presented in Figure 7.8 is reasonably representative of the reality, and it makes sense to use it for CA models evaluation.

7.6.2. Standard and improved averaging methods

7.6.2.1. The standard method

We start by introducing the “standard” averaging method, defined as the “diminished mean” we introduced in Equation 7.1. Its defining adjective, “standard”, is in my view well chosen as it corresponds to the way wind statistics are computed in the vast majority meteorological stations.

In word, it can be described by the mean of all anemometer *readings*, including the ones whose value is zero due to the real wind being lower than the activation speed U_a .

Averages are computed by the data logger, and would these be totally inflexible, then the problem would not meet the definition of the English word “problem”, as something one can solve. Rather, we would accept things as they are, and maybe cry in despair would them not be exactly as we would desire.

But in the real World, the degree of flexibility of data loggers is a variable. Typically, the professional and research grade models offer a level of flexibility ranging from “fair yet useable” to “extreme”, while the amateur grade models tend to be entirely hard-wired.

Professional and research grade data loggers (and *a fortiori* eddy covariance systems) cost much more, and are then less diffused than amateur grade systems.

So: in case you have an amateur grade data logger in your hand, you may read this section and get a lot of cultural profit from it, but will have serious difficulties to put in practice the suggestions given here. (There

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is of course the possibility of using a DIY (Do-It-Yourself) data logger: I anticipate this could be a fascinating subject for a hobby or personal research project, but will also tax you a lot of devotion, trials-and-errors, time, money and “technical bravery” - measuring wind reliably using an off-the-shelf microcontroller development board, like for example an Arduino, is less trivial than it seems at first glance).

But if you are lucky enough to have a professional-grade data logger, or an eddy covariance system, then you *can* experiment in your specific situation.

7.6.2.2. The “improved” method

Using the “standard” averaging method we are sure the averages may contain a contamination “towards the zero”, and this will be more and more evident as the mean speed decreases.

The consequence is an underestimation of slow wind speeds.

But suppose to consider a different approach. Let’s for a while change our professional life, and enter a physics laboratory.

We’re measuring something difficult, say the concentration of radon in air. “Difficult” means that the instrument we’re using has a detection limit r_0 , below which no measurement is really possible, but whose value is close to the range the concentration of radon has in our experiment.

If an instrument measurement has a value $r \geq r_0$, then we (are very happy and) accept it.

But if $r < r_0$, then we automatically know the concentration is “too low to detect it”, and we do *not* accept it as it is (depending on the instrument the reading r may be zero, but also not).

The most commonly adopted convention, then, is to replace the below-limit reading r with half the detectability limit, $r_0/2$.

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The rationale behind this is something in the line “Ok, I don’t know what the real value it, but I’m sure it’s very small: so small in fact, that it’s below the range where I could see interesting things; the error I’ll get assuming a value in the middle of the non-detectable range is so also small, and I can safely neglect it”.

That the mid of the below-detectability range $0 \leq r < r_0$ is really a good approximation of reality is, of course, just an assumption, whose mathematical hope is that the real values below the detectability limit follow a statistical distribution which is not too far from uniform, so that the median will be precisely $r_0/2$: the chance of being wrong on the left is then, by the definition of median, the same as being wrong on right, and as readings below the detectability limit accumulate, the associated errors we’ll do choosing half the detectability limit will cancel out.

This assumption, on the other side, in many cases is quite sensible: the below-detectability interval is so small that, whatever statistical distribution will be for the full data range, the part of it below the detectability limit will not be too far from a uniform distribution.

Just an assumption, as I said.

Now. In case of a CA, its activation threshold is something very alike a detectability limit.

So, why not assuming the value $U_a/2$ instead of 0 when a below- U_a wind speed value is encountered?

Values generated this way, once 10-minutes averaged, form what we could name the “improved” method. Of course we still not know whether this method works really better than the “standard” one: allow me, however, to adopt this little form of wishful thinking.

As I described the “improved” method, I imagine you have considered a possible objection: with physics laboratory instruments we find detectability limits which are *very small* respect to the values we’ll commonly measure.

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But the activation threshold of real-world CAs can well be *large* respect to the winds commonly found at that precise site. So, the assumption that the distribution of real below- U_a speeds is uniform is *another* form of wishful thinking.

So, instead of assuming a value $U_a/2$ for below-threshold speed, we would better assign some U_ξ which better respects the *real* distribution. Maybe, the average, or the median, of below-threshold values. This leads, after 10 minutes averaging, to what I've called the “extended” method.

It is worth noticing (and I'm quite confident you have made already) that this “extended” definition assumes in advance a precise knowledge of the distribution of data *below the observability limit* of the CA. But this, unless you have a UA available together with the CA as happens in the very uncommon SHAKEUP case, is surely not the case.

You can only approximate it in normal cases, provided you set up a characterization campaign using a temporary UA. In this case you may, after some days, obtain a constant $U_\xi, 0 \leq U_\xi < U_a$ which is better than U_a . The same campaign could allow you to estimate a confidence limit for U_ξ , as an indication of how solid your value assumption will be.

7.6.2.3. Are “improved” and “extended” methods really better than the “standard”?

As we've defined them, we have little doubts the “improved” and “extended” method will behave *differently* from the standard one. The questions are, how much, and whether the change occurs in the “better” direction.

As a starting point we try to ascertain what happens with the various methods as U_a assumes different values. We can see them in Figure 7.9.

What Figure 7.9 says is, in my opinion, quite interesting and articulate. If U_a is 0.5 m/s, that is, “small”, then the improved and extended methods

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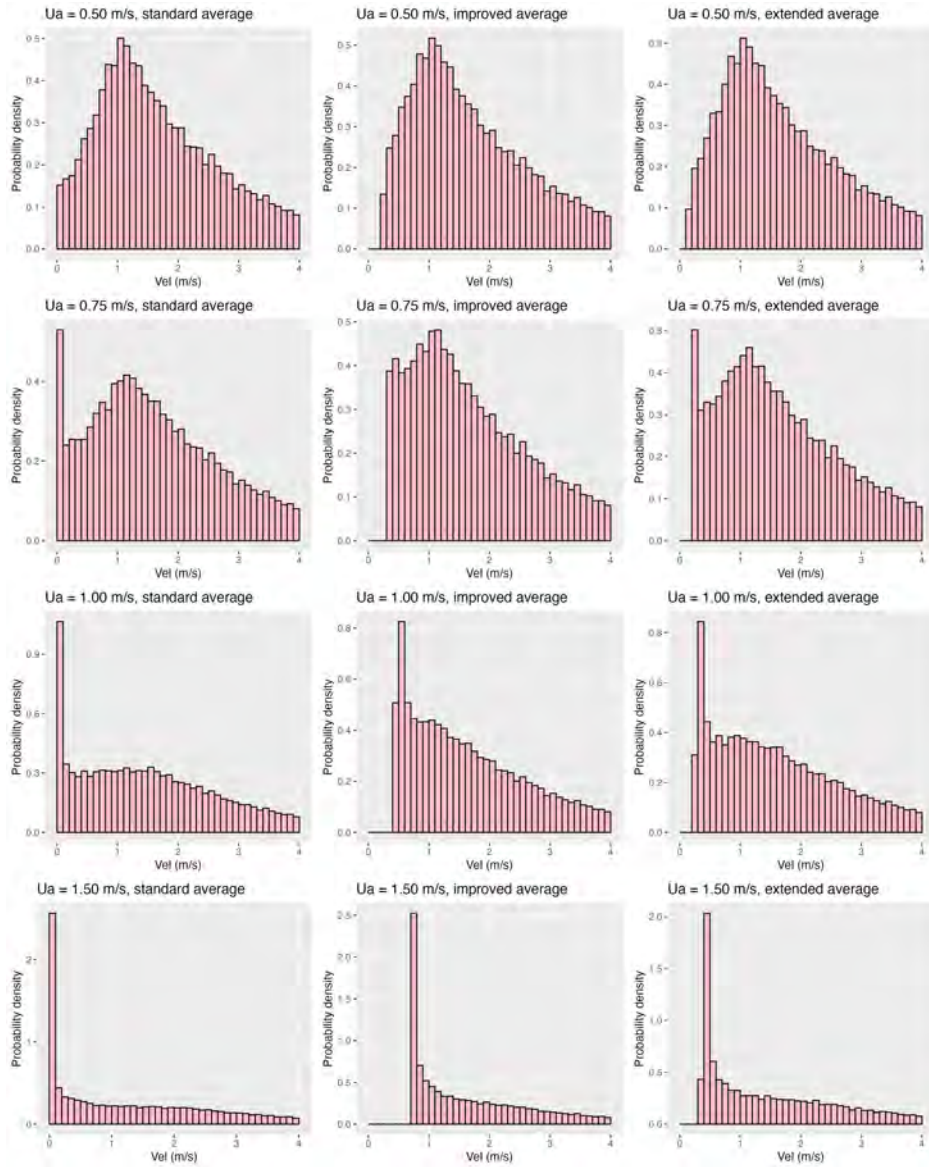


Figure 7.9.: Distributions generated as CA data are simulated using the standard, improved and extended methods for various values of the activation threshold (data from the 2016 sample already used; extended method implemented using the mean of below-threshold values, as it better harmonizes with the arithmetic mean based 10 minutes averaging)

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are both able to recover the original wind distribution shown in Figure 7.8. This does not happen for the standard method however.

But it is sufficient U_a grows a little bit for the distribution *not* being recovered any more. This happens as soon as the activation threshold reaches 0.75 m/s: in these cases, the extended method behaves apparently worse than the improved, but honestly none of the three methods provide a precise approximation of the UA distribution.

The morale is that at low U_a values the improved and extended methods behave better than the standard method in dealing with low speed data. But at higher U_a values there is nothing one can make: the speed data have been already corrupted by the CA to a level beyond any possibility of reasonable recovery.

7.6.2.4. Direct assessment of the averaging methods respect to original data

The models developed so far allow to go beyond the simple comparison of histograms: data are indeed available “coupled”, and we can perform paired comparisons.

The result is visible in Figure 7.10.

What Figure 7.10 tells can be summarized this way:

- The standard method underestimates UA averages, and the amount of the underestimation is larger at lower speed values.
- The improved and extended method, instead, show a less biased correspondence.
- In the mean time, the improved and extended methods significantly *overestimate* UA speeds at the lowest speed values.
- The latter overestimation is larger for the extended method.

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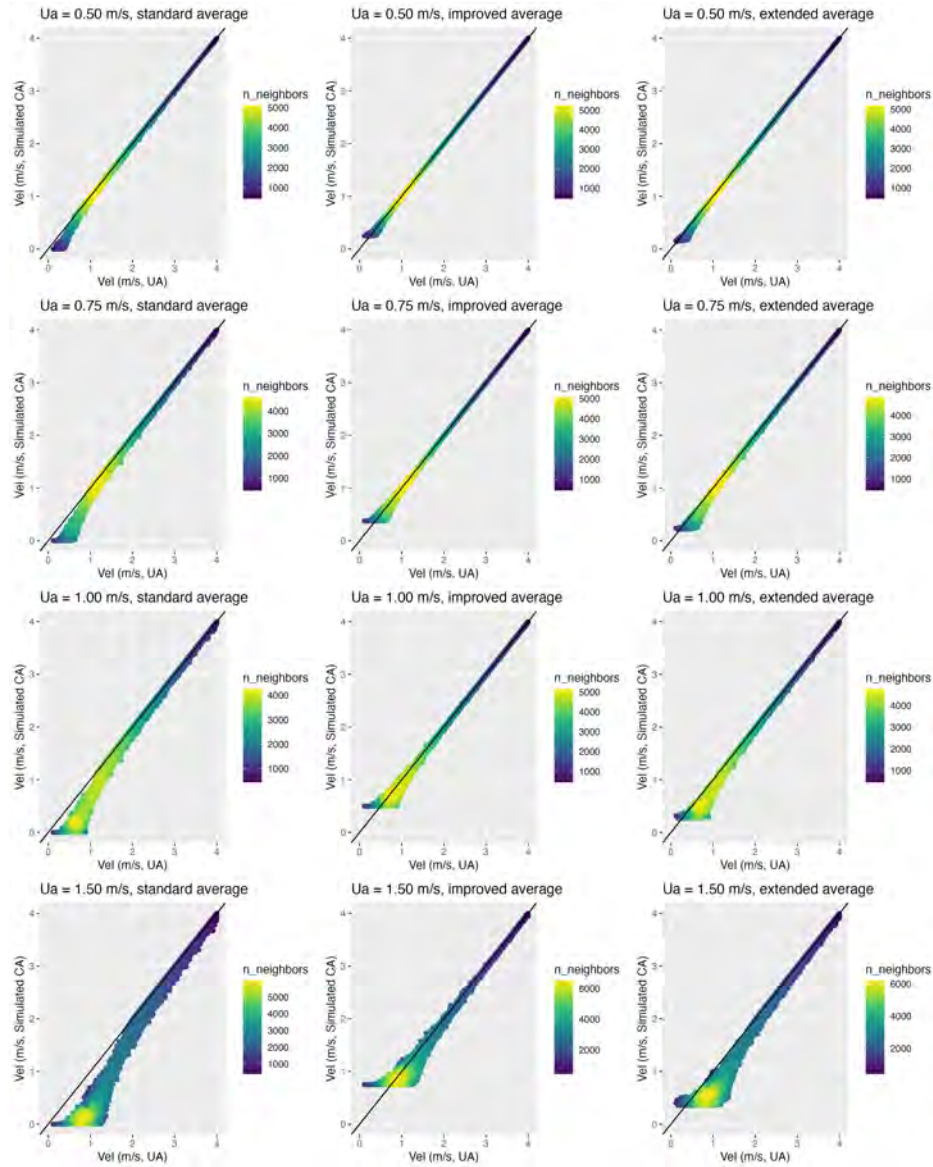


Figure 7.10.: Comparison between UA 10 minutes averages and the CA homologous averages simulated adopting different averaging methods and activation threshold values (data sample from 2016 at SHAKEUP Cinisello Parco Nord site)

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- The tendency towards underestimations by standard method, and overestimation by improved and extended methods, gets worse at increasing activation threshold values.

In extreme synthesis, high values of the activation threshold distort wind speed to a level which cannot be recovered.

7.6.2.5. Possible site dependencies

The conclusions we have drawn depend of course on the distribution of *instantaneous* UA data, which varies from site to site.

Then it is more than reasonable to assume some degrees of site dependency occurs: this is an interesting research subject, but I leave it for the moment unexplored.

7.6.2.6. The variability of sub-activation-threshold means

In the previous sections I've mentioned the usefulness of confidence limits for the means of raw UA data below the activation threshold.

This can be evaluated on every 10 minute averaging blocks, and histograms built for specific activation threshold values.

In doing the actual calculations it is worth noticing that for some 10 minutes block *no* below-threshold raw UA speeds may exist: in these cases the lower speed mean will be not defined, and should then be excluded from the histograms. The fraction of undefined lower speed means will get larger as U_a increases.

Figure 7.11 shows the mean below-threshold speed follow a distribution which is far from being uniform, or normal. However, mean and median can be calculated, and we can see they are always to the right of mid-threshold. This means two things:

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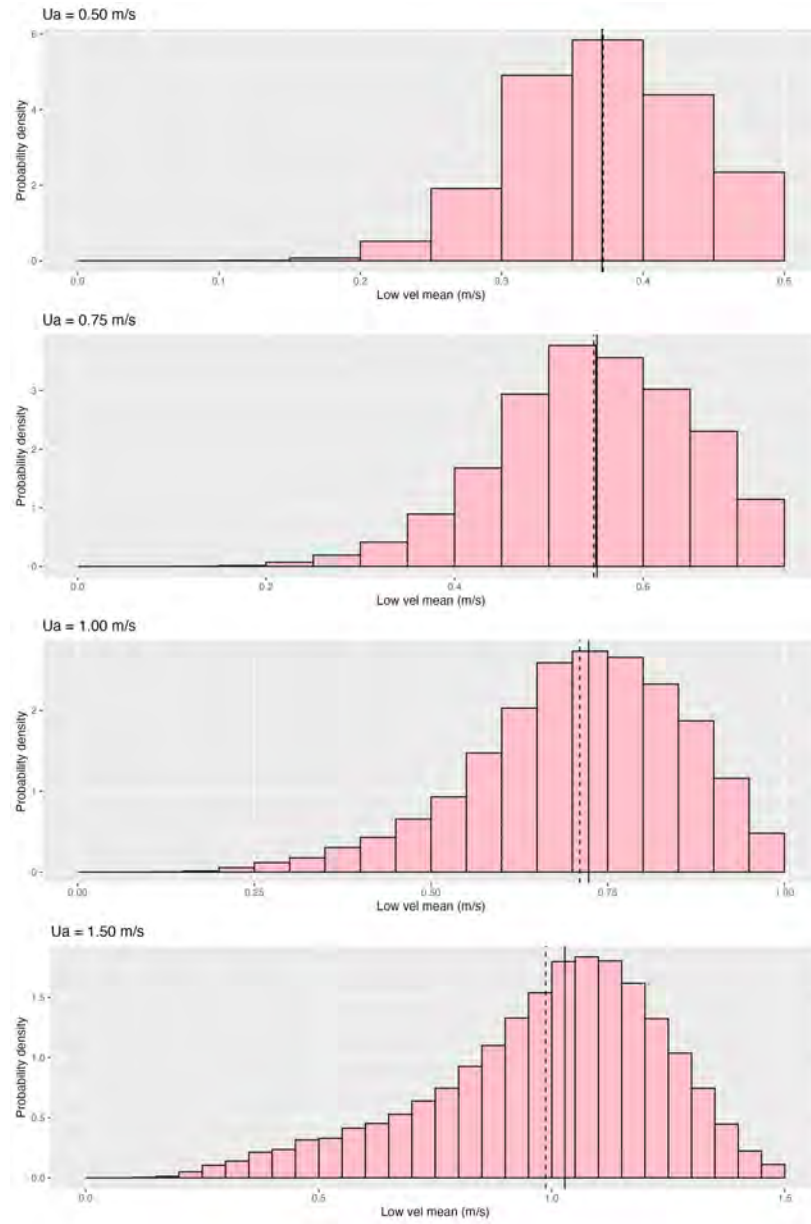


Figure 7.11.: Histograms of 10 minutes below-threshold mean speeds (2016 sample at SHAKEUP Cinisello Parco Nord station; continuous vertical line: median, dashed vertical line: mean)

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- The convention of using mid-threshold $U_a/2$ value instead of zero is in many cases not ideal.
- A better value can be identified by an appropriate statistical analysis of raw ultrasonic anemometer data.

This whole-campaign value can be used to define a new method intermediate between the improved and the extended, as part of a site characterization phase.

7.6.3. Integration of averaging method and predictive maintenance

As we have seen, if the U_a is small then the improved, extended, or even the extended-by-characterization averaging methods have a good chance to give somewhat better results than the standard method.

But for this to happen, the CA should be chosen so that U_a starts small, and the maintenance process defined so that *it stays so*.

In other words, use of predictive maintenance and averaging methods other than the standard one should go hand in hand.

7.6.4. Nothing ideal with CAs at low speed

Anyway. All the discussion made shows that CA, with their low-speed behavior, in any case corrupt the data.

In applications where an accurate low-speed behavior is of paramount importance (for example chemical hazard management, defense, or air quality modeling), and the site wind statistics make the chance of finding low speed high enough, it should be taken into consideration to replace the CA with a three-dimensional UA and its associated data acquisition and processing system.

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Other solutions may be explored as well, like for example hot wire 3D precision anemometers instead of UAs.

In any case, when designing a meteorological station located at variable sites, or at sites characterised by low wind speed, then the option of *not* using a CA should be taken seriously. If a CA is anyway used, then the designer should provide proper engineering and scientific justifications, along with a clear indication of the nominal U_a value. A maintenance plan allowing to guarantee the real U_a value remains close to the nominal is also due.

7.6.5. Is “averaging” appropriate?

In this chapter I’ve used means as indications of the central location of the wind speed. But in previous chapters we saw the wind speed distribution is not normal, and even symmetric: so, in principle, the mean is not “statistically appropriate”.

A better indicator could be the *median*. But:

- Evaluating the median demands wind speed data to be sorted, which at best requires $O(n \log n)$ instead of the $O(n)$ needed to calculate a mean. The consequence is the need of a larger computing power than typically available in data loggers (that could not be a problem with the powerful computers used in eddy covariance systems).
- For data to be sorted, they must be held in random access memory within the data logger: this means a relatively large memory capacity. Again, no problem with an eddy covariance system, but possibly a hardness to a data logger.

Part V.

This part describes the experimental campaign necessary to characterize the site where a cup anemometer will be installed, or to check an already existing site has retained the measurement context it had at installation.

Wind measurements indeed are almost never conducted at ideal measurement sites, that is perfectly uncontaminated flat lands where no change occurs to vegetation whatever the observation direction is, and where there are no other obstructions.

The deviation of a site from ideality has visible consequences on the wind pattern one could observe and, consequently, their measurements.

Assessing these differences is not possible using only the instrument under investigation, but demands the use of a higher class sensor to which the data from the instrument under test could be compared, and in the same time can indicate the extent and location of the non-uniformities.

8. Site Characterization

8.1. Introduction

This chapter deals with the “characterization” of sites where wind speed will be measured using a cup anemometer.

The purpose of site characterization is to establish the micro-meteorological environment in which a prospective cup anemometer will operate: as we have seen in the preceding chapters the behavior of cup anemometers, and consequently the quality of their wind speed data, depends on site and conditions, whose knowledge permits to evaluate the kind of stresses the cup anemometer will have to withstand.

As mentioned in previous chapters, the micro-meteorological characterization of a site may be done before a cup anemometer is installed, or also after it was. In the latter case, the purpose of the characterization is to check the conditions assumed for station design have changed, and whether the change may interfere with the cup anemometer functioning in a way which is not equal to the one originally assumed.

8.2. Method

The characterization described in this chapter is based on a simplified form of eddy covariance (Aubinet, Vesala, and Papale 2012; Sozzi and Favaron 1996)], applied to the ultrasonic anemometer raw data collected by the SHAKEUP stations already considered (Favaron et al., n.d.).

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More specifically, the available raw ultrasonic anemometer data have been first converted from their storage form, as binary SSB files (Favaron, n.d.a), to the textual form used by SonicLib.

Data have then been averaged on a 30 minutes basis excluding all invalid and suspect data, but using all the data set available, spanning non-overlapping time intervals. Eddy covariance processing was then performed on these “non-rotated” averages using an identical set of standard assumptions:

- Linear detrending enabled.
- Spike detection not performed.
- 2 rotations.

Of the many indicators made available by SonicLib, only those relevant to characterization will be discussed.

The following table shows the time span of the various data sets: they reflect the contents of the raw data collection in possess of the author. As mentioned, and as the table shows, they do not overlap completely, so no rigorous station to station comparison could be made from them.

Table 8.1.: Time span of raw data available for the SHAKEUP stations; some data within the time interval may not be available

Station	Id	Height (m above msl)	From	To
Arconate	AC	183	2012-09-17	2015-04-22
Caiolo	CA	278	2013-01-01	2013-05-28
Passo del Foscagno	FC	2310	2013-06-01	2013-11-23
Mantova Lunetta	ML	26	2012-12-04	2013-07-26
Parco Nord	PN	146	2013-01-01	2016-12-31

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Station	Id	Height (m above msl)	From	To
Pieve San Giacomo	SG	36	2012-09-25	2013-06-25

8.3. Wind speed distribution

8.3.1. The cumulate wind distribution plot

What is the fraction of wind speeds below the activation threshold of the cup anemometer we are evaluating?

How many wind speed data will likely be contaminated by the existence of below-threshold speeds?

A very useful tool for answering these, and related, questions is given by the cumulate speed plot, that is the plot of the inverse function of

$$p(U_s) = 100 \frac{|\{U \leq U_s\}|}{n}$$

where $\{U\}$ designates the set of available (valid) wind speeds, $n = |\{U\}|$ its cardinality (number of elements), and U_s a given reference speed.

The cumulate speed plots for the SHAKEUP stations are shown in Figure 8.1.

Figure 8.1 clearly shows the situation is different from a site to another, with Passo del Foscagno (FC) site presenting the higher speeds.

The cumulate speed plot immediately allows to see how many wind speed data will fall below the sensitivity of the cup anemometer under evaluation. For example, the stations CA (Caiolo), AC (Arconate) and PN (Parco Nord) show the fraction of wind speeds below the activation speed will be

8. Site Characterization

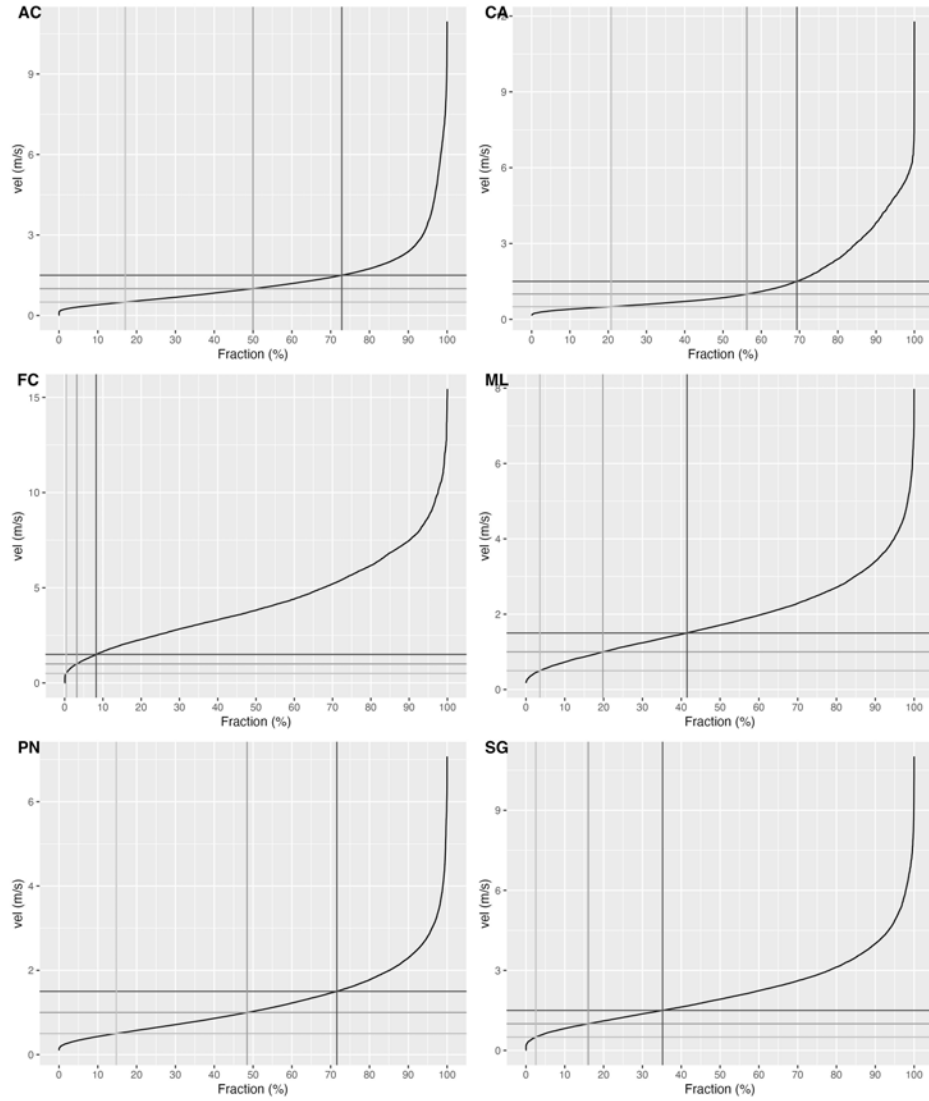


Figure 8.1.: Cumulate speed plots of SHAKEUP stations; the grey lines correspond to the activation threshold, 0.5 m/s, and the two heavier grey lines indicate the 1.0 and 1.5 m/s limits (two and three times the cup anemometer's speed threshold)

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about 21%, 17% and 14% respectively: a severe fraction, inducing one to consider whether a sensor with an activation threshold significantly smaller than 0.5 m/s could be a better solution (or whether adopting a wind speed sensor altogether different from a cup anemometer is a more sensible and cost-effective choice).

As we have seen in Chapter 5, mean wind speed data smaller than three times the activation threshold present some contamination from below-threshold values. In this regard the situation presented by the SHAKEUP stations is quite flashy, with about the 72% of likely contaminated data for the two stations of Cinisello Parco Nord and Arconate.

The cumulate speed plot can also be read in reverse: suppose one wants to determine the activation threshold permitting a given, desired fraction of uncontaminated data.

For user convenience, the cumulate speed plot can be plotted in a slightly different way, in this case. We can see an example in Figure 8.2.

Figure 8.2 is, basically, a design tool: given the maximum tolerable fraction of data which are contaminated by below-threshold individual cup anemometer readings, it allows to determine a sensible activation threshold allowing to fulfill it (or, would the indicated activation speeds be too small for current technology at a specified budget level, it yields an argument against use of a cup anemometer; in this case, other kind of sensors, namely ultrasonic, hot wire or hot film anemometers, could be taken into account).

The maximum tolerable fraction of contaminated wind speed data depends on the application, but the cumulate speed curve depends on the measurement site: it cannot be obtained but with an experimental site characterization campaign.

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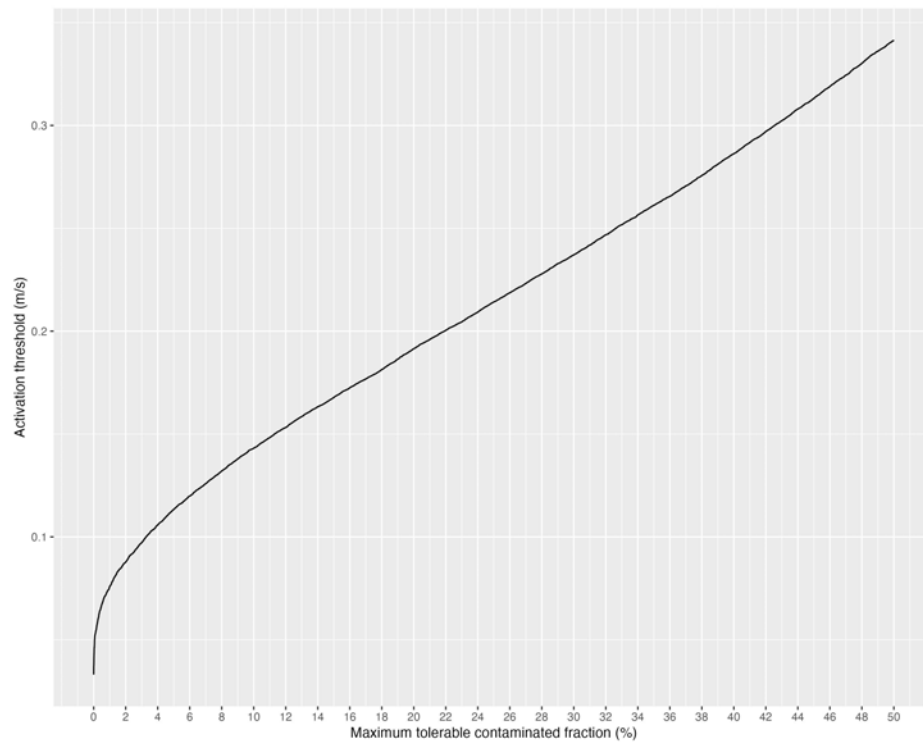


Figure 8.2.: Cumulate speed plot at Cinisello Parco Nord (PN) site, with the maximum contaminated fraction indicated

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8.3.2. Wind speed and provenance direction

Given the fact, proven in previous chapters, that cup anemometers respond differently to slow and faster wind speeds, it is legitimate to expect a larger effect from provenance directions for which the imbalance is larger.

The study of wind speed by their provenance direction can also indicate whether blind spots exist at a specific site: their occurrence, easy to detect using an ultrasonic anemometer, is often masked by resonance effects by wind vanes (Sozzi, Georgiadis, and Valentini 2002), with the possible adverse effects on the overall wind data. It may also happen that blind spots are so large and widespread to be *not* masked by an electro-mechanical anemometer, in which case use of the site for certain applications (e.g. dispersion modelling) may be questionable.

Usually, the study of the dependence on speed versus direction is conducted by using statistical plots like wind roses and similar graphs. In these cases, wind speed and direction data are classified by lookup speed and direction tables, and a simplified polar presentation is made.

In order to be meaningful, wind roses must be computed using a high number of data. This may not be compatible with the characterization campaigns, whose duration is often short. In addition, they make the detection of blind spots below the angular width of direction classes difficult and uncertain.

A better alternative is represented by density scatter plots of actual wind means, shown for example in Figure 8.3.

Density scatter plots demand a high computing load, but are much more informative than conventional wind roses.

For example, they inform us immediately of the presence of blind spots (none visible in SHAKEUP stations), a strongly channeled circulation (FC, located at a saddle point within a relatively narrow valley, and characterized by a circulation mostly directed along the valley axis), or a preferential

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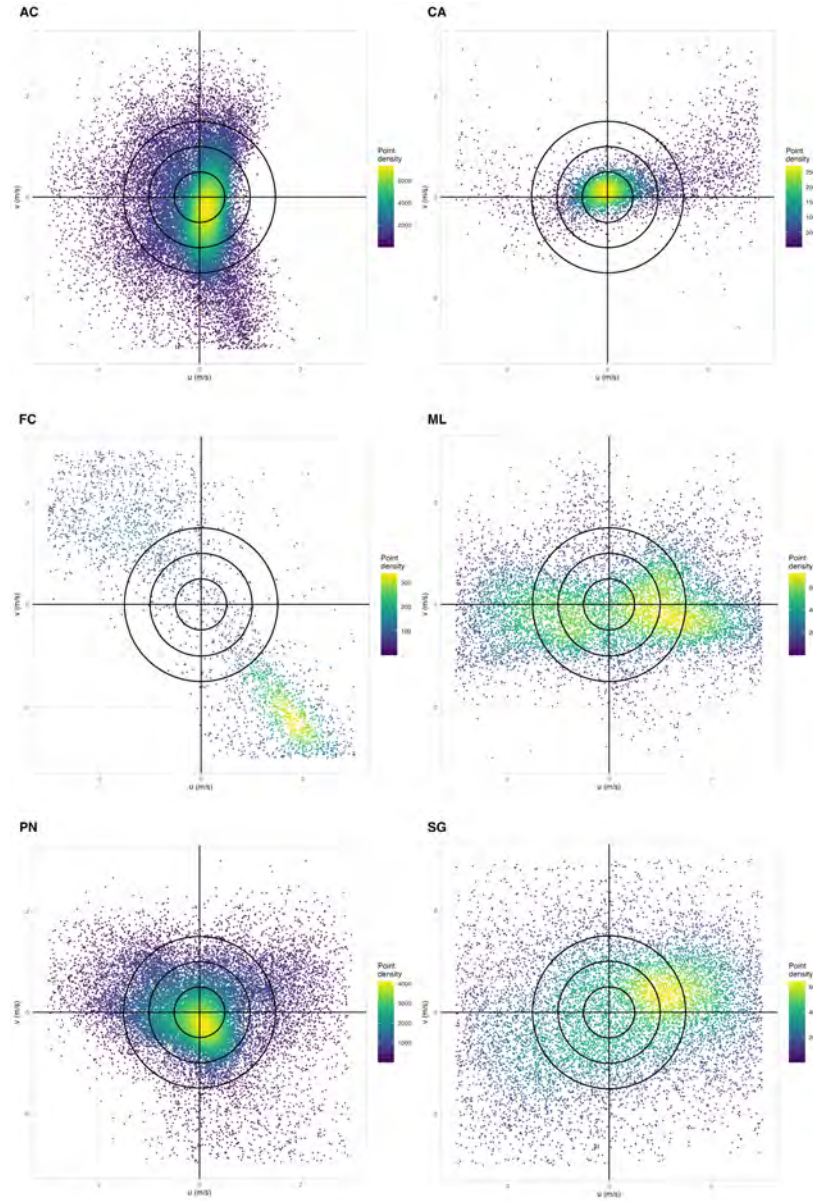


Figure 8.3.: Density scatter plots of wind at the SHAKEUP stations (wind values with components beyond -3 and $+3$ m/s are not shown; the three black circles have radii of 0.5 , 1.0 and 1.5 m/s)

8. Site Characterization

direction (ML and SG in our case, with a circulation more or less aligned to the West-East axis followed by synoptic winds). Even the *absence* of a preferential direction, like at sites AC and PN, is informative: it tells (at least in Lombardy) the station, maybe located on what appears a flatland, in reality is affected by the breeze winds induced by the South-facing Alps nearby (they may be considered “pre-alpine” stations).

Another precious information Figure 8.3 gives at a glance is the campaign size: the two cases of AC and PN, characterized by large available data sets, appear visually as more densely populated than those of all other stations.

8.3.3. Wind speed and stability

The static (in)stability of the lower atmosphere (Stull 1988; Foken 2008; Arya 2001) is known to be correlated with wind speed and direction. This may be expected to be stronger at “pre-alpine” sites, where the local circulation effects are stronger.

This correlation is visible whenever “stable” and “convective” roses are compared: in many cases they are so different that a statistical “same distribution” test is not necessary.

But as shown in Section 8.3.2 wind roses demand larger data sets than often available in site characterization studies, and once again the use of density scatter wind plots may be better.

In Figure 8.4 to Figure 8.9 we can see the side-to-side comparison of stable and convective for the SHAKEUP stations. Stable conditions are defined in a crude way by the condition $H_0 \leq 0$, and convective by $H_0 > 0$ where H_0 is the turbulent flux of sensible heat. This is quite an abuse of language, as neutral conditions are aggregated to stable or convective based on sign of H_0 , but prevents the proliferation of plots.

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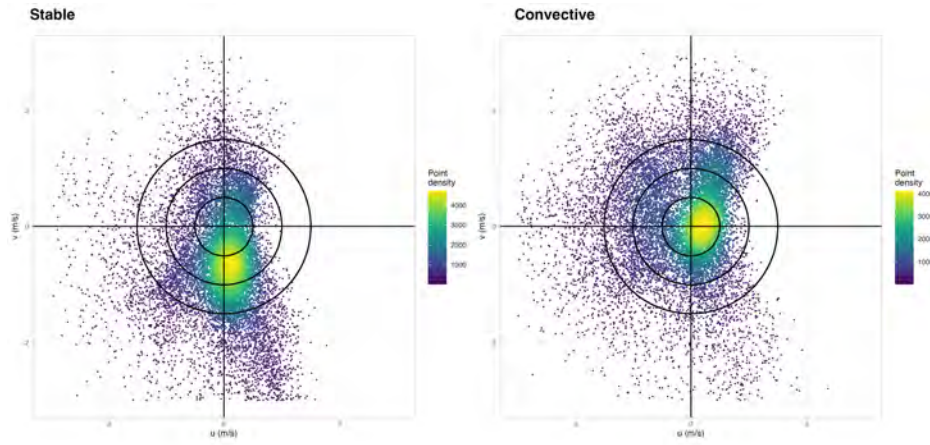


Figure 8.4.: Density scatter plots for stable and convective cases at Arconate (AC)

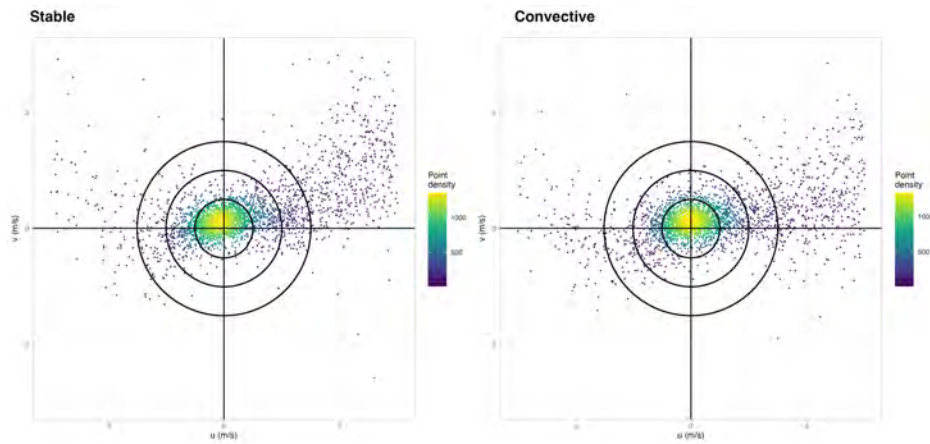


Figure 8.5.: Density scatter plots for stable and convective cases at Caiolo (CA)

8. Site Characterization

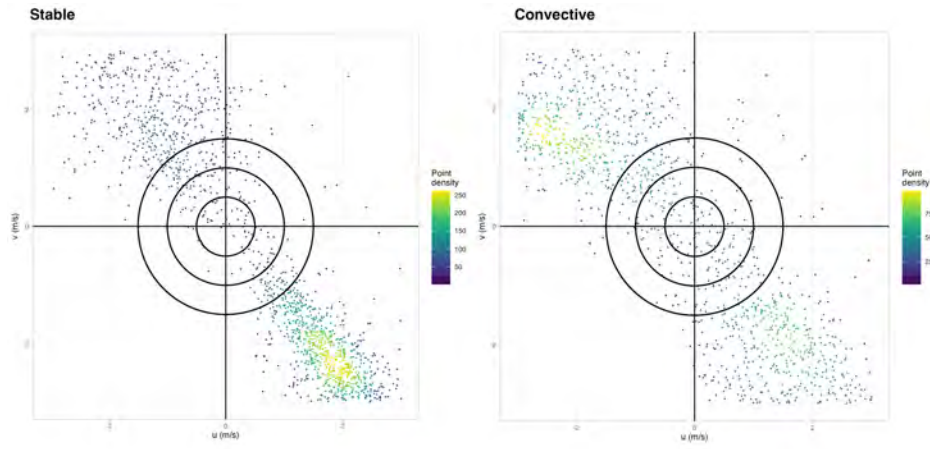


Figure 8.6.: Density scatter plots for stable and convective cases at Passo del Foscagno (FC)

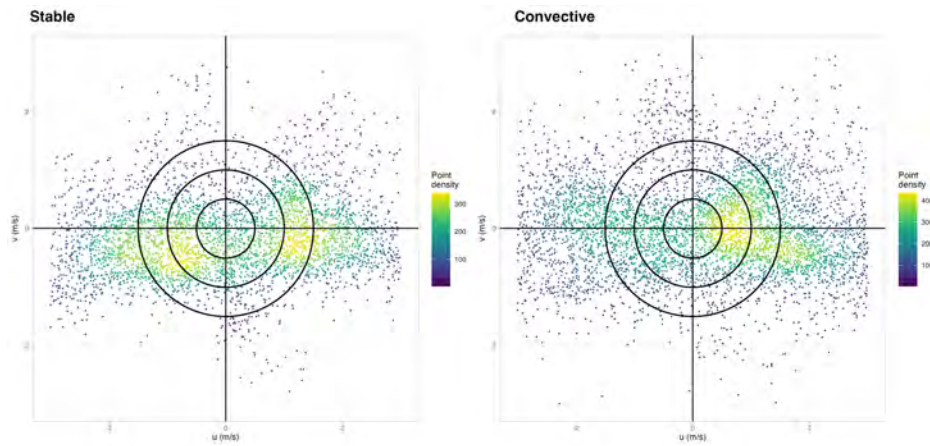


Figure 8.7.: Density scatter plots for stable and convective cases at Mantova Lunetta (ML)

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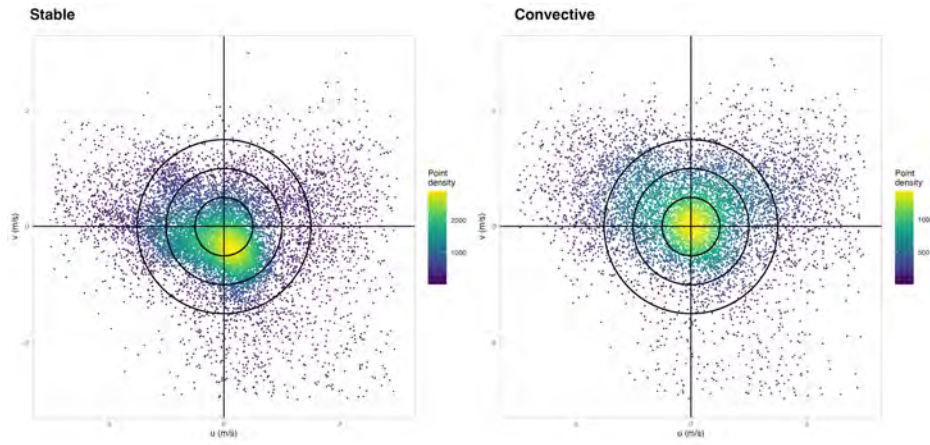


Figure 8.8.: Density scatter plots for stable and convective cases at Cinisello Parco Nord (PN)

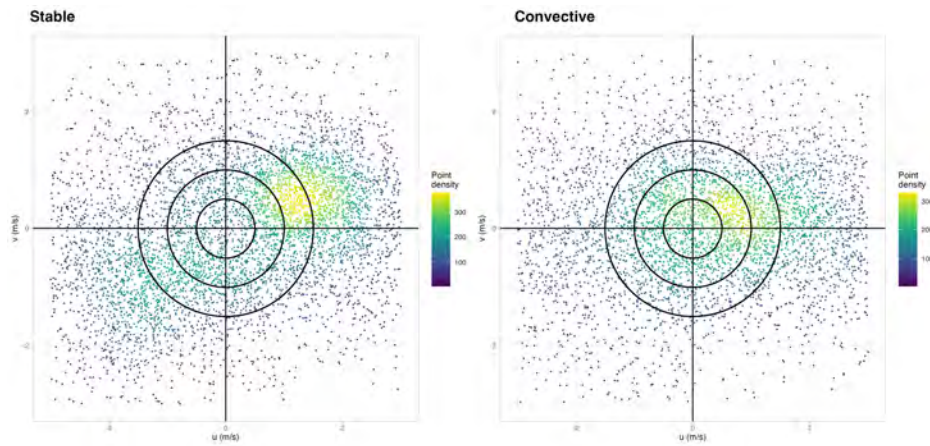


Figure 8.9.: Density scatter plots for stable and convective cases at Pieve San Giacomo (SG)

8. Site Characterization

Figure 8.4 and Figure 8.8 show visually detectable differences between the stable and convective data, suggesting a strong effect due to pre-alpine breezes. Figure 8.7 and Figure 8.9 present very similar data on convective and stable conditions, but this is not unexpected as both stations are placed deep in the Po Valley, close to mean sea level, so a combination of the mostly West-East synoptic circulation and the valley axis, also West to East, can provide a possible explanation. Finally, Figure 8.5 and Figure 8.6 show the effect of channeling through narrow valleys.

Assuming these explains are correct, there is no special reason to expect the conditions at one of the station site will change appreciably in the next future (on geological time scale). The dependence is strong on site, however, with details which cannot be predicted even for stations (like AC versus PN, or ML versus SG) which can not be predicted.

8.4. Vertical wind and related quantities

8.4.1. Mean vertical wind

Wind tunnel and theoretical studies shortly summarized in this book confirm the presence of a vertical wind component may alter the speed measurements taken using a cup anemometer.

It is then important to check whether a nonzero vertical speed is actually present at the site under investigation.

In theory, if the terrain is flat and uniform (that is, no obstructions occur, and the roughness elements - e.g. crops - have identical properties whatever the wind provenance direction) the mean of the wind vertical component is zero, or close to zero (Arya 2001).

Even in the Po Valley, a perfectly flat terrain cannot be detected: the slope may be quite small or large, but in most cases can be appreciated by

8. Site Characterization

the naked eye, without support of a total station or a theodolite. In addition obstructions exist, and roughness elements may seldom be considered uniform, especially in densely populated areas.

The possibility of significant departure cannot then be ruled out, and one of the key points of characterization is to assess it at the site under investigation.

Figure 8.10 to Figure 8.15 show the mean vertical component per direction at the SHAKEUP sites.

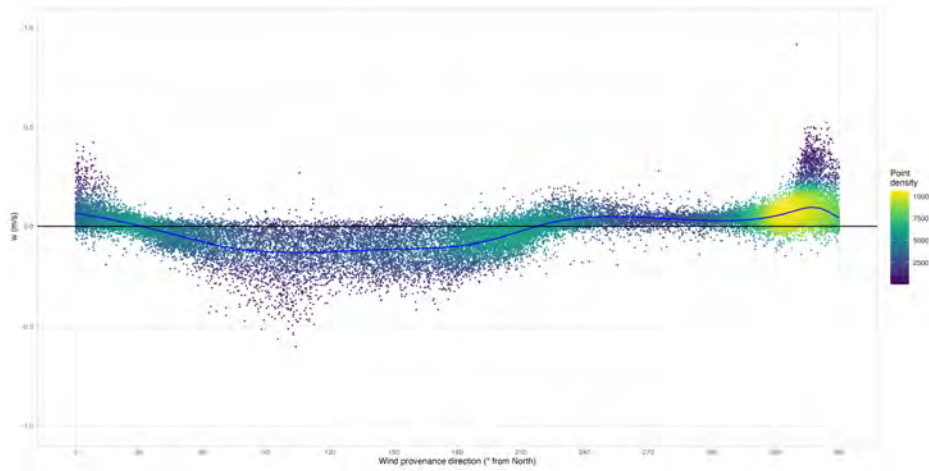


Figure 8.10.: Directional diagram of the vertical component means at Arconate (AC) (the blue line indicates the trending value)

As seen for other indicators, also the vertical wind shows site dependency. Some stations “see” a vertical component larger than others, with the high mountain site FC demonstrating the highest values. This is consistent with, for FC, the fact that the station is located in a place characterized by a significant slope, while all others stand on apparently flat terrain.

8. Site Characterization

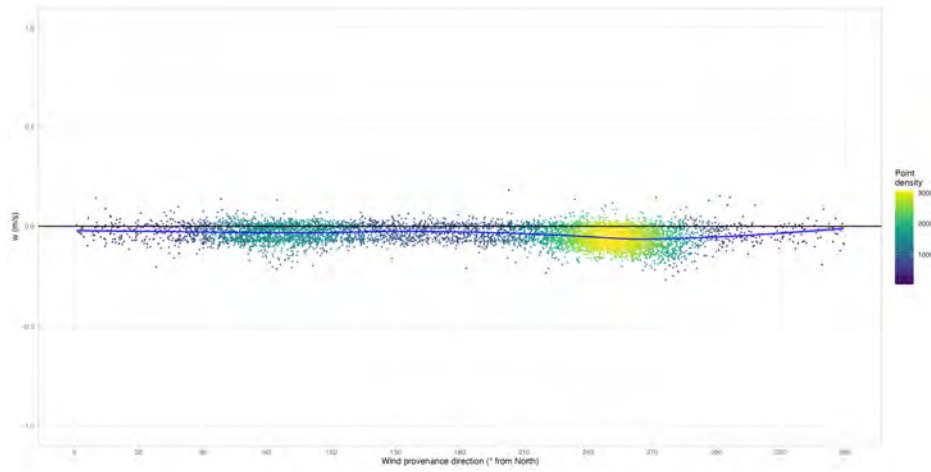


Figure 8.11.: Directional diagram of the vertical component means at Caiolo (CA) (the blue line indicates the trending value)

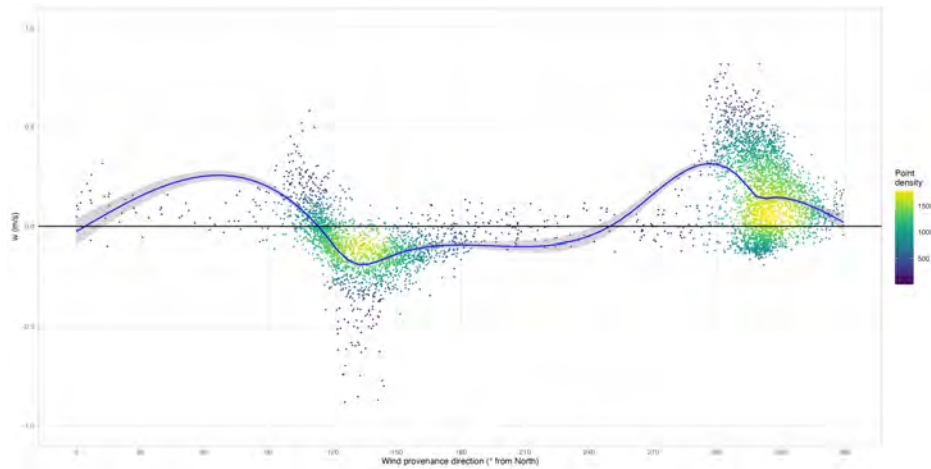


Figure 8.12.: Directional diagram of the vertical component means at Passo del Foscagno (FC) (the blue line indicates the trending value)

8. Site Characterization

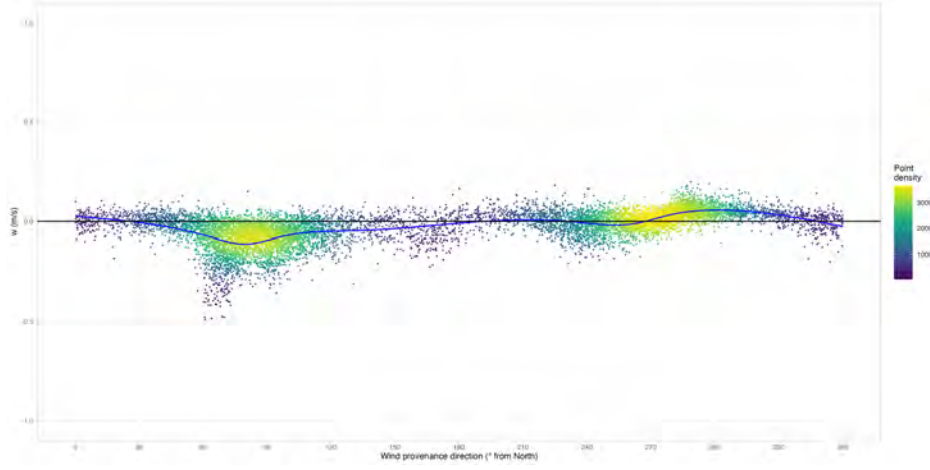


Figure 8.13.: Directional diagram of the vertical component means at Mantova Lunetta (ML) (the blue line indicates the trending value)

8.4.2. Wind angle from the horizon

One important thing the figures in this section do not say is whether the wind vector is inclined more or less from the horizon: if the vertical speed magnitude can in itself induce changes in the indicated horizontal speed, a large (positive or negative) angle of wind from the horizon may indicate, in association with strong winds, to accelerated wear-out and premature failure of the cup anemometer.

Given a vertical wind speed value, the angle from the horizon may assume any value depending on the magnitude of the horizontal speed. This can be seen from the mathematical expression of the slope angle of wind,

$$\phi = \tan^{-1} \frac{\bar{w}}{\sqrt{\bar{u}^2 + \bar{v}^2}}$$

8. Site Characterization

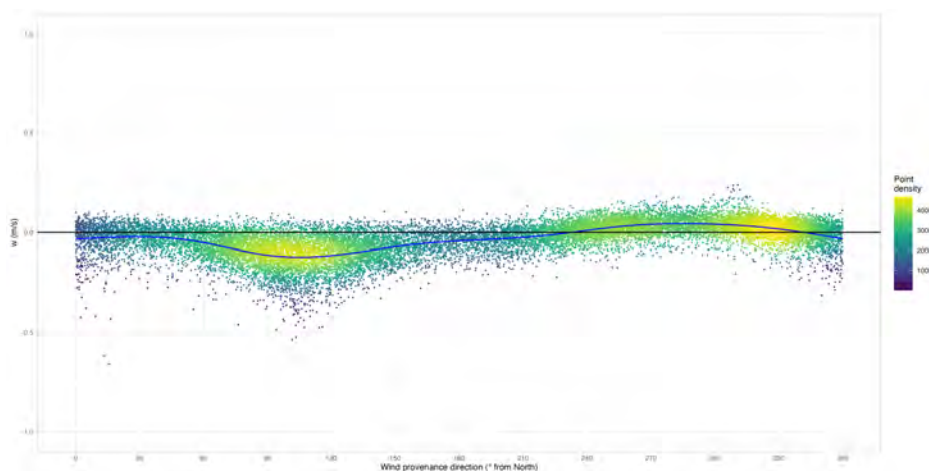


Figure 8.14.: Directional diagram of the vertical component means at Cinisello Parco Nord (PN) (the blue line indicates the trending value)

In Figure 8.16 to Figure 8.21 we can see the directional distribution of angle to horizon data.

Plots in this section resemble somewhat those in Section 8.4.1. This is not surprising, the two-arguments arc-tangent retaining the sign of the first argument if the second, as in our case, is positive.

yet, and this is the important thing, the y scale is different. relatively small values in the mean vertical velocity may give rise, under slow wind, to large angles.

Overall, the two pre-alpine stations located at Arconate and Cinisello Parco Nord show larger angular changes with direction, with the sub-urban site of Arconate largest.

8. Site Characterization

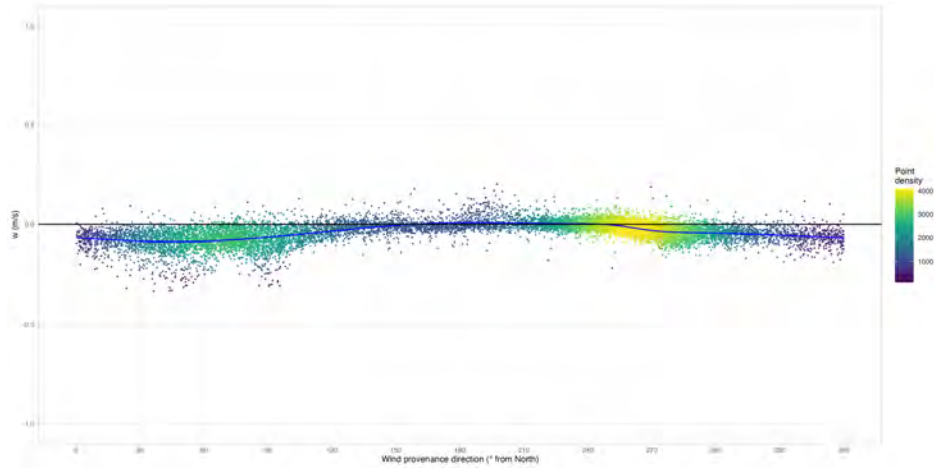


Figure 8.15.: Directional diagram of the vertical component means at Pieve San Giacomo (SG) (the blue line indicates the trending value)

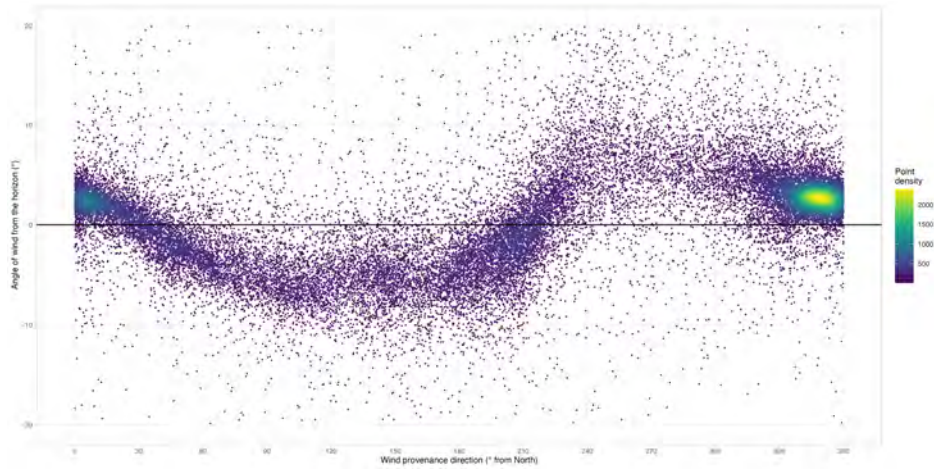


Figure 8.16.: Angle of wind from the horizon at Arconate (AC)

8. Site Characterization

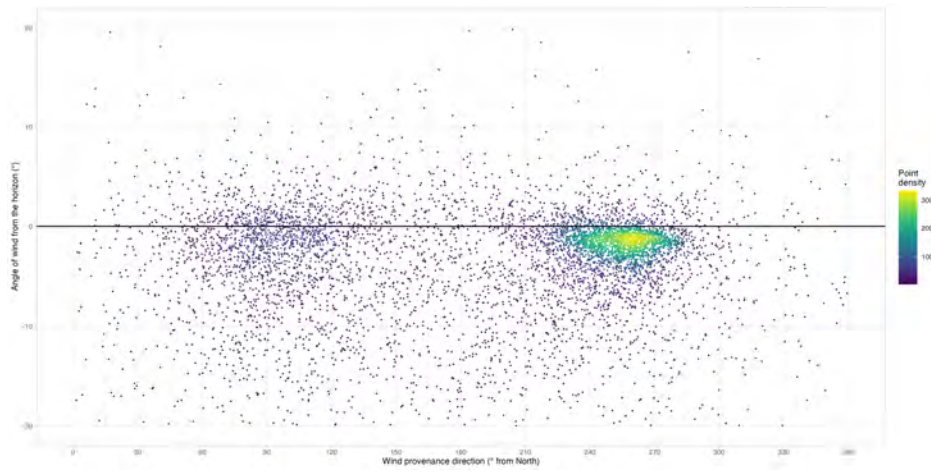


Figure 8.17.: Angle of wind from the horizon at Caiolo (CA)

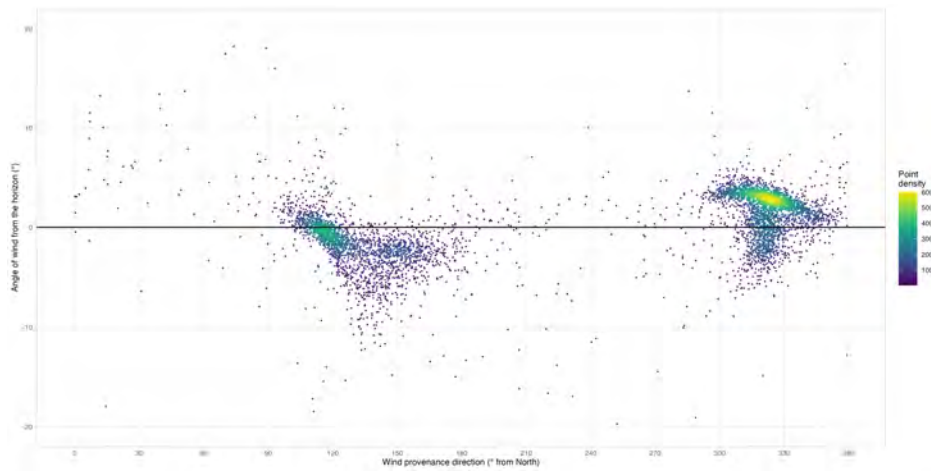


Figure 8.18.: Angle of wind from the horizon at Passo del Foscagno (FC)

8. Site Characterization

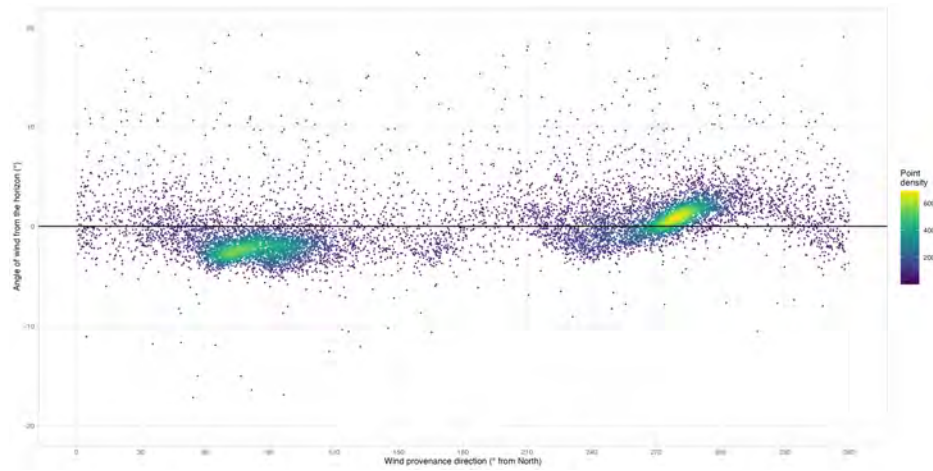


Figure 8.19.: Angle of wind from the horizon at Mantova Lunetta (ML)

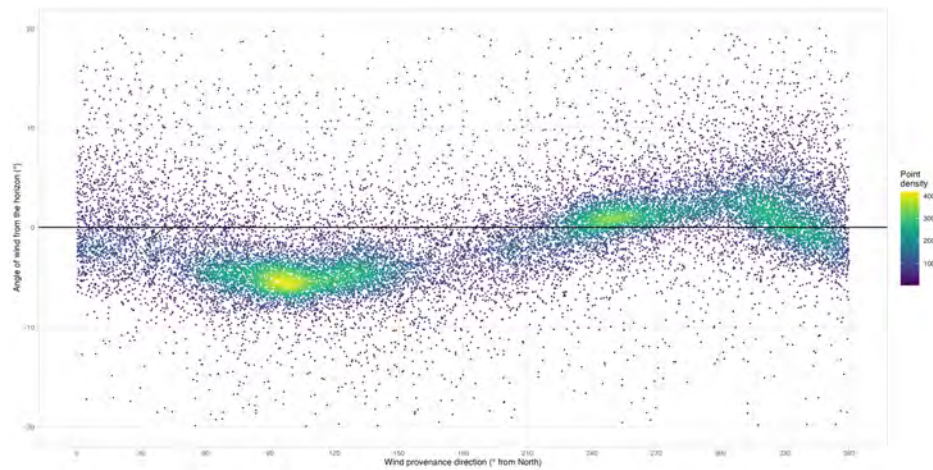


Figure 8.20.: Angle of wind from the horizon at Cinisello Parco Nord (PN)

8. Site Characterization

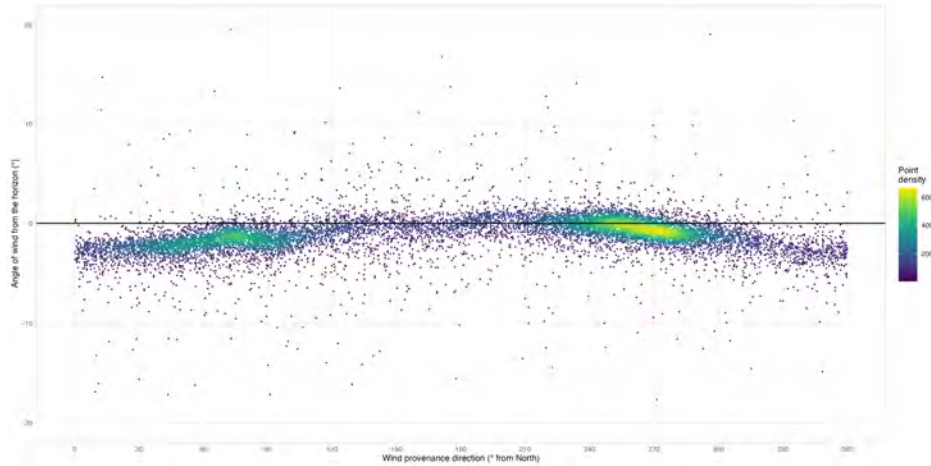


Figure 8.21.: Angle of wind from the horizon at Pieve San Giacomo (SG)

8.5. Temperature statistics

As any mechanical object, cup anemometers contract and dilate as the temperature changes. Very low or high temperatures may, in addition, accelerate the degrade of ball bearings and lubricants. Consequences may range from null to catastrophic, depending on the application and its safety integrity level (D. J. Smith and Simpson 2004).

Knowing the typical seasonal range of temperature is a precious design tool, allowing to discard or lower a cup anemometer whose environmental requirements are not compatible with the ones found on field.

In the specific case of temperature the extreme values are of most interest, together with their frequency. The possibly short campaigns used in characterization studies may not be enough to yield a reliable estimation of temperature extremes and overall distribution.

Fortunately, however, temperature is routinely measured by many public and private meteorological stations and networks, and extensive climato-

8. Site Characterization

logical data exist, at least in Europe. The answer to temperature-related questions may then be found by this way.

As we have seen indirectly in Section 8.3.3, characterization demands evaluating the turbulent flux of sensible heat H_0 , and this implies temperature, in addition to wind, is measured.

If the characterization sensor is in particular an ultrasonic anemometer, then temperature is guaranteed to be collected at high resolution.

Sonic temperature is not the usual dry-bulb temperature, but rather an approximation to virtual temperature. Its value can be used nevertheless, and give interesting indications especially whether the campaign duration is long enough.

If this is the case, then a cumulate plot of sonic temperatures can be useful. We can see an example, given for illustrative purposes only, in Figure 8.22.

8.6. Turbulence

8.6.1. Importance

On a first glance turbulence may seem uninteresting for cup anemometers: if they react approximately as first-order instruments, then they will smooth out any irregular fluctuations from the speed signal, leaving only its mean part.

This naive view was questioned by investigators (Frenzen 1988; Kristensen 1998), who found turbulence to have an effect on measurements, with potential maintenance indications.

This, of course, if we consider only the direct consequences turbulence may have on cup anemometers. There are some also indirect, however, allowing

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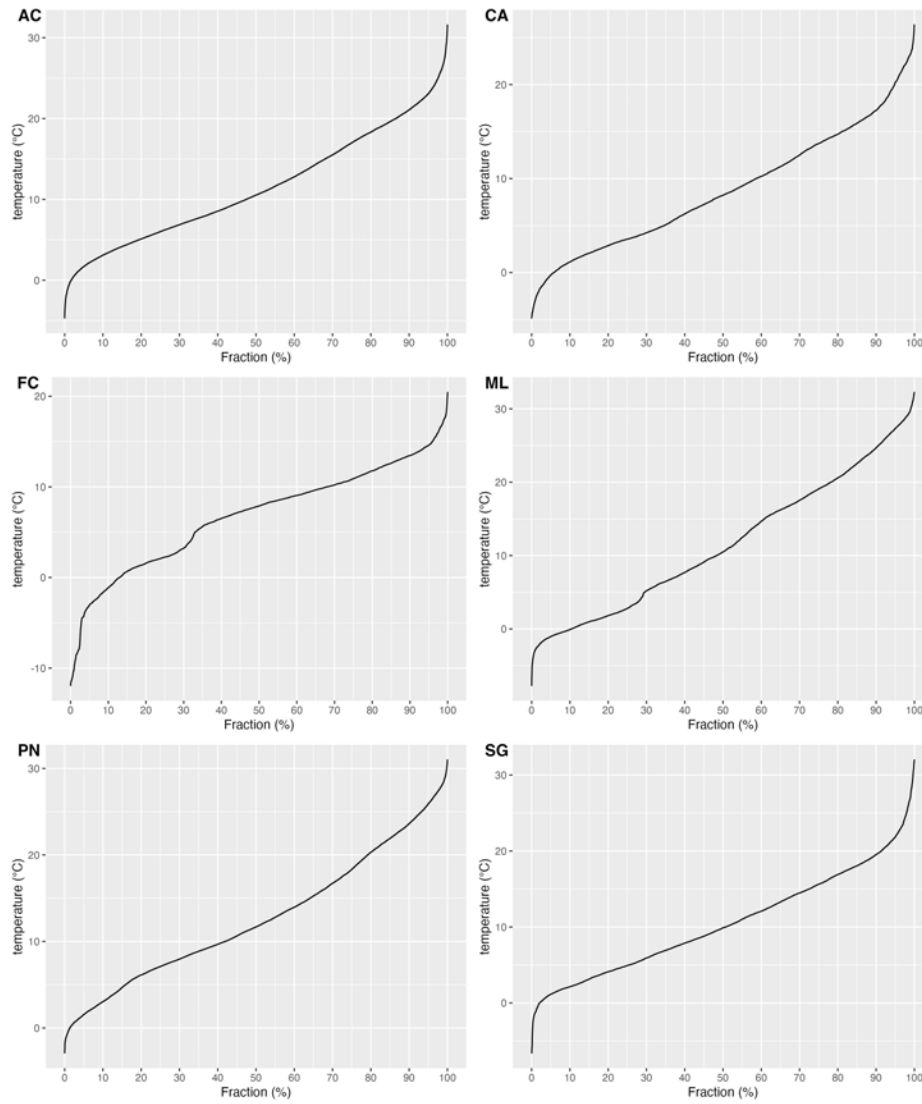


Figure 8.22.: Cumulate plot of temperatures recorded by the ultrasonic anemometer at SHAKEUP sites

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to improve the knowledge of sites where cup anemometer measurements are or will be taken.

To date, the study of turbulence can benefit of fast-sampling, high-resolution wind and temperature sensors (one example is the three-dimensional ultrasonic anemometer, but others exist) and advanced computing procedures (for example eddy covariance).

As already noticed by Leonardo da Vinci in his Codex Atlanticus, turbulence is a complex phenomenon, whose point-wise description is not realistically possible. This is true also to date, the level of detail in principle necessary to just represent turbulence would extend down to the molecular scale.

This is similar to what happened for common meteorological quantities like wind intensity and temperature, whose detailed definition relies on statistical mechanics: the state of turbulence is represented at macroscopic level by a set of indicators.

Among these, three are of special interest to characterization studies:

- The *Turbulent Kinetic Energy*, defined as $TKE = \frac{1}{2} (\sigma_u^2 + \sigma_v^2 + \sigma_w^2)$ and giving information about the total amount of turbulence whatever its origin.
- The *friction velocity*, $u_* = \sqrt[4]{\overline{u'v'^2} + \overline{v'w'^2}}$, a proxy of turbulence of mechanical origin induced by the friction obstructions exert on wind.
- The turbulent flux of sensible heat H_0 already met in Section 8.3.3, a proxy of turbulence of thermal origin.

The precise definitions, and theory accompanying them, can be found in any micro-meteorology textbook, like for example (Stull 1988).

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The reason to consider all these indicators instead of the *TKE* alone is that turbulent structures originating from different (mechanical or thermal) causes tend to differ qualitatively, and may impact not equally cup anemometer performance.

In the past, outside the controlled environment of wind tunnels by big research institutions, the numerical indicators of turbulence mentioned above were only accessible through estimation, starting from meteorological measurements readily available on these times (Favaron 2022) (namely horizontal wind, temperature, relative humidity, precipitation and, possibly, global and net radiation).

Many estimation paths did exist, and one among them, but “van Ulden - Holtslag” approach, gained special popularity. In very short this approach starts from the measurements just mentioned and proceed by chaining various semi-empirical relations.

These chains of estimates were (and are) available in the form of “meteorological processors”, computer programs transforming the available meteorological data in the form needed for practical uses (most often to build the “meteorological input files” used in dispersion modeling).

Use of meteorological processors is still common to date, as stations hosting turbulence profilers like those in the SHAKEUP and similar networks are still not the norm.

The nature of the meteorological processor estimates, and turbulence indicators in particular, allows for cautious use in applications like evaluating the population exposure to pollutants with long-term toxic effects, but lacks the fine granularity and accuracy necessary for a technical assessment of the cup anemometer operational environment: hence the author’s recommendation to avoid using them.

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8.6.2. Turbulent Kinetic Energy

The TKE, defined as

$$TKE = \frac{1}{2} (\overline{u'u'} + \overline{v'v'} + \overline{w'w'})$$

(Aubinet, Vesala, and Papale 2012; Stull 1988), indicates the “total” amount of turbulence.

The TKE is half the trace of the covariance matrix, and so it is guaranteed to be invariant by many matrix transformations (Landi and Zampini 2018), and rotations in particular, so eddy covariance axis adjustment is not necessary when computing it.

Turbulent kinetic energy constitutes a part of the global kinetic energy,

$$KE = \frac{1}{2} (u^2 + v^2 + w^2)$$

In itself this quantity, instantaneous, is of little utility. But upon Reynolds-decomposing, taking the mean and applying the Reynolds conditions (Monin and Yaglom 2007), we get

$$\overline{KE} = \frac{1}{2} \overline{(u^2 + v^2 + w^2)} = \frac{1}{2} [\overline{(\bar{u} + u')^2} + \overline{(\bar{v} + v')^2} + \overline{(\bar{w} + w')^2}]$$

The rightmost member can, by linearity, be rewritten as

$$\frac{1}{2} [\overline{(\bar{u} - u')^2} + \overline{(\bar{v} - v')^2} + \overline{(\bar{w} - w')^2}]$$

that is, a linear combination of expressions having the form $\overline{(\bar{s} + s')^2}$. But

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$$\overline{(\bar{s} + s')^2} = \overline{\bar{s}^2 + 2\bar{s}s' + s'^2} = \overline{\bar{s}^2} + 2\overline{\bar{s}s'} + \overline{s'^2}$$

Now, again applying Reynolds conditions,

$$\overline{\bar{s}^2} = \bar{s}^2$$

and

$$\overline{\bar{s}s'} = \overline{\bar{s}s'} = \bar{s}0 = 0$$

In conclusion, we may write

$$\overline{(\bar{s} + s')^2} = \bar{s}^2 + \overline{s's'}$$

and finally

$$\overline{KE} = \frac{1}{2} (\overline{u^2} + \overline{v^2} + \overline{w^2}) + \frac{1}{2} (\overline{u'u'} + \overline{v'v'} + \overline{w'w'}) = \frac{1}{2} (\overline{u^2} + \overline{v^2} + \overline{w^2}) + TKE$$

The quantity

$$MKE = \frac{1}{2} (\overline{u^2} + \overline{v^2} + \overline{w^2})$$

is known as *mean kinetic energy* (Oke et al. 2017). In characterization studies both the TKE and the dimensionless ratio

$$R_{KE} = \frac{TKE}{MKE + TKE}$$

are important.

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In Figure 8.23 we can see the cumulate plots for all the SHAKEUP stations.

High values of TKE tend to occur seldom at all the SHAKEUP sites considered, yet they do occur.

Some site dependency occurs, as observed for other parameters, and these differences may contribute to local changes in cup anemometer data quality. More importantly, however, the situations shown in Figure 8.23 for the optimal SHAKEUP sites location may not be representative of the bulk of meteorological stations, whose placement is often the result of technical and administrative compromises: TKE, in these sub-optimal sites, is likely to be higher, and differently distributed.

An important point, not addressed by Figure 8.23, is that at least part of turbulence, and hence of its kinetic energy, is directly caused by friction, and then it may be interesting to compare it to the mean kinetic energy. This is done in Figure 8.24.

In Figure 8.24 most of the data points are close to low MKE and TKE values. But if we consider the higher values, at some sites (AC, CA, PN and SG) it is possible to figure out the overlap of data points in two clusters, with an apparent linear shape and with two different slopes. No specific investigation has been made about this phenomenon, and its possible relation with the two cases of mechanical only and mechanical plus convective turbulence patterns allowed by stable and convective PBL structures respectively: this subject is left open to anyone desiring to analyze the subject.

Of more interest to this book focus is the relationship between MKE and TKE at low MKE (and then speed) values: this is shown by Figure 8.25.

At Figure 8.25's magnification it is possible to see a dominant, site dependent MKE vs TKE area, whose position informs about the most recurring cases, and to see whether it is more associated with smaller or higher MKE. In particular, the two "pre-alpine" sites of AC and PN show a prevailing

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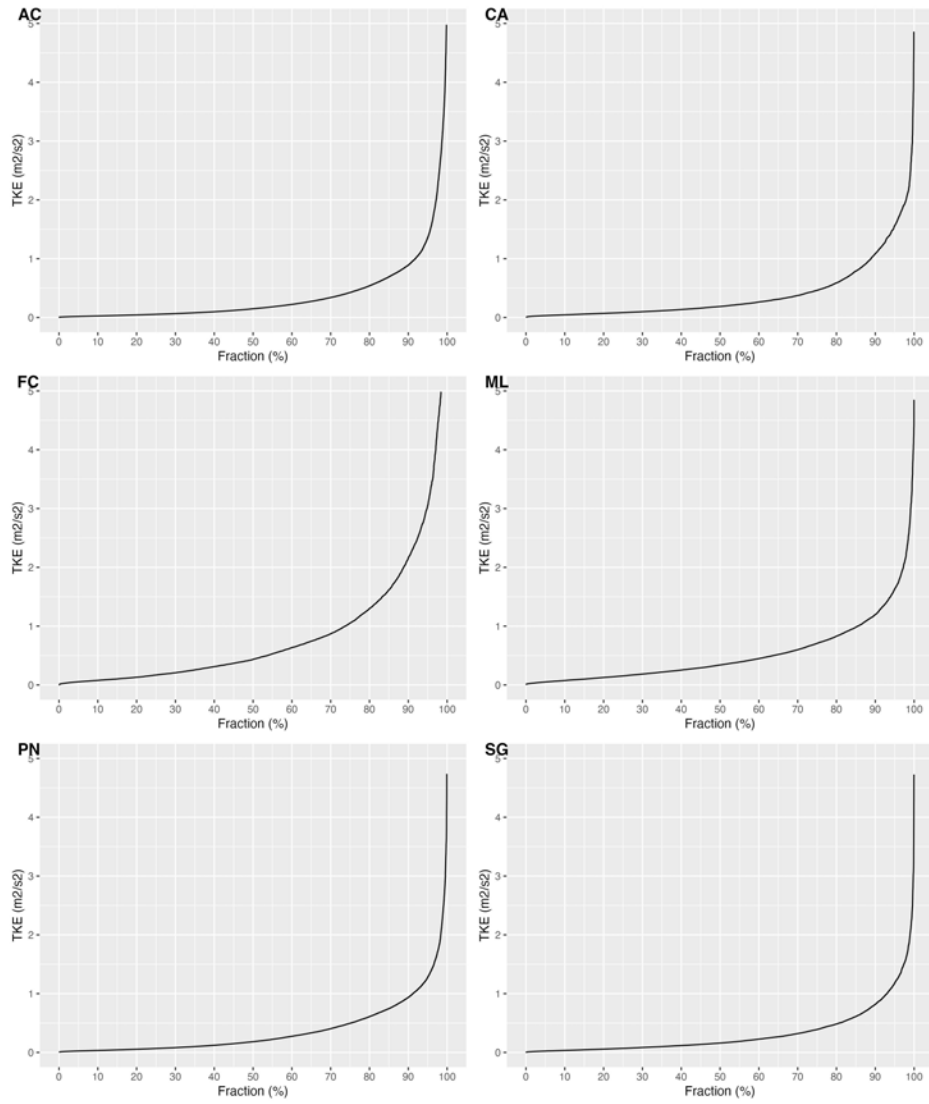


Figure 8.23.: Cumulate TKE frequency plots for the SHAKEUP stations

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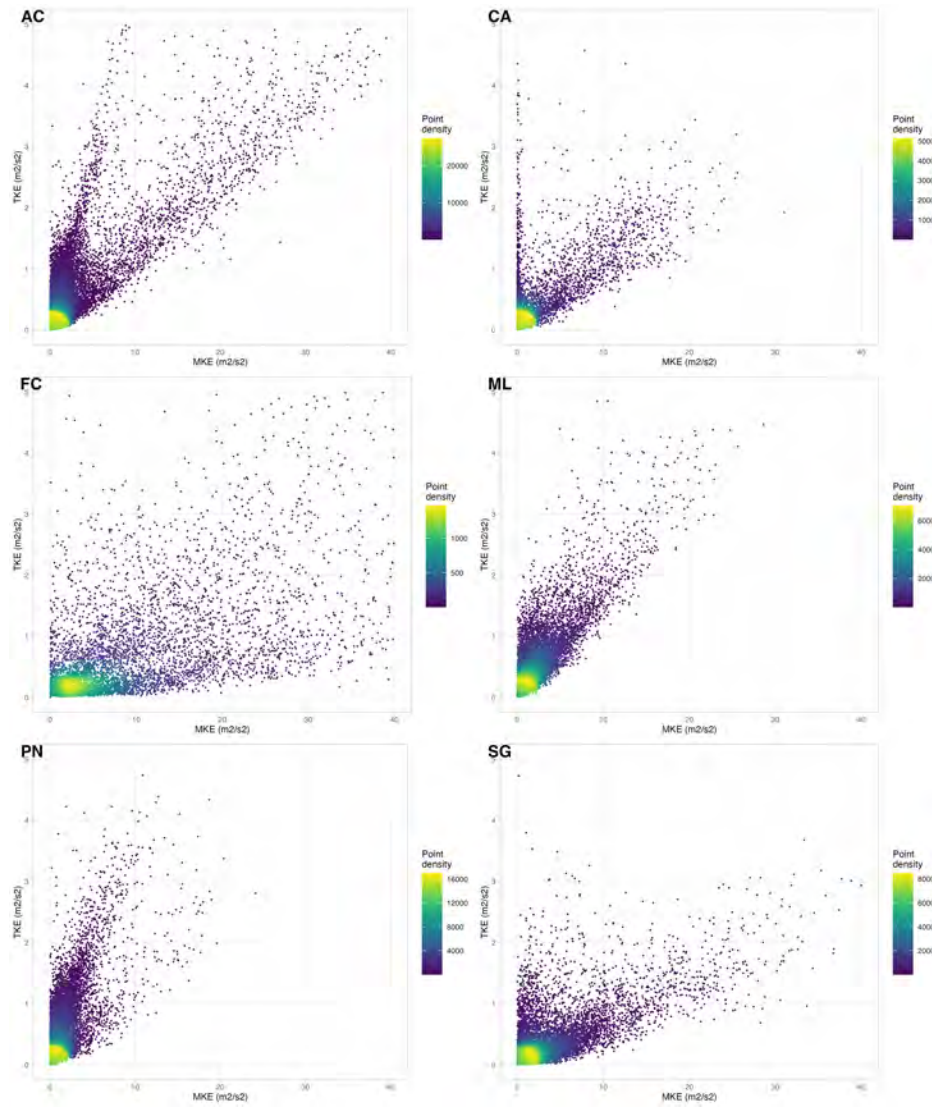


Figure 8.24.: Comparison plots of TKE and MKE at SHAKEUP sites

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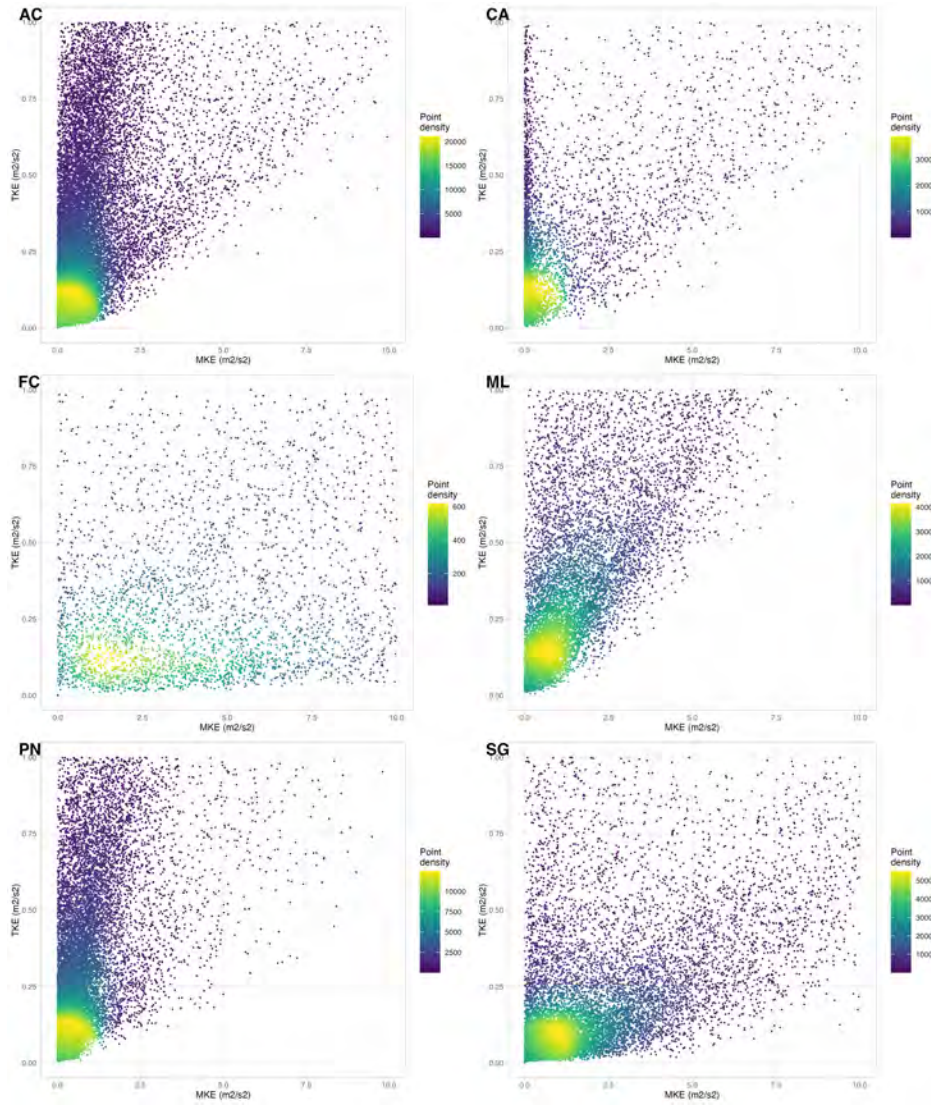


Figure 8.25.: MKE vs TKE, with values restricted to low MKE cases

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of low MKE cases, suggesting a relatively larger importance of convection-associated turbulence.

Last, as anticipated, the R_{KE} ratio is shown (@fig-r-ke).

@fig-r-ke shows various things, all relevant to site characterization. First, and most striking, is the different behavior of the two “pre-alpine” sites AC and PN, with huge changes in R_{KE} at very low MKE values, from all others, in which the bulk of data points appears more concentrated in the R_{KE} axis.

The second interesting thing is that at MKE (and wind speed) increases the values of R_{KE} tend to remain around a constant level, about 10%, in all cases where high wind speed was common enough to visually appreciate this phenomenon. At the two pre-alpine sites a second cluster about 30% is somewhat visible. But, more importantly, as the MKE (and wind speed) decreases this clustering gets less and less evident, until disappearing entirely at very low MKE values.

Overall, this behaviour suggests to not using the wind speed (and its associated MKE) as a proxy of R_{KE} , with very large or very small values being possible at low MKE values.

8.6.3. Friction velocity

The friction velocity, u_* (Stull 1988), is a proxy of “mechanical” turbulence, that is the part of turbulence directly induced by the friction of wind against or close to the surface irregularities it finds along its path to a measurement point, or in contact with air masses moving at different velocity (“wind shear”).

Because of this multiple origin, it is possible that friction velocity is non-zero also at a distance from the Planetary Boundary Layer.

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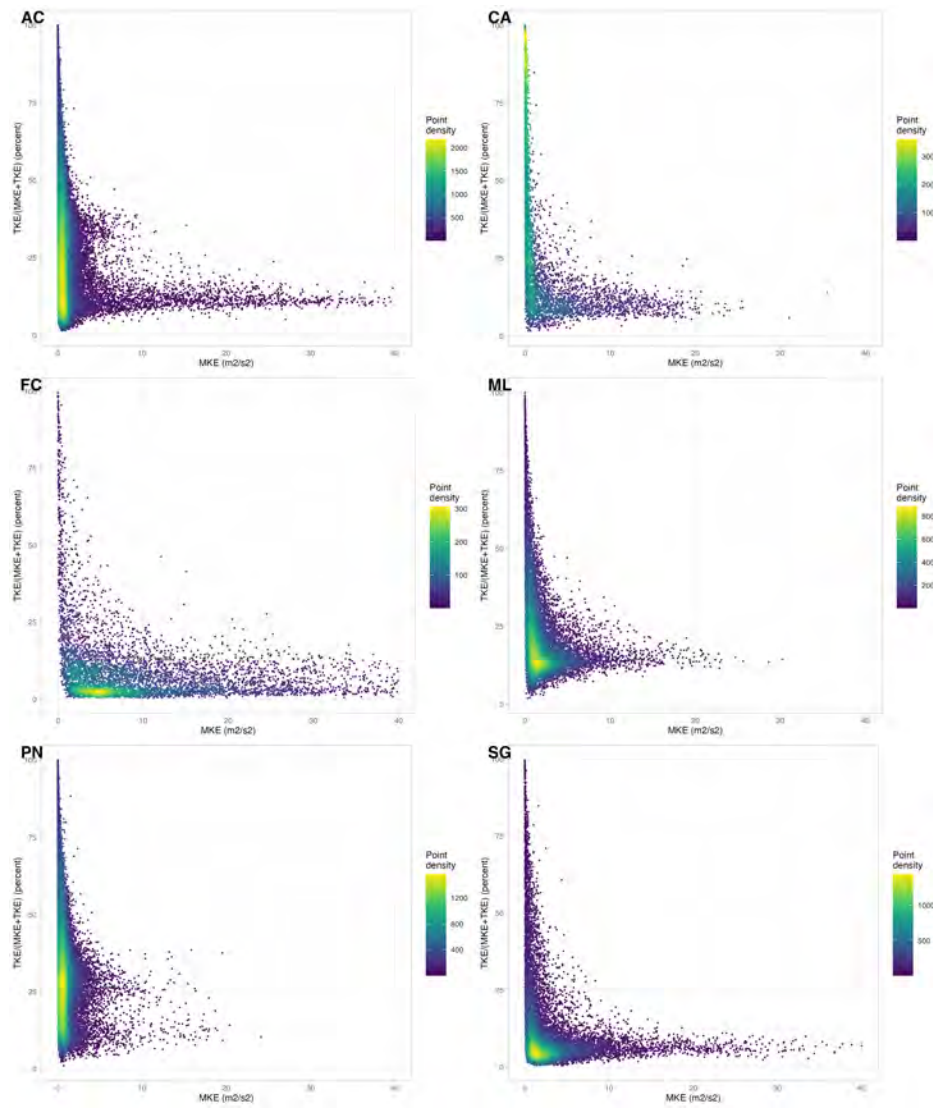


Figure 8.26.: Ratio of TKE to the sum of MKE and TKE at SHAKEUP sites

8. Site Characterization

It is not even excluded that the mechanical generation of turbulence is more effective above the measurement point than below (Monin and Yaglom 2007), in which case some of its numerical definitions assign it a negative value.

Close to the Earth surface, and beyond the plant canopy, the most common case is however that of friction velocity being generated below the anemometer: in this case, its magnitude is related to the *aerodynamic roughness length*, a parameter having the dimensions of a length and describing the effect the “roughness elements” (constructions, trees, orchards, forests, mountains, and whatever may change the path of wind by its sole presence) air encounters along its trajectory to the anemometer.

In general, the value of u_* depends on what happened along the entire wind trajectory

$$\mathbf{x}(t, \tau) = [x(t, \tau), y(t, \tau), z(t, \tau)]$$

where t represents the time backwards from the instant τ when the wind was measured. Of course, if $t = \tau$ then we have $\mathbf{x}(\tau, \tau) = \mathbf{x}_0$, the Eulerian frame position of the measurement point. As $t < \tau$, the position of $\mathbf{x}(t, \tau)$ will in general differ from $\mathbf{x}_0$, and this is quite an intuitive fact if we imagine the measurement time instant τ fixed.

But as we allow τ to change, we will have

$$\mathbf{x}(\tau_1 - \delta t, \tau_1) \neq \mathbf{x}(\tau_2 - \delta t, \tau_2)$$

whenever $\tau_1 \neq \tau_2$. The degree two trajectories “photographed” at instants τ_1 and τ_2 differ will depend on many factors, turbulent fluctuations included, and one is practically assured no easy prediction is likely, at least with the technology available in characterization studies.

In addition, the “friction” turbulence one could measure at a point may be affected by the transport of turbulence (caused or not by friction) from

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remote sites, a thing more likely to happen during stable hours, when “strange” phenomena not immediately related to friction alone occur (Zilitinkevich and Baklanov 2002).

All that given, it could be clear u_* contains information from many phenomena, and so it is likely to result in a complex statistical pattern.

The part of this pattern most relevant to characterization studies is its dependency on the provenance direction: for the reasons explained above, however, this dependency will have a complex form whose description is only possible statistically.

In addition: as mentioned above, the friction velocity u_* has the dimensions of a velocity, and so it is tempting to compare it with wind speed.

Many early studies conducted above flat lands suggest

$$u_* = \alpha U_s$$

with $\alpha > 0$ a proportionality constant (Stull 1988). In ideal site, it is known from practice that $u_*/U_s \approx 10\%$, but it is not uncommon to find values higher or smaller - the author has found measurement points slightly above the plant canopy where this ratio exceeded 60% (unpublished private data).

High values may also be common in urban and peri-urban sites (Oke et al. 2017), if measurements are taken nearby the roof canopy of within street canyons.

In Figure 8.27 we can see the situation for the SHAKEUP stations.

Plots in Figure 8.27 show the ratio is significantly larger than the 10% at the “pre-alpine” sites AC and PN, and at the deep Po valley ML, indicating obstructions may have a larger effect there - sites CA, FC and SG show a “cleaner” view from inbound wind directions.

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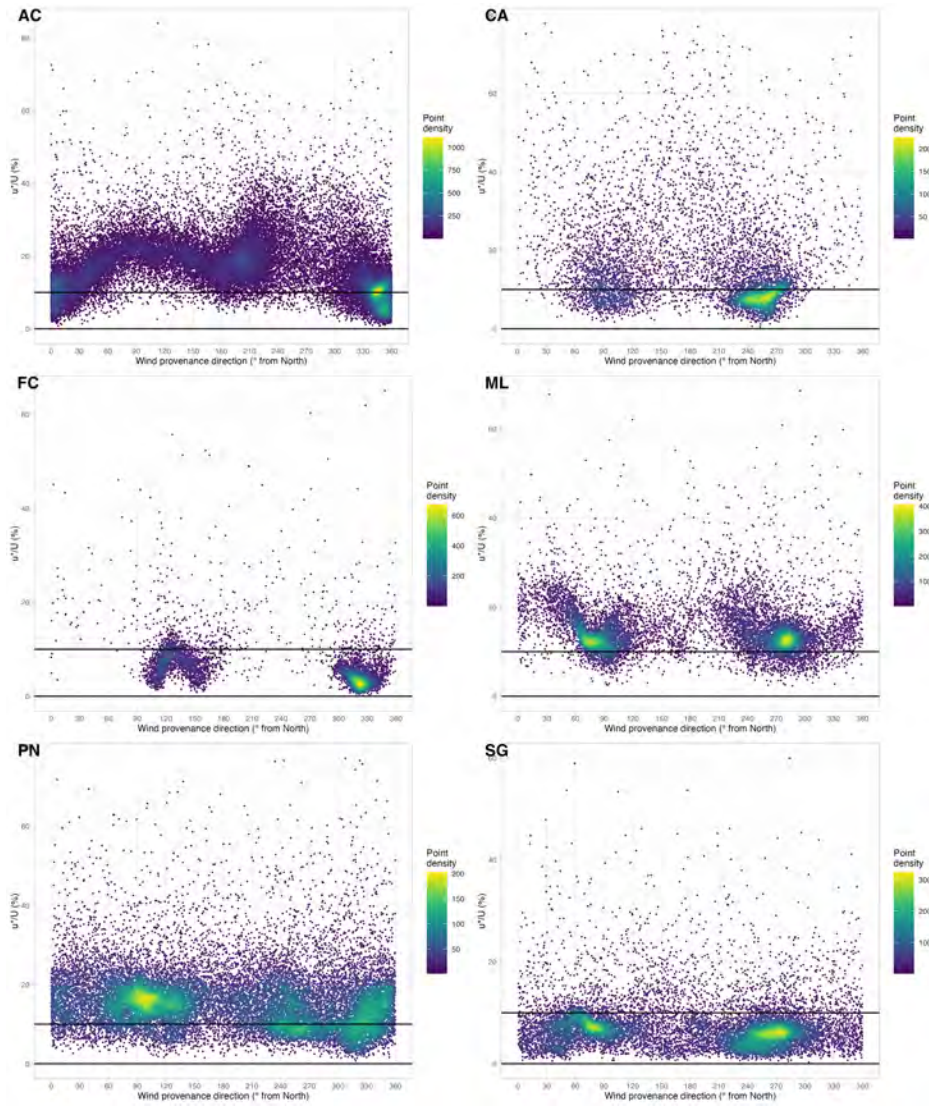


Figure 8.27.: Directional density scatter plot for the ratio of friction velocity to scalar speed at SHAKEUP sites

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The causes are quite local: at AC values of the ratio are higher when wind blows approximately from the sectors about NW-S-NE, characterized by an urban texture, while the narrow N sector sees an open field; the interesting side of this is the buildings in the urban sector are low (about 6m from ground at roof tops) respect to the 10m anemometer height above ground.

At PN, the anemometer is located at about the center of a circular artificial clearing having about 160m wide within a woodland (also artificial) about 6 meters high at measurements time.

At ML, the measurement site is located in a sport field surrounded by buildings taller than the station at N, and lower but close nearby at S. The view in other directions is less disturbed, as suggested by the smaller values.

The case of the high mountain site FC is peculiar: obstructions exist, namely the two slopes perpendicular to the valley axis, but they are so high to impede any significant flow along them. The only represented values correspond to the two wind directions made possible by topography, where the view very is clean.

A similar consideration applies to CA site, located on the flat surface of Valtellina, there wide enough, and with a West-East axis allowing anabatic and katabatic winds. This explains why data are not as directionally concentrated as at FC, but also whatever the wind direction is: the view is clean from all directions - which is in line with the site use, as the airport of Sondrio, mostly used by helicopter-ambulances.

Last, the deep Po Valley site SG shows an essentially clean view on all directions.

Overall, Figure 8.27 shows which directions a cup anemometer located at these sites are most mechanical turbulence prone, a valuable information when deciding which type of sensor to select at a site.

8. Site Characterization

It also has another implication: the cleaner a direction is, the less wind will be slowed down by obstructions. This is an intuitive fact, justified by the fact the generation of mechanical turbulence originates from a transformation of overall wind energy into TKE.

This helps to understand the apparent paradox of SG “higher” wind speeds, if compared to ML, placed similarly within the Po Valley. It also helps making sense of the effect buildings and other roughness elements, even apparently “too small” to have effects, still have demonstrable effects.

Just for comparison, in Figure 8.28 we can see the directional density scatter plots for friction velocity alone.

What’s interesting in this case is the plots, taken alone, give an entirely different impression than Figure 8.27. By the author’s own experience, using u_* alone in characterization studies often yield misleading results, and is positively not recommended.

Nor it is recommended to use estimates of the friction velocity obtained by meteorological processors starting from conventional (cup-vane or propeller) anemometers: the modelling assumptions made might give statistically sensible results, provided the geognostic information necessary is assigned correctly, but cannot in any way reproduce the complexities of reality.

A direct measurement of u_* is then preferable, and its critical evaluation by studying the aerodynamic roughness length spatial distribution is highly recommended; this is the subject of the later Section 8.7.

8.6.4. Vertical turbulent flux of sensible heat

The vertical turbulent flux of sensible heat, H_0 , determines whether the atmosphere in contact with the Earth surface is characterized by a stable, neutral or unstable static equilibrium (Arya 2001; Stull 1988).

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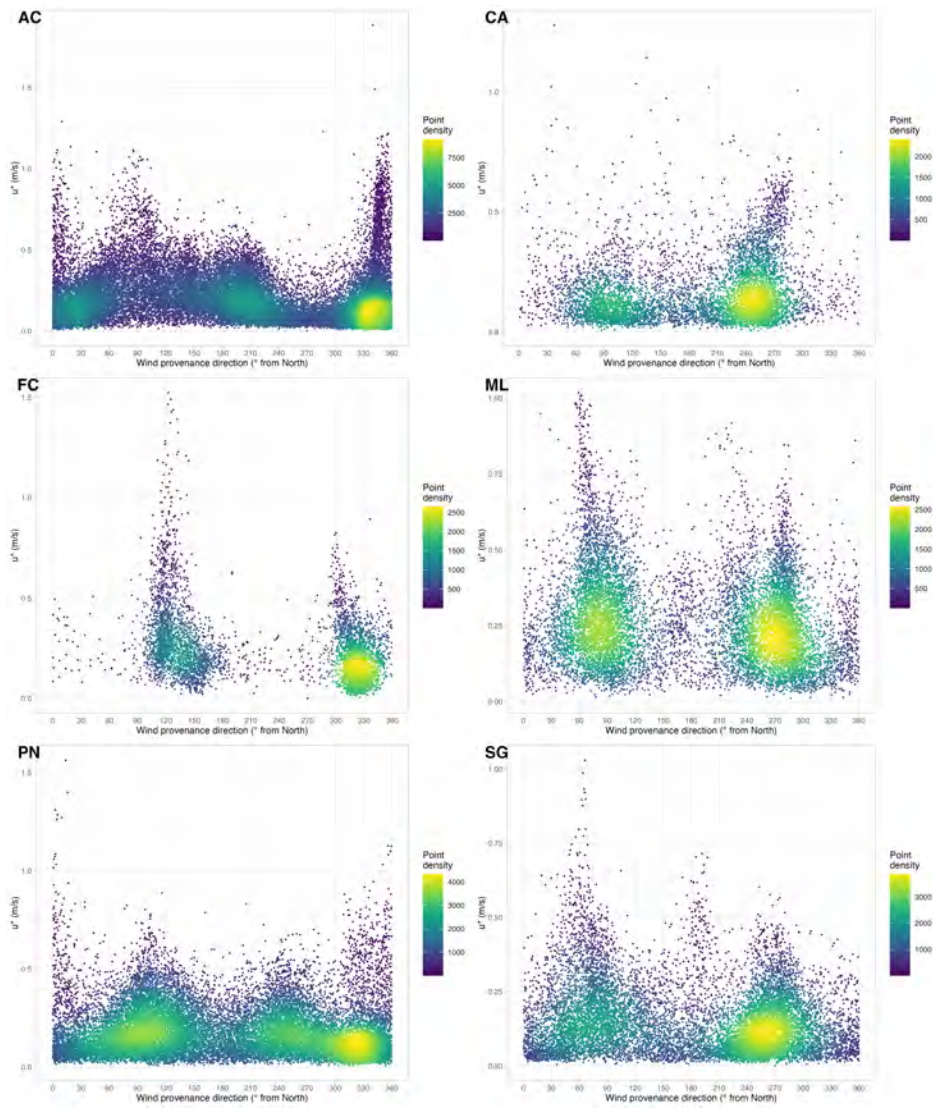


Figure 8.28.: Directional density scatter plot of friction velocity at SHAKEUP sites

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This is an important information in respect to cup anemometer field exposure to turbulence, because the atmosphere' stable or unstable static equilibrium is associated to different, specific circulation pattern at the micro-scale, with a specially important impact when wind is slow.

Broadly speaking, when H_0 is positive and "high" (say, larger than 15W/m^2) and the wind is slow (the actual limit may change with authors; (Oke 1987) proposes 4.5 m/s, larger than most speeds observed at the SHAKEUP sites) thermally-driven convection cells form in correspondence of any obstructions to wind. If wind speed is really slow, say 1.5 m or less, then a condition named "free convection" may develop and an intense turbulence, almost entirely due to thermal effects, develops.

When H_0 is negative and large, on the contrary, a stable stratification develops accompanied by a ground-based thermal inversion: this has the effect to inhibit the vertical development of turbulence, and as a consequence the "mechanical" effects, due to friction, will prevail.

When H_0 is close to zero, the equilibrium is neutral or almost neutral and again turbulence develops typically by mechanical only effects.

So H_0 , jointly with wind speed, controls the frequency of occurrence of higher (thermal or mechanical) or lower (mechanical) turbulence.

In addition to the effects turbulence has on cup anemometer readings, another very harmful effect presents when high H_0 is associated to low U_s . Turbulence, and especially under free convection, may be thought as the overlap of vortices at scales from the fully developed PBL thickness (in the order of 1km at Lombardy latitudes) down to 1 cm and less. The production of vortices tends to occur at the upper end on this length scale, but as vortices interact and coalesce they also split in smaller vortices. The process continues until all the energy contained in the larger vortices transforms to a minuscule amount of heat when the vortex sizes become so small that viscosity effects become important.

8. Site Characterization

This process has even been described by a little poem, attributed to the famous meteorologist Lewis Fry Richardson in 1922 (Nature-editorial 2016):

Big whorls have little whorls

Which feed on their velocity,

And little whorls have lesser whorls

And so on to viscosity.

The consequence of this is that if we consider a small volume (comparable to the sphere having the same diameter of a cup anemometer), then very seldom will it happen the speed will be zero.

Wind direction may be completely indeterminate in these cases, because of second-order induced resonances and other causes (Sozzi, Georgiadis, and Valentini 2002), but the instantaneous wind speed may exceed at some time the activation threshold, and be then counted contributing to a non-zero measured speed even in presence of air which, would not because of the endless turbulent agitation, would be stand still.

This overestimate, whose characteristics are still an open problem by the author knowledge, may give in extreme cases the misleading impression of a more favorable situation than really is when the non-turbulent U_s is very small.

The problem is then to understand, during characterization, how frequent the association of high H_0 and low U_s is. This can be seen in Figure 8.29.

@fig-h0-u implies that at some sites (namely AC and PN) the fraction of data possibly affected by free convection cannot be neglected. The following table presents the actual counts, expressed in relative units.

8. Site Characterization

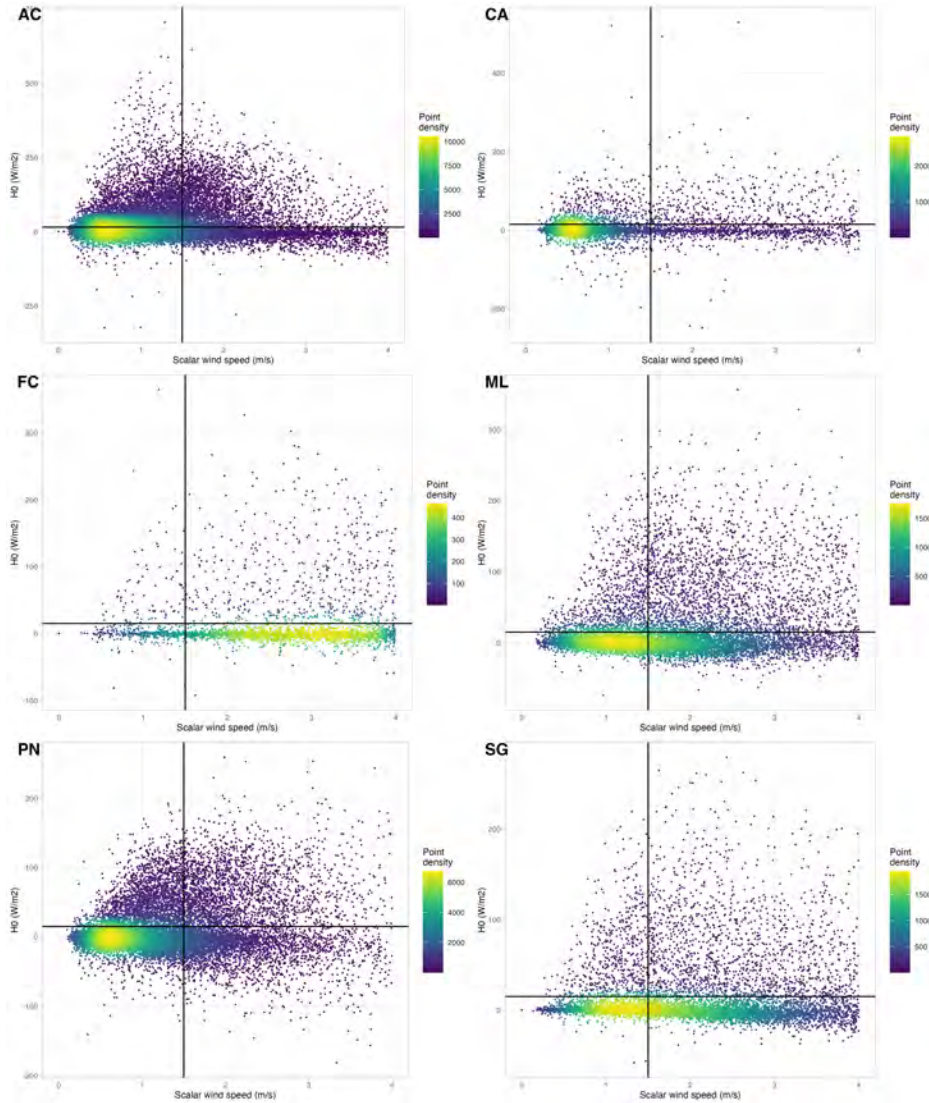


Figure 8.29.: Density scatter plot of H0 versus wind speed; the top-left rectangle identifies situations likely affected by free convection; speed data limits artificially limited to slow values in professor Oke's sense

8. Site Characterization

Table 8.2.: Fraction of data potentially under free convection at SHAKEUP sites

Site	Fraction of data potentially under free-convection conditions
AC	17.42
CA	8.81
FC	1.46
ML	9.04
PN	11.55
SG	4.96

Whether the amount of data potentially under free convection is a problem or not depends on the application.

8.7. Empirical data supporting site characterization

8.7.1. Introduction

The characterization elements described in the preceding sections are based on data, collected using sensors of higher sensitivity and resolution than cup anemometers.

Data have the great advantage of being easy to communicate, and in our culture are attributed an objectivity and authoritativeness qualitative observations do not rival.

Taken alone, however, they may lead to dangerous misunderstandings.

In the author's experience, a careful set of field activities may provide the context information necessary to critically evaluate the numerical data and, to some extent, help in their validation.

8. *Site Characterization*

The technology available to date allows to execute these supporting activities in fast and easy way, and in the following sub-sections some glances to the possibilities available are given.

Unlike the data-oriented descriptions used in the preceding section of this chapter, in this section and sub-section the level of presentation will be more theoretical and schematic. This is due to the fact that field operations, of whatever nature, may involve the collection of information pertaining to people and their properties, whose disclosure may infringe privacy laws and regulations.

The description, however, will be detailed enough to allow you, the reader, to conduce these activities on yourself - and distribute the results with the proper level of confidentiality.

8.7.2. Satellite imagery

To date, satellite imagery is readily and easily accessible through well-known sources.

In many cases the publicly available data may lag behind the current time, with latest updates being in some cases years old.

In addition, use of imagery may not be allowed by their providers' licensing policies without a written permission: it is the user's responsibility to check which exact conditions do apply, and to take the appropriate measures. The grant of permission is not automatic, and may be conceded only after the payment of a fee.

This is especially true for characterization studies, which despite the deep scientific contents qualify as professional works, rather than research activities. Rules are changing fast however, and the author's advice to readers is to check themselves.

8. *Site Characterization*

All this given, it is the author's opinion that satellite imagery should be used with extreme caution for characterization studies. The kind of information they most likely can reliably give is representing the land cover, but the level of detail is in most cases not sufficient to get useful indications.

In addition, one should take into account the fact that commonly available satellite imagery has been almost always more or less heavily edited, for aesthetic and legal purposes. One can easily check this by accessing the most important imagery distribution site and find that people is routinely removed from images. This editing may make the imagery unrealistic, and not up to the standards required by characterization studies.

8.7.3. Technical and cadastral maps

In urban areas, technical maps (typically at scale 1:10000 in Italy) and cadastral extracts (at scales ranging possibly from 1:1000 to 1:200) may provide useful supporting information.

In Italy, technical (and many other) maps can be downloaded for free from institutional sites - for instance (Lombardia, n.d.) for Lombardy.

8.7.4. Drone flights

8.7.4.1. Introduction

More detailed imagery can be gathered through drone flights. Use of drones in support to micro-meteorology has been already investigated, and a downloadable poster was presented on this subject by the author at 2022 AISAM national congress (Favaron n.d.b).

Use of drone allows collecting information similar to satellite imagery, but up to date and at much higher resolution.

These advantages are balanced in part by some factors:

8. Site Characterization

- Weather conditions can impair the ability to collect meaningful data (and in case of hail, snow, rain, drizzle or heavy fog use of drones is positively not recommended: the electrical motor might fail, even temporarily, and the drone might crash from altitude causing harm to people or property; besides, imagery quality may be scant if weather conditions are not clear).
- Local regulation may not allow flight missions at some places (for example in urban areas, over nature conservation sites, or close to airports, hospitals and other facilities); in some other cases flight missions may be allowed, but altitude restrictions apply: in Italy, for example, “free” flight is allowed up to 110m above ground where “no restrictions” apply, but lower altitudes are often prescribed; this is specially concerning in high population density areas like Milan and its outskirts (on the Internet it is easy to find footage of people sending drones at altitudes of 500m and above: this is technically possible, but is an *extremely dangerous* practice, and a show of criminal irresponsibility).
- Temporary restrictions may come active from time to time.

In Italy, special permissions may be issued under motivated requests, but they are given only if the caller has a drone license and after significant bureaucratic work. By the way, the authorization may be denied - this is more than reasonable: even sub-250g drones are aircraft heavier than air with hard and sharp propeller blades moving at high speed, which can cause serious damage in case of fall or impact.

In addition, some old-good-sense rules are always mandatory, and prescribed by regulations. For example, it is always forbidden to fly at less than 50m by a people gathering (in case of system failure, escape could be impossible even if the awry drone is located in time). And of course, photographing private property may not be allowed for privacy reasons.

Last point: even a sub-249g drone with a camera (a type more than sufficient for collecting imagery in support of characterization studies) is legally

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not considered a “toy drone”. It can be flown without license, in Italy, but the operator must register as a drone pilot and an insurance activated. Please take your information before deciding to use drone missions. Or, of course, hire specialized companies.

What precedes suggests drone missions must be planned carefully. The author recommends a two-phase method:

1. Feasibility and preliminary planning (for example when formulating budget estimates): check regulations and maps, and verify a drone mission is actually allowed at site; if it is, record the maximum allowed flight altitude above ground, and determine how large on terrain your images will be given your drone camera angle.
2. Pre-flight final check (on the mission day): check NOTAM message and verify the area where the mission is to take place is still accessible (as explained, it might not be).

Once on site, before take-off and landing ensure no other people but you is on site. Perform take-off and landing on a vertical path, then, once reached the desired altitude, ensure the horizontal movement is the least possible; during the flight mission always be sure you can actually see the air-bound drone.

Almost all lightweight photographic-movie drones on the market allow “first-person view” operation, which is useful for precise alignment, but regulations mandate you maintain visual contact with the drone, and the author suggestion is to not rely on it exclusively. This apparently restrictive regulation is more than sensible: the view solid angle allowed by the camera is just a tiny fraction of the whole sphere, and the situation awareness FPV gives you is, by today standards, not even comparable to that allowed from the cockpit of an F35 (nor from the windows on a passenger flight, for what matters). This is important: even if you maintain a route ahead (drones can move laterally and backwards too) you cannot see whether, for example, a bird was attracted by your equipment and is

8. Site Characterization

mobbing in an potentially deadly attempt to divert it from their nest with little birds inside waiting their mom and dad.

8.7.4.2. Fast evaluation of aerodynamic roughness length

A drone mission provides invaluable data for estimating the value of z_0 around the wind measurement point.

Reference values for z_0 can be found in literature (Oke 1987), where tables relate aerodynamic roughness length values with roughness elements.

The position (and geo-referencing) of these roughness elements can be obtained by locating features in photo shot with camera down-pointing also visible in satellite imagery or technical maps and use them to determine approximate terrain coordinates. Once done, roughness elements can be located, their z_0 value estimated by table inspection, and distance-azimuth from station point estimated.

Tabular assignment of z_0 is not a high-accuracy method, and use of advanced geospatial processing to obtain precision orthomosaics is then not strictly needed: results will be in any case indicative, although of high value.

An example of z_0 tabular estimation can be found in the poster (Favaron n.d.b).

Other estimation methods, of geometrical nature, can be applied to the case of constructed areas (provided flight missions are allowed there), and accounts may be find in literature (Borghi, Favaron, and Frustaci 2011; Favaron et al. 2012).

8.7.4.3. Documenting line view obstructions

In Section 8.8 techniques are shown for determining the angle obstructions have over the horizon when seen from the anemometer level.

8. Site Characterization

These are numerically based, however, and often do not give prospective users and decision makers a clear indication of how obstructions would appear to an observer placed at anemometer height.

If drone flight are possible at measurement site, then photos can be taken from anemometer height with the camera leveled to horizontal, as we can see in Figure 8.30.



Figure 8.30.: Example photo of obstructions at anemometer level (taken at the SHAKEUP site of Vertemate con Minoprio, not part of this book's data set)

Images like Figure 8.30 are usually taken in correspondence of the cardinal sectors, and yield a quick, although imprecise, impression of which directions are potentially or actually obstructed.

As mentioned in other sections of this chapter obstructions may be both artificial and natural, and in the latter case their size may be enough to channel the airflow along a limited set of allowed directions.

8. Site Characterization

Large-scale obstructions, and in particular mountains, are typically so high that measuring their angle from the horizon would yield almost identical results if the reading is taken at eye- or station-level. Yet, station-level photos are immediate and highly suggestive, as we can see in Figure 8.31.



Figure 8.31.: Station-level view of Valsassina from Prato Buscante (Barzio); the camera is pointed approximately towards the Valsassina axis, northwards

8.7.4.4. Diagnostic images

It happens from time to time that data collected at some measurement sites show something unusual, and investigations are made in these cases to understand what the cause is.

In some cases anomalous measurements are the result of sensor or data logger malfunction, but there are situations in which the site itself “did fail”,

8. Site Characterization

and in such cases a drone flight may help providing useful troubleshooting information.

Figure 8.32 shows one of these cases: an anemometer (in this case ultrasonic) recorded growing u_*/U_s values over its operational time, and the author was eventually asked to do a maintenance tour for diagnosing the problem.



Figure 8.32.: Example of site failure: the plant canopy engulfed by the anemometer height

The drone flight revealed the anemometer performed its “wind measurements” inside the plant canopy.

8.8. Site survey

8.8.1. Introduction

The main purpose of site survey in site characterization studies is to collect information which, once properly evaluated, allows to anticipate or confirm potential or actual data quality issues, and other problems whose solution is essential to permit the station to meet its application requirements.

Characterization site survey is essentially a data collection, organization and reporting activity. Site survey has *not* the purpose to formulate a judgement about a site adequacy to some application, but rather to collect the objective evidence allowing application experts to take their decisions on a rational basis.

The deliverable of site survey is the *obstructions map*, with its accompanying notes. In corporate settings it may be packaged as a technical report or other official document, but this is a detail outside of this book scope.

8.8.2. Types of obstructions maps

The obstructions map can be drafted in many ways, all conveying the same basic information: the angle from the anemometer to obstructions, measured with reference to the horizontal plane.

In Figure 8.33 and Figure 8.34 we can see two basic examples.

Of the two forms, polar and Cartesian, the second is easier to prepare and to interpret. In the polar obstructions plot we see a hypothetical case in which the anemometer is located into a pit, and all obstructions have a positive angle to the horizon. In the Cartesian, almost all direction see the free horizon, at an angle which is close to zero (it is not exactly because of the Earth curvature).

8. Site Characterization

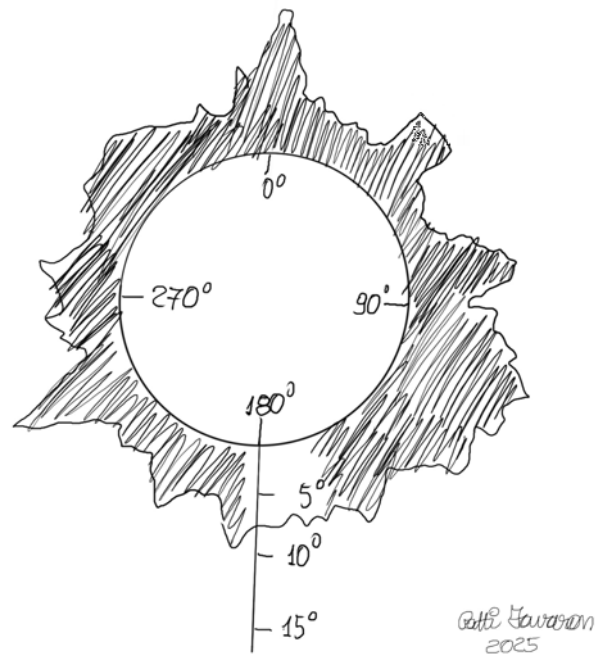


Figure 8.33.: A fictive obstruction plot in polar form

8. Site Characterization

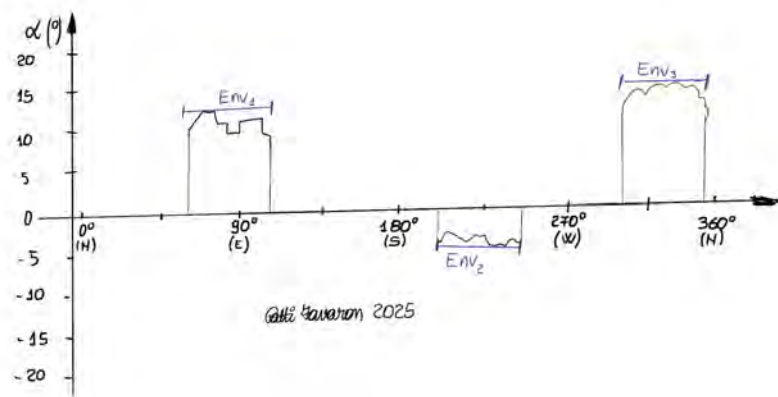


Figure 8.34.: A fictive obstructions plot in Cartesian form

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But on some directions we can nevertheless see obstructions characterized by a negative angle, that is, below the horizon when measured from the anemometer level.

As we have seen in this chapter even obstruction below the horizon can distort the airflow enough for effects manifesting at the measurement level.

8.8.3. The WMO CIMO no.8 report

The famous WMO CIMO no.8 report contains a chapter (WMO 2023a) in which requirements are given for measuring meteorological variables.

One of its sections deal with anemometers, and indicates a measurement height above ground of 10m or more. The same section also gives anemometer siting requirements, one of which stating obstructions being placed at a distance no less than ten, and possibly fifteen, times their height above ground.

The CIMO Report no.8 indications are often assumed as requirements for institutional meteorological stations, but can seldom be fulfilled in amateur-level stations due to space, safety and budget constraints.

Following the CIMO Report no.8 indications allows a considerable degree of comparability among stations located at some distance, and contributes to improve the value of data in view of synoptic meteorology applications.

Focus, in synoptic meteorology, is on assimilating data from diverse sources, surface-bound meteorological stations being just one of them (Li and Kalnay 2010; Kalnay 2003).

Requirements for general meteorology may, by the way, not be comparable with those of other applications. For example, air quality and pollution risk studies place focus on long term statistics and slow wind, while general meteorology is more interested to “medium to high” wind speeds, able to

8. *Site Characterization*

transport significant amounts of mass and energy among continents and oceans.

As characterization is application-agnostic, it given no neglect to cases “within requests”: all obstructions are potentially interesting whatever their height above ground.

Because of this, in this book no effort was made to enforce or retain compatibility with “WMO indications”.

8.8.4. Safety considerations

The vast area of safety in field activities is the subject of extensive literature and regulations, and this is not the place where to make a detailed review.

The author assumes you, the reader, have or may get the relevant information in your specific sector, and have a basic understanding of field activities and their most commonly associated risks. In case you don’t, you can find the minimum information in primers like (Lisle, Brabham, and Barnes 2011) and similar books.

In this section you can find information the author, as an applied mathematician often involved in field activities, has gained from direct experience or observation of colleagues, and considers worth sharing.

The first and foremost thing worth saying is, any field activity would be better imagined using a military mind-set rather than a tourist attitude.

Sure, in most cases one will not find in the middle of a war zone, and there typically is no enemy fire targeting you. But:

- You are there for a very specific reason, with a defined set of mission objectives which must be reached for the mission to be successful.
- In many cases site surveys occur in natural landscape, in contact with wild animals and plants.

8. *Site Characterization*

- Also in many cases site survey occurs in inhabited and constructed areas, in contact with people and domestic animals.
- Any site looking remote to your eyes is after all not that remote, if you were able to reach it: it is safe to expect someone else is in the surrounding, and possibly in visual contact with you. This may be good in most cases, if these people can provide assistance in case of incident. But they might be hostile as well.
- Karst and other potentially dangerous terrain features may be present, which are not easily detectable in the maps and imagery used in planning the survey.
- Whatever your plan, it is safe you expect some level of “friction”: things may quickly deviate from the conditions you considered during planning, and in case you will have to improvise.

The most general suggestion is to always plan for teamwork, with two or more people operating in cooperation and, in case of emergency, provide immediate assistance.

In addition:

- It makes sense to define during planning a set of minimum objectives, and concentrate on achieving them first of all.
- The faster the overall execution, the least the exposure to site specific risks.
- Transfer to and from survey site should not be overlooked, and should be planned as carefully as possible.
- It is generally a very good idea to act in kind and respective way, avoiding unnecessary displays of bravado or aggressivity.
- Be prepared to explain what you are doing and why. Planning would better include the level of information to disclose to field personnel so that they can, in case of necessity, provide the context information.

8. *Site Characterization*

Usually installing meteorological stations does not cause great concern, but if the station still does not exist people may imagine you are oil or mining prospectors or some other kind of people residents may not like being there.

Last, it is wise to consider people differences when planning a survey tour. Some are weaker, some other stronger, some are more resistant, some are less, some need a lot of water, some other can survive one day of intense field work or mountaineering with just a tea cup as lunch. It is also wise to take into account these characteristics may not match commonsense stereotypes (for instance, the record-mission with only one teacup as lunch belong to me and a female friend during a long excursion to Grigna).

8.8.5. What to look for

In order to draw the obstruction map, the interesting subjects are “obstruction envelopes”, that is the topmost point the tallest obstruction has at any given direction.

If we look at a panoramic photo taken from a generic point, like we can see in Figure 8.35, we have to admit the actual obstructions envelope may be extremely complex, to the point of being almost intractable by means of a lightweight operation.

Such a level of detail is however much more than needed, and a coarser resolution would be just enough.

On the other side, just indicating where the envelope line is does not provide all the information necessary: the angle value seen by the anemometer is also required, and estimating it visually from imagery may be a difficult and error prone task.

Before to show the details related to measuring the actual obstructions view angles, however, it makes sense to establish some rules for reducing

8. Site Characterization



Figure 8.35.: Panoramic photo of a random urban location, the horizontal yellow line at about the center of photo indicates the level from which the image was taken, while the yellow discontinuous line above the roofs and plants indicates the obstruction envelope

the obstructions map resolution to a manageable level. We begin, with indicating what angle should we measure in ambiguous cases.

Figure 8.36 shows what the “envelope” is, in presence of more obstructions.

Instead of following the obstructions profiles exactly, it is better to measure a representative altitude (the tallest). But, which angular sector should it cover? It of course depends on the obstruction overall width, but this may be ambiguous in many cases, as we can see on the left image in Figure 8.37.

The right image in the same Figure 8.37 gives another suggestion: consider the angle of the tallest point, and not of some visually representative level.

In case of plants or forests, it is better to act the same, and consider the top-most point of the canopy as angle measurement point. But this, if the canopy is visible. It may indeed happen one finds the prospective station installation site laying within a woodland, with no clear view angle to the plant canopy. If this is the case, then the author suggestion is to choose

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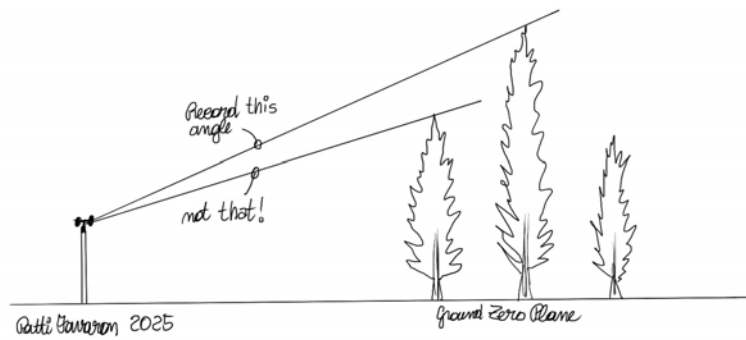


Figure 8.36.: The envelope angle refers to the tallest obstruction visible from anemometer level along a given direction

8. Site Characterization

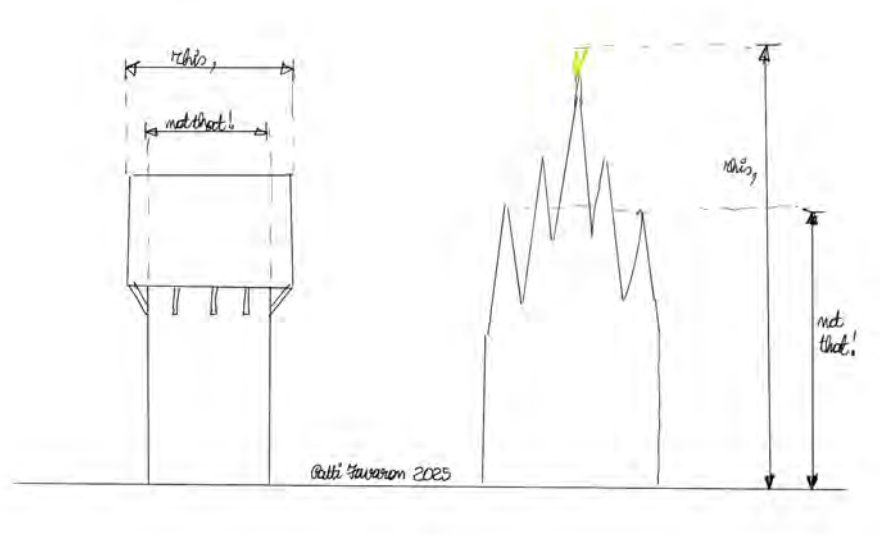


Figure 8.37.: Suggestions for resolving ambiguous cases related to buildings in obstruction maps

8. Site Characterization

another location: one could succeed in installing the station (at the cost of killing many plants), but:

- The station “altitude above ground” should be expressed in reality as “altitude above local plant canopy”, which usually is many meters less (possibly obliging to increase the mast length with all the safety and cost concerns of the case).
- Plants regrow very fast at temperate latitudes, and tend to occupy any previously cleared area within few years; canopy growth may be as fast as 2 meters per year when plants are still young and striving to reach more illuminated heights; they then tend to grow at a slower rate once they have reached the canopy level, but never stop rising. The most likely consequence is a long list of repeated cuts, with the cost, safety and ethical consequences one can imagine.

Is it necessary to include all visible obstructions in an obstructions map? It depends: in principle, object transparent to wind can not be measured. Few objects on field are transparent to wind however, and Figure 8.38 shows one frequent exception: power lines.

Before to begin the data collection, it may be useful to draw a preliminary map in which the relevant terrain features are shown, and their azimuth ranges determined using a compass indicated. For each range it is convenient to define a symbol identifying it, with information allowing a simple obstruction classification. An example is shown in Figure 8.39.

It cannot be overlooked that plots like Figure 8.39, which may be physically drawn on a copy of a technical topographic map prepared for the occasion, can only be filled with *field* information: neither satellite imagery, nor topographic maps, are updated enough to allow producing reliable plots by paperwork only.

8. Site Characterization

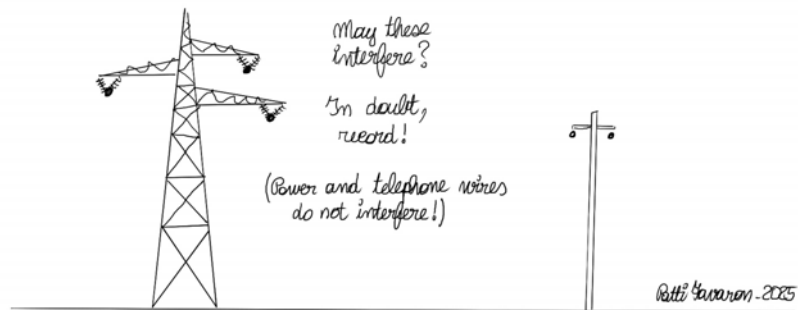


Figure 8.38.: Power masts and high voltage pylons usually interfere with wind, but power wires almost do not

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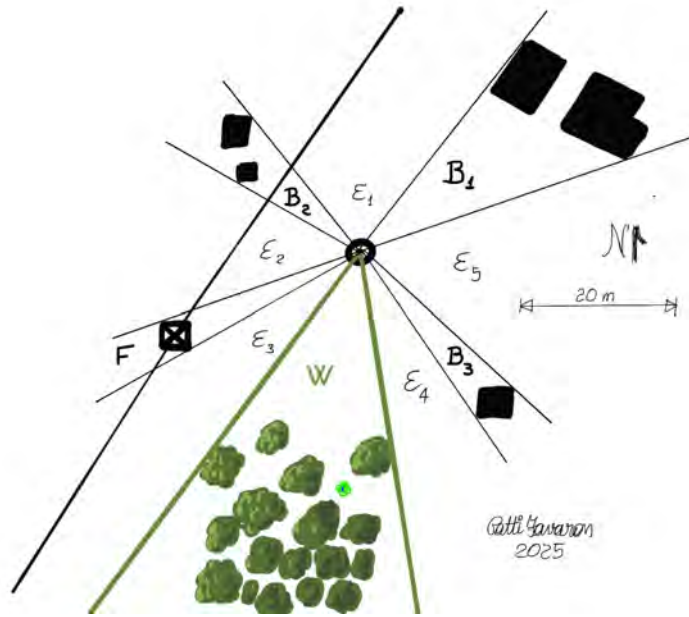


Figure 8.39.: Azimuth classification of the obstructions to measure (B indicates buildings, E the clear view sectors, W the woodlands, F the power pylons, and so on; this classification was invented by the author, feel free to choose another if you like)

8. Site Characterization

8.8.6. The case of mountain sites

In Figure 8.40 we can see a photo of a portion of quite a narrow alpine valley.



Figure 8.40.: View of Valsassina from Prato Buscante, at anemometer level; the photo faces North, towards the narrow gorge of Chiuso

Figure 8.40 shows a common and important case: a site characterized by a complex topography nearby, but placed on the flat bottom of a valley (the area now occupied by Prato Buscante was, until quite a recent past, occupied by a natural lake progressively buried by clay and silt sediments).

Obstructions are visible along the flat valley bottom, but their height above the ground is minimal respect to the mountains.

In cases like these it may be better to draw two obstructions map, one showing the mountain profiles, and another the smaller scale obstructions.

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8.8.7. Foliage

In case one sector is occupied by trees or a woodland, it is important to record whether the vegetation is deciduous or evergreen.

In the former case it is not infrequent the cup anemometer will sense interference from trees which are different depending whether foliage is developed or not.

Evergreen forests may seem less than a problem in this regard, but the kind of evergreen forests most common in Lombardy (coniferous) shows a needle density dynamics, which cannot be entirely neglected (Margolis et al. 1995). The reader is then encouraged to take site specific information, in case of necessity.

8.8.8. Expressing obstruction angles from anemometer viewpoint

Obstructions affect airflow and its measurements depending on the angle they form from the horizontal plane through the anemometer, which in general differs from the same angle measured at eye level. Figure 8.41 shows the point.

In Figure 8.41 we see a common situation. Angle measurements are taken at eye level, h_0 , resulting in the value α_0 . The same angle measured at anemometer height, h_1 , would yield the value α_1 , in this case smaller, but with same sign.

In Figure 8.41 the observer and the anemometer mast base seem to be at a different distance from the obstruction, but this is not unavoidable, as the observer can move making these two distanced identical, and equal to d .

The height of the obstruction, H , can then be expressed in two ways as

8. Site Characterization

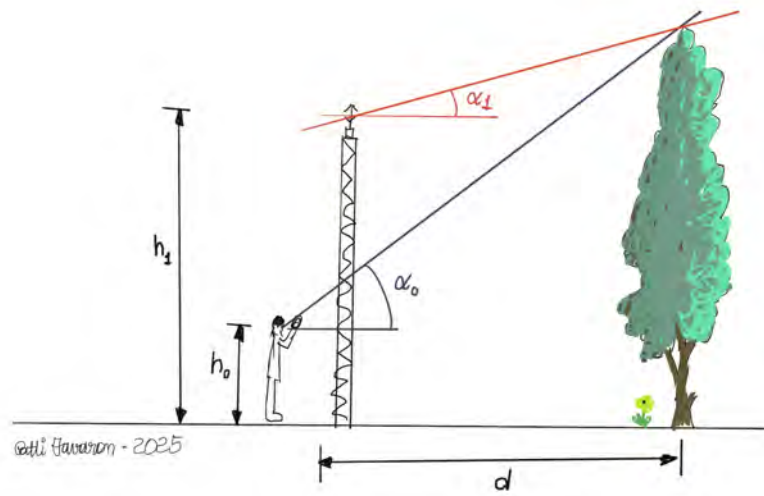


Figure 8.41.: Obstruction angles from different view points

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$$H = h_0 + d \tan \alpha_0$$

and

$$H = h_1 + d \tan \alpha_1$$

By equating the two and solving for α_1 we get

$$\alpha_1 = \arctan \left(\frac{h_0 - h_1}{d} + \tan \alpha_0 \right) \quad (8.1)$$

Note the sign of $h_0 - h_1$ is negative in typical situations. If the distance d is small and the eye-level angle α_0 is positive, Equation 8.1 allows for a negative α_1 . This means the anemometer is located at a height larger than the obstruction's, which may well happen.

As mentioned before in this section also obstructions with negative angles may have an effect on wind speed measurements and the generation of turbulence arriving at the cup anemometer.

Equation 8.1 also implies the angle seen at anemometer level tends to the eye-level angle as the distance of observation, d , tends to infinity. It is indeed possible to calculate a critical distance at which $|\alpha_1 - \alpha_0| < 10\% \cdot |\alpha_0|$. The critical distance is shown in for typical values $h_0 = 1.7\text{m}$ and $h_1 = 10\text{m}$.

8.8.9. Use of panoramic site photographs

8.8.10. Use of smartphones and tablets

8.9. Choice of campaign sensors

Past literature suggests the possibility of using an ultrasonic anemometer as reference sensor in characterization studies, and this is the path followed by the author for this book.

Technology however today allows other choices, and the final decision is up to the study designers.

For example, three-axial hot wire anemometers could be used. Hot wire anemometers have quite a noteworthy size advantage over sonics, and allow pushing the sampling rate to the kHz, very much inside the “K41” turbulence (Benzi and Frisch 2010), of course provided a fast and capable enough data acquisition system is available; hot wire anemometers are however delicate, and tend to fail often in case the atmosphere is “dirty” enough.

Two-dimensional ultrasonic anemometers might also be used, and some of them are now really inexpensive. Caution might be taken however, and excessive optimism is best avoided: mostly two-axial ultrasonic anemometers are designed as replacements to cup anemometers, and although less sensitive to wind calms they may have been calibrated so that their response resembles that of conventional anemometers. Among two-axial ultrasonic anemometers some models exist which allow very precise and high resolution measurements, suitable to get a good grasp of the *horizontal* wind. But the cost of these advanced models is much larger than ordinary two-axial sonic anemometers, and a little larger investment would allow then to get a *three*-axial model, with an opportunity to get detailed information about the important vertical component.

8. *Site Characterization*

Characterization data will be collected in raw form, and processed as such: use of a sophisticated data processing equipment like an eddy covariance system is then not essential. Take in any case into account that a hefty data acquisition system will be anyway needed: ultrasonic (or hot wire) anemometers can produce megabytes of data per hour, at rates orders of magnitude beyond the capabilities of standard data loggers.

Last, and of course, the UA should be placed as close as possible to where the CA will be or is, and at the same height above ground. Large distances (even 5m) could easily make the UA and CA data sets not comparable.

8.10. Campaign articulation and duration

As we have seen in previous chapters the wind statistics tends to change not extensively from year to year, provided the immediate neighborhood of the station are not subject to large changes in building position and size, orchard development and other obstructions modifications.

However, sub-yearly changes in mean wind speed occur, at some site particularly important. This may depend on many site-dependent causes, as for example the development of foliage in deciduous trees.

If performing a full-year campaign may be impractical, then the problem arises on when in the year to perform it, and how long.

Given the maximum seasonal variation tends to occur between about Winter and Summer, it is reasonable to articulate characterization campaigns in two phases:

- One to be programmed in Winter or late Autumn.
- Another in late Spring or Summer.

8. Site Characterization

For the phases to be useful each of them should be long enough to draw reasonably strong conclusions.

In this regard various well known criteria have been proposed in general statistics. For example, a sample size

$$n_s = 8 \left(\frac{\sigma}{\varepsilon} \right)^2$$

may be used, where σ is the data standard deviation and ε the desired “error margin”, defined as half the confidence interval width (Whitlock and Schluter 2020). Other criteria exist as well, and are widely reported in practically-inclined statistical literature (Montgomery 2001).

All these objective criteria are founded in the assumption the data are normally distributed, but in Chapter 5 we have seen already this hypothesis is typically violated. Wind speed data distribution in addition changes with time, so a better approach could be to include information on this variability into the experimental design.

In principle, the two phases of a characterization campaign would greatly benefit from their time span to be as long as possible, for example the three months covering each of the two interesting stations.

Shorter periods could also be used, but in this case the conclusions will be less statistically significant and, hence, more affected by variability.

Just for illustrative purposes, the plots in Figure 8.42 indicate what could happen if a full-Winter campaign (January to March) is replaced by shorter, monthly campaigns.

Figure 8.42 shows the fraction of contaminated data could be about 57% if March data are considered, 72% if January data are used, and 65% if February or whole-winter data are employed. Interestingly, if all data available are used the evaluation is about 72%.

8. Site Characterization

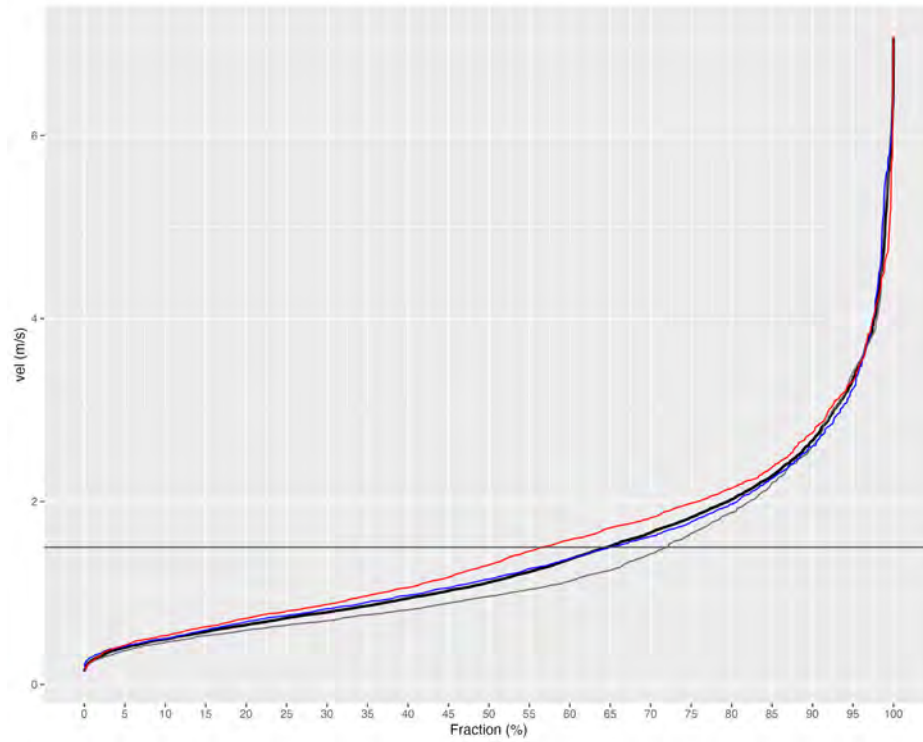


Figure 8.42.: Cumulate speed plot of winter (thick black line, Jan to Mar), January (thin grey line), February (thin blue line), March (thin red line) on Cinisello Parco Nord data

8. *Site Characterization*

This strong time dependency is due to important changes in wind speed on a monthly scale, and cannot be considered entirely incidental: there are moments in the year, namely those characterized by lower wind speed, in which data contamination is more severe.

9. Summary

9.1. What we have discovered

The statistical analysis of cup anemometer and ultrasonic anemometer speed data, along with corresponding ultrasonic anemometer data, showed important differences, at both slow and high wind speed.

The consequences of these differences on specific applications have not been addressed here, but could be estimated by field experts.

Nonetheless something can be said, at quite a general level:

- The statistical distribution of cup anemometer speeds shows a concentration of data towards the lowest speed, compared with homologous ultrasonic anemometer data.
- In particular many cup anemometer mean speeds are indicated as 0 m/s, causing a peak in the speed histograms. These zero-speed peaks do not occur with ultrasonic anemometer speed data.
- The extent of the difference between the cup and ultrasonic distributions depends on various factors. Among them, the site dependent wind climatology (in particular how often the instantaneous wind falls below the cup anemometer activation speed U_a), along with cup anemometer design characteristics and maintenance history.
- The statistical distribution of wind speed from cup and ultrasonic anemometer differs from a Weibull distribution by a level which is

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statistically significant. However, the departure from a Weibull distribution is much larger for cup anemometer data.

- The time variability observed on yearly medians is small (although in principle statistically significant due to the large sample size involved) according to both UA and CA data, but appears to be larger if CA data are used.
- On the monthly and daily time scales the CA overestimates wind speed at high wind velocity, and underestimates at low wind speed; the absolute magnitude of the misestimation is site-dependent and often large enough to be readily noticed even in a small scale plots; in any case it is on occasions quite large, and surely cannot be neglected.
- Daily scale changes exhibit strong site dependence, possibly related to the different intertwining of local and synoptic scale factors. This possible cause is not certain however, and needs further investigations.
- Daily scale changes show also a strong seasonal variation; the magnitude and pattern of this variation depends on the specific site characteristics.
- Any conclusions tend to be very site-specific, and this suggest to not draw any general (site-independent) conclusions on cup anemometer performance and suitability for some application. Rather, in case data quality plays an important role, a site characterization of wind by ultrasonic campaigns is recommended.
- Site and time variations are so important that, in general, a campaign for characterizing a site in view of installing a new or assessing an existing CA based station cannot be too short: unless happy cases the recommended campaign duration amounts to “some months”; if the station exists already, its data can be used to identify the most interesting time periods, maybe allowing a reduction of the total campaign duration (and, hence, cost).

9.2. Limitations of this work

This book, and the associated study, have been developed using data from a small sample of stations, where a cup and ultrasonic anemometer operate in close proximity.

The findings reported in this study are strongly site-dependent, consistently with the known fact that wind is distributed unevenly on the Earth surface (Davis et al. 2023), and consequently the occurrence of slow wind episodes can be smaller or larger depending on location. Because of this the findings reported here cannot replace documents like WMO’s CIMO report and its sensor- and measurement-specific parts (in particular (WMO 2023a), (WMO 2023b), (WMO 2023c)).

9.3. Suggestion for cup anemometer usage

9.3.1. Sensor laboratory testing

Literature on CA sensors deals extensively with cup anemometer wind tunnel calibration (Hansen, Hansen, and Kristensen 2012). Wind tunnel testing has also been presented as a way for classifying cup anemometers based on their performance (Dahlberg, Friis Pedersen, and Busche 2006).

A major contributing community in this effort was the wind energy industry, and this is reflected in the attention given to measurement ranges which tend to not include the slow velocity cases. For example in (MEAS-NET 2009) the testing range extends from 7 to 13 m/s, where wind turbines on common use are productive (Burton et al. 2011).

Deciding whether these commonly used procedures are adequate to CA sensors intended for use also in low wind speed conditions is not among the objectives of this work.

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However, results of the analyses made in preparation of this book provide to cup anemometers manufacturers the suggestion of adapting the standard procedures to speeds from 0 m/s upwards.

Another suggestion is to take into account during testing and calibration the factors potentially affecting cup anemometers accuracy. Among these:

- Different tilt conditions, in order to simulate the effect of the non-zero (positive or negative) vertical wind component component which are likely to occur on field.
- Diverse motion regimes could also be reproduced. In particular, laminar and friction induced turbulent airflow could be generated, and the cup anemometer checked to see whether differences are visible under same mean speed.

I realize this could be additional work on manufacturer side. Yet doing and documenting appropriately it would allow prospective users and buyers to verify in advance the suitability of cup anemometers to their specific applications, and not discover it later with possible legal or trust consequences.

Standards like (“Wind Measurement - Application of Meteorological Mast, Nacelle and Spinner Mounted Instruments,” n.d.) may provide useful inspiration to the case at hand.

9.3.2. Site characterization

9.3.2.1. A-priori characterization campaigns

This section is most useful in case a cup anemometer has not yet been installed, and its use is still under evaluation.

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In this situation executing a preliminary characterization campaign would be of great benefit in places where one could suspect a high incidence of low wind speed data (Davis et al. 2023).

The main purpose of this exploratory campaign is to assess the site-specific statistical distribution of wind speed as measured by an appropriate “reference” sensor (for example a three-dimensional ultrasonic or a 3D hot-wire anemometer). Specific details are given in Chapter 8, and further inspiration may come from (“Wind Measurement - Application of Meteorological Mast, Nacelle and Spinner Mounted Instruments,” n.d.).

The main outcomes of an a-priori characterization study could include:

- Making sure the site is exposed to a wind speed distribution compatible with cup anemometers, that is, in which the frequency of wind speeds below a minimum activation threshold in the order to 0.25 m/s as of current technology is not too large.
- In case the wind speed distribution is compatible with cup anemometer technology, indication of a maximum activation threshold in correspondence of which the measurements are not too adversely affected. Knowing this target activation threshold may in turn be useful as a guidance to cup anemometer selection.

9.3.2.2. A-posteriori characterization campaigns

If a-priori characterization apply essentially to *prospective measurement sites*, a-posteriori characterization applies to existing cup anemometer installations.

The planning and conduction of the campaign could be similar to the a-priori case, but the outcome changes somewhat.

Objectives of the characterization study may now include the following:

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- Determining whether the speed distribution has changed significantly from the past known one, and, if in case, whether the existing sensor is still compatible with the site statistics.
- Estimating possible increases of the cup anemometer’s actual activation threshold, possibly indicative of sensor degradation and conducive to maintenance interventions.
- Evaluating the effect, or suggesting the necessity, of “preventive maintenance”.

9.4. Cup anemometer modelling

In Chapter 7 we have seen the definition and practical application of a model of cup anemometer. Compared to literature examples, like the one in (Dahlberg, Friis Pedersen, and Busche 2006), the model presented in this book is decidedly simplified: its purpose, however, is not to provide a general purpose indication of the anemometer response over a broad speed range, but rather to allow making sense of the sensors behavior at low wind speed.

So, although less sophisticate than deep models presented in literature (an example is (Pedersen and Dahlberg 2024)), the simplified model allows to predict the effect on readings of a given activation threshold, focusing on this parameter exclusively.

Its application, as described in Chapter 7, can for example be in an a-posteriori campaign to derive an objective and easy estimation of the real activation threshold, thus allowing a mean to identify possible sensor wear-out before a major failure occurs.

The model also allowed to compare different averaging schemes, and two new averaging methods have been introduced and evaluated.

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Of course the averaging methods introduced in this book are only a small sample of the many existing. The same approach presented here may also be used on these additional averaging schemes not discussed in this book.

9.5. Next developments

As this book has a research side, it is not completely out of kilter to indicate some future developments.

The first, directly applicable to the case of cup anemometers, consists in filling the gaps, and in particular addressing what I've indicated as interesting research work:

- Investigating the behavior of other electro-mechanical wind speed sensors, namely *propeller* anemometers.
- Extending the scope from wind speed alone to the couple speed-and-direction.
- Extending investigations to other sites, and different operational conditions.
- Assessing the effect of the wind speed distortion induced by cup anemometers on key applications and use cases.
- Building a new, anemometer-savvy data logger able to cope with some of the problems found with cup anemometer reactions to low wind speed.
- Checking the effect of a moderate time base misalignment on cup vs ultrasonic data comparisons.

9. Summary

Quite a huge program, and almost surely longer than my residual life. But nothing mandates *I* must do it all! I count on you.

Sure this work is not perfect (and besides, sorry so sloppy). Yet, someone had to do, and now here it is!

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